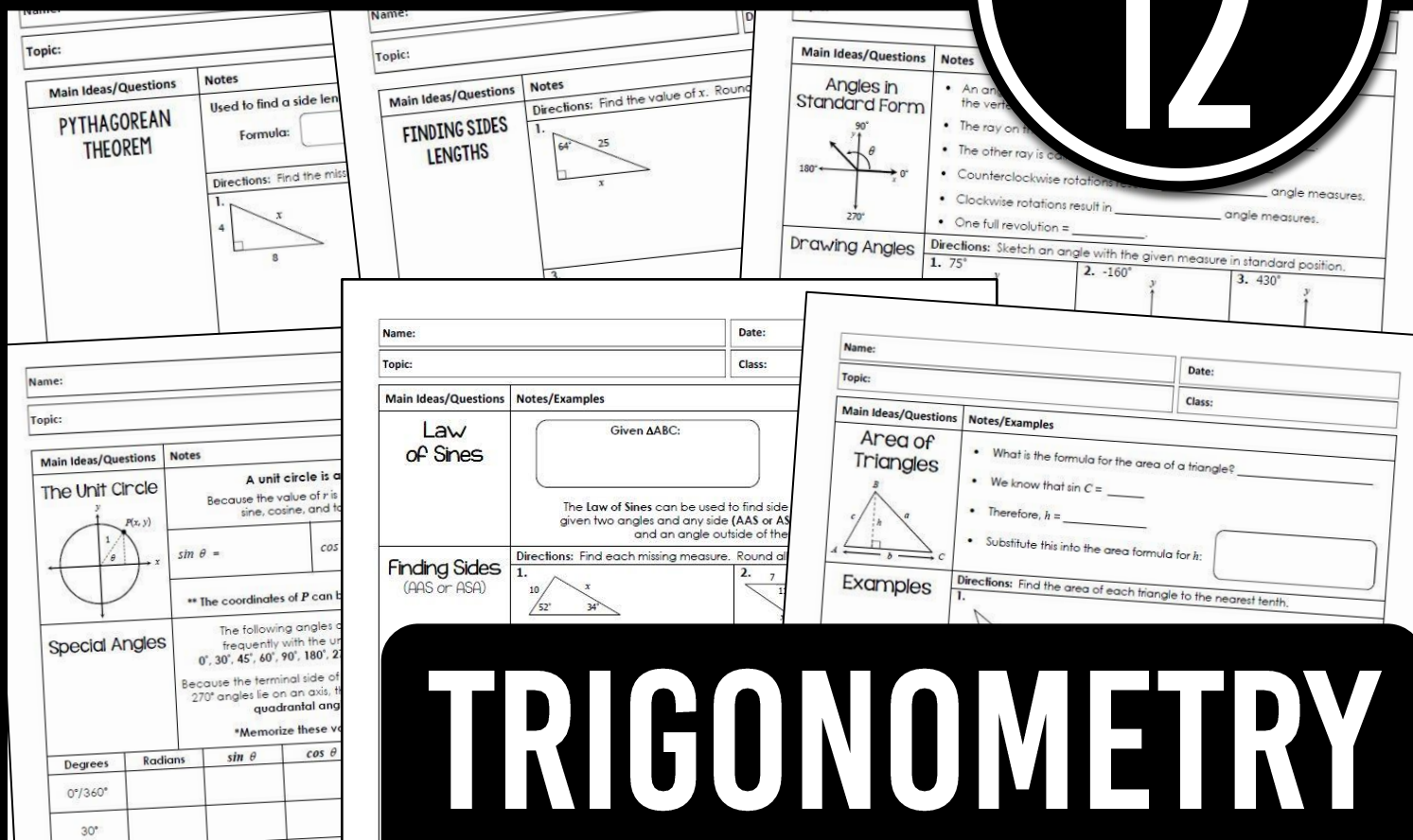


ALGEBRA 2

Unit 12



TRIGONOMETRY

NOTES • HOMEWORK • QUIZZES • TEST

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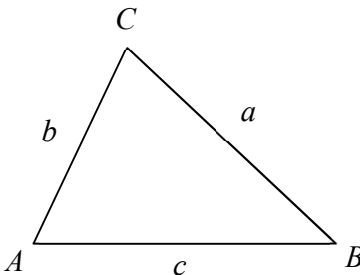
Unit 12 - Trigonometry: Sample Unit Outline

	TOPIC	HOMEWORK
DAY 1	Pythagorean Theorem, Special Right Triangles, Six Trigonometric Functions	HW #1
DAY 2	Finding Side and Angle Measures; Applications	HW #2
DAY 3	Angles in Standard Position, Converting Degrees and Radians, Coterminal Angles, Reference Angles	HW #3
DAY 4	The Unit Circle	HW #4
DAY 5	Quiz 12-1	None
DAY 6	Law of Sines; Ambiguous Case	HW #5
DAY 7	Law of Cosines	HW #6
DAY 8	Mixed Law of Sines and Law of Cosines; Area of Triangles	HW #7
DAY 9	Unit 12 Review (Part I)	Study for Test
DAY 10	UNIT 12 TEST (Part I)	None
DAY 11	Graphing Trigonometric Functions: $\sin$, $\cos$, $\tan$	HW #8
DAY 12	Graphing Trigonometric Functions: $\cot$, $\csc$, $\sec$	HW #9 (reviews all)
DAY 13	Translating Trigonometric Graphs: Phase/Vertical Shifts *with sine, cosine, and tangent graphs only*	HW #10
DAY 14	Quiz 12-2	None
DAY 15	Fundamental Trigonometric Identities	HW #11
DAY 16	Sum and Difference of Angle Identities	HW #12
DAY 17	Double-Angle and Half-Angle Identities	HW #13
DAY 18	Quiz 12-3	None
DAY 19	Solving Trigonometric Equations	HW #14
DAY 20	Unit 12 Review (Part II)	Study for Test
DAY 21	UNIT 12 TEST (Part II)	None

See sample images of the pages on the next page.

TRIGONOMETRY

Formulas Reference

Law of Sines	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$		
Law of Cosines	$a^2 = b^2 + c^2 - 2bc \cos A$ $b^2 = a^2 + c^2 - 2ac \cos B$ $c^2 = a^2 + b^2 - 2ab \cos C$		
Area of Triangle	$A = \frac{1}{2} ab \sin C$		
Reciprocal Identities	$\sin \theta = \frac{1}{\csc \theta}$ $\csc \theta = \frac{1}{\sin \theta}$	$\cos \theta = \frac{1}{\sec \theta}$ $\sec \theta = \frac{1}{\cos \theta}$	$\tan \theta = \frac{1}{\cot \theta}$ $\cot \theta = \frac{1}{\tan \theta}$
Quotient Identities	$\tan \theta = \frac{\sin \theta}{\cos \theta}$	$\cot \theta = \frac{\cos \theta}{\sin \theta}$	
Pythagorean Identities	$\cos^2 \theta + \sin^2 \theta = 1$	$\tan^2 \theta + 1 = \sec^2 \theta$	$\cot^2 \theta + 1 = \csc^2 \theta$
Sum of Angles Identities	$\sin(A + B) = \sin A \cdot \cos B + \cos A \cdot \sin B$ $\cos(A + B) = \cos A \cdot \cos B - \sin A \cdot \sin B$ $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$		
Difference of Angles Identities	$\sin(A - B) = \sin A \cdot \cos B - \cos A \cdot \sin B$ $\cos(A - B) = \cos A \cdot \cos B + \sin A \cdot \sin B$ $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \cdot \tan B}$		
Double-Angle Identities	$\sin 2\theta = 2 \sin \theta \cdot \cos \theta$	$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ $\cos 2\theta = 1 - 2 \sin^2 \theta$ $\cos 2\theta = 2 \cos^2 \theta - 1$	$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$
Half-Angle Identities	$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$	$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$	$\tan \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$

Name:

Class:

Topic:

Date:

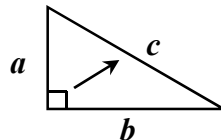
Main Ideas/Questions

Notes

PYTHAGOREAN THEOREM

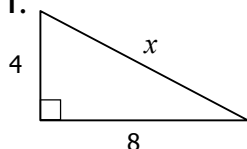
Used to find a side length on a right triangle.

Formula:

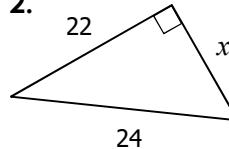


Directions: Find the missing side. Give your answer in simplest radical form.

1.

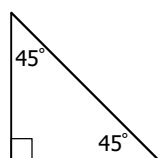


2.



SPECIAL RIGHT TRIANGLES

45°-45°-90°

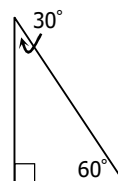


Leg = _____

Leg = _____

Hypotenuse = _____

30°-60°-90°



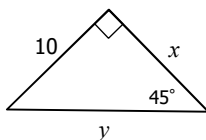
Shorter Leg = _____

Longer Leg = _____

Hypotenuse = _____

Directions: Find each missing side. Give your answer in simplest radical form.

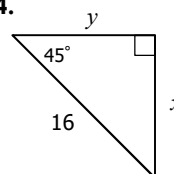
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$x =$ _____

$y =$ _____

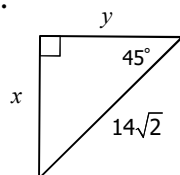
4.



$x =$ _____

$y =$ _____

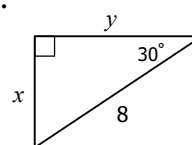
5.



$x =$ _____

$y =$ _____

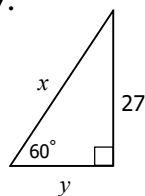
6.



$x =$ _____

$y =$ _____

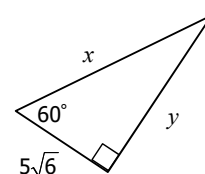
7.



$x =$ _____

$y =$ _____

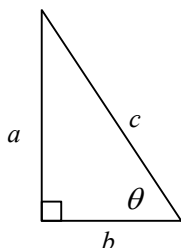
8.



$x =$ _____

$y =$ _____

TRIGONOMETRIC FUNCTIONS



- A **trigonometric function** is a function whose rule is defined by a trigonometric ratio.
- A **trigonometric ratio** compares the lengths of two sides of the triangle.
- The Greek letter θ is used to represent the measure of an acute angle in a right triangle.

SINE

$$\sin \theta =$$

COSINE

$$\cos \theta =$$

TANGENT

$$\tan \theta =$$

RECIPROCAL FUNCTIONS

$$\text{COSECANT} \left(\frac{1}{\sin \theta} \right)$$

$$\csc \theta =$$

$$\text{SECANT} \left(\frac{1}{\cos \theta} \right)$$

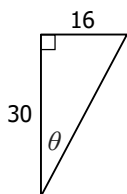
$$\sec \theta =$$

$$\text{COTANGENT} \left(\frac{1}{\tan \theta} \right)$$

$$\cot \theta =$$

EXAMPLES

9. Find all six trig ratios for θ shown in the triangle below.



10. Given the ratio for $\cos \theta$, find the remaining ratios.

$$\sin \theta =$$

$$\csc \theta =$$

$$\sin \theta =$$

$$\csc \theta =$$

$$\cos \theta =$$

$$\sec \theta =$$

$$\cos \theta = \frac{2}{3}$$

$$\sec \theta =$$

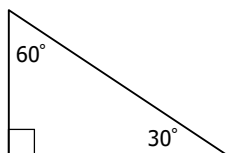
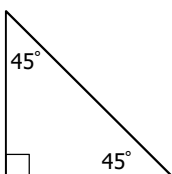
$$\tan \theta =$$

$$\cot \theta =$$

$$\tan \theta =$$

$$\cot \theta =$$

TRIG RATIOS OF SPECIAL ANGLES



Angles of 30° , 60° , and 45° are used frequently in trigonometry. You can use your knowledge of the side relationships in special right triangles to find the values of the trigonometric ratios.

$$\sin 45^\circ =$$

$$\cos 45^\circ =$$

$$\tan 45^\circ =$$

$$\sin 30^\circ =$$

$$\cos 30^\circ =$$

$$\tan 30^\circ =$$

$$\sin 60^\circ =$$

$$\cos 60^\circ =$$

$$\tan 60^\circ =$$



Name: _____

Unit 12: Trigonometry

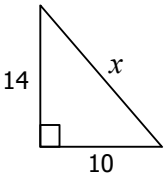
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Homework 1: Pythagorean Theorem, Special Right Triangles, & Trig Functions

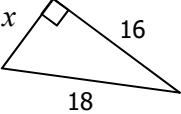
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Directions: Find each missing length. Give all answers in simplest radical form.

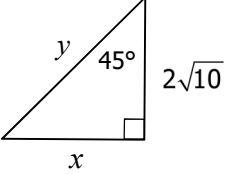
1.



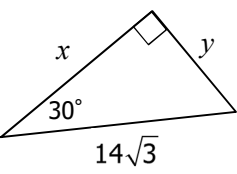
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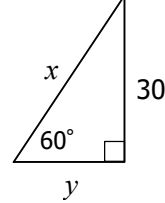
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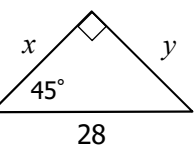
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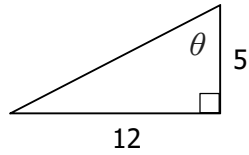
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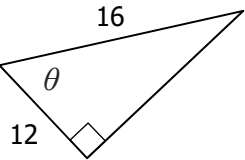
6.



7. Find the values of the six trigonometric functions for θ .



8. Find the values of the six trigonometric functions for θ .



$\sin \theta =$

$\csc \theta =$

$\sin \theta =$

$\csc \theta =$

$\cos \theta =$

$\sec \theta =$

$\cos \theta =$

$\sec \theta =$

$\tan \theta =$

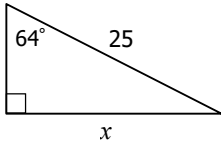
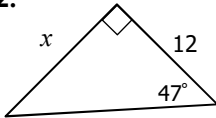
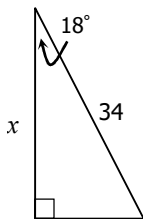
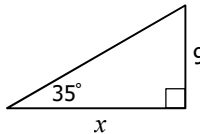
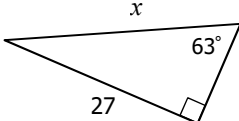
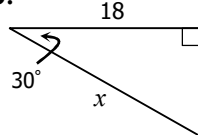
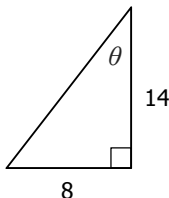
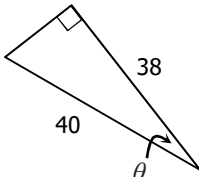
$\cot \theta =$

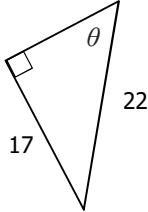
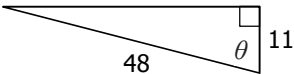
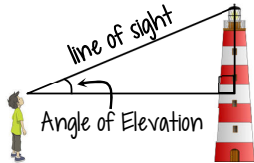
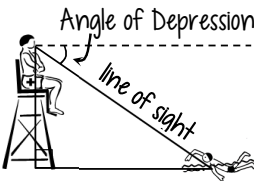
$\tan \theta =$

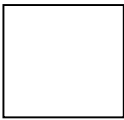
$\cot \theta =$

9. Given the ratio for $\tan \theta$, find the remaining ratios.		10. Given the ratio for $\csc \theta$, find the remaining ratios.	
$\sin \theta =$	$\csc \theta =$	$\sin \theta =$	$\csc \theta = \frac{\sqrt{7}}{2}$
$\cos \theta =$	$\sec \theta =$	$\cos \theta =$	$\sec \theta =$
$\tan \theta = \frac{4}{5}$	$\cot \theta =$	$\tan \theta =$	$\cot \theta =$
Directions: Give the exact ratios for each function below.			
11. $\sin 45^\circ$	12. $\tan 60^\circ$	13. $\cos 30^\circ$	
14. $\cot 60^\circ$	15. $\sec 45^\circ$	16. $\csc 60^\circ$	

Name:	Class:
Topic:	Date:

Main Ideas/Questions	Notes					
FINDING SIDES LENGTHS	Directions: Find the value of x . Round to the nearest tenth.					
	1. 	2. 				
	3. 	4. 				
	5. 	6. 				
	FINDING ANGLES MEASURES					
	If you know the sin, cosine, or tangent ratio of an angle, you can use the inverse of the ratio ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) to find the measure of the angle.					
If $\sin \theta = x$, then			If $\cos \theta = x$, then		If $\tan \theta = x$, then	
Directions: Find the missing angle. Round to the nearest tenth.						
7. 			8. 			

	9. 	10. 
ANGLE OF ELEVATION 	11. The angle of elevation from a fishing boat to the top of a lighthouse is 5°. If the top of the lighthouse is 161 feet above sea level, how far is the boat from the lighthouse? 12. The angle of elevation from the bottom of a slide to the top of the slide is 44°. If the top of the slide sits 16 feet above the ground, find the length of the slide. Assume it is a straight slide. 13. A 10-foot ramp is used to unload furniture from a truck. If the back of the truck sits 3 feet above the ground, find the measure of the angle that the ramp makes with the ground.	
ANGLE OF DEPRESSION 	14. A fire is spotted from the top of a 125-foot tall fire tower. If the angle of depression to the fire is 19°, how far is the fire from the base of the tower? 15. An airplane is flying at an altitude of 8,900 feet over water. The pilot spots a raft floating in the water at a 28° angle of depression. What is the horizontal distance from the airplane to the raft? 16. Max is sitting in the stands at the baseball stadium. He catches a foul ball and decides to throw it back to a player standing on first base. If the horizontal distance from Max to the player is 61 feet and the ball travels 76 feet, what is the angle of depression from Max to the player?	



Name: _____

Unit 12: Trigonometry

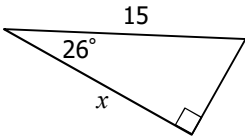
Date: _____ Bell: _____

Homework 2: Finding Side and Angle Measures

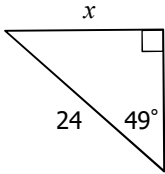
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Directions: Find each missing measure. Round all answers to the nearest tenth.

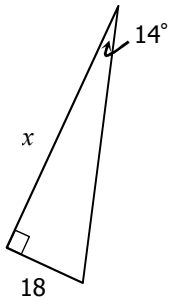
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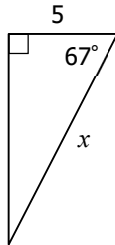
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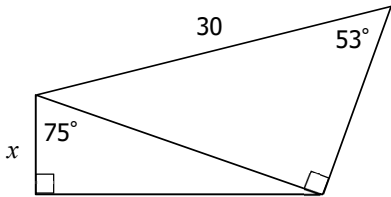
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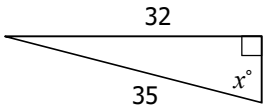
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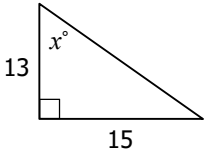
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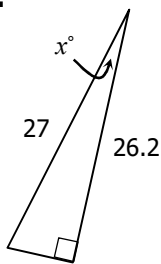
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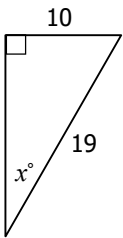
7.



8.



9.



10. David is standing 500 feet away from the base of a cell phone tower. The angle of elevation from his eyes to the top of the tower is 78° . If David's eyes are 6 feet above the ground, how tall is the tower?

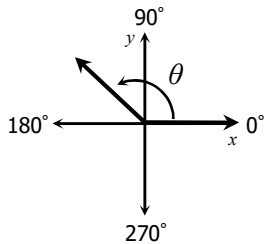
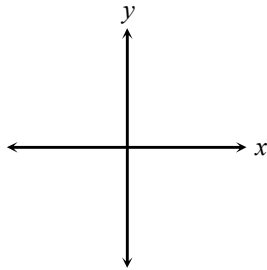
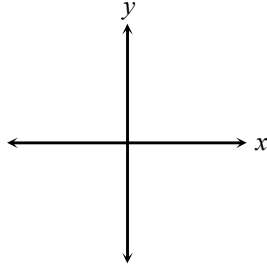
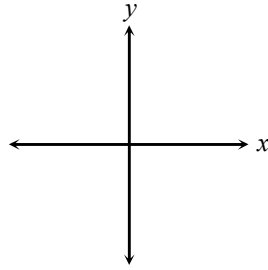
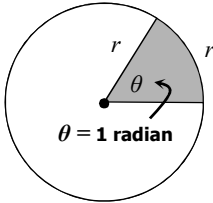
11. The angle of elevation from the top of one building to another is 26° . If the shorter building is 400 feet tall and the buildings are 1500 feet apart, how tall is the other building?

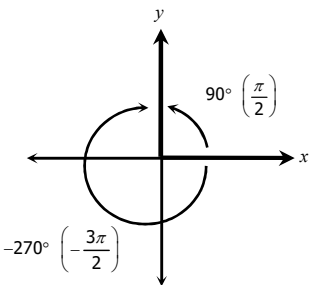
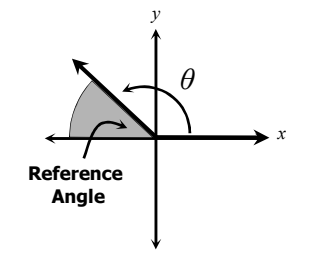
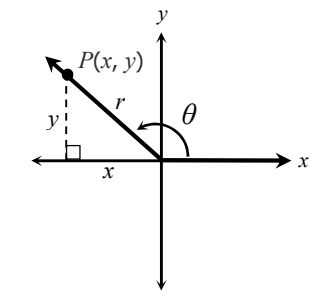
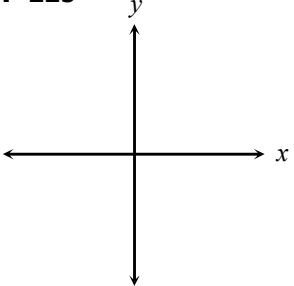
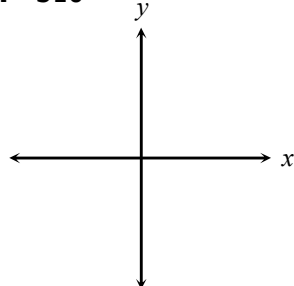
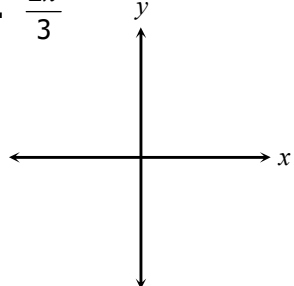
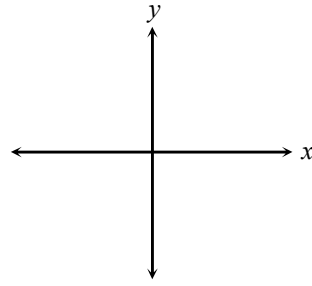
12. A 200-foot supporting cable is tied to the top of a cell phone tower then secured to a point on the ground. If the angle of elevation from the point on the ground to the top of the tower is 38° , find the height of the tower.

13. The angle of depression from the top to the bottom of a ski lift is 43° . If the vertical distance between the top and bottom of the lift is 800 feet, determine the length of the cable between them.

14. An airplane is descending to a runway. At an altitude of 15.2 miles, its horizontal distance to the runway is 60 miles. Find the angle of descent.

Name:	Class:
Topic:	Date:

Main Ideas/Questions	Notes		
Angles in Standard Form 	- An angle on the coordinate plane is in standard form when the vertex is on the origin and one ray lies on the positive x-axis. - The ray on the x-axis is called the _____. - The other ray is called the _____. - Counterclockwise rotations result in _____ angle measures. - Clockwise rotations result in _____ angle measures. - One full revolution = _____.		
Drawing Angles	Directions: Sketch an angle with the given measure in standard position.		
	1. 75° 	2. -160° 	3. 430° 
Radians vs. Degrees 	A radian is a unit of angle measure based on arc length. One radian is defined as the measure of the angle formed when the radius is equivalent to the length of the intercepted arc. Recall that the circumference of a circle is $2r\pi$, therefore: $360^\circ =$ _____ ; $180^\circ =$ _____		
	Converting Degrees → Radians		Converting Radians → Degrees
	Radians = Degrees $\cdot \left(\frac{\pi \text{ radians}}{180} \right)$		Degrees = Radians $\cdot \left(\frac{180}{\pi \text{ radians}} \right)$
Degrees → Radians	Directions: Convert each measure to radians.		
	4. 30°	5. 150°	6. -220°
Radians → Degrees	Directions: Convert each measure to degrees.		
	7. $\frac{4\pi}{3}$	8. $-\frac{5\pi}{36}$	9. $\frac{7\pi}{4}$

Coterminal Angles 	Angles in standard position with the same terminal side are coterminal angles. Give two coterminal angles for each given angle, one positive and one negative:		
Reference Angles 	10. 65°	11. 540°	
	12. $\frac{13\pi}{18}$	13. $\frac{14\pi}{9}$	
Trig Functions 	For an angle θ in standard form, the reference angle is the positive acute angle form by the terminal side and the x-axis. Sketch and find the reference angles for each angle:		
	14. 225° 	15. -310° 	16. $\frac{2\pi}{3}$ 
	$\sin \theta =$	$\cos \theta =$	$\tan \theta =$
	Let θ be an angle in standard form and $P(x, y)$ be a point on the terminal side of θ. The distance from P to the the origin, r, can be found using the formula: _____ (The Pythagorean Theorem).		
	$\csc \theta =$	$\sec \theta =$	$\cot \theta =$
	$\sin \theta =$	$\cos \theta =$	$\tan \theta =$



Name: _____

Unit 12: Trigonometry

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Homework 3: Angles and Angle Measure

**** This is a 2-page document! ****

Directions: Convert each measure to radians.

1. 225°

2. 20°

3. -255°

4. -140°

5. 75°

6. -300°

Directions: Convert each measure to degrees.

7. $\frac{23\pi}{12}$

8. $-\frac{31\pi}{36}$

9. $\frac{\pi}{12}$

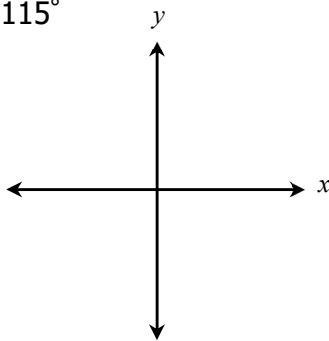
10. $\frac{5\pi}{9}$

11. $-\frac{\pi}{2}$

12. $-\frac{7\pi}{6}$

Directions: Sketch each angle. Then give two coterminal angles (one positive and one negative) and the reference angle.

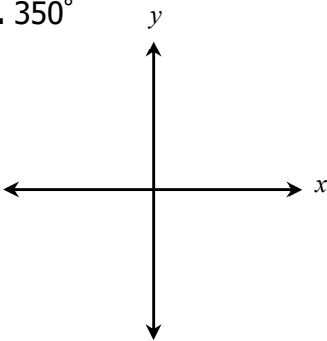
13. 115°



Coterminal $\angle$'s:

Reference $\angle$:

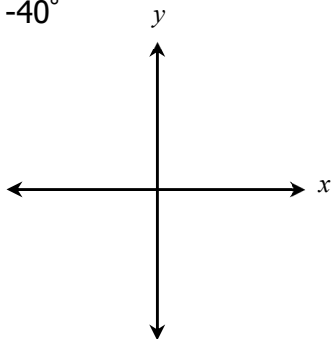
14. 350°



Coterminal $\angle$'s:

Reference $\angle$:

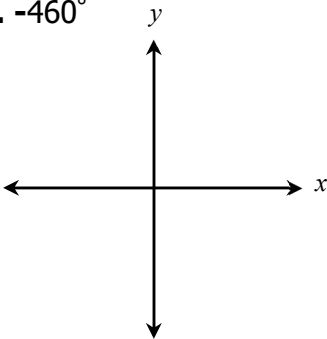
15. -40°



Coterminal $\angle$'s:

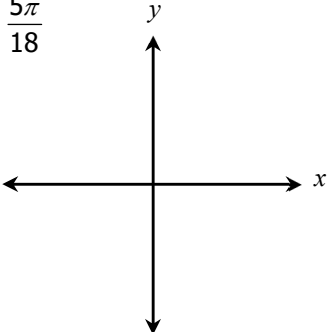
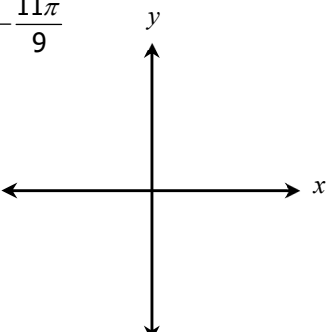
Reference $\angle$:

16. -460°

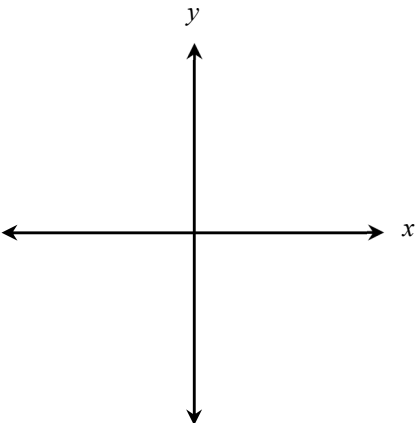
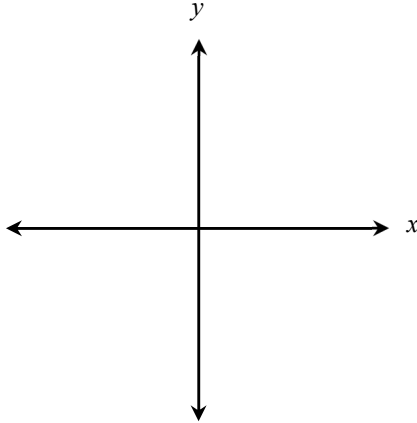
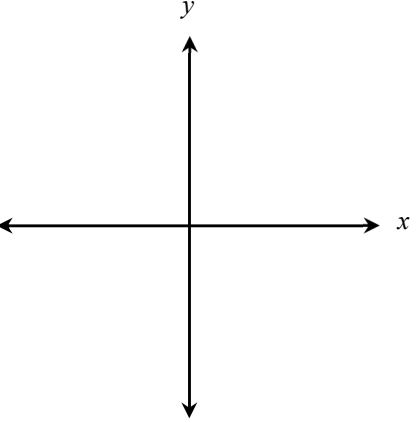


Coterminal $\angle$'s:

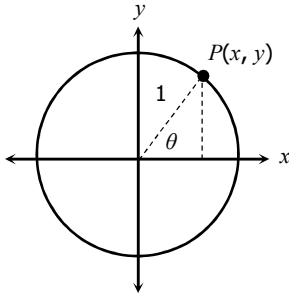
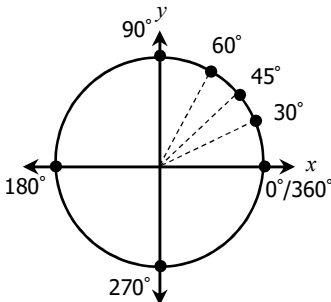
Reference $\angle$:

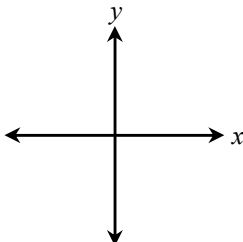
17. $\frac{5\pi}{18}$  Coterminal $\angle$'s: Reference $\angle$:	18. $-\frac{11\pi}{9}$  Coterminal $\angle$'s: Reference $\angle$:
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Directions: Sketch the angle in standard form whose terminal side passes through the given point. Find the exact value for each trigonometric function.

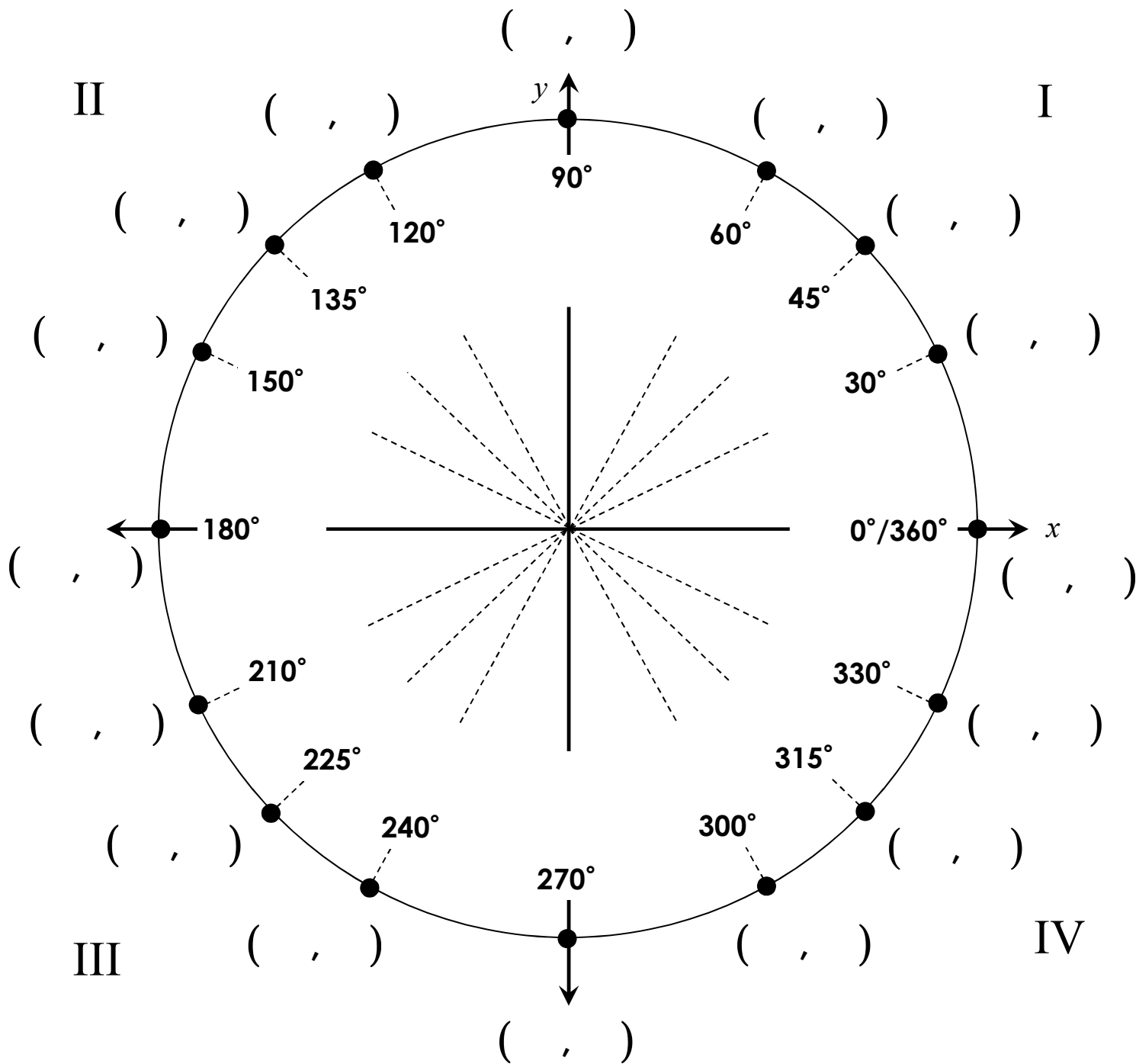
19. $(-5, 12)$ 	$\sin \theta =$	$\csc \theta =$
	$\cos \theta =$	$\sec \theta =$
	$\tan \theta =$	$\cot \theta =$
20. $(2, 8)$ 	$\sin \theta =$	$\csc \theta =$
	$\cos \theta =$	$\sec \theta =$
	$\tan \theta =$	$\cot \theta =$
21. $(3, -6)$ 	$\sin \theta =$	$\csc \theta =$
	$\cos \theta =$	$\sec \theta =$
	$\tan \theta =$	$\cot \theta =$

Name:	Class:
Topic:	Date:

Main Ideas/Questions		Notes				
The Unit Circle 		A unit circle is a circle with a radius of 1 unit. Because the value of r is 1 for each point $P(x, y)$ on the circle, the sine, cosine, and tangent values for θ are defined as:				
		$\sin \theta =$		$\cos \theta =$		$\tan \theta =$
		** The coordinates of P can be written as _____ . **				
Special Angles		The following angles are used frequently with the unit circle: $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ, 180^\circ, 270^\circ$, and 360° Because the terminal side of $0^\circ, 90^\circ, 180^\circ$ and 270° angles lie on an axis, they are called quadrantal angles. *Memorize these values!*				
Degrees	Radians	$\sin \theta$	$\cos \theta$	$\tan \theta$	$P(x, y)$	
$0^\circ/360^\circ$						
30°						
45°						
60°						
90°						
180°						
270°						
Labeling the Unit Circle		- Label the ordered pairs for each angle above. - Use these angles as reference angles for the remaining angles around the circle. - Adjust your coordinates by looking at which quadrant the point is located in.				

Signs of Trig Functions	Which trig functions are POSITIVE in:			
	- Quadrant I? _____ - Quadrant II? _____ - Quadrant III? _____ - Quadrant IV? _____			
	Remember the phrase "All Students Take Calculus!"			
Using Reference Angles to find Trig Values	1. If $\tan \theta > 0$, which quadrant(s) could the terminal side of θ lie?		2. If $\sec \theta < 0$, which quadrant(s) could the terminal side of θ lie?	
	3. If $\sin \theta > 0$, which quadrant(s) could the terminal side of θ lie?		4. If $\cos \theta < 0$ and $\cot \theta > 0$, which quadrant(s) could the terminal side of θ lie?	
	Step 1: Find the measure of the reference angle.			
	Step 2: Identify the trig function for the reference angle.			
	Step 3: Adjust the sign, if needed.			
	Directions: Give the exact value for each trigonometric function.			
5. $\sin 150^\circ$		6. $\tan 300^\circ$		7. $\cos 225^\circ$
8. $\cot 315^\circ$		9. $\tan 240^\circ$		10. $\sec 300^\circ$
11. $\cos \frac{11\pi}{6}$		12. $\csc \frac{7\pi}{4}$		13. $\sin \frac{4\pi}{3}$
14. $\tan \frac{3\pi}{4}$		15. $\sin \frac{7\pi}{6}$		16. $\sec \frac{\pi}{2}$

THE UNIT CIRCLE



Trig Values:

$\sin \theta =$
$\cos \theta =$
$\tan \theta =$

Summary of Signs in the Quadrants:

	I	II	III	IV
$\sin \theta$				
$\cos \theta$				
$\tan \theta$				



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Unit 12: Trigonometry

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Homework 4: The Unit Circle

1. Which trig functions are positive for angles terminating in Quadrant IV?	2. Which trig functions are negative for angles terminating in Quadrant II?
3. If $\cos \theta < 0$, which quadrant(s) could the terminal side of θ lie?	4. If $\csc \theta > 0$, which quadrant(s) could the terminal side of θ lie?
5. If $\tan \theta < 0$ and $\sec \theta > 0$, which quadrant(s) could the terminal side of θ lie?	6. If $\sin \theta < 0$ and $\cot \theta < 0$, which quadrant(s) could the terminal side of θ lie?

Directions: Find the exact value for each trigonometric function. Use reference angles when necessary.

7. $\sin 60^\circ$	8. $\cos 225^\circ$	9. $\tan 330^\circ$
10. $\cot 150^\circ$	11. $\sec 135^\circ$	12. $\tan 210^\circ$
13. $\sec 270^\circ$	14. $\csc 315^\circ$	15. $\sin 240^\circ$
16. $\tan \frac{7\pi}{6}$	17. $\cos \frac{3\pi}{4}$	18. $\csc \frac{5\pi}{3}$
19. $\sin \frac{\pi}{6}$	20. $\cot \frac{5\pi}{6}$	21. $\sec \frac{11\pi}{6}$

Name: _____

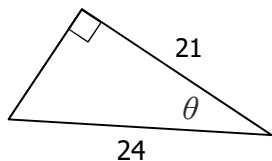
Algebra 2

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Unit 12: Trigonometry

Quiz 12-1: Introduction to Trig; The Unit Circle

1. Find the values of the six trigonometric functions for angle θ . Give answers in simplest form.



$$\sin \theta = \underline{\hspace{2cm}} \quad \csc \theta = \underline{\hspace{2cm}}$$

$$\cos \theta = \underline{\hspace{2cm}} \quad \sec \theta = \underline{\hspace{2cm}}$$

$$\tan \theta = \underline{\hspace{2cm}} \quad \cot \theta = \underline{\hspace{2cm}}$$

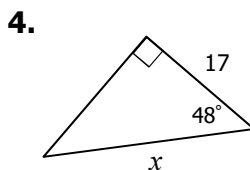
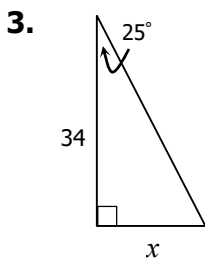
2. If $\cot \theta = \frac{\sqrt{3}}{5}$, find the remaining trigonometric functions.

$$\sin \theta = \underline{\hspace{2cm}} \quad \csc \theta = \underline{\hspace{2cm}}$$

$$\cos \theta = \underline{\hspace{2cm}} \quad \sec \theta = \underline{\hspace{2cm}}$$

$$\tan \theta = \underline{\hspace{2cm}}$$

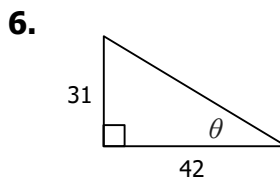
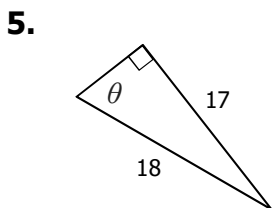
Find the missing side lengths. Round all answers to the nearest tenth.



3. _____

4. _____

Find the measure of θ . Round all answers to the nearest tenth.



5. _____

6. _____

7. Beth is flying a kite with 300 feet of string. The kite gets stuck in a tree, so she ties the other end of the string to a rock on the ground making the string taut. The angle of elevation from the rock to kite is 35° . How far up the tree is the kite? Round your answer to the nearest tenth.

7. _____

Give each degree measure in radians and each radian measure in degrees.

8. 130°

9. -285°

8. _____

9. _____

10. $-\frac{17\pi}{9}$

11. $\frac{7\pi}{36}$

10. _____

11. _____

Give two coterminal angles (one positive and one negative) and the reference angle.

12. 290°

12. Coterminal $\angle$'s: _____

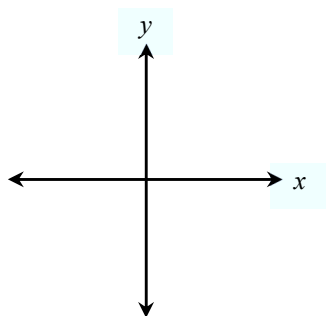
Reference $\angle$: _____

13. $-\frac{4\pi}{5}$

13. Coterminal $\angle$'s: _____

Reference $\angle$: _____

14. The terminal side of an angle in standard form passes through the point (6, -4). Make a sketch, then find each function value.



$\sin \theta =$ _____ $\csc \theta =$ _____

$\cos \theta =$ _____ $\sec \theta =$ _____

$\tan \theta =$ _____ $\cot \theta =$ _____

15. If $\tan \theta < 0$ and $\sec \theta < 0$ which quadrant(s) could the terminal side of θ lie?

15. _____

Find the exact value for each trigonometric function.

16. $\tan 135^\circ$

17. $\csc 240^\circ$

18. $\sin 120^\circ$

16. _____

17. _____

18. _____

19. $\cos \frac{7\pi}{4}$

20. $\sec \frac{2\pi}{3}$

21. $\cot \frac{\pi}{6}$

19. _____

20. _____

21. _____

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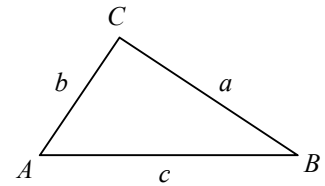
Class:

Main Ideas/Questions

Notes/Examples

Law of Sines

Given $\triangle ABC$:

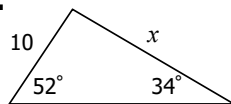


The **Law of Sines** can be used to find side and angle measures given two angles and any side (**AAS or ASA**) or two side lengths and an angle outside of them (**SSA**).

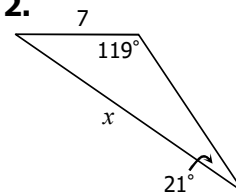
Finding Sides (AAS or ASA)

Directions: Find each missing measure. Round all answers to the nearest tenth.

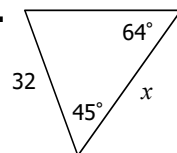
1.



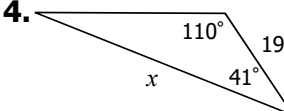
2.



3.



4.

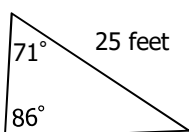


Use for questions 5-6: Kevin, Adam, and Dan are passing a hockey puck.

5. How far is Kevin from Adam?

6. How far is Dan from Kevin?

Adam



Kevin

Dan

Finding Angles (SSA)

Watch out! When finding angles, it's possible to have one solution, two solutions, or no solution! This is called the **ambiguous case**.

Important Facts:

- The sine function always has a range between ____ and ____.
- If the sine function is positive, then the angle can lie in quad ____ or ____.
- To find the other possible angle, subtract from ____.
- The sum of the angle measures in a triangle is always ____.
- No triangle can have two _____ angles.

Directions: Information about $\triangle ABC$ is given below. Check your answers and give all possibilities.

7. $m\angle A = 50^\circ$, $AB = 4$, $BC = 9$, find $m\angle C$.

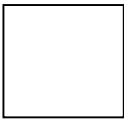
8. $m\angle B = 37^\circ$, $AB = 47$, $AC = 29$, find $m\angle C$.

9. $m\angle C = 75^\circ$, $AB = 28$, $AC = 19$, find $m\angle B$.

10. $m\angle A = 114^\circ$, $AB = 34$, $BC = 12$, find $m\angle C$.

11. $m\angle B = 47^\circ$, $AB = 33$, $AC = 26$, find $m\angle A$.

12. $m\angle C = 73^\circ$, $AB = 8$, $BC = 17$, find $m\angle B$.



Name: _____

Unit 12: Trigonometry

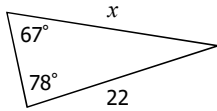
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Homework 5: Law of Sines

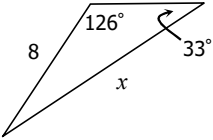
**** This is a 2-page document! ****

Directions: Find each side length to the nearest tenth.

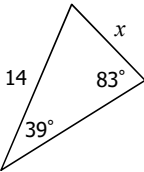
1.



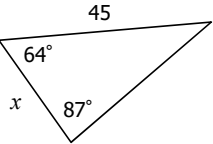
2.



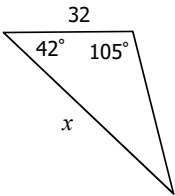
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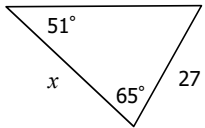
4.



5.



6.



Directions: Information about $\triangle ABC$ is given below. Give all possible answers..

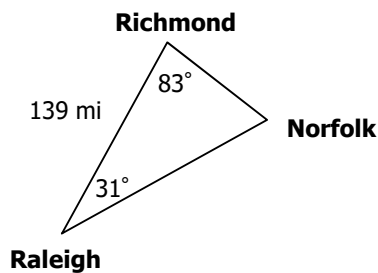
7. $m\angle C = 63^\circ$, $AB = 28$, $AC = 29$, find $m\angle B$.

8. $m\angle C = 36^\circ$, $AB = 31$, $BC = 24$, find $m\angle A$.

9. $m\angle A = 46^\circ$, $BC = 9$, $AC = 24$, find $m\angle B$.	10. $m\angle C = 105^\circ$, $AB = 32$, $AC = 29$, find $m\angle B$.
11. $m\angle C = 36^\circ$, $AB = 21$, $AC = 30$, find $m\angle A$.	12. $m\angle A = 55^\circ$, $AB = 27$, $BC = 12$, find $m\angle B$.

Directions: Read each problem, then solve for the missing part(s).

- 13.** A pilot flying over the Norfolk airport is told he will have to redirect and land in either Richmond or Raleigh. Due to air traffic, he is then told he must land in Raleigh. Find the distance he will travel.



- 14.** Part of a marching band formed a triangle with the trumpet players on one side, clarinet players on one side, and saxophone players on the third side. The angle formed by the trumpet and saxophone players measured 65° . If the clarinet line is 20 feet long and the saxophone line is 8 feet long, find the angle formed by the trumpet and clarinet players.

Name:

Date:

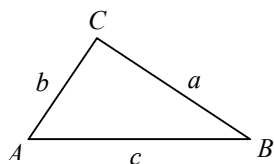
Topic:

Class:

Main Ideas/Questions

Notes/Examples

Law of Cosines



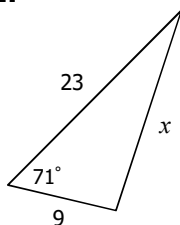
The **Law of Cosines** can be used to find side and angle measures given two sides and the included angle (**SAS**) or three sides (**SSS**).

Given $\triangle ABC$:

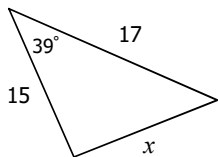
- _____
- _____
- _____

Finding Sides: Find each missing measure. Round all answers to the nearest tenth.

1.



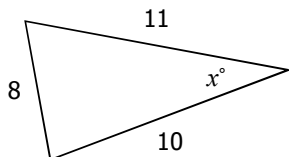
2.



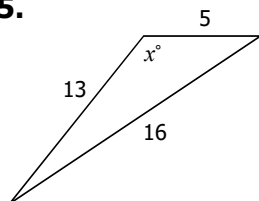
3. In $\triangle ABC$, $m\angle B = 106^\circ$, $AB = 10$, and $BC = 7$. Find AC .

Finding Angles: Find each missing measure. Round all answers to the nearest tenth.

4.



5.



6. In $\triangle LMN$, $LM = 25$, $MN = 20$, and $LN = 24$. Find $m\angle L$.

Applications: Give side measures to the nearest tenth and angle measures to the nearest degree.

7. Maple Street, Oak Drive, and West Ave intersect to form a triangular piece of land. The angle formed by Maple Street and Oak Drive is 79° . If the Maple Street section of the triangle is 390 feet and the Oak Drive section is 250 feet, find the length of the West Ave section.

8. Jack and Tom start at a tree and begin running apart with an angle of 23° between them. If Jack runs 50 meters and Tom runs 38 meters, find the distance between them.

9. Two radar stations 2.5 miles apart are tracking an airplane. The straight-line distance between Station A and the plane is 7 miles. The straight-line distance between Station B and the plane is 9 miles. What is the angle of elevation from Station B to the plane?



Name: _____

Unit 12: Trigonometry

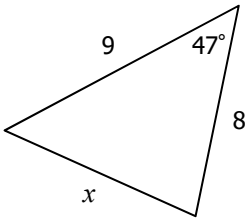
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Homework 6: Law of Cosines

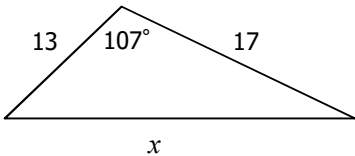
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Directions: Find each missing measure. Round all answers to the nearest tenth.

1.

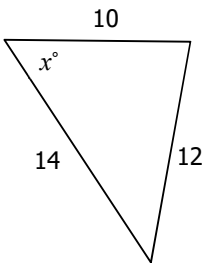


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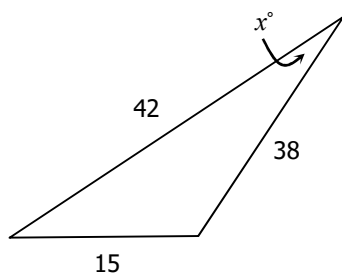


3. In $\triangle QRS$, $m\angle Q = 73^\circ$, $QR = 21$, and $QS = 30$. Find RS .

4.



5.



6. In $\triangle DEF$, $DE = 18$, $EF = 16$, and $DF = 12$. Find $m\angle F$.

7. Three boats are out at sea: Boat A, Boat B, and Boat C. The crew of Boat A can see both Boat B and Boat C. The angle between the line of sight to Boat B and the line of sight to Boat C is 49° . If the distance between Boat A and Boat B is 1.8 miles and the distance between Boat A and Boat C is 3.2 miles, what is the distance between Boat B and Boat C?

8. A farmer plants a 16-foot row of tomatoes, 13-foot row of carrots, and a 21-foot row of green peppers in a triangle. Find the angle formed by the tomatoes and carrots.

Name:

Date:

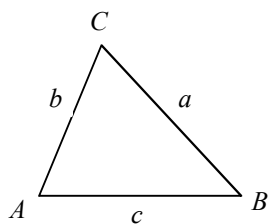
Topic:

Class:

Main Ideas/Questions

Notes/Examples

LAW OF SINES *wo.* **LAW OF COSINES**



Law of Sines

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Use when given...

Law of Cosines

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

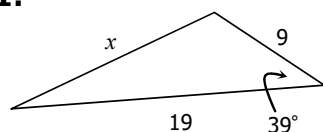
$$b^2 = a^2 + c^2 - 2ac \cdot \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cdot \cos C$$

Use when given...

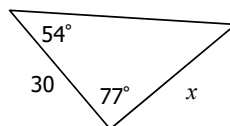
EXAMPLES: Indicate the law, then find each missing measure to the nearest tenth.

1.



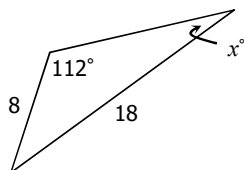
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2.



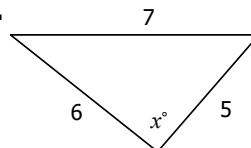
☐ S ☐ C

3.



☐ S ☐ C

4.



☐ S ☐ C

SOLVING TRIANGLES: To **solve a triangle** means to **find ALL missing side and angle measures**.
Use the information given to solve each triangle. Round all answers to the nearest tenth.

5. In $\triangle ABC$, $m\angle A = 47^\circ$, $m\angle B = 52^\circ$ and $AC = 13$.

$$m\angle C = \underline{\hspace{2cm}}$$

$$AB = \underline{\hspace{2cm}}$$

$$BC = \underline{\hspace{2cm}}$$

6. In $\triangle XYZ$, $XY = 29$, $YZ = 22$, and $XZ = 21$.

$$m\angle X = \underline{\hspace{2cm}}$$

$$m\angle Y = \underline{\hspace{2cm}}$$

$$m\angle Z = \underline{\hspace{2cm}}$$

7. In $\triangle DEF$, $m\angle F = 72^\circ$, $DE = 15$, and $EF = 12$.

$$m\angle D = \underline{\hspace{2cm}}$$

$$m\angle E = \underline{\hspace{2cm}}$$

$$DF = \underline{\hspace{2cm}}$$

8. In $\triangle RST$, $m\angle R = 55^\circ$, $RS = 17$, and $RT = 9$.

$$m\angle S = \underline{\hspace{2cm}}$$

$$m\angle T = \underline{\hspace{2cm}}$$

$$ST = \underline{\hspace{2cm}}$$

Name:

Date:

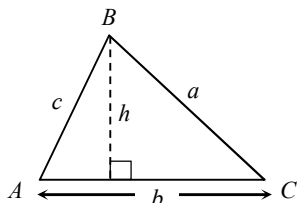
Topic:

Class:

Main Ideas/Questions

Notes/Examples

Area of Triangles

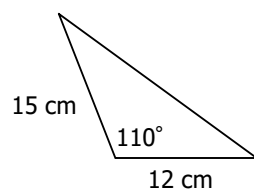


- What is the formula for the area of a triangle? _____
- We know that $\sin C =$ _____
- Therefore, $h =$ _____
- Substitute this into the area formula for h :

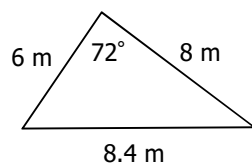
Examples

Directions: Find the area of each triangle to the nearest tenth.

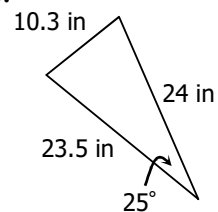
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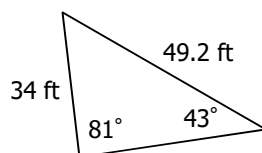
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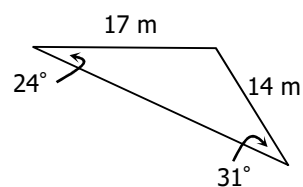
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4.



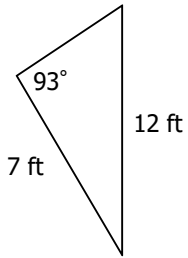
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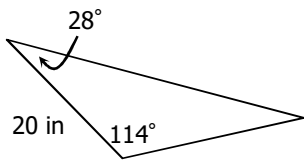
Using the Laws of Sines & Law of Cosines

In order to find the area, you will need two sides and an included angle (SAS).

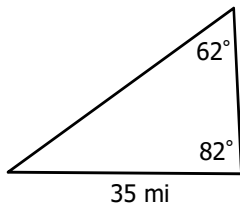
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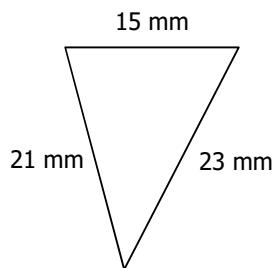
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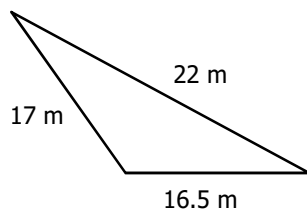
8.



9.



10.





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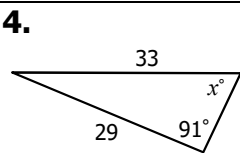
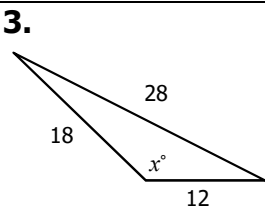
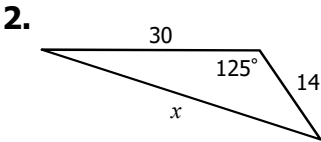
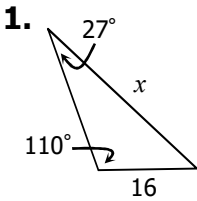
Unit 12: Trigonometry

Date: _____ Bell: _____

Homework 7: Law of Sines & Law of Cosines,
Area of Triangles

**** This is a 2-page document! ****

Directions: Find each missing measure to the nearest tenth.



Directions: Use the information given to solve each triangle. Round to the nearest tenth.

5. In $\triangle PQR$, $m\angle Q = 76^\circ$, $PQ = 18$, and $QR = 27$.

$m\angle P =$ _____
 $m\angle R =$ _____
 $PR =$ _____

6. In $\triangle JKL$, $m\angle K = 117^\circ$, $m\angle L = 41^\circ$, and $JK = 14$.

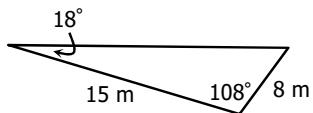
$$m\angle J = \underline{\hspace{2cm}}$$

$$JL = \underline{\hspace{2cm}}$$

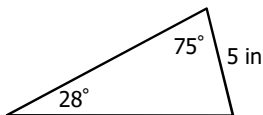
$$KL = \underline{\hspace{2cm}}$$

Directions: Find the area of each triangle to the nearest tenth.

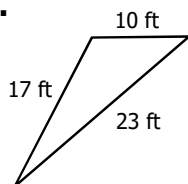
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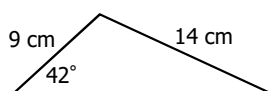
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9.



10.



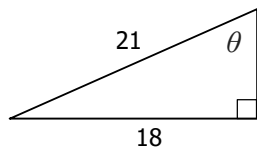
Unit 12 Test Study Guide (Trigonometry - Part I)

Name: _____

Date: _____ Block: _____

Topic 1: Trigonometric Functions

1. Find the values of the six trigonometric functions for angle θ . Give answers in simplest form.



$$\sin \theta = \underline{\hspace{2cm}} \quad \csc \theta = \underline{\hspace{2cm}}$$

$$\cos \theta = \underline{\hspace{2cm}} \quad \sec \theta = \underline{\hspace{2cm}}$$

$$\tan \theta = \underline{\hspace{2cm}} \quad \cot \theta = \underline{\hspace{2cm}}$$

2. If $\tan \theta = \frac{2}{7}$, find the remaining trigonometric functions.

$$\sin \theta = \underline{\hspace{2cm}} \quad \csc \theta = \underline{\hspace{2cm}}$$

$$\cos \theta = \underline{\hspace{2cm}} \quad \sec \theta = \underline{\hspace{2cm}}$$

$$\tan \theta = \underline{\hspace{2cm}} \quad \cot \theta = \underline{\hspace{2cm}}$$

3. If $\sec \theta = \frac{4\sqrt{2}}{3}$, find the remaining trigonometric functions.

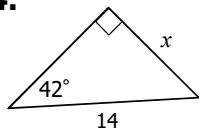
$$\sin \theta = \underline{\hspace{2cm}} \quad \csc \theta = \underline{\hspace{2cm}}$$

$$\cos \theta = \underline{\hspace{2cm}} \quad \sec \theta = \underline{\hspace{2cm}}$$

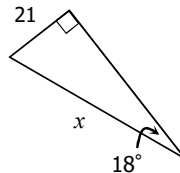
$$\tan \theta = \underline{\hspace{2cm}} \quad \cot \theta = \underline{\hspace{2cm}}$$

Directions: Find each missing measure to the nearest tenth.

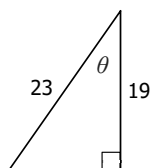
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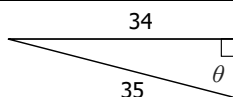
5.



6.



7.



Topic 2: Angles & Angle Measures

Directions: Give each degree measure in radians and radian measure in degrees.

8. 165°	9. -54°	10. 243°
11. $\frac{9\pi}{5}$	12. $-\frac{13\pi}{20}$	13. $-\frac{4\pi}{45}$

Directions: Give two coterminal $\angle$'s (one positive and one negative) and a reference $\angle$ for each.

14. 305° Coterminal $\angle$'s: _____ Reference $\angle$: _____	15. $\frac{7\pi}{10}$ Coterminal $\angle$'s: _____ Reference $\angle$: _____
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Directions: The terminal side of θ in standard position contains each point. Find the exact values of each trigonometric function of θ .

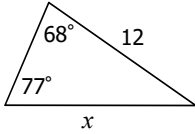
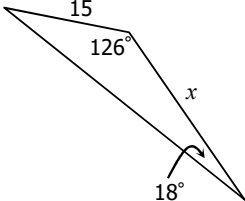
16. (6, -8)	$\sin \theta = \underline{\hspace{2cm}}$	$\csc \theta = \underline{\hspace{2cm}}$
	$\cos \theta = \underline{\hspace{2cm}}$	$\sec \theta = \underline{\hspace{2cm}}$
	$\tan \theta = \underline{\hspace{2cm}}$	$\cot \theta = \underline{\hspace{2cm}}$
17. (9, 3)	$\sin \theta = \underline{\hspace{2cm}}$	$\csc \theta = \underline{\hspace{2cm}}$
	$\cos \theta = \underline{\hspace{2cm}}$	$\sec \theta = \underline{\hspace{2cm}}$
	$\tan \theta = \underline{\hspace{2cm}}$	$\cot \theta = \underline{\hspace{2cm}}$

Topic 3: The Unit Circle

18. Which trigonometric functions are negative in quadrant IV?	19. If $\cos \theta < 0$ and $\csc \theta < 0$ which quadrant(s) could the terminal side of θ lie?
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Directions: Give the exact value of each trigonometric function.		
20. $\sin 45$	21. $\cot 135^\circ$	22. $\sec 330^\circ$
23. $\tan \frac{4\pi}{3}$	24. $\csc \frac{11\pi}{6}$	25. $\cos \frac{2\pi}{3}$
26. $\tan \frac{7\pi}{6}$	27. $\sec \frac{5\pi}{4}$	28. $\cot \pi$

Topic 4: Law of Sines

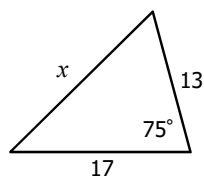
Directions: Find each missing length to the nearest tenth.	
29. 	30. 
Directions: Find each missing angle to the nearest tenth. Give all possible solutions.	
31. In $\triangle ABC$, $m\angle A = 40^\circ$, $AB = 26$, and $BC = 20$. Find $m\angle C$.	
32. In $\triangle PQR$, $m\angle P = 53^\circ$, $PQ = 16$, and $QR = 10$. Find $m\angle Q$.	

33. In $\triangle JKL$, $m\angle L = 113^\circ$, $JK = 21$, and $JL = 10$. Find $m\angle K$.

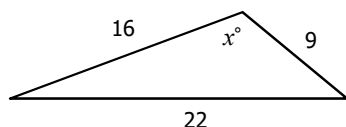
Topic 5: Law of Cosines

Directions: Find each missing measure to the nearest tenth.

34.



35.



36. In $\triangle XYZ$, $m\angle X = 96^\circ$, $XZ = 15$, and $XY = 24$. Find YZ .

Topic 6: Solving Triangles

Directions: Solve each triangle. Round all answers to the nearest tenth.

37. In $\triangle DEF$, $DE = 9$, $EF = 12$, and $DF = 13$.

$$m\angle D = \underline{\hspace{2cm}}$$

$$m\angle E = \underline{\hspace{2cm}}$$

$$m\angle F = \underline{\hspace{2cm}}$$

38. In $\triangle MNP$, $m\angle M = 26^\circ$, $m\angle P = 48^\circ$, and $MP = 22$.

$$m\angle N = \underline{\hspace{2cm}}$$

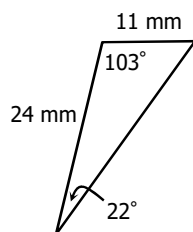
$$MN = \underline{\hspace{2cm}}$$

$$NP = \underline{\hspace{2cm}}$$

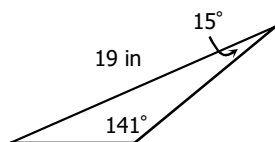
Topic 7: Area of Triangles

Directions: Find the area of each triangle to the nearest tenth.

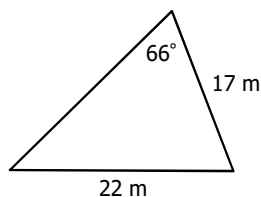
39.



40.



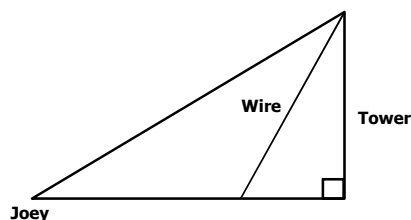
41.



Topic 8: Applications

42. From the top of a building, the angle of depression to a car parked on the street below is 48° . If the building is 75 feet tall, how far is the car parked from the base of the building?

- 43.** A 95-foot wire attached from the top of a cell phone tower makes a 62° angle with the ground. Joey is standing 150 feet from the end of the wire, looking up at the tower. Find the angle of elevation from the point on the ground where Joey is standing to the top of the tower.



- 44.** Two trains leave from the same point and travel along straight tracks that differ in direction by 72° . If their speeds are 64 miles per hour and 52 miles per hour, approximate how far apart they will be after 30 minutes.

- 45.** The angles of elevation to a balloon in the air from two points on level ground are 27° and 45° . If the points are 9 miles apart and the balloon is between the points, in the same vertical plane, approximate, to the nearest tenth of a mile, the height of the balloon above the ground.

- 46.** A triangular lot has sides of 98 feet, 137 feet, and 84 feet. Find the area of the lot to the nearest tenth.

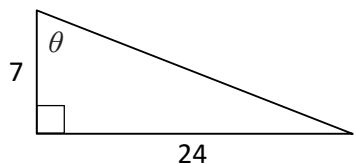
Name: _____

Unit 12 Test (Part I)

Date: _____ Bell: _____

Trigonometry

For questions 1-2, use the diagram below to give the exact trigonometric ratios.



1. $\tan \theta$

- A. $\frac{7}{24}$ B. $\frac{24}{7}$ C. $\frac{7}{25}$ D. $\frac{25}{7}$

2. $\sec \theta$

- A. $\frac{7}{25}$ B. $\frac{25}{7}$ C. $\frac{24}{25}$ D. $\frac{25}{24}$

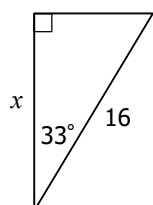
3. If $\sin \theta = \frac{8}{9}$, find $\cot \theta$.

- A. $\frac{9\sqrt{17}}{17}$ C. $\frac{8\sqrt{17}}{17}$
B. $\frac{\sqrt{17}}{9}$ D. $\frac{\sqrt{17}}{8}$

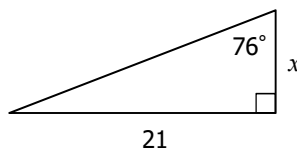
4. If $\tan \theta = \frac{\sqrt{7}}{5}$, find $\csc \theta$.

- A. $\frac{4\sqrt{14}}{7}$ C. $\frac{5\sqrt{2}}{8}$
B. $\frac{\sqrt{14}}{8}$ D. $\frac{4\sqrt{2}}{5}$

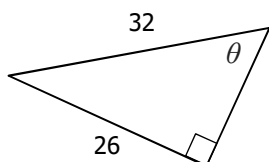
5. Find x to the nearest tenth.



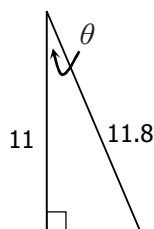
6. Find x to the nearest tenth.



7. Find the measure of θ to the nearest tenth.



8. Find the measure of θ to the nearest tenth.



9. Convert 63° to radians.	10. Convert -280° to radians.
11. Convert $\frac{13\pi}{18}$ to degrees.	12. Convert $\frac{19\pi}{10}$ to degrees.
13. The terminal side of an angle measuring $\frac{13\pi}{12}$ radians lies in which quadrant? A. I B. II C. III D. IV	14. The terminal side of an angle measuring $-\frac{9\pi}{5}$ radians lies in which quadrant? A. I B. II C. III D. IV
15. Which angle is coterminal to 128°? A. -52° B. 308° C. 232° D. 488°	16. Which angle is coterminal to $\frac{2\pi}{5}$? A. $-\frac{3\pi}{5}$ C. $-\frac{8\pi}{5}$ B. $\frac{4\pi}{5}$ D. $\frac{7\pi}{5}$
17. Name the reference angle to 205°. A. 15° B. 25° C. 65° D. 75°	18. Name the reference angle to $\frac{7\pi}{10}$. A. $\frac{\pi}{5}$ C. $\frac{7\pi}{20}$ B. $\frac{3\pi}{10}$ D. $\frac{3\pi}{20}$
Use for questions 19-20: The terminal side of θ in standard position contains the point $(-4, -2)$. Find the exact value for each trigonometric function.	19. $\sin \theta$ A. $\frac{2\sqrt{5}}{5}$ B. $-\frac{2\sqrt{5}}{5}$ C. $\frac{\sqrt{5}}{5}$ D. $-\frac{\sqrt{5}}{5}$
	20. $\cot \theta$ A. $\frac{1}{2}$ B. $-\frac{1}{2}$ C. 2 D. -2

21. If $\cot \theta < 0$, in which quadrant(s) could the terminal side of angle θ lie?

- A. II and IV
- B. II and III
- C. III only
- D. IV only

22. If $\sin \theta = \cos \theta$, in which quadrant(s) could the terminal side of angle θ lie?

- A. I only
- B. III only
- C. I and III
- D. II and IV

For questions 23-26, give the exact value for each trigonometric function.

23. $\sin 120^\circ$

- A. $-\frac{\sqrt{3}}{2}$
- B. $-\frac{1}{2}$
- C. $\frac{\sqrt{3}}{2}$
- D. $\frac{1}{2}$

24. $\csc 225^\circ$

- A. $-\frac{\sqrt{2}}{2}$
- B. $-\sqrt{2}$
- C. $\frac{\sqrt{2}}{2}$
- D. $\sqrt{2}$

25. $\tan \frac{11\pi}{6}$

- A. $\frac{\sqrt{3}}{3}$
- B. $-\frac{\sqrt{3}}{3}$
- C. $\sqrt{3}$
- D. $-\sqrt{3}$

26. $\sec \frac{\pi}{2}$

- A. 0
- B. 1
- C. -1
- D. undefined

27. What function has a value of $-\sqrt{3}$?

- A. $\tan \frac{\pi}{3}$
- B. $\cot \frac{5\pi}{6}$
- C. $\tan \frac{7\pi}{6}$
- D. $\cot \frac{4\pi}{3}$

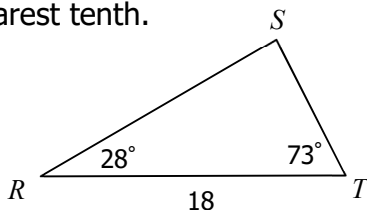
28. What function has a value of 2?

- A. $\csc \frac{11\pi}{6}$
- B. $\csc \frac{7\pi}{6}$
- C. $\sec \frac{5\pi}{3}$
- D. $\sec \frac{4\pi}{3}$

29. Which functions have a value of -1? Check all that apply.

☐ $\cos \pi$	☐ $\tan \frac{3\pi}{4}$	☐ $\sin \frac{3\pi}{2}$	☐ $\csc \frac{\pi}{2}$
☐ $\sec \frac{3\pi}{2}$	☐ $\tan \frac{\pi}{2}$	☐ $\cot \pi$	☐ $\cot \frac{7\pi}{4}$

30. Find ST to the nearest tenth.



$ST =$

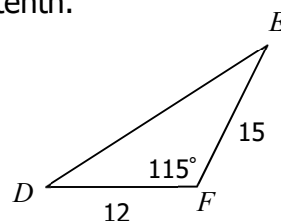
31. In $\triangle ABC$, $m\angle A = 58^\circ$, $AC = 22$, and $BC = 25$. Find $m\angle B$.

- A. 48°
- B. 74°
- C. 48° or 132°
- D. Not Possible

32. In $\triangle XYZ$, $m\angle X = 49^\circ$, $XY = 18$, and $YZ = 14$. Find $m\angle Y$.

- A. 55°
- B. 76°
- C. 76° or 104°
- D. 55° or 27°

33. Find DE to the nearest tenth.



$DE =$

34. Given $\triangle JKL$ with $JK = 16$, $KL = 14$, and $JL = 10$, solve the triangle. Round all answers to the nearest tenth.

$m\angle J =$

$m\angle K =$

$m\angle L =$

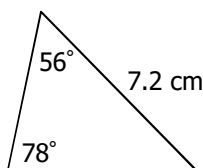
- 35.** Given $\triangle PQR$ with $m\angle P = 42^\circ$, $m\angle R = 26^\circ$, and $PQ = 19$, solve the triangle. Round all answers to the nearest tenth.

$$m\angle Q =$$

$$QR =$$

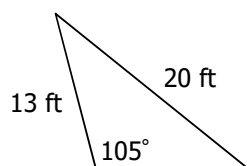
$$PR =$$

- 36.** Find the area of the triangle below to the nearest tenth.



- A.** 13.7 cm^2 **C.** 18.4 cm^2
B. 12.2 cm^2 **D.** 15.8 cm^2

- 37.** Find the area of the triangle below to the nearest tenth.



- A.** 80.5 ft^2 **C.** 76.6 ft^2
B. 71.3 ft^2 **D.** 68.1 ft^2

- 38.** A 24-foot ladder is placed against a building to reach a window on the second floor. The angle of depression from the top of the ladder to the bottom of the ladder is 62° . Find the distance between the bottom of the ladder and the base of the building to the nearest tenth of a foot.

- 39.** Alex is standing 60 feet from the base of a flagpole. He measures the angle of elevation to the top of the pole as 35° . Vera is 36 feet closer to the base of the flagpole on a straight level path. Find the angle of elevation from the point Vera is standing to the top of the flagpole to the nearest tenth of a degree.

- 40.** A course in the shape of a triangle consists of a running path, bike path, and skateboarding path. The bike path is 51 meters long and the skateboarding path is 64 meters long. If the angle between the running and bike paths is 78° , find the angle between the running and skateboarding paths to the nearest tenth of a degree.

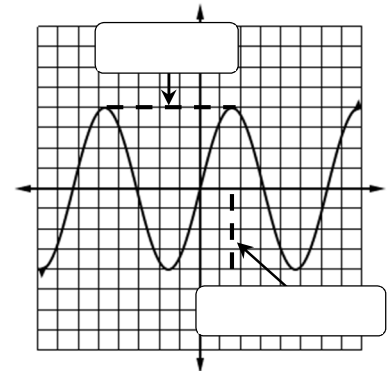
- 41.** The distances from a boat to two buoys are 75 meters and 52 meters. If the angle between the two lines of sight is 67° , find the distance between the buoys. Round to the nearest tenth of a meter.

- 42.** A triangular sail on a boat has side lengths that measure 15 feet, 36 feet, and 38 feet. Find the area of the sail to the nearest tenth of a foot.

Name: _____	Date: _____
Topic: _____	Class: _____

Main Ideas/Questions	Notes/Examples
----------------------	----------------

PERIODIC FUNCTIONS	- Periodic functions are functions that repeat exactly in regular intervals, called cycles. - Trigonometric functions are periodic! Period: the length of a cycle. Periods can start at any point on the graph. Amplitude: Half the distance between the minimum and maximum values.
--	--



GRAPHING SINE & COSINE	Use the unit circle values! x represents the angle θ in standard form and y represents the corresponding trigonometric value.		
	SINE $(y = \sin \theta)$		Amplitude: Period:
	COSINE $(y = \cos \theta)$		Amplitude: Period:
	The graphs of sine and cosine are called “waves” because they have a wave-like rolling appearance.		

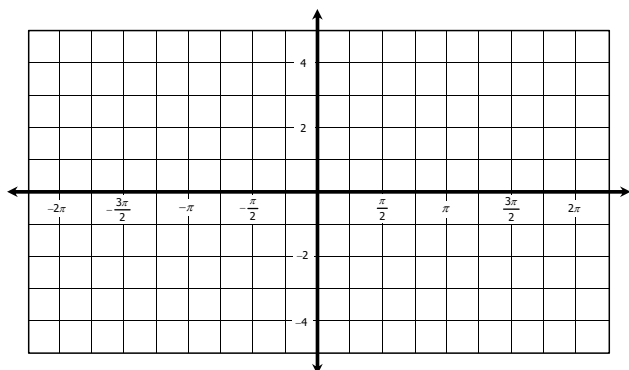
<i>Changing the Amplitude & Period</i>	Given $y = a \cdot \sin b\theta$ or $y = a \cdot \cos b\theta$ where θ is in radians: the amplitude is _____ and the period is _____.
--	---

Directions: Identify the amplitude and period of each function.

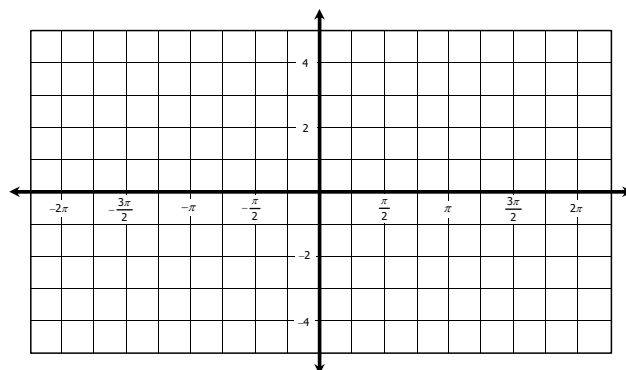
1. $y = 3 \cdot \sin 2\theta$	2. $y = \frac{3}{2} \cdot \sin \frac{1}{3} \theta$
Amp: _____ Period: _____	Amp: _____ Period: _____

Directions: Identify the amplitude and period, then graph each function.

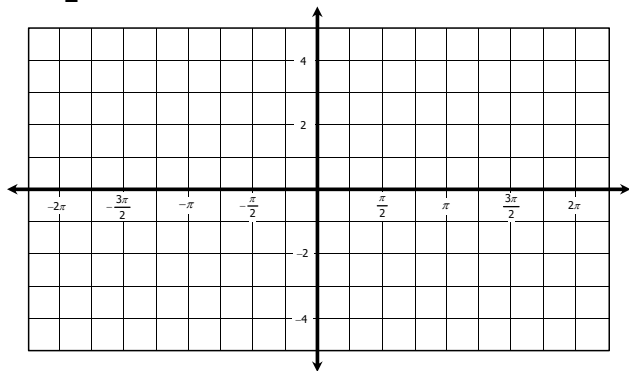
3. $y = 4 \cdot \sin \theta$ Amp: _____ Period: _____



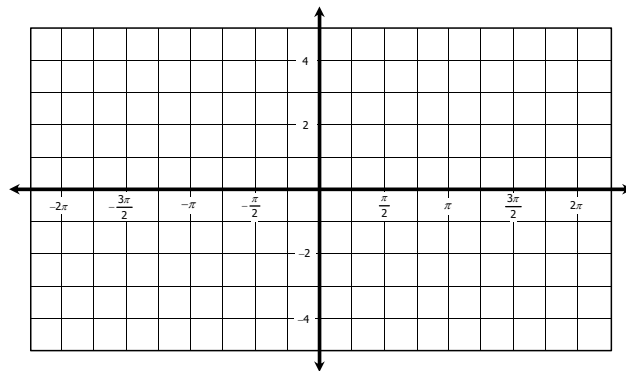
4. $y = \cos \frac{1}{2} \theta$ Amp: _____ Period: _____



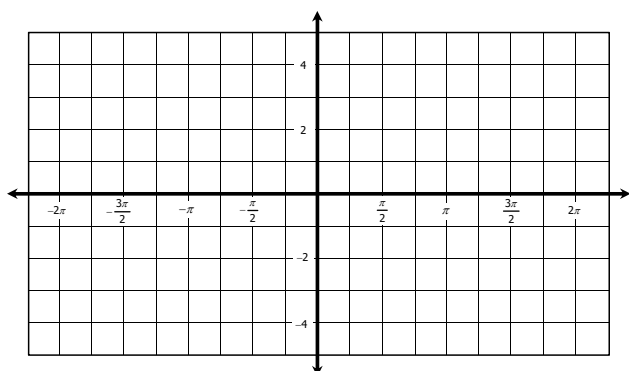
5. $y = \frac{1}{2} \cdot \cos 3\theta$ Amp: _____ Period: _____



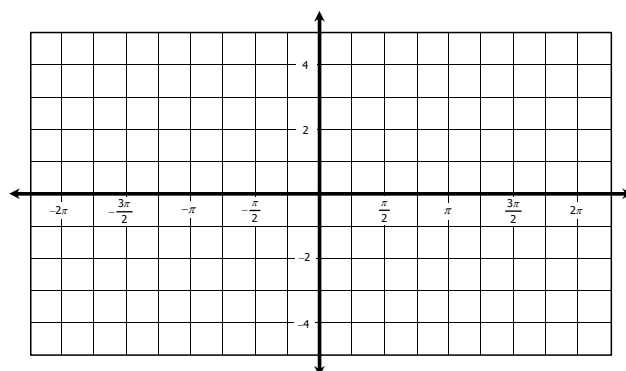
6. $y = 5 \cdot \sin \frac{1}{3} \theta$ Amp: _____ Period: _____



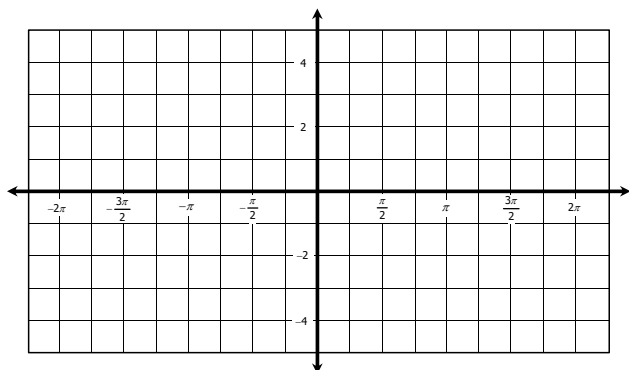
7. $y = \cos \frac{3}{4} \theta$ Amp: _____ Period: _____



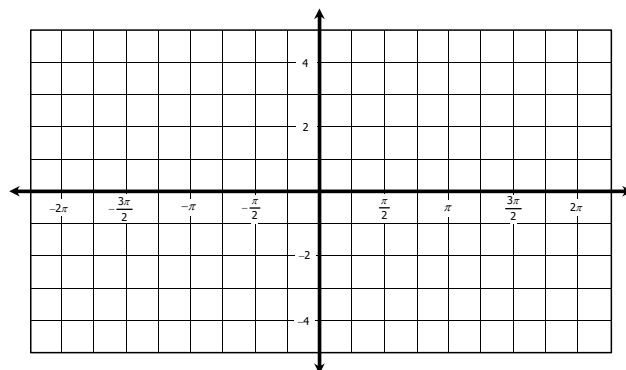
8. $y = \frac{5}{2} \cdot \sin \theta$ Amp: _____ Period: _____



9. $y = 3 \cdot \sin 2\theta$ Amp: _____ Period: _____



10. $y = 2 \cdot \cos \frac{3}{2} \theta$ Amp: _____ Period: _____



Name:

Date:

Topic:

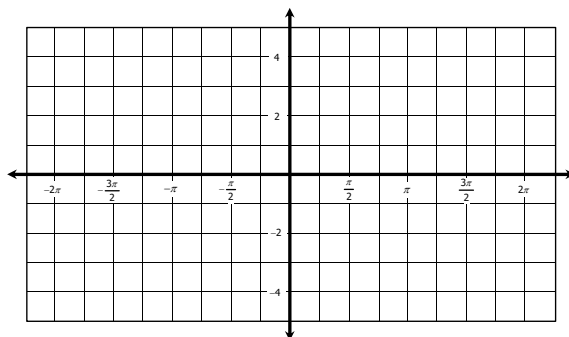
Class:

Main Ideas/Questions

Notes/Examples

TANGENT
GRAPHS

The tangent function is one of the trigonometric functions whose graphs have asymptotes.
Graph the function below and show the asymptotes.

TANGENT
($y = \tan \theta$)

Amplitude:

Period:

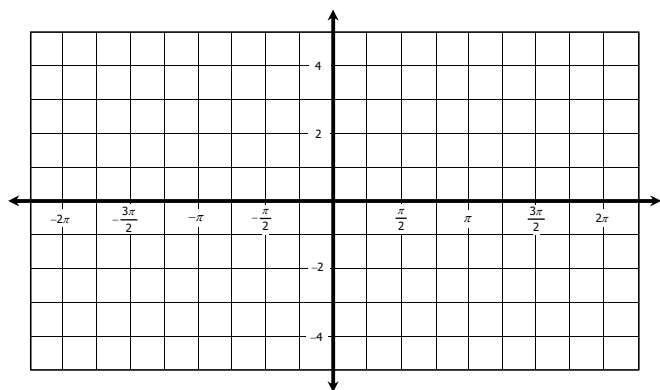
*Changing the
Period & Asymptotes*

Given $y = a \cdot \tan b\theta$ where θ is in radians, the period is _____,
and the asymptotes are _____ multiples of _____.

Directions: Identify the period, then graph each function.

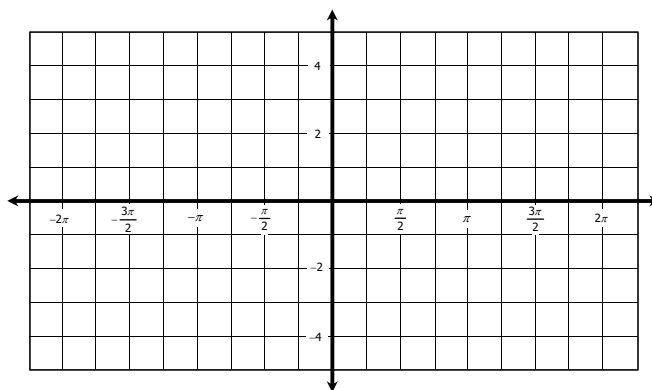
1. $y = 2 \cdot \tan \theta$

Period: _____



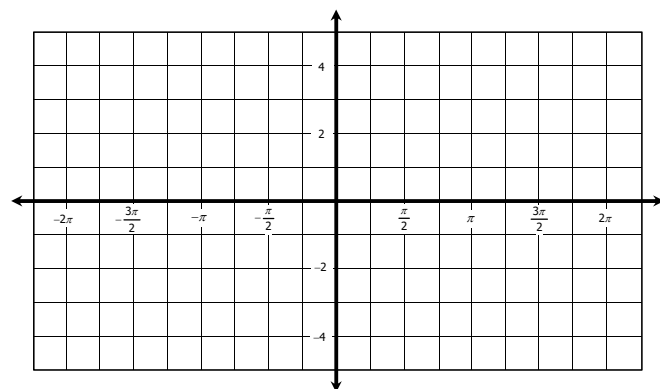
2. $y = \tan \frac{1}{3}\theta$

Period: _____



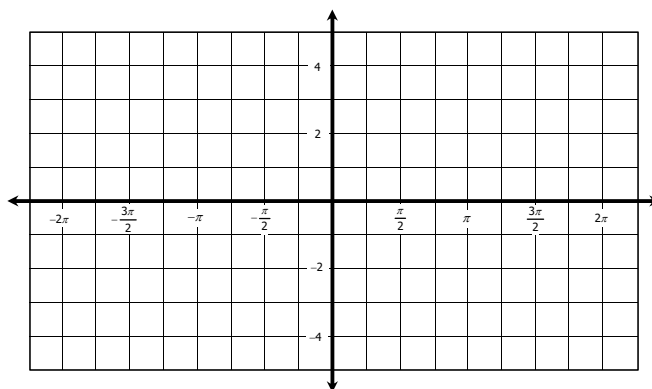
3. $y = \frac{1}{2} \cdot \tan 2\theta$

Period: _____



4. $y = 4 \cdot \tan \frac{1}{2}\theta$

Period: _____



Name: _____

Unit 12: Trigonometry

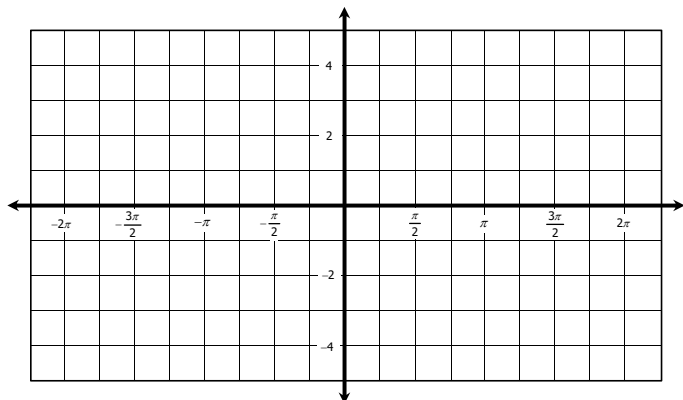
Date: _____ Bell: _____

Homework 8: Graphing Trigonometric Functions:
Sine, Cosine, and Tangent

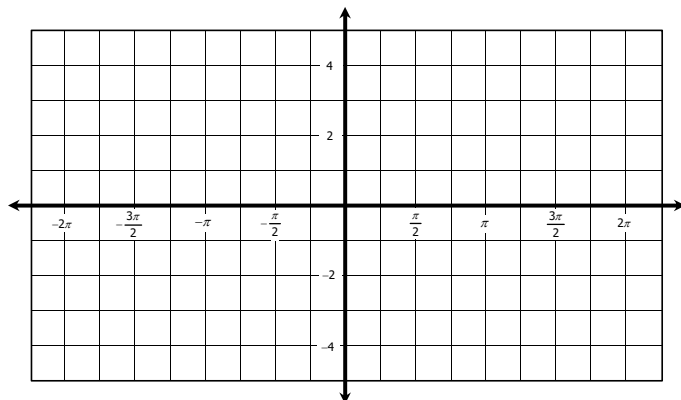
**** This is a 2-page document! ****

Directions: Identify the amplitude, if possible, and the period of each function, then graph.

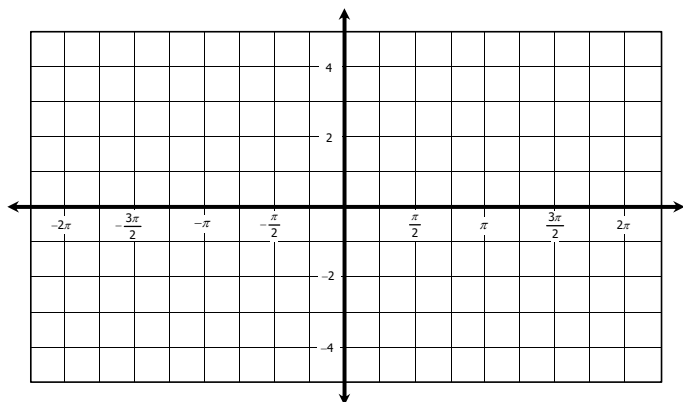
1. $y = 3 \cdot \tan \theta$ **Amp:** _____ **Period:** _____



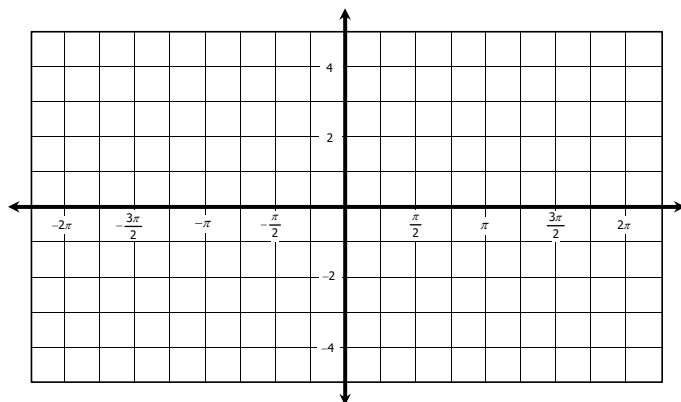
2. $y = \sin 4\theta$ **Amp:** _____ **Period:** _____



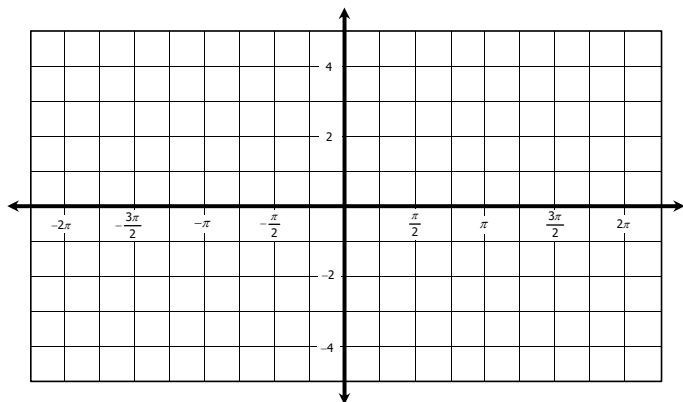
3. $y = \frac{1}{2} \cdot \cos 2\theta$ **Amp:** _____ **Period:** _____



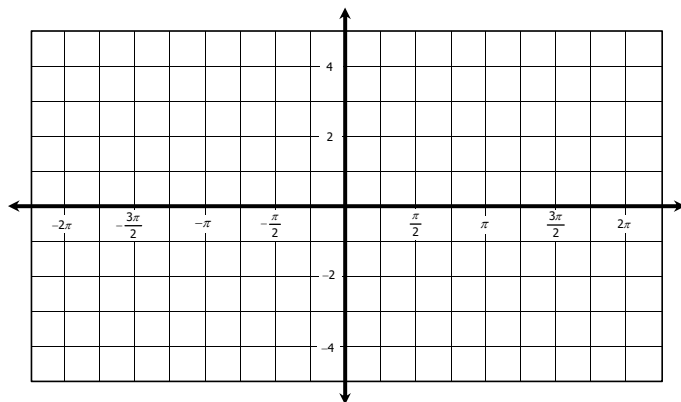
4. $y = \frac{7}{2} \cdot \sin \theta$ **Amp:** _____ **Period:** _____



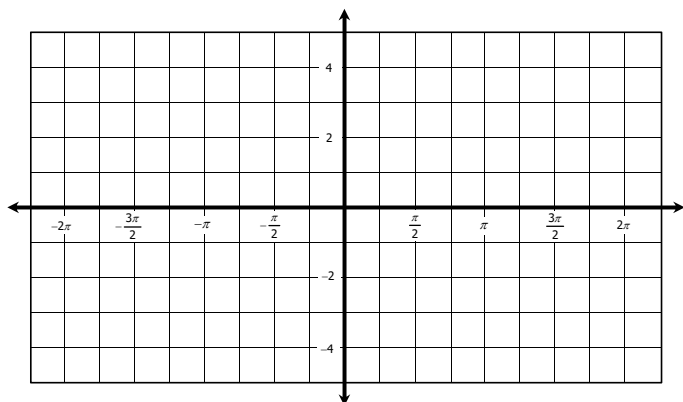
5. $y = 4 \cdot \cos 3\theta$ **Amp:** _____ **Period:** _____



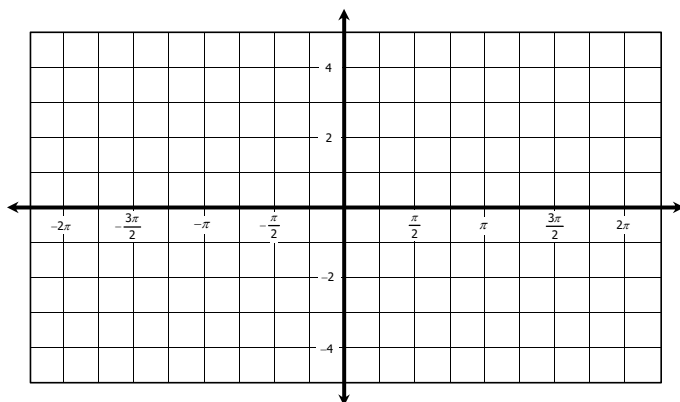
6. $y = \tan \frac{1}{2} \theta$ **Amp:** _____ **Period:** _____



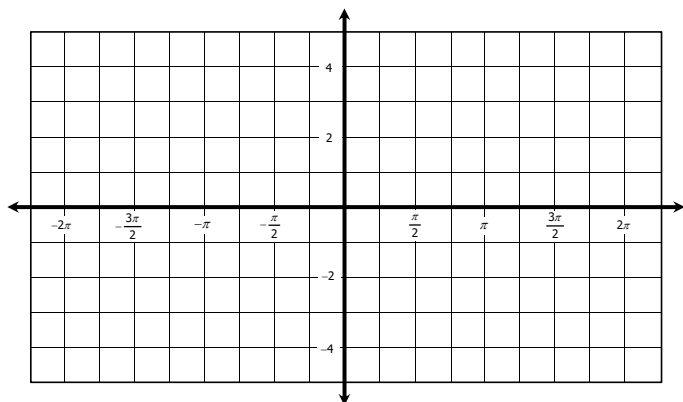
7. $y = \frac{3}{2} \cdot \cos \frac{1}{2} \theta$ Amp: _____ Period: _____



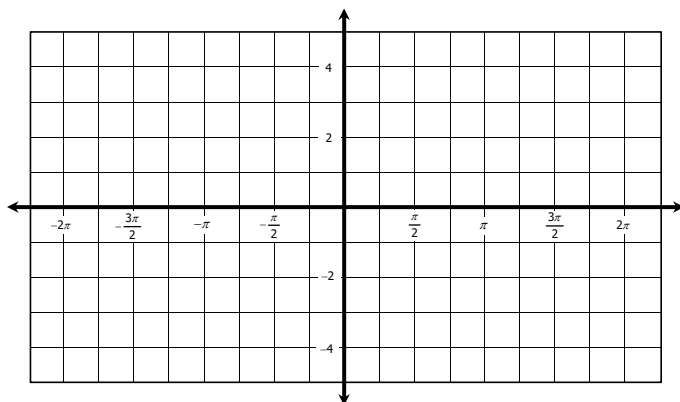
8. $y = \sin \frac{1}{3} \theta$ Amp: _____ Period: _____



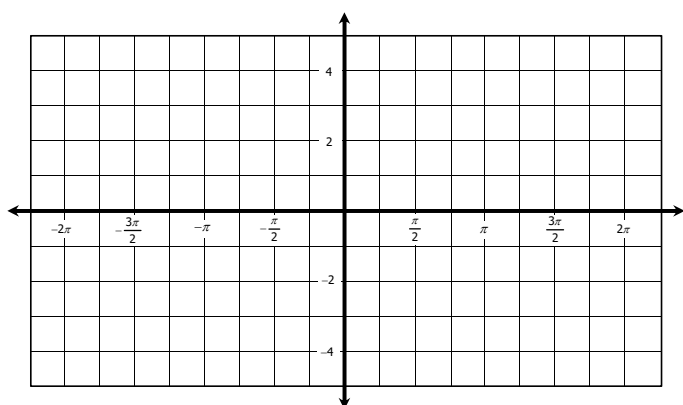
9. $y = 2 \cdot \cos 2 \theta$ Amp: _____ Period: _____



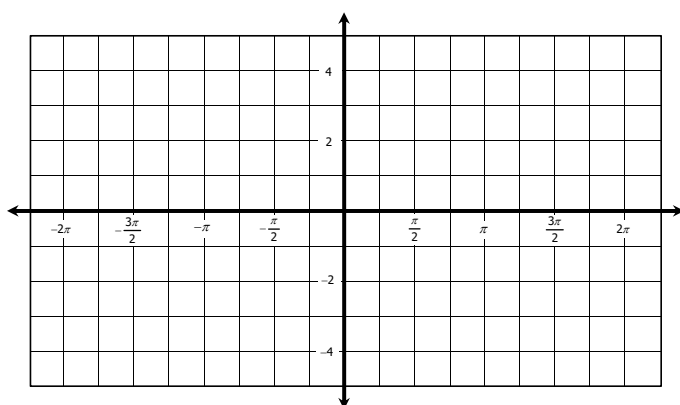
10. $y = \tan 3 \theta$ Amp: _____ Period: _____



11. $y = \frac{1}{2} \cdot \tan \frac{1}{3} \theta$ Amp: _____ Period: _____



12. $y = 5 \cdot \sin \theta$ Amp: _____ Period: _____



Name:

Date:

Topic:

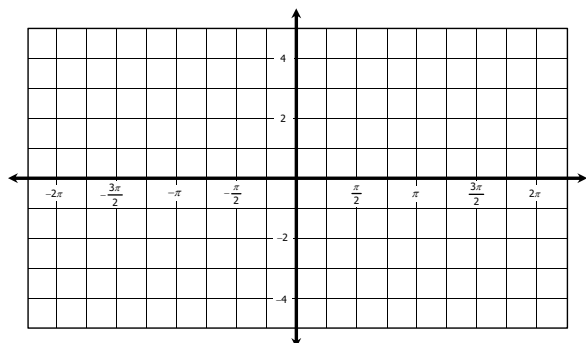
Class:

Main Ideas/Questions	Notes/Examples		
GRAPHING RECIPROCAL FUNCTIONS	COSECANT ($y = \csc \theta$)		Amplitude: Period:
	SECANT ($y = \sec \theta$)		Amplitude: Period:
	COTANGENT ($y = \cot \theta$)		Amplitude: Period:
	If given $y = a \cdot \csc b\theta$, $y = a \cdot \sec b\theta$, or $y = a \cdot \cot b\theta$, apply the changes to the amplitude, period, and/or asymptotes the same way you did for the other trigonometric functions.		

Directions: Identify the period, then graph each function.

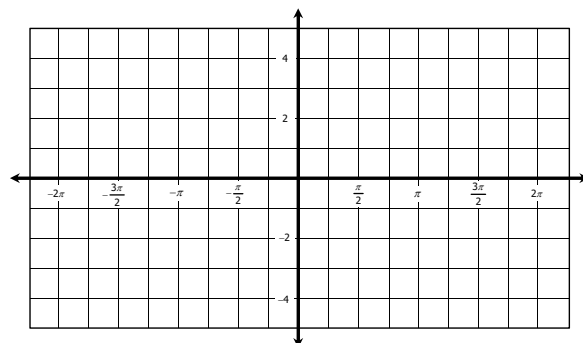
1. $y = 3 \cdot \csc \theta$

Period: _____



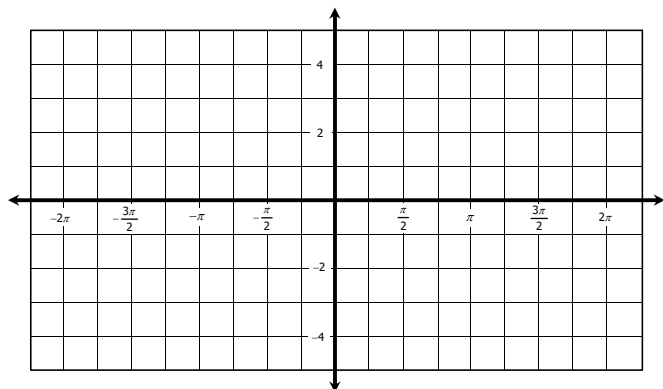
2. $y = \sec \frac{1}{2} \theta$

Period: _____



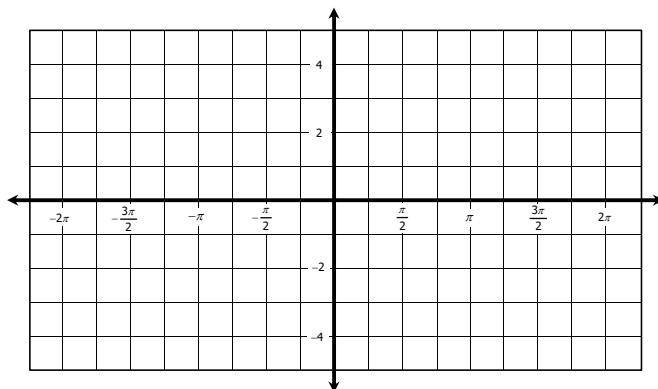
3. $y = \cot 2\theta$

Period: _____



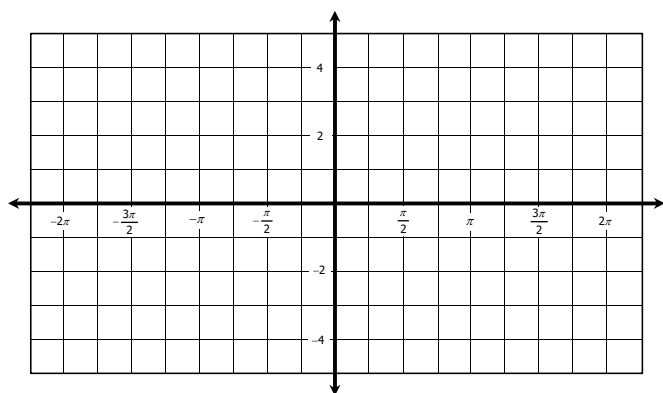
4. $y = 2 \cdot \sec \theta$

Period: _____



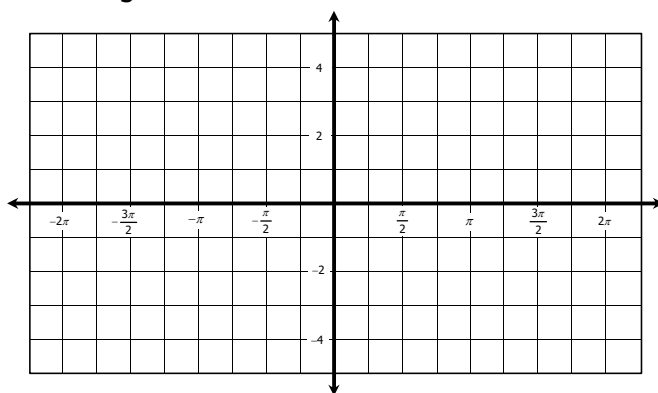
5. $y = \frac{1}{2} \cdot \cot \theta$

Period: _____



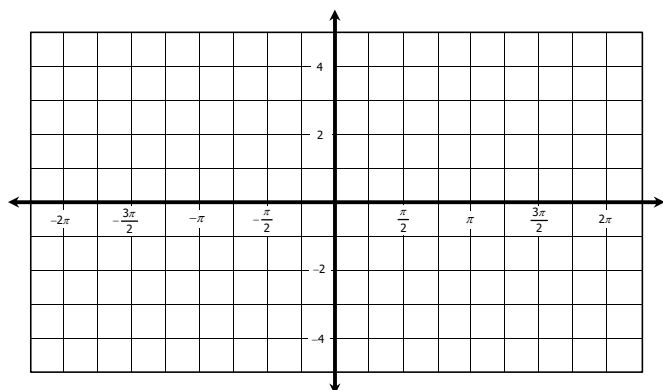
6. $y = \csc \frac{4}{3} \theta$

Period: _____



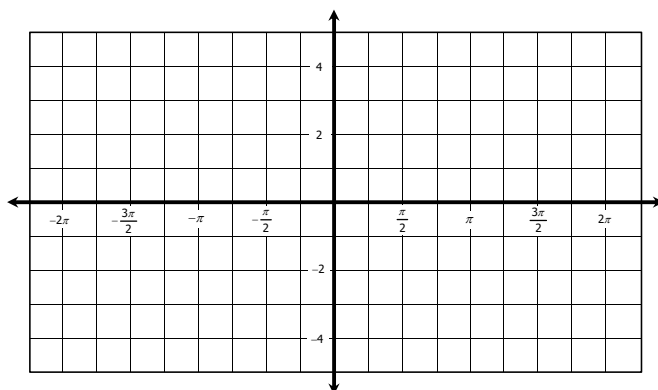
7. $y = 2 \cdot \sec 2\theta$

Period: _____



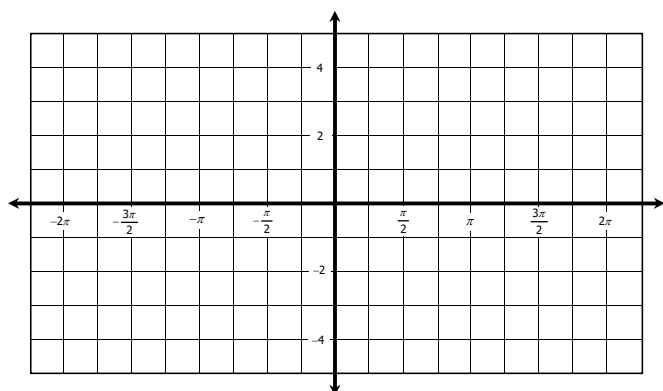
8. $y = 3 \cdot \csc \frac{1}{2} \theta$

Period: _____



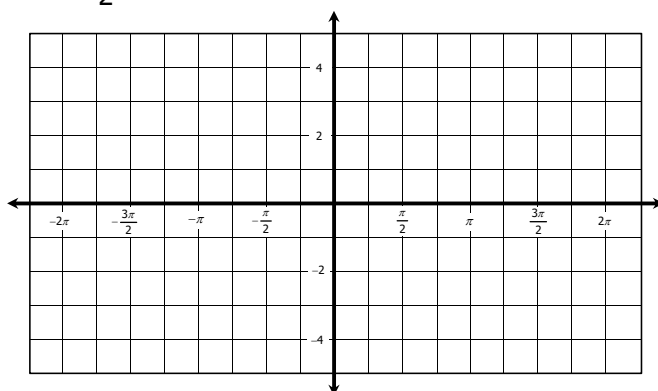
9. $y = 4 \cdot \cot \frac{1}{2} \theta$

Period: _____



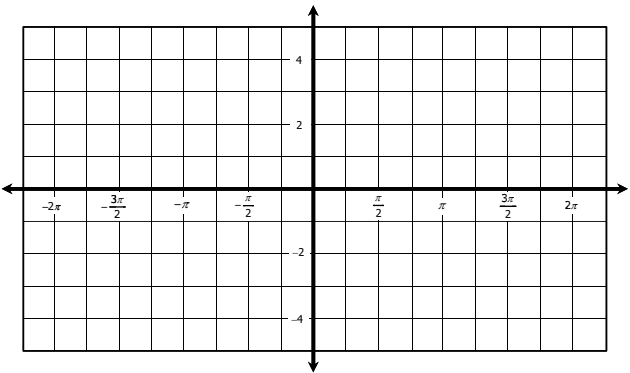
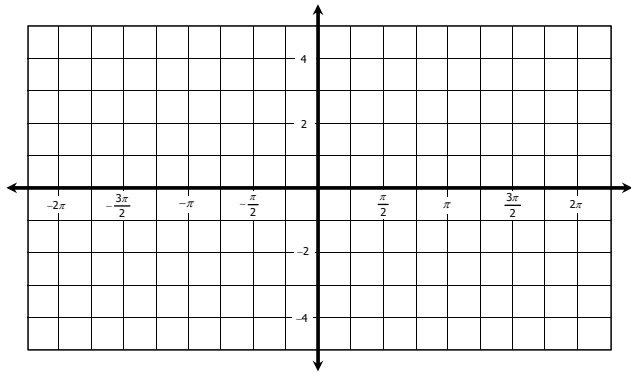
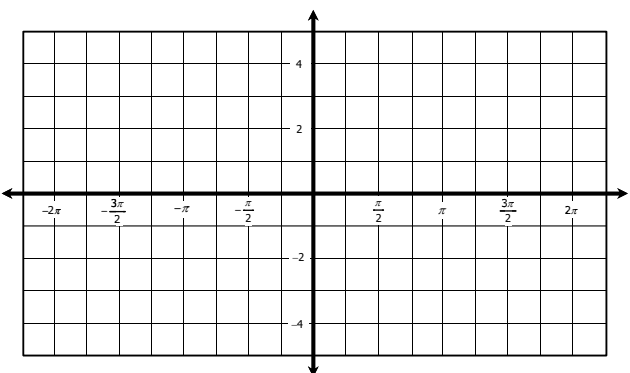
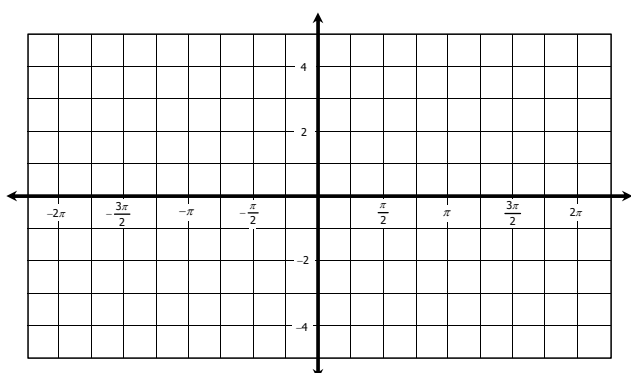
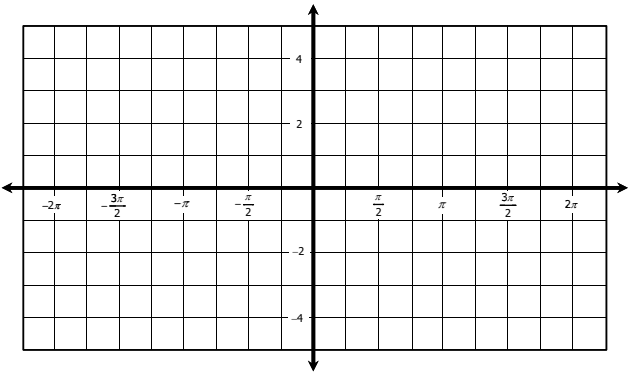
10. $y = \frac{5}{2} \cdot \csc 3\theta$

Period: _____



Graphing Trigonometric Functions

	Function & Graph	Amplitude & Period	Example
SINE		Amplitude = _____ Period = _____	$y = 4 \cdot \sin \frac{1}{2} \theta$ Amp: _____ Period: _____
		Amplitude = _____ Period = _____	$y = \frac{3}{2} \cdot \cos 2 \theta$ Amp: _____ Period: _____
		Amplitude = _____ Period = _____	$y = 2 \cdot \tan \frac{1}{3} \theta$ Amp: _____ Period: _____
COSINE		Amplitude = _____ Period = _____	$y = 4 \cdot \sin \frac{1}{2} \theta$ Amp: _____ Period: _____
		Amplitude = _____ Period = _____	$y = \frac{3}{2} \cdot \cos 2 \theta$ Amp: _____ Period: _____
		Amplitude = _____ Period = _____	$y = 2 \cdot \tan \frac{1}{3} \theta$ Amp: _____ Period: _____
TANGENT		Amplitude = _____ Period = _____	$y = 4 \cdot \sin \frac{1}{2} \theta$ Amp: _____ Period: _____
		Amplitude = _____ Period = _____	$y = \frac{3}{2} \cdot \cos 2 \theta$ Amp: _____ Period: _____
		Amplitude = _____ Period = _____	$y = 2 \cdot \tan \frac{1}{3} \theta$ Amp: _____ Period: _____

COSECANT		Amplitude = _____ Period = _____	$y = \frac{5}{2} \cdot \csc \frac{2}{3} \theta$ Amp: _____ Period: _____ 	
	SECANT		Amplitude = _____ Period = _____	$y = 2 \cdot \sec \frac{4}{3} \theta$ Amp: _____ Period: _____ 
		COTANGENT		Amplitude = _____ Period = _____

Name: _____

Unit 12: Trigonometry

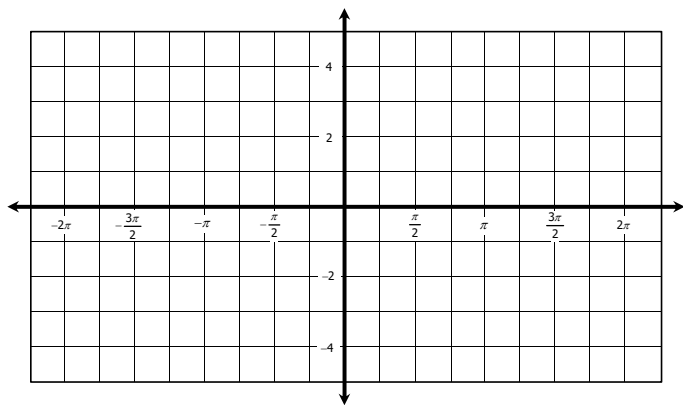
Date: _____ Bell: _____

Homework 9: Graphing Trigonometric Functions:
(All Six Functions!)

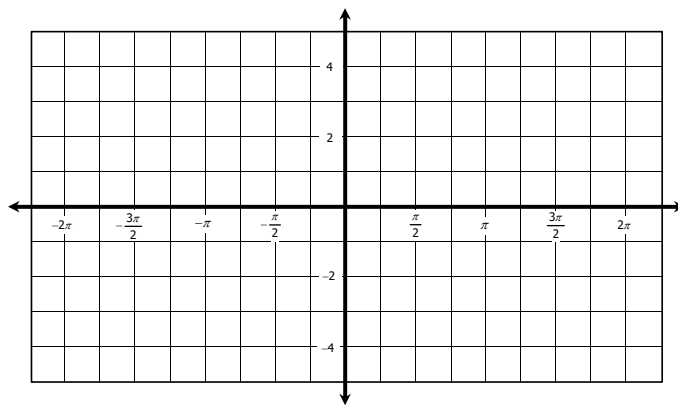
**** This is a 2-page document! ****

Directions: Identify the amplitude, if possible, and the period for each function. Then graph.

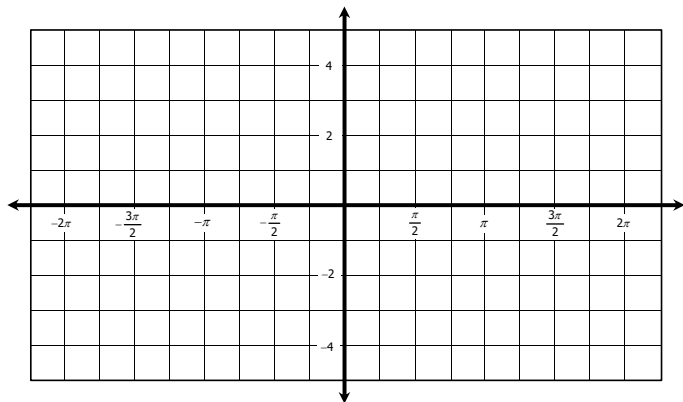
1. $y = 2 \cdot \csc \theta$ Amp: _____ Period: _____



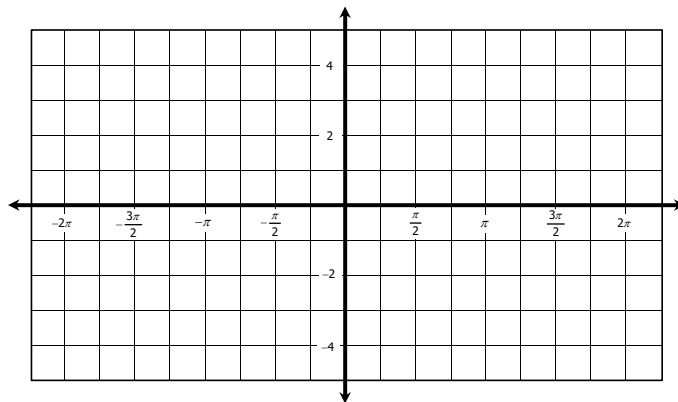
2. $y = \tan \frac{2}{3} \theta$ Amp: _____ Period: _____



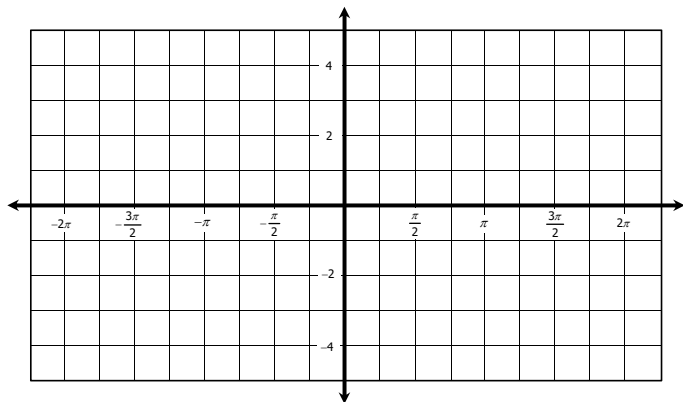
3. $y = \sec \frac{8}{3} \theta$ Amp: _____ Period: _____



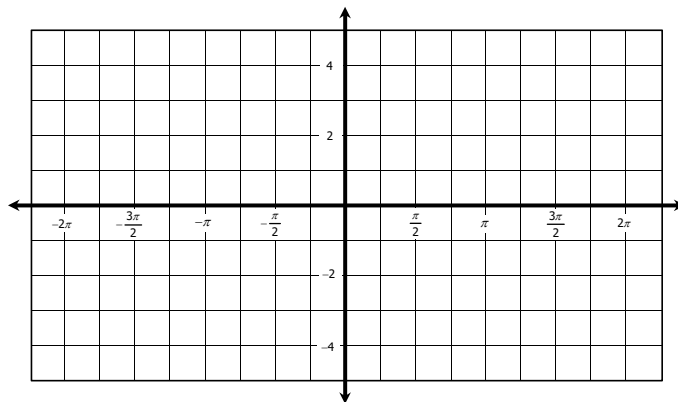
4. $y = 5 \cdot \cos \theta$ Amp: _____ Period: _____



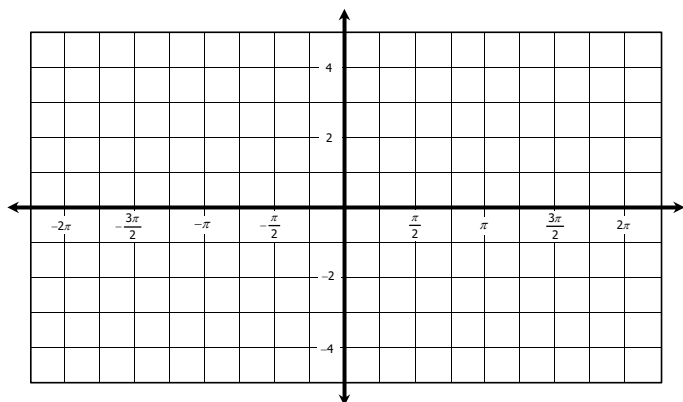
5. $y = \frac{1}{4} \cdot \cot \theta$ Amp: _____ Period: _____



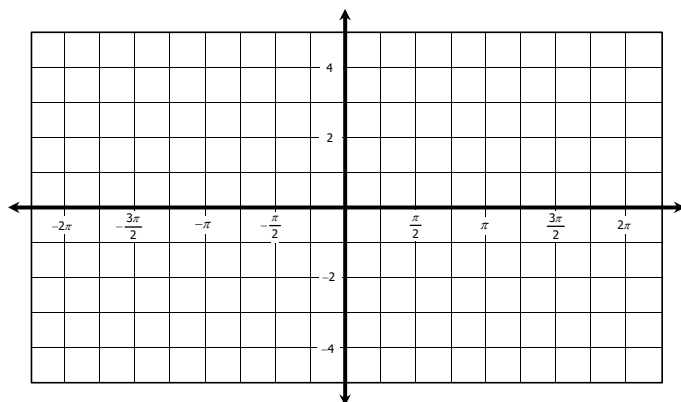
6. $y = 2 \cdot \sin \frac{8}{3} \theta$ Amp: _____ Period: _____



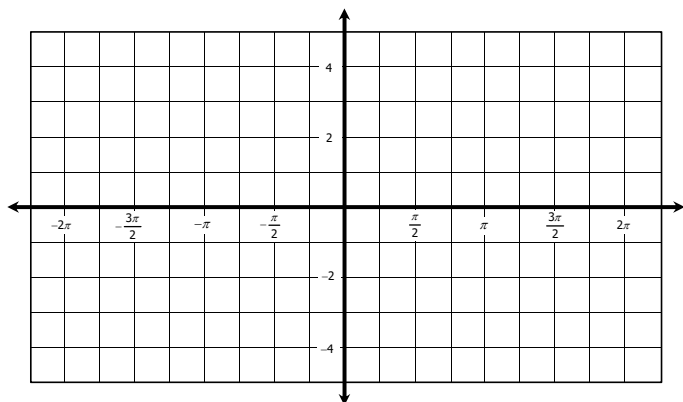
7. $y = 3 \cdot \csc 2\theta$ Amp: _____ Period: _____



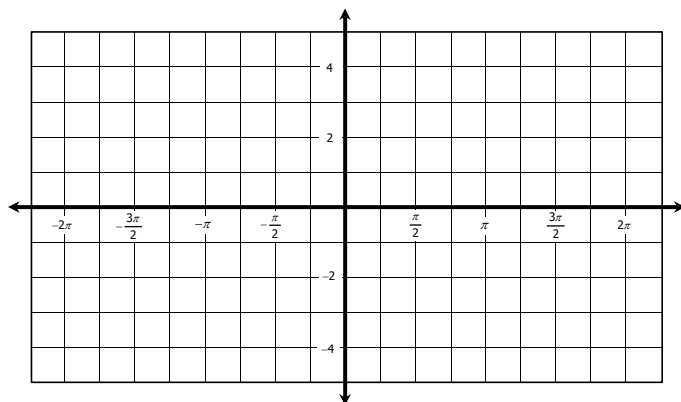
8. $y = 4 \cdot \tan \frac{1}{3}\theta$ Amp: _____ Period: _____



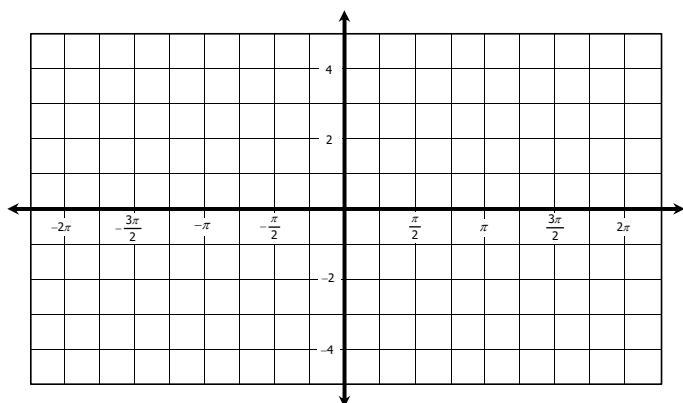
9. $y = \frac{3}{2} \cdot \sin 4\theta$ Amp: _____ Period: _____



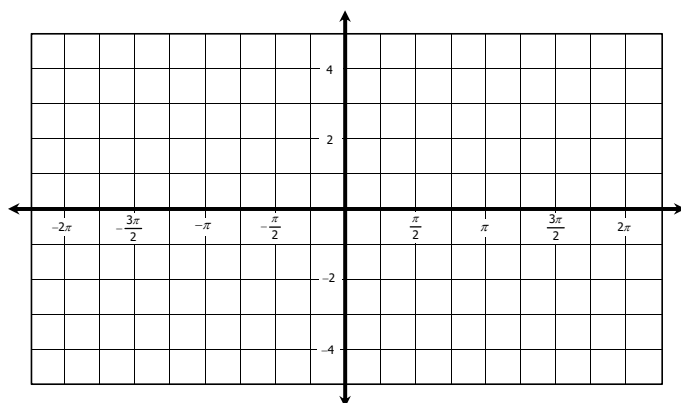
10. $y = \frac{1}{2} \cdot \cot 2\theta$ Amp: _____ Period: _____



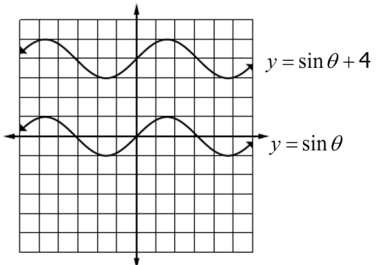
11. $y = \frac{5}{2} \cdot \sec \frac{4}{3}\theta$ Amp: _____ Period: _____



12. $y = 3 \cdot \cos \frac{1}{2}\theta$ Amp: _____ Period: _____

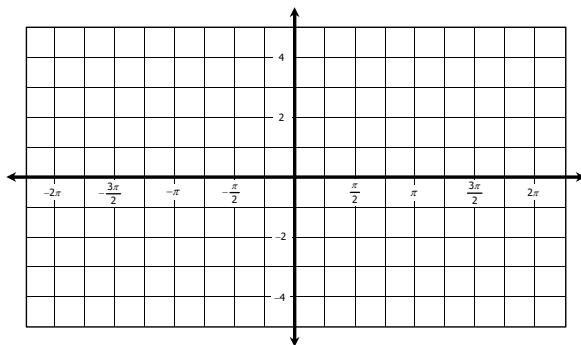


Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
TRANSFORMATIONS: Phase Shifts & Vertical Shifts $y = a \cdot \sin b(\theta - h) + k$ $y = a \cdot \cos b(\theta - h) + k$ $y = a \cdot \tan b(\theta - h) + k$	Like other functions, trigonometric functions can be translated. ➤ Phase Shifts: (Horizontal Shifts) $+h$ shifts _____ and $-h$ shifts _____. - Examples: (a) $y = \sin(\theta - \pi)$ shifts _____ units to the _____. (b) $y = \tan\left(\theta + \frac{\pi}{4}\right)$ shifts _____ units to the _____. ➤ Vertical Shifts: $+k$ shifts _____ and $-k$ shifts _____. - Examples: (c) $y = \tan \theta + 2$ shifts _____ units _____. (d) $y = \cos \theta - 4$ shifts _____ units _____.	
PUTTING IT ALL TOGETHER	Using the sine function as an example, label each part: $y = a \cdot \sin b(\theta - h) + k$	
MIDLINE	- Shifting a graph vertically will create a new horizontal axis, called the _____. - This is the reference line to which a graph oscillates. - The equation for the midline is _____.	
DESCRIBING TRANSFORMATIONS	Directions: State the amplitude, period, phase shift, vertical shift, and midline for each function.	
	1. $y = \frac{1}{2} \sin 3\theta - 5$ Amplitude: _____ Period: _____ Phase Shift: _____ Vertical Shift: _____ Midline: _____	2. $y = 4 \cdot \tan 2(\theta + 2\pi) + 1$ Amplitude: _____ Period: _____ Phase Shift: _____ Vertical Shift: _____ Midline: _____

Directions: State the amplitude, period, phase shift, vertical shift, and midline for each function. Then graph each function.

3. $y = \sin 2\left(\theta + \frac{\pi}{2}\right)$



Amplitude: _____

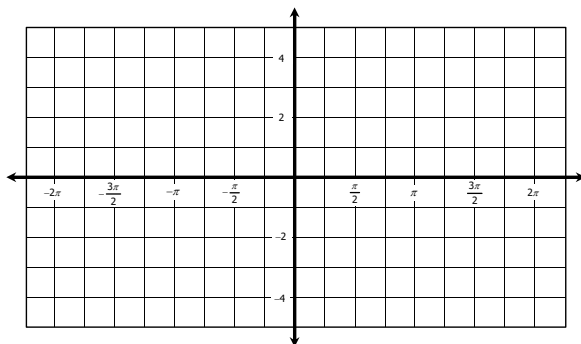
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

4. $y = \frac{1}{2} \cdot \tan \theta - 3$



Amplitude: _____

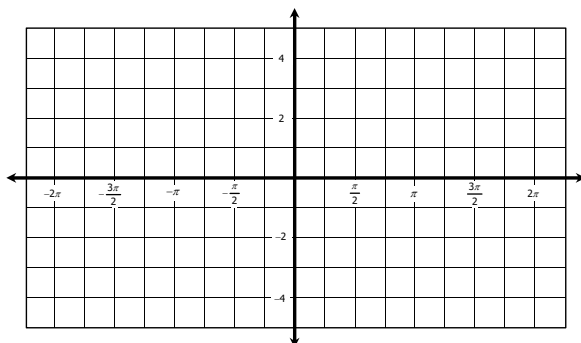
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

5. $y = 4 \cdot \cos\left(\theta - \frac{3\pi}{4}\right) - 1$



Amplitude: _____

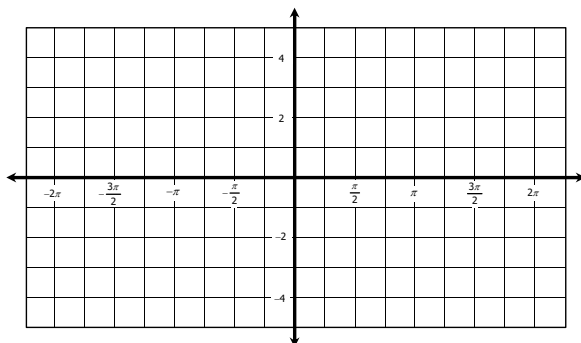
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

6. $y = 2 \cdot \tan\left(\theta + \frac{\pi}{4}\right) - 3$



Amplitude: _____

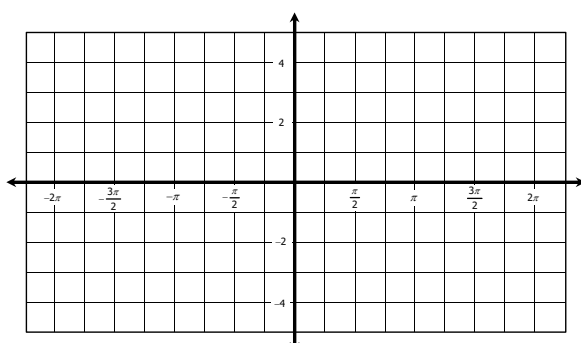
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

7. $y = \frac{5}{2} \cdot \sin(\theta + \pi) + 2$



Amplitude: _____

Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

Name: _____

Unit 12: Trigonometry

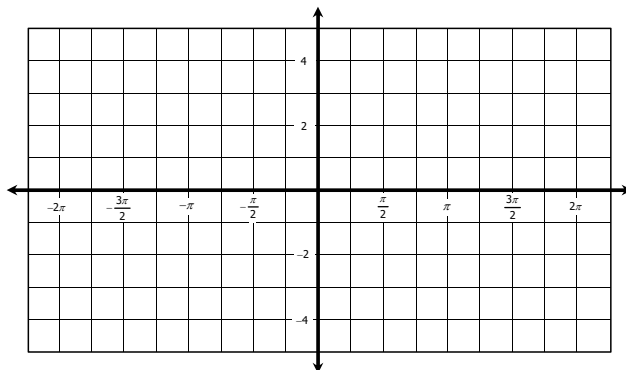
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Homework 10: Translating Trigonometric Graphs

**** This is a 2-page document! ****

Directions: State the amplitude, period, phase shift, vertical shift, and midline for each function. Then graph.

1. $y = \sin(\theta + \pi)$



Amplitude: _____

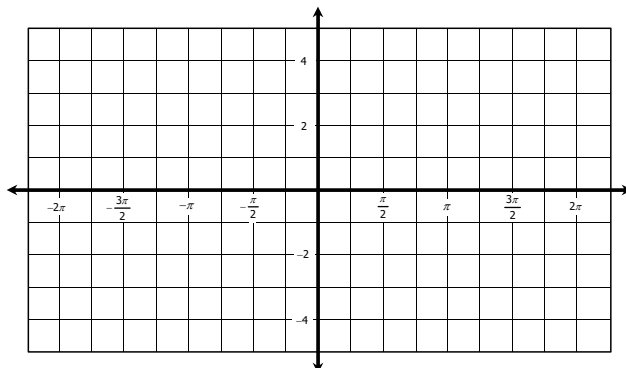
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

2. $y = \tan \theta - 4$



Amplitude: _____

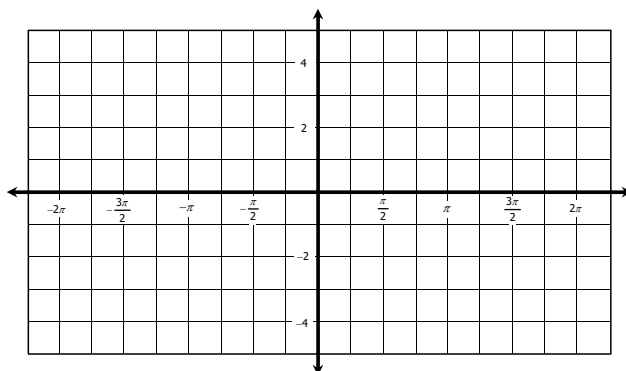
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

3. $y = 2 \cdot \cos\left(\theta - \frac{\pi}{4}\right)$



Amplitude: _____

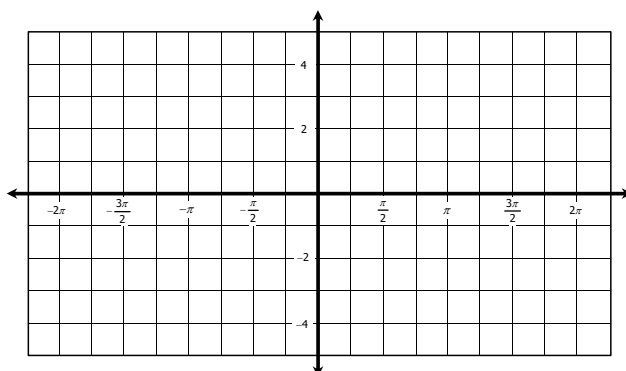
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

4. $y = 3 \cdot \sin 2\theta - 1$



Amplitude: _____

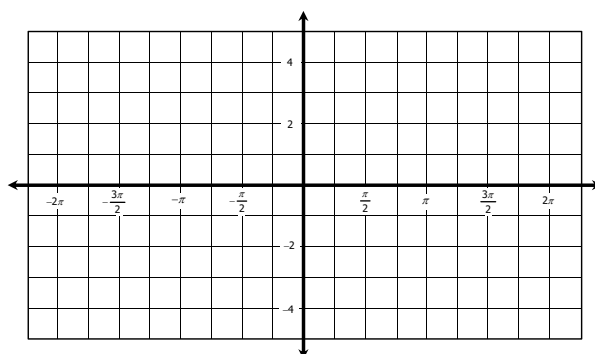
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

5. $y = \frac{1}{2} \cdot \tan\left(\theta - \frac{\pi}{2}\right)$



Amplitude: _____

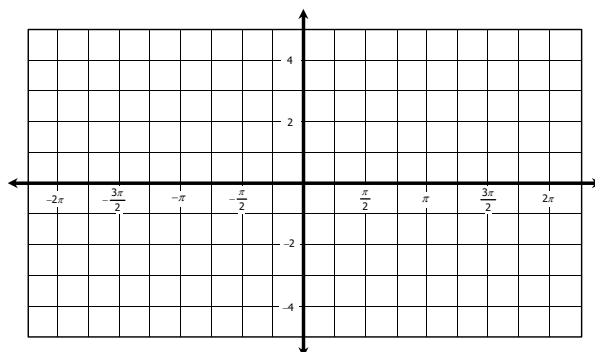
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

6. $y = 2 \cdot \cos\frac{1}{2}\theta + 3$



Amplitude: _____

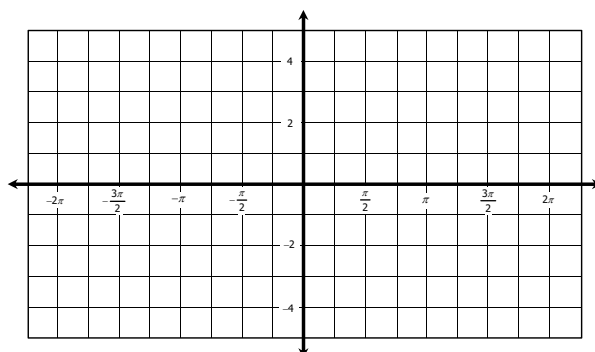
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

7. $y = 3 \cdot \tan 2\left(\theta + \frac{5\pi}{4}\right) - 2$



Amplitude: _____

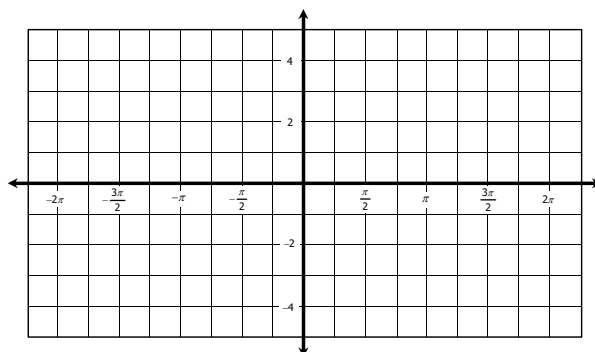
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

8. $y = 4 \cdot \sin 2(\theta - \pi) + 1$



Amplitude: _____

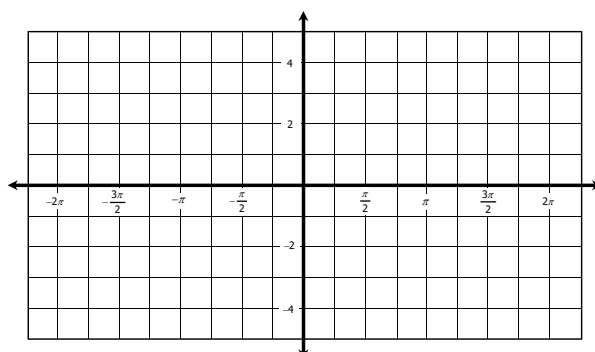
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

9. $y = \frac{5}{2} \cdot \cos\frac{1}{2}\left(\theta + \frac{3\pi}{4}\right) - 2$



Amplitude: _____

Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

Name: _____

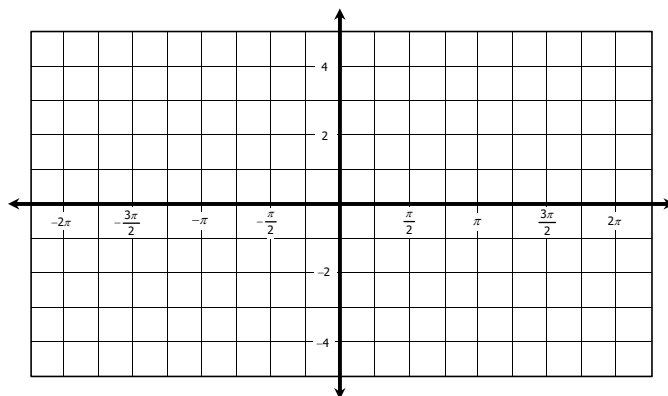
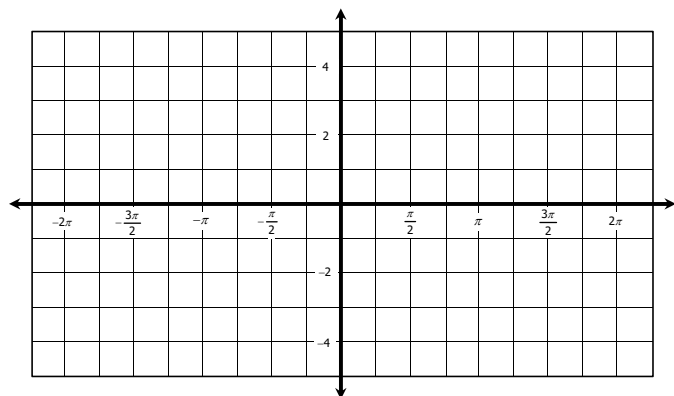
Algebra 2

Date: _____ Per: _____

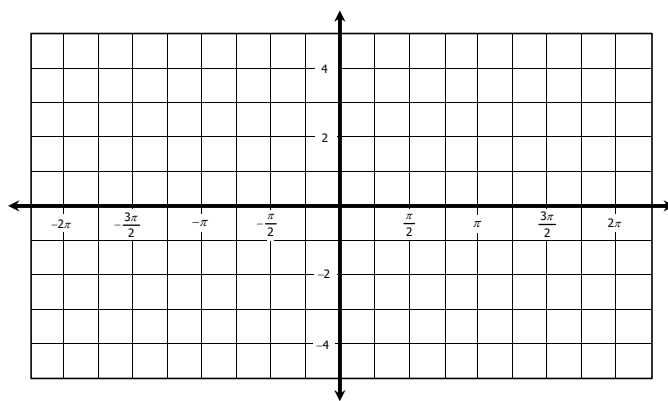
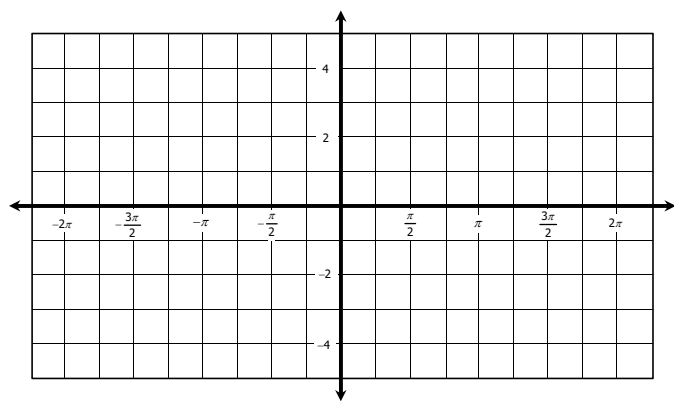
Unit 12: Trigonometry

Quiz 12-2: Graphing Trigonometric Functions**Directions:** Identify the amplitude, if possible, and period for each function. Then graph.

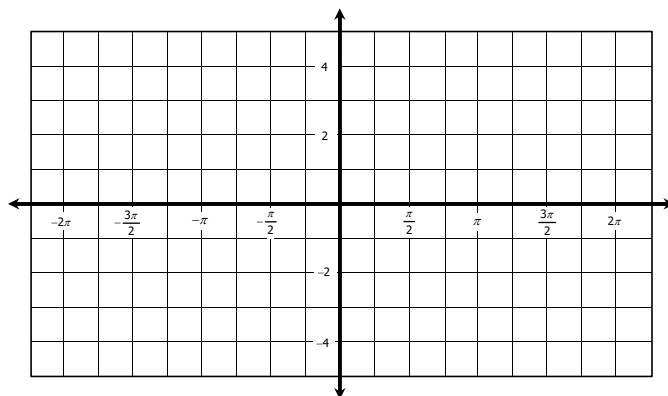
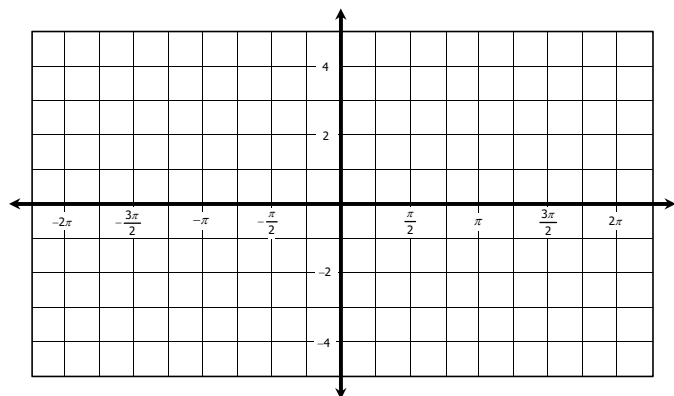
1. $y = \sin 2\theta$ Amp: _____ Per: _____ 2. $y = 4 \cdot \cos \theta$ Amp: _____ Per: _____



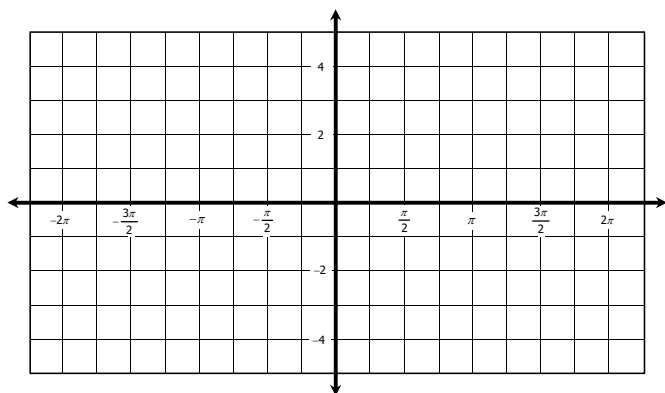
3. $y = \tan \frac{4}{3}\theta$ Amp: _____ Per: _____ 4. $y = \frac{1}{2} \cdot \sec \theta$ Amp: _____ Per: _____



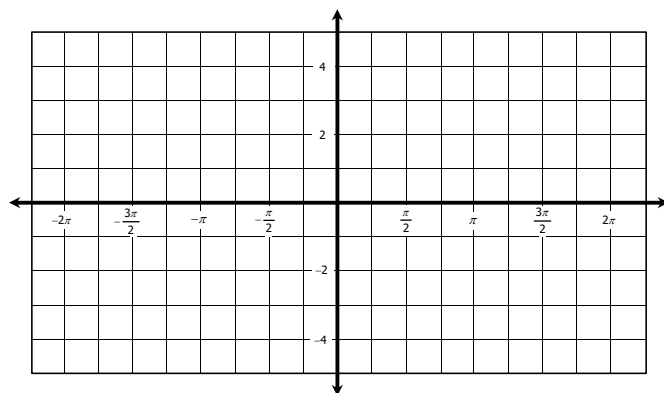
5. $y = 2 \cdot \csc \frac{8}{5}\theta$ Amp: _____ Per: _____ 6. $y = 2 \cdot \cot \frac{1}{2}\theta$ Amp: _____ Per: _____



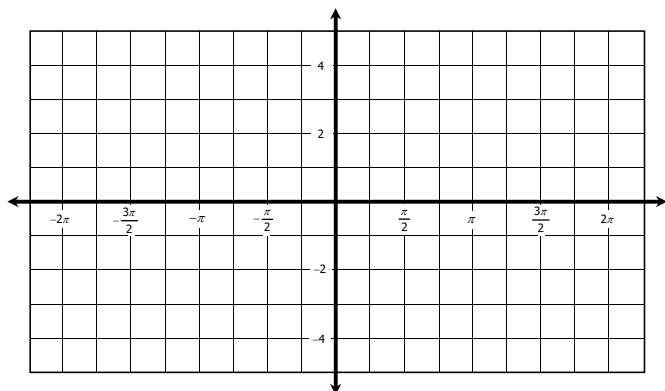
7. $y = \frac{1}{4} \cdot \tan \frac{1}{3} \theta$ Amp: _____ Per: _____



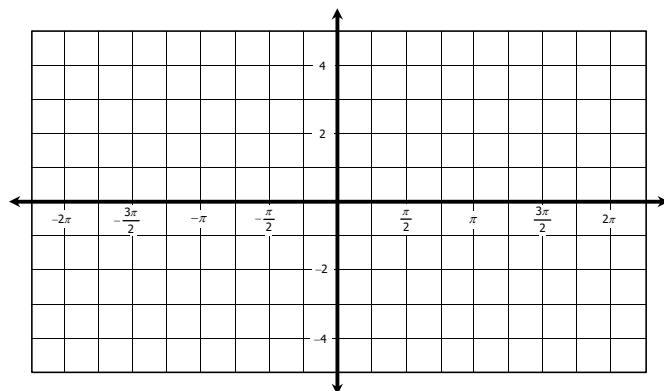
8. $y = \frac{7}{2} \cdot \sec 2\theta$ Amp: _____ Per: _____



9. $y = 5 \cdot \sin \frac{1}{2} \theta$ Amp: _____ Per: _____

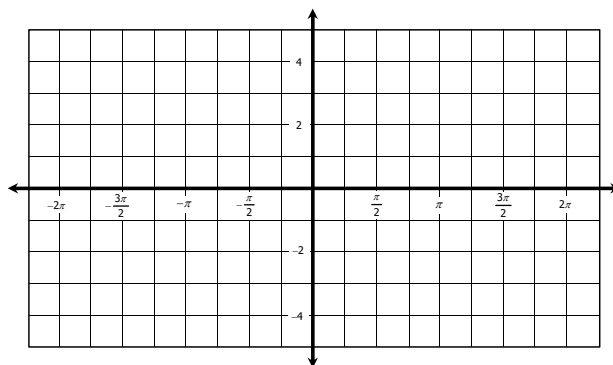


10. $y = \frac{3}{2} \cdot \csc \frac{2}{3} \theta$ Amp: _____ Per: _____



Directions: Identify the amplitude, if possible, period, translations, and midline. Then graph.

11. $y = 3 \cdot \cos(\theta - \pi) + 1$



Amplitude: _____

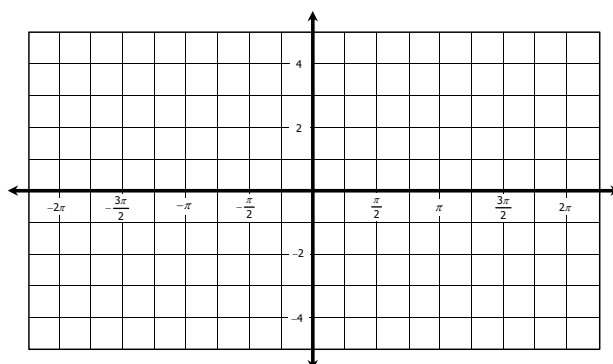
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

12. $y = \frac{1}{2} \cdot \tan\left(\theta + \frac{\pi}{4}\right) - 3$



Amplitude: _____

Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

Name:	Class:
Topic:	Date:

Main Ideas/Questions	Notes	
Trigonometric Identities	- A trigonometric identity is an equation involving trigonometric functions that is true for all values for which every expression in the equation is defined. - The identities help to evaluate functions, simplify expressions, prove identities, and solve equations.	
Fundamental Trigonometric Identities	Quotient Identities	$\tan \theta = \frac{\sin \theta}{\cos \theta}; \cos \theta \neq 0$ $\cot \theta = \frac{\cos \theta}{\sin \theta}; \sin \theta \neq 0$
	Reciprocal Identities	$\sin \theta = \frac{1}{\csc \theta}; \csc \theta \neq 0$ $\csc \theta = \frac{1}{\sin \theta}; \sin \theta \neq 0$ $\cos \theta = \frac{1}{\sec \theta}; \sec \theta \neq 0$ $\sec \theta = \frac{1}{\cos \theta}; \cos \theta \neq 0$ $\tan \theta = \frac{1}{\cot \theta}; \cot \theta \neq 0$ $\cot \theta = \frac{1}{\tan \theta}; \tan \theta \neq 0$
	Pythagorean Identities	$\cos^2 \theta + \sin^2 \theta = 1$ $\tan^2 \theta + 1 = \sec^2 \theta$ $\cot^2 \theta + 1 = \csc^2 \theta$
Finding Trig Function Values	1. Find the exact value of $\sin \theta$ if $\cos \theta = -\frac{4}{7}$ and $\frac{\pi}{2} < \theta < \pi$.	
	2. Find the exact value of $\cos \theta$ if $\cot \theta = \frac{2}{5}$ and $180 < \theta < 270^\circ$.	
	3. Find the exact value of $\tan \theta$ if $\csc \theta = -2$ and $\frac{3\pi}{2} < \theta < 2\pi$.	

Simplify Expressions

To **simplify a trigonometric expression** means to rewrite it as a numerical expression or in terms of a single trigonometric function, if possible. Use the identities to help you simplify the expressions below.

4. $\sin \theta \cdot \sec \theta$

5. $\tan \theta \cdot \cos \theta$

6. $2(\sec^2 \theta - \tan^2 \theta)$

7. $\cos^2 \theta - \sin^2 \theta$

8. $\frac{\sin \theta \cdot \cos \theta}{\tan \theta}$

9. $\frac{2 - 2\sin^2 \theta}{\cos \theta}$

10. $\frac{\sec \theta \cdot \sin^2 \theta}{1 + \sec \theta}$

11. $\sin^2 \theta - \cos^2 \theta \cdot \sin^2 \theta$

Name:	Class:
Topic:	Date:

Main Ideas/Questions	Notes
PROVING IDENTITIES	- Proving an identity means to show that both sides of the equation are the same. - Simplify one side using the basic trigonometric identities of the equation until it is the same as the other side. Ideally, pick the side that is more complicated to work with.
Directions: Prove each identity below.	
1. $\frac{\tan \theta}{\sec \theta} = \sin \theta$	2. $1 + \sec^2 \theta \cdot \sin^2 \theta = \sec^2 \theta$
3. $\csc \theta = \frac{\cot \theta}{\sec \theta} + \sin \theta$	4. $\csc \theta - \cot \theta \cdot \cos \theta = \sin \theta$
5. $\cos^2 \theta = \frac{\csc \theta \cdot \cos \theta}{\tan \theta + \cot \theta}$	6. $\frac{\sec \theta}{\cos \theta} - \frac{\tan \theta}{\cot \theta} = 1$

7. $\cos^2 \theta - \sin^2 \theta = 1 - 2\sin^2 \theta$

8. $\sec^2 \theta \cdot \cot^2 \theta - 1 = \cot^2 \theta$

9. $\sec^2 \theta = \frac{\csc^2 \theta}{\csc^2 \theta - 1}$

10. $\cot \theta + 1 = \csc \theta \cdot (\cos \theta + \sin \theta)$

11. $\cos \theta + \sin \theta \cdot \tan \theta = \sec \theta$

12. $\frac{\cot \theta - \tan \theta}{\sin \theta \cdot \cos \theta} = \csc^2 \theta - \sec^2 \theta$

Name: _____

Unit 12: Trigonometry

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Homework 11: Fundamental Trig Identities

**** This is a 2-page document! ******Directions:** Find the exact value of each expression if $90^\circ < \theta < 180^\circ$.

1. If $\cos \theta = -\frac{1}{2}$, find $\cot \theta$.

2. If $\csc \theta = \frac{4}{3}$, find $\sec \theta$.

Directions: Find the exact value of each expression if $270^\circ < \theta < 360^\circ$.

3. If $\tan \theta = -\frac{1}{4}$, find $\sin \theta$.

4. If $\sec \theta = \frac{\sqrt{15}}{3}$, find $\csc \theta$.

Directions: Simplify each expression.

5. $\csc \theta \cdot \tan \theta$

6. $\frac{(1 + \sin \theta)(1 - \sin \theta)}{\cot^2 \theta}$

7. $\frac{1 - \cot^2 \theta \cdot \sec^2 \theta}{\cot^2 \theta}$

8. $\frac{\cos \theta \cdot \sin^2 \theta + \cos^3 \theta}{\sin \theta}$

9. $\csc^2 \theta \cdot \tan^2 \theta \cdot \cos \theta$

10. $\frac{\sec^2 \theta - \tan^2 \theta}{\tan \theta \cdot \cos \theta}$

Directions: Prove each identity.

11. $\cos \theta \cdot \cot \theta + \sin^2 \theta \cdot \csc \theta = \csc \theta$

12. $\csc \theta - \sin \theta = \frac{\cot^2 \theta}{\csc \theta}$

13. $\frac{\cos^2 \theta + \sin^2 \theta}{1 + \tan^2 \theta} = \cos^2 \theta$

14. $\tan^2 \theta = \csc^2 \theta \cdot \tan^2 \theta - 1$

15. $\frac{\csc \theta \cdot \cos \theta}{\tan \theta + \cot \theta} = \cos^2 \theta$

16. $\frac{\cos \theta}{1 - \sin \theta} - \tan \theta = \sec \theta$

Name:	Class:
Topic:	Date:

Main Ideas/Questions	Notes	
Sum and Difference of Angles Identities	Sum of Angles	Difference of Angles
	$\sin(A + B) = \sin A \cdot \cos B + \cos A \cdot \sin B$ $\cos(A + B) = \cos A \cdot \cos B - \sin A \cdot \sin B$ $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$	$\sin(A - B) = \sin A \cdot \cos B - \cos A \cdot \sin B$ $\cos(A - B) = \cos A \cdot \cos B + \sin A \cdot \sin B$ $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \cdot \tan B}$
Finding Trig Values	Directions: Find the exact value of each expression. Use the sum and difference identities to break the value apart into the unit circle angles!	
	1. $\sin 165^\circ$	2. $\cos (-75^\circ)$
	3. $\tan \frac{\pi}{12}$	4. $\cos \left(-\frac{19\pi}{12} \right)$
	5. $\sin \frac{7\pi}{12}$	6. $\tan \left(-\frac{13\pi}{12} \right)$

Directions: Find the exact value of each expression by applying the identity.

7. Given angles A and B in Quadrant I with $\tan A = \frac{7}{24}$ and $\tan B = \frac{15}{8}$, find $\tan(A + B)$.

8. Given angles A and B in Quadrant I with $\sin A = \frac{5}{13}$ and $\cos B = \frac{4}{5}$, find $\sin(A + B)$.

9. Given angle A in Quadrant I with $\sin A = \frac{12}{13}$ and angle B in Quadrant II with $\sin B = \frac{4}{5}$, find $\cos(A - B)$.

Proving Identities

Directions: Use the sum and difference identities to verify each identity.

10. $\sin(\theta - \pi) = -\sin \theta$

11. $\cos\left(\frac{\pi}{2} + \theta\right) = -\sin \theta$

12. $\tan(\pi + \theta) = \tan \theta$

13. $\cos\left(\frac{3\pi}{2} - \theta\right) = -\sin \theta$

Name: _____

Unit 12: Trigonometry

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Homework 12: Sum and Difference of Angles
Identities

**** This is a 2-page document! ****

Directions: Find the exact value of each function.

1. $\cos 195^\circ$

2. $\sin(-345^\circ)$

3. $\tan \frac{5\pi}{12}$

4. $\sin\left(-\frac{17\pi}{12}\right)$

5. $\cos \frac{\pi}{12}$

6. $\tan\left(-\frac{11\pi}{12}\right)$

7. Given angles A and B in Quadrant I with $\tan A = 8$ and $\tan B = \frac{1}{2}$, find $\tan(A + B)$.

8. Given angles A and B in Quadrant I with $\sin A = \frac{8}{17}$ and $\cos B = \frac{3}{5}$, find $\sin(A + B)$.

9. Given angles A and B in Quadrant I with $\tan A = \frac{2}{3}$ and $\sin B = \frac{\sqrt{5}}{5}$, find $\tan(A - B)$.

10. Given angle A in Quadrant II with $\sin A = \frac{4}{5}$ and angle B in Quadrant III with $\cos B = -\frac{5}{13}$, find $\cos(A - B)$.

Directions: Prove each identity.

11. $\sin\left(\frac{3\pi}{2} - \theta\right) = -\cos \theta$

12. $\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$

13. $\tan(\pi - \theta) = -\tan \theta$

14. $\sin\left(\frac{\pi}{2} + \theta\right) = \cos \theta$

Name:	Class:
Topic:	Date:

Main Ideas/Questions	Notes		
Double-Angle Identities	SINE	COSINE	TANGENT
	$\sin 2\theta = 2 \cdot \sin \theta \cdot \cos \theta$	$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ $\cos 2\theta = 1 - 2 \cdot \sin^2 \theta$ $\cos 2\theta = 2 \cdot \cos^2 \theta - 1$	$\tan 2\theta = \frac{2 \cdot \tan \theta}{1 - \tan^2 \theta}$
Finding Trig Values	1. If $\sin \theta = \frac{5}{13}$ and $0 < \theta < \frac{\pi}{2}$, find $\cos 2\theta$.		
	2. If $\cos \theta = -\frac{7}{25}$ and $\pi < \theta < \frac{3\pi}{2}$, find $\tan 2\theta$.		
	3. If $\tan \theta = -\frac{20}{21}$ and $\frac{\pi}{2} < \theta < \pi$, find $\cos 2\theta$.		
Half-Angle Identities	SINE	COSINE	TANGENT
	$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$	$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$	$\tan \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$
Finding Trig Values	4. If $\cos \theta = -\frac{4}{5}$ and $\frac{\pi}{2} < \theta < \pi$, find $\sin \frac{\theta}{2}$.		

	5. If $\tan \theta = 2\sqrt{6}$ and $0 < \theta < \frac{\pi}{2}$, find $\cos \frac{\theta}{2}$.	
	6. If $\sin \theta = -\frac{8}{17}$ and $\pi < \theta < \frac{3\pi}{2}$, find $\tan \frac{\theta}{2}$.	
Proving Identities	Directions: Prove each identity.	
	7. $\cos^2 \theta - \cos 2\theta = \sin^2 \theta$	8. $(\cos \theta + \sin \theta)^2 = 1 + \sin 2\theta$
	9. $\tan \theta = \frac{1 - \cos 2\theta}{\sin 2\theta}$	10. $\sin 2\theta = \frac{2 \cdot \tan \theta}{1 + \tan^2 \theta}$

Name: _____

Unit 12: Trigonometry

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Homework 13: Double- and Half-Angle Identities

**** This is a 2-page document! ****

Directions: Use the information given to find the exact value of each trigonometric function.

1. $\cos \theta = -\frac{3}{5}$ and $\frac{\pi}{2} < \theta < \pi$; find $\sin 2\theta$.

2. $\tan \theta = \frac{12}{13}$ and $\pi < \theta < \frac{3\pi}{2}$; find $\cos 2\theta$.

3. $\sin \theta = -\frac{20}{29}$ and $\frac{3\pi}{2} < \theta < 2\pi$; find $\tan 2\theta$.

4. $\sin \theta = \frac{7}{25}$ and $0 < \theta < \frac{\pi}{2}$; find $\cos \frac{\theta}{2}$.

5. $\cos \theta = -\frac{4}{5}$ and $\frac{\pi}{2} < \theta < \pi$; find $\tan \frac{\theta}{2}$.

6. $\tan \theta = \sqrt{35}$ and $\pi < \theta < \frac{3\pi}{2}$; find $\sin \frac{\theta}{2}$.

Directions: Prove each identity.

7. $\cot \theta = \frac{\sin 2\theta}{1 - \cos 2\theta}$

8. $\sin \theta \cdot \cos \theta + \sin 2\theta = 3 \cdot \sin \theta \cdot \cos \theta$

9. $2 \sin^2 \theta + \cos 2\theta = 1$

10. $\sin 2\theta = 2 \cdot \cot \theta \cdot \sin^2 \theta$

11. $\sin 2\theta - 2 \cdot \sin \theta = (2 \cdot \sin \theta)(\cos \theta - 1)$

12. $\frac{2 \cdot \cos \theta}{\sin 2\theta} = \csc \theta$

Name: _____

Algebra 2

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Unit 12: Trigonometry

Quiz 12-3: Trigonometric Identities

Pythagorean Identities:

1. Given $\sin \theta = \frac{9}{41}$ and $0 < \theta < \frac{\pi}{2}$, find $\cos \theta$.

1. _____

2. Given $\sec \theta = -\frac{6}{5}$ and $\pi < \theta < \frac{3\pi}{2}$, find $\cot \theta$.

2. _____

Sum and Differences of Angles Identities:

3. Find the exact value of $\tan \frac{17\pi}{12}$.

3. _____

4. Find the exact value of $\sin \left(-\frac{5\pi}{12} \right)$.

4. _____

5. Given angles A and B in Quadrant I with $\sin A = \frac{12}{13}$ and $\cos B = \frac{4}{5}$, find:

a) $\cos(A + B)$

5a. _____

b) $\tan(A - B)$

b. _____

Double-Angle and Half-Angle Identities:

6. Given $\cos \theta = \frac{1}{4}$ and $\frac{3\pi}{2} < \theta < 2\pi$, find $\sin 2\theta$.

6. _____

7. Given $\sin \theta = \frac{\sqrt{5}}{5}$ and $0 < \theta < \frac{\pi}{2}$, find $\tan 2\theta$.

7. _____

8. Given $\sin \theta = -\frac{7}{25}$ and $\pi < \theta < \frac{3\pi}{2}$, find $\cos \frac{\theta}{2}$.

8. _____

9. Given $\tan \theta = -\frac{5}{12}$ and $\frac{\pi}{2} < \theta < \pi$, find $\tan \frac{\theta}{2}$.

9. _____

Simplifying Trigonometric Expressions:

10. $\frac{\sec \theta \cdot \sin \theta}{\tan \theta + \cot \theta}$

10. _____

$$11. \frac{\sec \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta}$$

11. _____

$$12. \frac{\cos \theta}{1 + \sin \theta} + \tan \theta$$

12. _____

Proving Trigonometric Identities. Show all steps!

$$13. \csc \theta = \sin \theta + \sin \theta \cdot \cot^2 \theta$$

$$14. \sec \theta + \tan \theta = \frac{\cos \theta}{1 - \sin \theta}$$

$$15. \frac{\tan \theta - \cot \theta}{\sin \theta \cdot \cos \theta} = \sec^2 \theta - \csc^2 \theta$$

$$16. \frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
TRIGONOMETRIC EQUATIONS	- A trigonometric equation can have several solutions, depending on the defined interval. - To solve a trigonometric equation, you must isolate the trigonometric function(s). - Watch out for extraneous solutions!	
BASIC EXAMPLES	Directions: Solve each equation for the given interval. For questions 1-4, give the solution(s) in degrees. For questions 5-6, give the solution(s) in radians.	
	1. $6 \sin \theta - 3 = 0; 0 \leq \theta \leq 360^\circ$	2. $4 = \cos \theta + 5; 0 \leq \theta \leq 180^\circ$
	3. $-2 \tan \theta = 2\sqrt{3}; 180^\circ \leq \theta \leq 360^\circ$	4. $5 \cos \theta + \sqrt{2} = 3 \cos \theta; 0^\circ \leq \theta \leq 270^\circ$
	5. $\sin \theta = 2 - \sin \theta; 0 \leq \theta \leq 2\pi$	6. $3 \tan \theta = \tan \theta + 2; \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$

EXAMPLES WITH FACTORING

Solve for the interval $0^\circ \leq \theta \leq 360^\circ$. Give all solutions in degrees.

7. $\sin^2 \theta + \sin \theta = 0$

8. $\cos \theta \cdot \tan \theta + \cos \theta = 0$

9. $\sin^2 \theta + 3 \sin \theta - 4 = 0$

10. $\cos^2 \theta - 2 \cos \theta - 3 = 0$

11. $4 \cos^2 \theta - 1 = 0$

12. $2 \sin^2 \theta + \sin \theta - 1 = 0$

13. $2 \cos^2 \theta - 5 \cos \theta = 3$

14. $3 \sin^2 \theta = 2 \sin \theta + 1$

USING IDENTITIES

Common Identities

Pythagorean
Reciprocal
Double Angle

Solve for the interval $0^\circ \leq \theta \leq 180^\circ$. Give all solutions in degrees.

15. $4(\cos 2\theta + \sin^2 \theta) = 3$

16. $\sin 2\theta - \cos \theta = 0$

17. $\cos \theta \cdot \tan \theta - \sin^2 \theta = 0$

18. $\cos^2 \theta - \cos \theta + 1 = \sin^2 \theta$

19. $\cos 2\theta + \cos \theta = 0$

20. $\cos 2\theta + 8\sin^2 \theta + 11\sin \theta + 3 = 0$

Name: _____

Unit 12: Trigonometry

Date: _____ Bell: _____

Homework 14: Trigonometric Equations

**** This is a 2-page document! ******Directions:** Solve each equation for the given interval. Give all solutions in **radians**.

1. $3 \tan \theta = -3; 0 \leq \theta \leq 2\pi$

2. $-4\sqrt{3} = 8 \cos \theta; 0 \leq \theta \leq \frac{3\pi}{2}$

3. $-2 \sin \theta = -\sqrt{2}; 0 \leq \theta \leq \pi$

4. $5 \tan \theta = 2\sqrt{3} + 3 \tan \theta; \pi \leq \theta \leq 2\pi$

5. $4 \cos \theta = \sqrt{2} + 2 \cos \theta; \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$

6. $6 \sin \theta - 2 = 4 \sin \theta; \frac{\pi}{2} \leq \theta \leq 2\pi$

Directions: Solve each equation for the interval $0^\circ \leq \theta \leq 360^\circ$. Give all solutions in **degrees**.

7. $2 \cos^2 \theta - \cos \theta = 0$

8. $3 \tan^2 \theta + \sqrt{3} \tan \theta = 0$

9. $3 \sin^2 \theta - 3 = 0$

10. $\cos^2 \theta + 2 \cos \theta = 3$

11. $\sin^2 \theta + 3\sin \theta + 7 = 5$

12. $2\cos^2 \theta - \cos \theta - 1 = 0$

13. $2\tan^2 \theta - \tan \theta = 3$

14. $4\cos^2 \theta + 3\cos \theta = 1$

15. $\sin^2 \theta = \cos \theta + 1$

16. $\sin 2\theta + \sin \theta = 0$

17. $\sec^2 \theta - \tan \theta \cdot \cot \theta = 1$

18. $3\cos 2\theta - 5\cos \theta = 1$

Unit 12 Test Study Guide (Trigonometry – Part II)

Name: _____

Date: _____ Block: _____

Topic 1: Graphing Trigonometric Functions

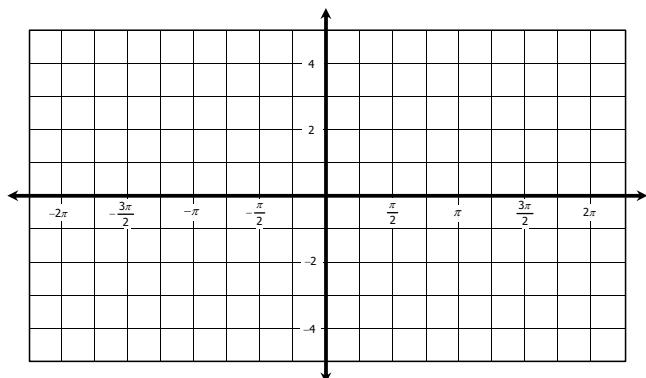
Given $y = a \cdot \sin b\theta$, $y = a \cdot \cos b\theta$, $y = a \cdot \tan b\theta$, $y = a \cdot \csc b\theta$, $y = a \cdot \sec b\theta$, or $y = a \cdot \cot b\theta$:

- The **amplitude** is _____.

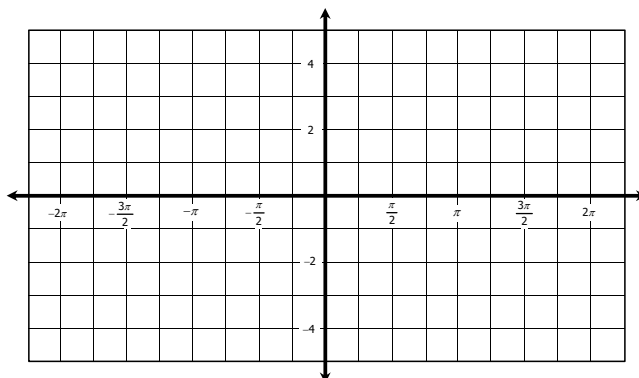
- For sin, cos, csc, and sec, the **period** is _____.
- For tan and cot, the **period** is _____.

Directions: Identify the amplitude, if possible, and period for each function. Then graph.

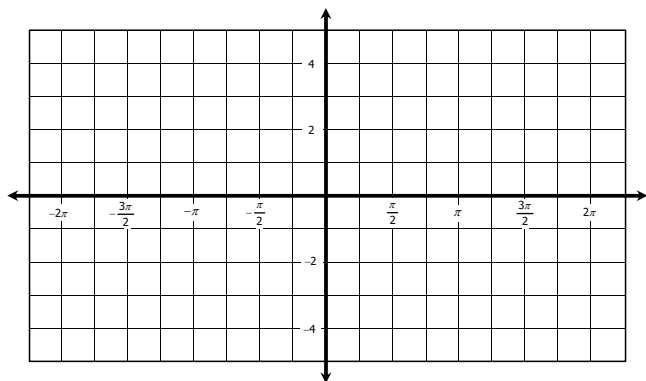
1. $y = 3 \cdot \sin \theta$ Amp: _____ Per: _____



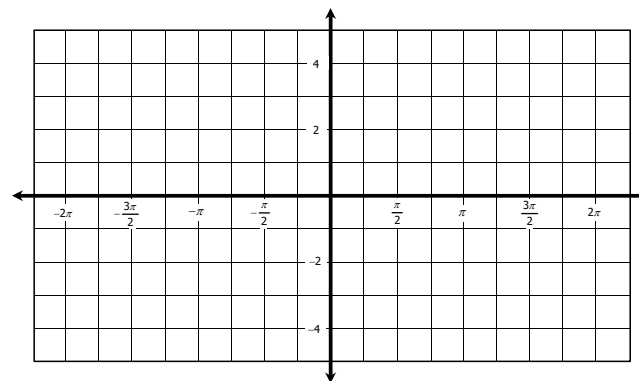
2. $y = 5 \cdot \cos \frac{1}{2} \theta$ Amp: _____ Per: _____



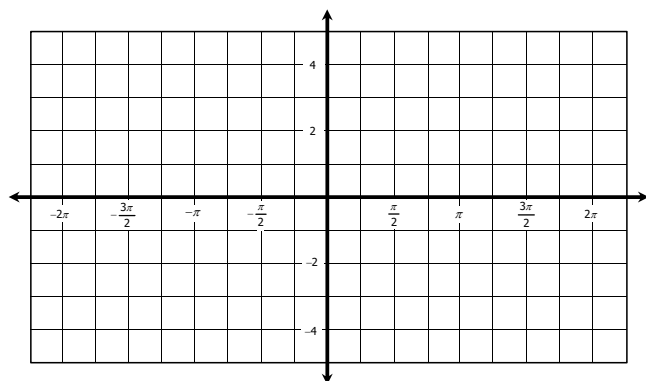
3. $y = \frac{1}{4} \cdot \tan 2\theta$ Amp: _____ Per: _____



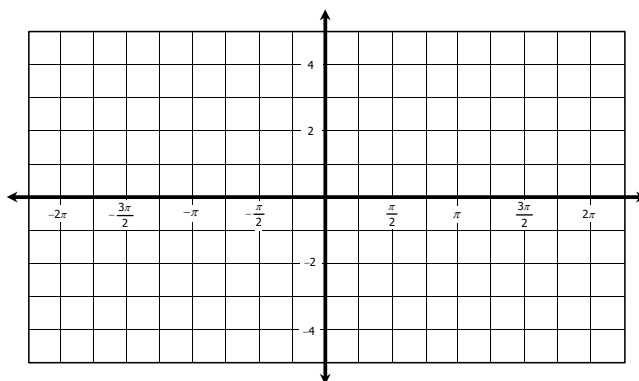
4. $y = \frac{1}{2} \cdot \csc \theta$ Amp: _____ Per: _____



5. $y = 2 \cdot \sec \frac{2}{3} \theta$ Amp: _____ Per: _____



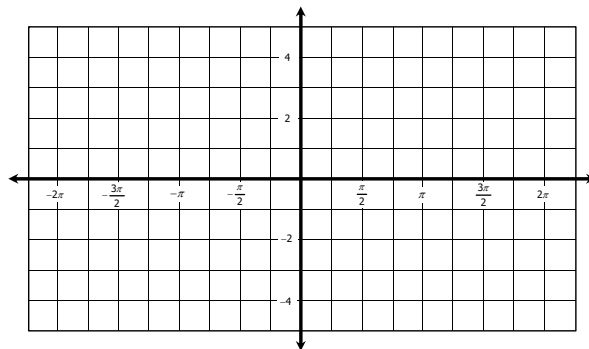
6. $y = \frac{1}{2} \cdot \cot \frac{1}{3} \theta$ Amp: _____ Per: _____



Topic 2: Translating Trigonometric Graphs

Directions: Identify the amplitude, if possible, period, translations, and midline. Then graph.

7. $y = \frac{3}{2} \cdot \sin 2\theta + 1$



Amplitude: _____

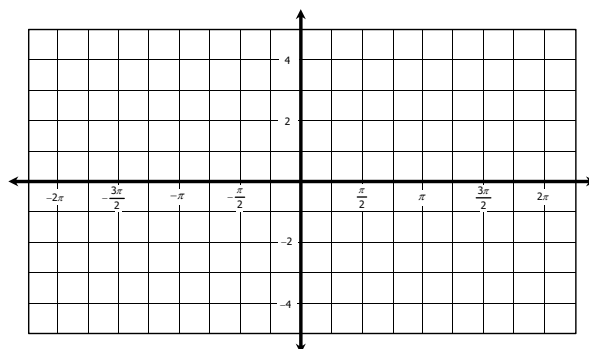
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

8. $y = \frac{1}{2} \cdot \tan\left(\theta - \frac{\pi}{4}\right)$



Amplitude: _____

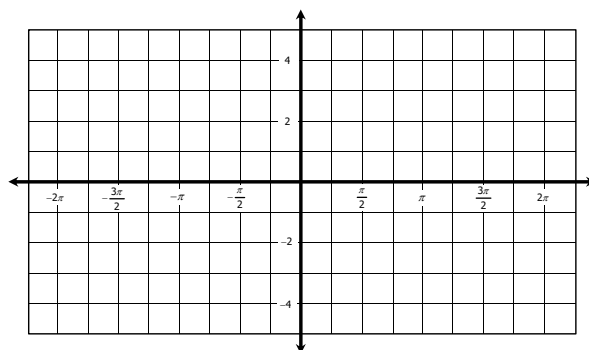
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

9. $y = 2 \cdot \cos \frac{1}{2}(\theta + \pi) - 3$



Amplitude: _____

Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

Topic 3: Fundamental Trigonometric Identities

Directions: Find the exact value of each function if $\frac{\pi}{2} < \theta < \pi$.

10. Find $\sec \theta$ if $\sin \theta = \frac{1}{3}$.

11. Find $\cot \theta$ if $\csc \theta = \frac{11}{5}$.

12. Find $\cos \theta$ if $\tan \theta = -\frac{4}{3}$.

13. Find $\tan \theta$ if $\sin \theta = \frac{3}{7}$.

Topic 4: Sum and Difference of Angles Identities**Directions:** Find the exact value of each function.

14. $\tan 345^\circ$

15. $\sin -195^\circ$

16. $\cos\left(-\frac{\pi}{12}\right)$

17. $\tan \frac{19\pi}{12}$

Directions: Find the exact value of each function if $\frac{3\pi}{2} < \theta < 2\pi$.

18. Find $\sin(A - B)$ if $\sin A = -\frac{3}{5}$ and $\cos B = \frac{12}{13}$.

19. Find $\cos(A + B)$ if $\cos A = \frac{8}{17}$ and $\tan B = -\frac{3}{4}$.

20. Find $\tan(A - B)$ if $\sin A = -\frac{7}{25}$ and $\sin B = -\frac{4}{5}$.

Topic 5: Double-Angle and Half-Angle Identities

Directions: Find the exact value of each function if $\pi < \theta < \frac{3\pi}{2}$.	
21. Find $\tan 2\theta$ if $\sin \theta = -\frac{\sqrt{2}}{4}$.	22. Find $\cos 2\theta$ if $\tan \theta = \frac{3}{5}$.
23. Find $\sin 2\theta$ if $\cos \theta = -\frac{24}{25}$.	24. Find $\cos \frac{\theta}{2}$ if $\sin \theta = -\frac{7}{25}$.
25. Find $\tan \frac{\theta}{2}$ if $\sin \theta = -\frac{5}{13}$.	26. Find $\sin \frac{\theta}{2}$ if $\tan \theta = \frac{4}{3}$.

Topic 6: Simplifying & Proving Trigonometric Identities

Directions: Simplify each expression.	
27. $\sec \theta - \sin \theta \cdot \tan \theta$	28. $\frac{\csc^2 \theta - 1}{\csc \theta}$

29. $\cos \theta \cdot \cot \theta \cdot \tan \theta \cdot \sec \theta$	30. $(1 + \cos \theta)(\csc \theta - \cot \theta)$
Directions: Prove each identity.	
31. $\frac{\csc \theta}{\sin \theta} - \frac{\cot \theta}{\tan \theta} = 1$	32. $\tan \theta + \cot \theta = \sec \theta \cdot \csc \theta$
33. $\cot \theta = \frac{\sin 2\theta}{1 - \cos 2\theta}$	34. $\tan \theta + \sin \theta = \frac{1 + \cos \theta}{\cot \theta}$

Topic 7: Trigonometric Equations

Directions: Solve each equation for the interval $0^\circ \leq \theta \leq 360^\circ$. Give all answers in **degrees**.

35. $9\sin \theta + 8 = 3\sin \theta + 5$	36. $4\cos^2 \theta - 3 = 0$
--	-------------------------------------

37. $2 \cos \theta \sin \theta + \sqrt{2} \sin \theta = 0$

38. $2 \tan^2 \theta + \sqrt{3} \tan \theta = \tan^2 \theta$

39. $\cos^2 \theta - 2 \cos \theta + 1 = 0$

40. $2 \sin^2 \theta - 9 \sin \theta = 5$

Directions: Solve each equation for the interval $0 \leq \theta \leq 2\pi$. Give all answers in **radians**.

41. $\sin \theta \cdot 2 \cot \theta = \sqrt{3}$

42. $4 \sin^4 \theta + 4 \sin^2 \theta \cdot \cos^2 \theta = 1$

43. $3(\sec^2 \theta - 1) = 1$

44. $\sin^2 \theta + \cos 2\theta - 3 \cos \theta = 4$

Name: _____

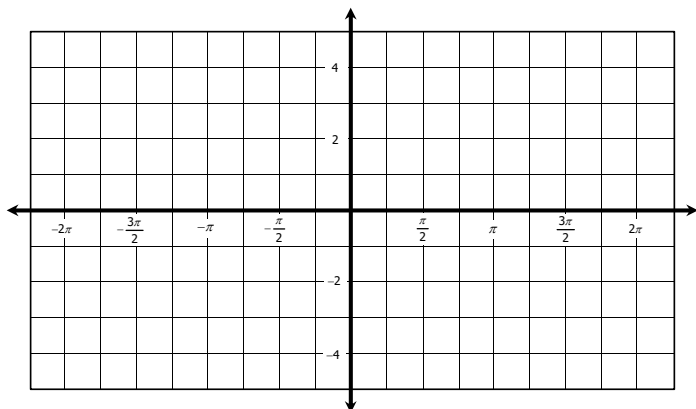
Unit 12 Test (Part II)

Date: _____ Bell: _____

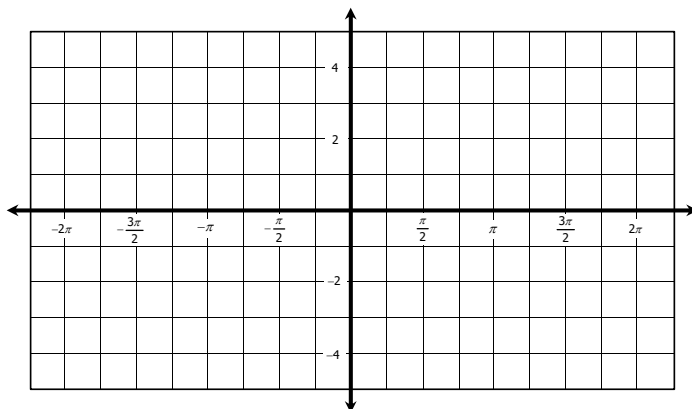
Trigonometry

Identify the amplitude (if possible) and period of each function, then graph.

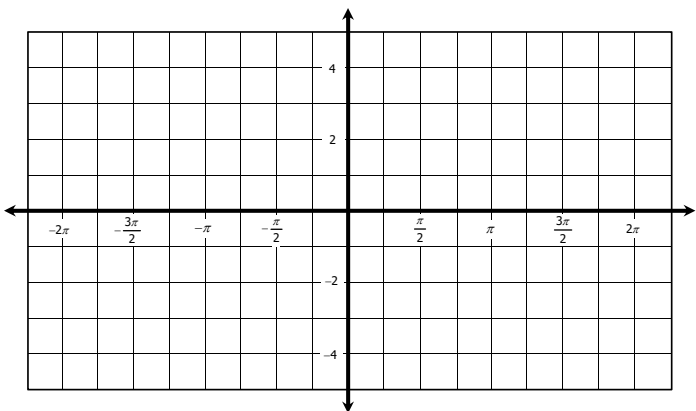
1. $y = \sin \frac{4}{3} \theta$ Amp: _____ Per: _____



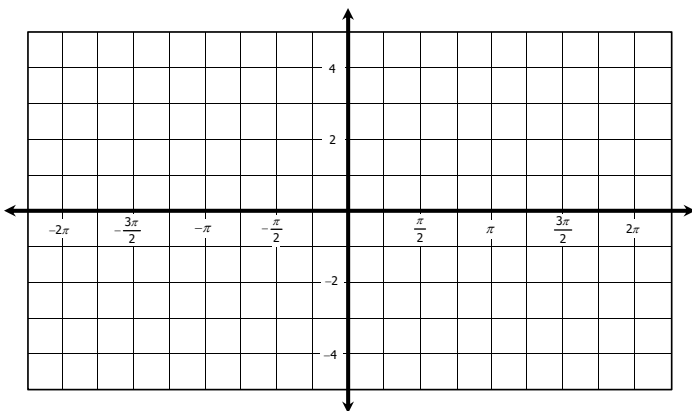
2. $y = \frac{1}{2} \cdot \tan \theta$ Amp: _____ Per: _____



3. $y = \frac{5}{2} \cdot \csc 4\theta$ Amp: _____ Per: _____

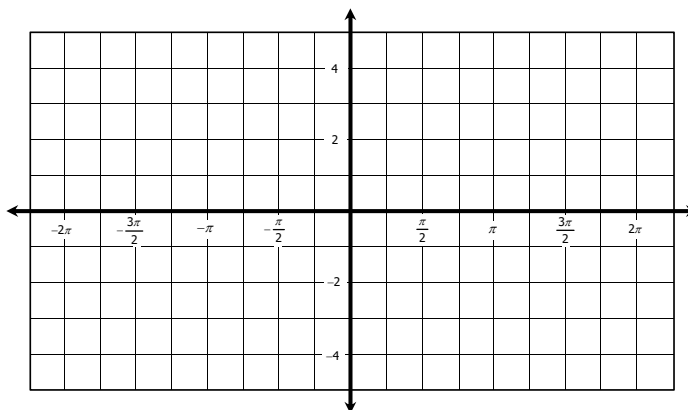


4. $y = 3 \cdot \sec \frac{1}{2} \theta$ Amp: _____ Per: _____



Identify the amplitude (if possible), period, translations, and midline. Then graph.

5. $y = 4 \cdot \cos 2 \left(\theta + \frac{\pi}{2} \right) - 1$



Amplitude: _____

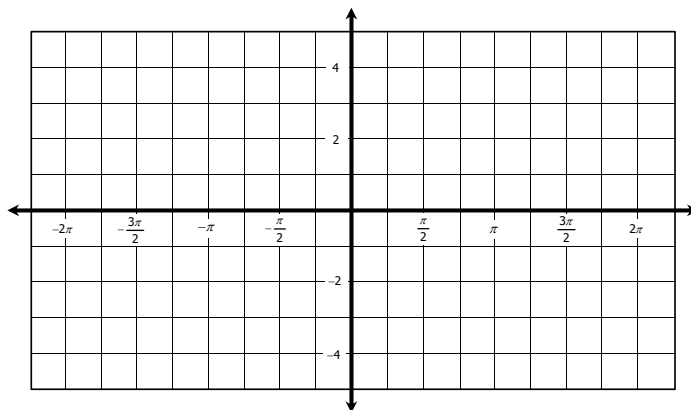
Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

6. $y = \tan \frac{2}{3}(\theta - \pi) + 3$



Amplitude: _____

Period: _____

Phase Shift: _____

Vertical Shift: _____

Midline: _____

7. Find $\sin \theta$ if $\cos \theta = \frac{5}{6}$ and $\frac{3\pi}{2} \leq \theta \leq 2\pi$.

A. $\frac{6\sqrt{11}}{11}$

C. $\frac{\sqrt{11}}{6}$

B. $-\frac{6\sqrt{11}}{11}$

D. $-\frac{\sqrt{11}}{6}$

8. Find $\sin \theta$ if $\cot \theta = -\frac{\sqrt{15}}{5}$ and $\frac{\pi}{2} \leq \theta \leq \pi$.

A. $\frac{2\sqrt{2}}{5}$

C. $\frac{\sqrt{10}}{4}$

B. $-\frac{2\sqrt{2}}{5}$

D. $-\frac{\sqrt{10}}{4}$

9. Find the exact value of $\sin 255^\circ$.

A. $\frac{\sqrt{6} + \sqrt{2}}{4}$

C. $\frac{\sqrt{2} - \sqrt{6}}{4}$

B. $\frac{\sqrt{6} - \sqrt{2}}{4}$

D. $\frac{-\sqrt{6} - \sqrt{2}}{4}$

10. Find the exact value of $\tan\left(\frac{5\pi}{12}\right)$.

A. $2 + \sqrt{3}$

C. $2 - \sqrt{3}$

B. $\sqrt{3} - 2$

D. $-2 - \sqrt{3}$

11. Find $(\cos A - B)$ if A and B are positive acute angles with $\sin A = \frac{12}{13}$ and $\cos B = \frac{3}{5}$.

A. $\frac{56}{65}$

C. $-\frac{33}{65}$

B. $\frac{63}{65}$

D. $-\frac{16}{65}$

12. Find $(\tan A + B)$ if A and B are in Quadrant II with $\tan A = -\frac{1}{3}$ and $\sin B = \frac{20}{29}$.

A. $-\frac{81}{83}$

C. $\frac{39}{43}$

B. $-\frac{81}{43}$

D. $\frac{39}{83}$

13. Find the exact value of $\sin 2\theta$ if $\tan \theta = -\frac{7}{24}$ and $\frac{\pi}{2} \leq \theta \leq \pi$.

A. $\frac{336}{625}$

C. $\frac{14}{25}$

B. $-\frac{336}{625}$

D. $-\frac{14}{25}$

14. Find the exact value of $\cos 2\theta$ if $\sin \theta = -\frac{7}{10}$ and $\pi \leq \theta \leq \frac{3\pi}{2}$.

A. $-\frac{1}{50}$

C. $-\frac{99}{50}$

B. $\frac{1}{50}$

D. $\frac{99}{50}$

15. Find the exact value of $\tan 2\theta$ if $\cos \theta = \frac{3}{5}$ and $0 \leq \theta \leq \frac{\pi}{2}$.

A. $\frac{7}{24}$

C. $-\frac{24}{7}$

B. $\frac{24}{25}$

D. $-\frac{25}{24}$

16. Find the exact value of $\cos \frac{\theta}{2}$ if $\sin \theta = -\frac{\sqrt{7}}{4}$ and $\pi \leq \theta \leq \frac{3\pi}{2}$.

A. $\frac{\sqrt{14}}{4}$

C. $\frac{\sqrt{2}}{4}$

B. $-\frac{\sqrt{14}}{4}$

D. $-\frac{\sqrt{2}}{4}$

17. Find the exact value of $\tan \frac{\theta}{2}$ if $\tan \theta = -\frac{12}{5}$ and $\frac{\pi}{2} \leq \theta \leq \pi$.

A. $\frac{3}{2}$

C. $-\frac{3}{2}$

B. 5

D. -5

Simplify each expression.

18. $\frac{\cot \theta \cdot \sin \theta}{\cot \theta \cdot \tan \theta - \sin^2 \theta}$

A. $\sin \theta$

B. $\cos \theta$

C. $\csc \theta$

D. $\sec \theta$

19. $\cos \theta \cdot \csc \theta (\sec^2 \theta - 1)$

A. $\tan \theta$

B. $\cot \theta$

C. $\cos \theta$

D. $\sec \theta$

20. $\frac{\tan \theta}{\tan \theta + \cot \theta}$

A. $\sec \theta$

B. $\csc \theta$

C. $\sin^2 \theta$

D. $\cos^2 \theta$

21. $\frac{\sin 2\theta \cdot \sec \theta}{1 - \cos^2 \theta}$

A. $2 \sec \theta$

B. $2 \csc \theta$

C. $\sec \theta - 1$

D. $\csc \theta - 1$

22. Which expressions are equivalent to $\cos \theta$? Check all that apply.

☐ $\frac{(\sec \theta + 1)(\sec \theta - 1)}{\sin \theta \cdot \sec \theta}$	☐ $\frac{\csc \theta}{\cot^2 \theta + 1}$	☐ $(\cos 2\theta + \sin^2 \theta) \cdot \sec \theta$
☐ $\sin \theta \cdot \cot \theta (\sin^2 \theta + \cos^2 \theta)$	☐ $\sin\left(\theta + \frac{\pi}{2}\right)$	☐ $\sin \theta + \cot \theta \cdot \cos \theta$

Prove each identity.

23. $\tan \theta (\cos \theta + \cot \theta) = \sin \theta + 1$

24. $\frac{1 + \cos 2\theta}{\sin 2\theta} = \cot \theta$

Solve each equation for the interval $0^\circ \leq \theta \leq 360^\circ$. Give all solutions in degrees.

25. $5 \tan \theta - \sqrt{3} = 2 \tan \theta$

26. $2 \sin \theta \cos \theta + \sqrt{2} \sin \theta = 0$

27. $\cos^2 \theta - 4\theta - 5 = 0$

28. $2\sin^2 \theta + \sin \theta + 1 = 4\sin \theta$

29. $\sin^2 \theta + \cos^2 \theta + 2\cos \theta = 1 + \sqrt{3}$

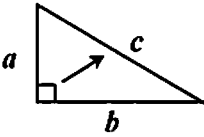
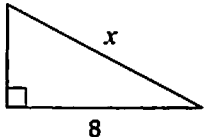
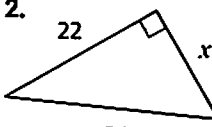
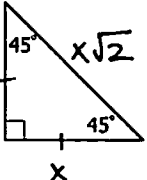
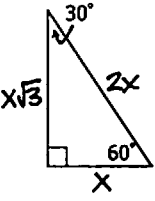
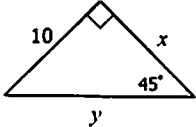
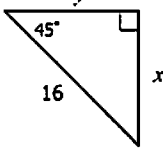
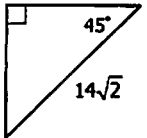
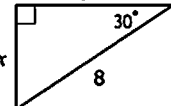
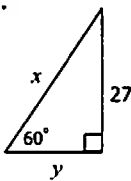
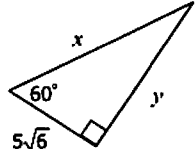
30. $6\sin^2 \theta + \cos 2\theta - 2 = 0$

Directions: Solve each equation for the given interval. Give all solutions in radians.

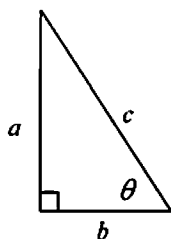
31. $\tan^2 \theta - \tan \theta = 0; 0 \leq \theta \leq \pi$

32. $2\sin \theta \cdot \cos \theta \cdot \cot = 1; \frac{\pi}{2} \leq \theta \leq 2\pi$

Name: _____	Class: _____
Topic: _____	Date: _____

Main Ideas/Questions	Notes
PYTHAGOREAN THEOREM	Used to find a side length on a right triangle. Formula: $a^2 + b^2 = c^2$ 
	Directions: Find the missing side. Give your answer in simplest radical form. 1.  $4^2 + 8^2 = x^2$ $16 + 64 = x^2$ $80 = x^2$ $x = 4\sqrt{5}$ 2.  $x^2 + 22^2 = 24^2$ $x^2 + 484 = 576$ $x^2 = 92$ $x = 2\sqrt{23}$
SPECIAL RIGHT TRIANGLES	45°-45°-90° 30°-60°-90°
	 Leg = <u>x</u> Leg = <u>x</u> Hypotenuse = <u>$x\sqrt{2}$</u>  Shorter Leg = <u>x</u> Longer Leg = <u>$x\sqrt{3}$</u> Hypotenuse = <u>2x</u>
	Directions: Find each missing side. Give your answer in simplest radical form. 3.  $x = \underline{10}$ $y = \underline{10\sqrt{2}}$ 4.  $\frac{16}{\sqrt{2}} = \frac{16\sqrt{2}}{2} = 8\sqrt{2}$ $x = \underline{8\sqrt{2}}$ $y = \underline{8\sqrt{2}}$
	5.  $\frac{14\sqrt{2}}{\sqrt{2}} = 14$ $x = \underline{14}$ $y = \underline{14}$ 6.  $x = \underline{4}$ $y = \underline{4\sqrt{3}}$
	7.  $\frac{27}{\sqrt{3}} = \frac{27\sqrt{3}}{3} = 9\sqrt{3}$ $x = \underline{18\sqrt{3}}$ $y = \underline{9\sqrt{3}}$ 8.  $x = \underline{10\sqrt{6}}$ $y = \underline{15\sqrt{2}}$ $5\sqrt{6} \cdot \sqrt{3} = 5\sqrt{18} = 15\sqrt{2}$

TRIGONOMETRIC FUNCTIONS



- A **trigonometric function** is a function whose rule is defined by a trigonometric ratio.
- A **trigonometric ratio** compares the lengths of two sides of the triangle.
- The Greek letter θ (theta) is used to represent the measure of an acute angle in a right triangle.

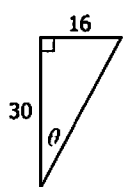
SINE	COSINE	TANGENT
$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{a}{c}$	$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{b}{c}$	$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{a}{b}$

RECIPROCAL FUNCTIONS

COSECANT $\left(\frac{1}{\sin \theta}\right)$	SECANT $\left(\frac{1}{\cos \theta}\right)$	COTANGENT $\left(\frac{1}{\tan \theta}\right)$
$\csc \theta = \frac{\text{hyp}}{\text{opp}} = \frac{c}{a}$	$\sec \theta = \frac{\text{hyp}}{\text{adj}} = \frac{c}{b}$	$\cot \theta = \frac{\text{adj}}{\text{opp}} = \frac{b}{a}$

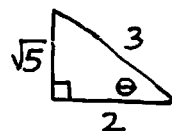
EXAMPLES

9. Find all six trig ratios for θ shown in the triangle below.



$$\begin{aligned} 16^2 + 30^2 &= x^2 \\ 256 + 900 &= x^2 \\ 1156 &= x^2 \\ 34 &= x \end{aligned}$$

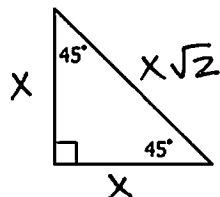
10. Given the ratio for $\cos \theta$, find the remaining ratios.



$$\begin{aligned} x^2 + 2^2 &= 3^2 \\ x^2 + 4 &= 9 \\ x^2 &= 5 \\ x &= \sqrt{5} \end{aligned}$$

$\sin \theta = \frac{16}{34} = \frac{8}{17}$	$\csc \theta = \frac{34}{16} = \frac{17}{8}$	$\sin \theta = \frac{\sqrt{5}}{3}$	$\csc \theta = \frac{3}{\sqrt{5}} = \frac{3\sqrt{5}}{5}$
$\cos \theta = \frac{30}{34} = \frac{15}{17}$	$\sec \theta = \frac{34}{30} = \frac{17}{15}$	$\cos \theta = \frac{2}{3}$	$\sec \theta = \frac{3}{2}$
$\tan \theta = \frac{16}{30} = \frac{8}{15}$	$\cot \theta = \frac{30}{16} = \frac{15}{8}$	$\tan \theta = \frac{\sqrt{5}}{2}$	$\cot \theta = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$

TRIG RATIOS OF SPECIAL ANGLES



$$\begin{aligned} \sin 45^\circ &= \frac{x}{x\sqrt{2}} = \frac{x\sqrt{2}}{2x} \\ &= \frac{\sqrt{2}}{2} \end{aligned}$$

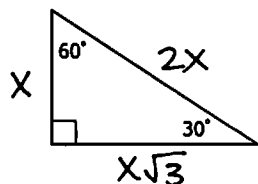
$$\begin{aligned} \cos 45^\circ &= \frac{x}{x\sqrt{2}} = \frac{x\sqrt{2}}{2x} \\ &= \frac{\sqrt{2}}{2} \end{aligned}$$

$$\tan 45^\circ = \frac{x}{x} = 1$$

$$\sin 30^\circ = \frac{x}{2x} = \frac{1}{2}$$

$$\cos 30^\circ = \frac{x\sqrt{3}}{2x} = \frac{\sqrt{3}}{2}$$

$$\begin{aligned} \tan 30^\circ &= \frac{x}{x\sqrt{3}} = \frac{x\sqrt{3}}{3x} \\ &= \frac{\sqrt{3}}{3} \end{aligned}$$



$$\sin 60^\circ = \frac{x\sqrt{3}}{2x} = \frac{\sqrt{3}}{2}$$

$$\cos 60^\circ = \frac{x}{2x} = \frac{1}{2}$$

$$\tan 60^\circ = \frac{x\sqrt{3}}{x} = \sqrt{3}$$

Name: _____

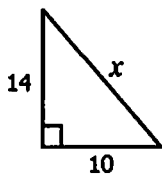
Unit 12 : Trigonometry

Date: _____ Bell: _____

Homework 1: Pythagorean Theorem, Special Right Triangles, & Trig Functions

**** This is a 2-page document! ******Directions:** Find each missing length. Give all answers in simplest radical form.

1.



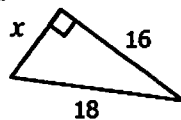
$$14^2 + 10^2 = x^2$$

$$196 + 100 = x^2$$

$$296 = x^2$$

$$2\sqrt{74} = x$$

2.



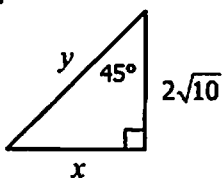
$$x^2 + 16^2 = 18^2$$

$$x^2 + 256 = 324$$

$$x^2 = 68$$

$$x = 2\sqrt{17}$$

3.



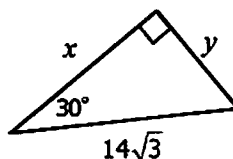
$$x = 2\sqrt{10}$$

$$y = 2\sqrt{10} \cdot \sqrt{2}$$

$$y = 2\sqrt{20}$$

$$y = 4\sqrt{5}$$

4.

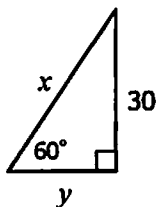


$$y = 7\sqrt{3}$$

$$x = 7\sqrt{3} \cdot \sqrt{3}$$

$$x = 21$$

5.

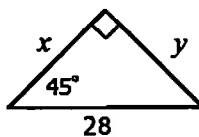


$$y = \frac{30}{\sqrt{3}} = \frac{30\sqrt{3}}{3}$$

$$y = 10\sqrt{3}$$

$$x = 20\sqrt{3}$$

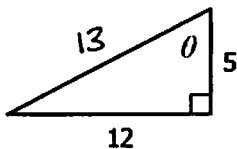
6.



$$x = \frac{28}{\sqrt{2}} = \frac{28\sqrt{2}}{2}$$

$$x = 14\sqrt{2}$$

$$y = 14\sqrt{2}$$

7. Find the values of the six trigonometric functions for θ .

$$5^2 + 12^2 = x^2$$

$$25 + 144 = x^2$$

$$169 = x^2$$

$$13 = x$$

$$\sin \theta = \frac{12}{13}$$

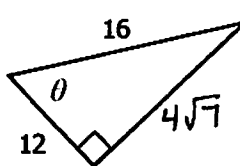
$$\csc \theta = \frac{13}{12}$$

$$\cos \theta = \frac{5}{13}$$

$$\sec \theta = \frac{13}{5}$$

$$\tan \theta = \frac{12}{5}$$

$$\cot \theta = \frac{5}{12}$$

8. Find the values of the six trigonometric functions for θ .

$$x^2 + 12^2 = 16^2$$

$$x^2 + 144 = 256$$

$$x^2 = 112$$

$$x = 4\sqrt{7}$$

$$\sin \theta = \frac{4\sqrt{7}}{16} = \frac{\sqrt{7}}{4}$$

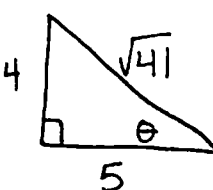
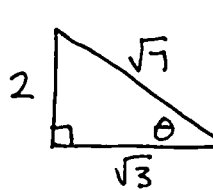
$$\csc \theta = \frac{16}{4\sqrt{7}} = \frac{4\sqrt{7}}{7}$$

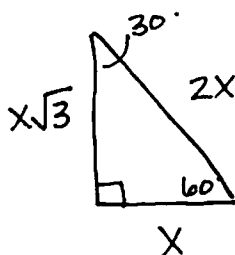
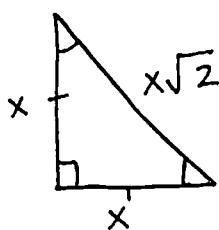
$$\cos \theta = \frac{12}{16} = \frac{3}{4}$$

$$\sec \theta = \frac{16}{12} = \frac{4}{3}$$

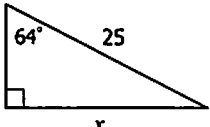
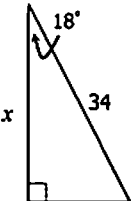
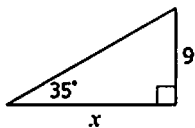
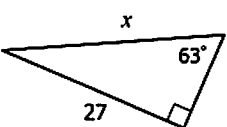
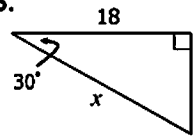
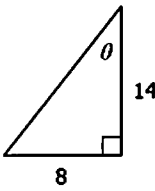
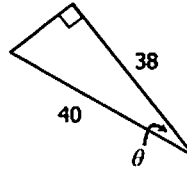
$$\tan \theta = \frac{4\sqrt{7}}{12} = \frac{\sqrt{7}}{3}$$

$$\cot \theta = \frac{12}{4\sqrt{7}} = \frac{3\sqrt{7}}{7}$$

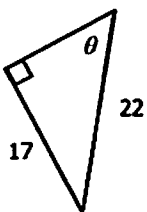
9. Given the ratio for $\tan \theta$, find the remaining ratios.  $4^2 + 5^2 = x^2$ $16 + 25 = x^2$ $41 = x^2$ $\sqrt{41} = x$	10. Given the ratio for $\csc \theta$, find the remaining ratios.  $2^2 + x^2 = \sqrt{7}^2$ $4 + x^2 = 7$ $x^2 = 3$ $x = \sqrt{3}$
$\sin \theta = \frac{4}{\sqrt{41}} = \frac{4\sqrt{41}}{41}$	$\sin \theta = \frac{2}{\sqrt{7}} = \frac{2\sqrt{7}}{7}$
$\cos \theta = \frac{5}{\sqrt{41}} = \frac{5\sqrt{41}}{41}$	$\cos \theta = \frac{\sqrt{3}}{\sqrt{7}} = \frac{\sqrt{21}}{7}$
$\tan \theta = \frac{4}{5}$	$\tan \theta = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$
Directions: Give the exact ratios for each function below.	
11. $\sin 45^\circ$ $\frac{x}{x\sqrt{2}} = \frac{x\sqrt{2}}{2x} = \frac{\sqrt{2}}{2}$	12. $\tan 60^\circ$ $\frac{x\sqrt{3}}{x} = \sqrt{3}$
13. $\cos 30^\circ$ $\frac{x\sqrt{3}}{2x} = \frac{\sqrt{3}}{2}$	14. $\cot 60^\circ$ $\frac{x}{x\sqrt{3}} = \frac{\sqrt{3}}{3}$
15. $\sec 45^\circ$ $\frac{x\sqrt{2}}{x} = \sqrt{2}$	16. $\csc 60^\circ$ $\frac{2x}{x\sqrt{3}} = \frac{2\sqrt{3}}{3}$



Name:	Class:
Topic:	Date:

Main Ideas/Questions	Notes				
FINDING SIDES LENGTHS	Directions: Find the value of x. Round to the nearest tenth.				
	1.  $\sin 64 = \frac{x}{25}$ $x = 22.5$	2.  $\tan 47 = \frac{x}{12}$ $x = 12.9$			
	3.  $\cos 18 = \frac{x}{34}$ $x = 32.3$	4.  $\tan 35 = \frac{9}{x}$ $x = 12.9$			
	5.  $\sin 63 = \frac{27}{x}$ $x = 30.3$	6.  $\cos 30 = \frac{18}{x}$ $x = 20.8$			
	If you know the sin, cosine, or tangent ratio of an angle, you can use the inverse of the ratio ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) to find the measure of the angle.				
	<table><tr><td>If $\sin \theta = x$, then $\theta = \sin^{-1}(x)$</td><td>If $\cos \theta = x$, then $\theta = \cos^{-1}(x)$</td><td>If $\tan \theta = x$, then $\theta = \tan^{-1}(x)$</td></tr></table>			If $\sin \theta = x$, then $\theta = \sin^{-1}(x)$	If $\cos \theta = x$, then $\theta = \cos^{-1}(x)$
If $\sin \theta = x$, then $\theta = \sin^{-1}(x)$	If $\cos \theta = x$, then $\theta = \cos^{-1}(x)$	If $\tan \theta = x$, then $\theta = \tan^{-1}(x)$			
FINDING ANGLES MEASURES	Directions: Find the missing angle. Round to the nearest tenth.				
	7.  $\tan \theta = \frac{8}{14}$ $\theta = \tan^{-1}\left(\frac{8}{14}\right)$ $\theta = 29.7^\circ$	8.  $\cos \theta = \frac{38}{40}$ $\theta = \cos^{-1}\left(\frac{38}{40}\right)$ $\theta = 18.2^\circ$			

9.

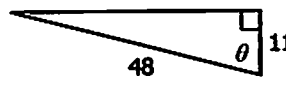


$$\sin \theta = \frac{17}{22}$$

$$\theta = \sin^{-1}(17/22)$$

$$\theta = 50.6^\circ$$

10.

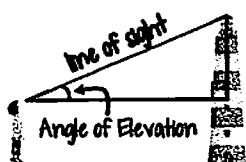


$$\cos \theta = \frac{11}{48}$$

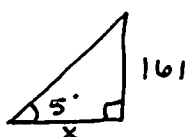
$$\theta = \cos^{-1}(11/48)$$

$$\theta = 76.8^\circ$$

ANGLE OF ELEVATION



11. The angle of elevation from a fishing boat to the top of a lighthouse is 5° . If the top of the lighthouse is 161 feet above sea level, how far is the boat from the lighthouse?

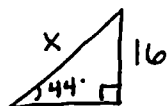


$$\tan 5 = \frac{161}{x}$$

$$x = \frac{161}{\tan 5}$$

$$x = 1840.2 \text{ ft}$$

12. The angle of elevation from the bottom of a slide to the top of the slide is 44° . If the top of the slide sits 16 feet above the ground, find the length of the slide. Assume it is a straight slide.

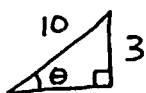


$$\sin 44 = \frac{16}{x}$$

$$x = \frac{16}{\sin 44}$$

$$x = 23 \text{ ft}$$

13. A 10 foot ramp is used to unload furniture from a truck. If the back of the truck sits 3 feet above the ground, find the measure of the angle that the ramp makes with the ground.

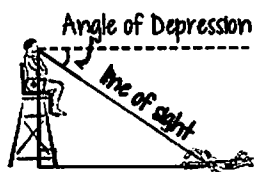


$$\sin \theta = \frac{3}{10}$$

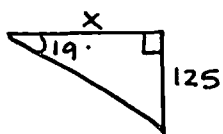
$$\theta = \sin^{-1}(3/10)$$

$$\theta = 17.5^\circ$$

ANGLE OF DEPRESSION



14. A fire is spotted from the top of a 125-foot tall fire tower. If the angle of depression to the fire is 19° , how far is the fire from the base of the tower?

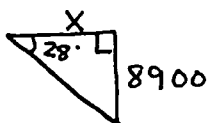


$$\tan 19 = \frac{125}{x}$$

$$x = \frac{125}{\tan 19}$$

$$x = 363 \text{ ft}$$

15. An airplane is flying at an altitude of 8,900 feet over water. The pilot spots a raft floating in the water at a 28° angle of depression. What is the horizontal distance from the airplane to the raft?

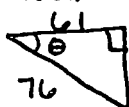


$$\tan 28 = \frac{8900}{x}$$

$$x = \frac{8900}{\tan 28}$$

$$x = 16738.5 \text{ ft}$$

16. Max is sitting in the stands at the baseball stadium. He catches a foul ball and decides to throw it back to a player standing on first base. If the horizontal distance from Max to the player is 61 feet and the ball travels 76 feet, what is the angle of depression from Max to the player?



$$\cos \theta = \frac{61}{76}$$

$$\theta = \cos^{-1}(61/76)$$

$$\theta = 36.6^\circ$$

Name: _____

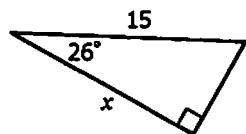
Unit 12: Trigonometry

Date: _____ Bell: _____

Homework 2: Finding Side and Angle Measures

**** This is a 2-page document! ******Directions:** Find each missing measure. Round all answers to the nearest tenth.

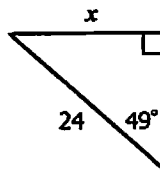
1.



$$\cos 26 = \frac{x}{15}$$

$$x = 13.5$$

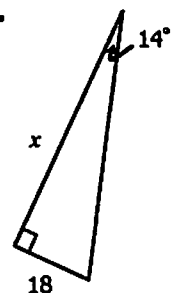
2.



$$\sin 49 = \frac{x}{24}$$

$$x = 18.1$$

3.

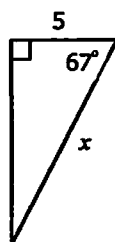


$$\tan 14 = \frac{18}{x}$$

$$x = \frac{18}{\tan 14}$$

$$x = 72.2$$

4.

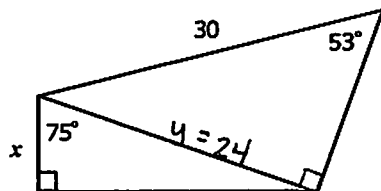


$$\cos 67 = \frac{5}{x}$$

$$x = \frac{5}{\cos 67}$$

$$x = 12.8$$

5.



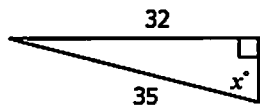
$$\sin 53 = \frac{y}{30}$$

$$y = 24$$

$$\cos 75 = \frac{x}{24}$$

$$x = 6.2$$

6.

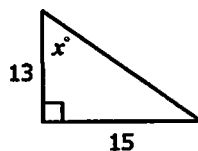


$$\sin x = \frac{32}{35}$$

$$x = \sin^{-1}(32/35)$$

$$x = 66.1^\circ$$

7.

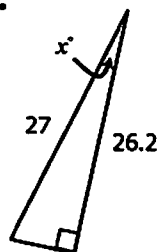


$$\tan x = \frac{15}{13}$$

$$x = \tan^{-1}(15/13)$$

$$x = 49.1^\circ$$

8.

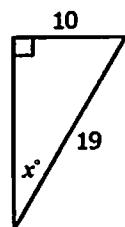


$$\cos x = \frac{26.2}{27}$$

$$x = \cos^{-1}(26.2/27)$$

$$x = 14^\circ$$

9.

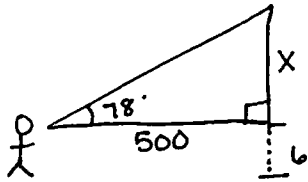


$$\sin x = \frac{10}{19}$$

$$x = \sin^{-1}(10/19)$$

$$x = 31.8^\circ$$

10. David is standing 500 feet away from the base of a cell phone tower. The angle of elevation from his eyes to the top of the tower is 78° . If David's eyes are 6 feet above the ground, how tall is the tower?

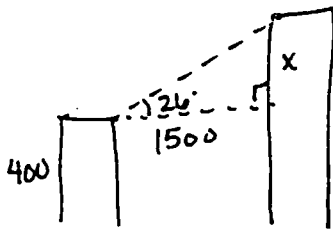


$$\tan 78 = \frac{x}{500}$$

$$x = 2352.3$$

$$2352.3 + 6 = \boxed{2358.3 \text{ ft}}$$

11. The angle of elevation from the top of one building to another is 26° . If the shorter building is 400 feet tall and the buildings are 1500 feet apart, how tall is the other building?

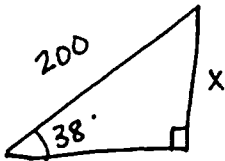


$$\tan 26 = \frac{x}{1500}$$

$$x = 731.6$$

$$731.6 + 400 = \boxed{1131.6 \text{ ft}}$$

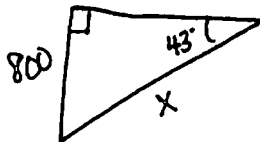
12. A 200-foot supporting cable is tied to the top of a cell phone tower then secured to a point on the ground. If the angle of elevation from the point on the ground to the top of the tower is 38° , find the height of the tower.



$$\sin 38 = \frac{x}{200}$$

$$x = 123.1 \text{ ft}$$

13. The angle of depression from the top to the bottom of a ski lift is 43° . If the vertical distance between the top and bottom of the lift is 800 feet, determine the length of the cable between them.

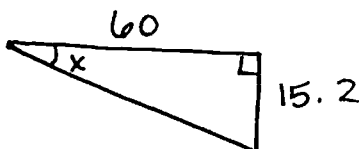


$$\sin 43 = \frac{800}{x}$$

$$x = \frac{800}{\sin 43}$$

$$x = 1173 \text{ ft}$$

14. An airplane is descending to a runway. At an altitude of 15.2 miles, its horizontal distance to the runway is 60 miles. Find the angle of descent.

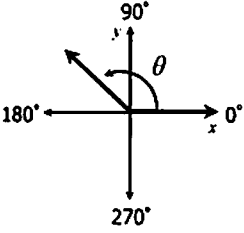
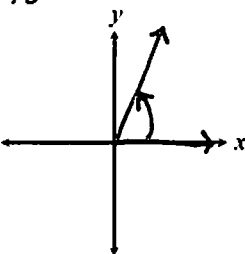
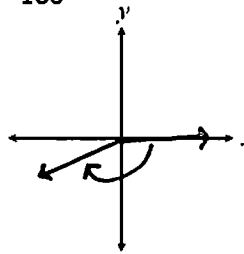
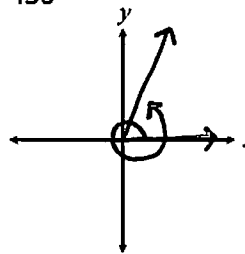
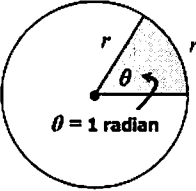


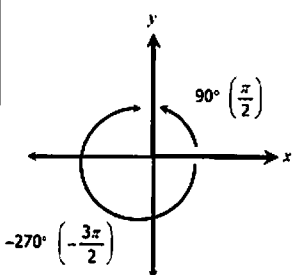
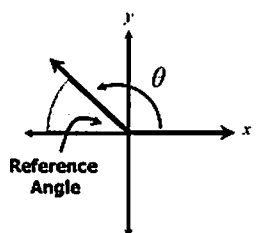
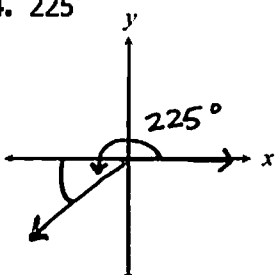
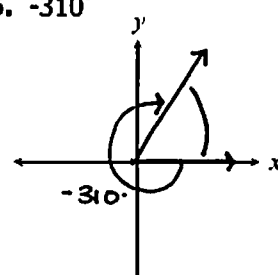
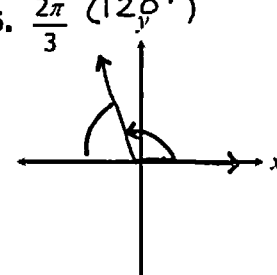
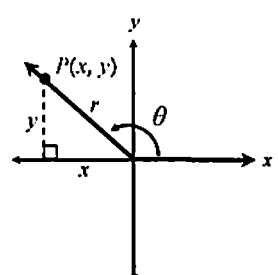
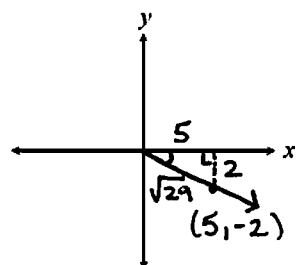
$$\tan x = \frac{15.2}{60}$$

$$x = \tan^{-1}(15.2/60)$$

$$x = 14.2^\circ$$

Name: _____	Class: _____
Topic: _____	Date: _____

Main Ideas/Questions	Notes		
Angles in Standard Form 	- An angle on the coordinate plane is in standard form when the vertex is on the origin and one ray lies on the positive x-axis. - The ray on the x-axis is called the <u>initial side</u>. - The other ray is called the <u>terminal side</u>. - Counterclockwise rotations result in <u>positive</u> angle measures. - Clockwise rotations result in <u>negative</u> angle measures. - One full revolution = <u>360°</u>.		
Drawing Angles	Directions: Sketch an angle with the given measure in standard position.		
	1. 75° 	2. -160° 	3. 430° 
Radians vs. Degrees 	A radian is a unit of angle measure based on arc length. One radian is defined as the measure of an angle formed when the radius is equivalent to the length of the intercepted arc. Recall that the circumference of a circle is $2\pi r$, therefore: $360^\circ = \underline{2\pi \text{ radians}}$; $180^\circ = \underline{\pi \text{ radians}}$		
	Converting Degrees → Radians		Converting Radians → Degrees
	Radians = Degrees $\cdot \left(\frac{\pi \text{ radians}}{180} \right)$		Degrees = Radians $\cdot \left(\frac{180}{\pi \text{ radians}} \right)$
Degrees → Radians	Directions: Convert each measure to radians.		
	4. 30° $30 \cdot \frac{\pi}{180} = \frac{30\pi}{180} = \boxed{\frac{\pi}{6}}$	5. 150° $150 \cdot \frac{\pi}{180} = \frac{150\pi}{180} = \boxed{\frac{5\pi}{6}}$	6. -220° $-220 \cdot \frac{\pi}{180} = \frac{-220\pi}{180} = \boxed{-\frac{11\pi}{9}}$
Radians → Degrees	Directions: Convert each measure to radians.		
	7. $\frac{4\pi}{3}$ $\frac{4\pi}{3} \cdot \frac{180}{\pi} = \frac{720}{3} = \boxed{240^\circ}$	8. $-\frac{5\pi}{36}$ $-\frac{5\pi}{36} \cdot \frac{180}{\pi} = \frac{-900}{36} = \boxed{-25^\circ}$	9. $\frac{7\pi}{4}$ $\frac{7\pi}{4} \cdot \frac{180}{\pi} = \frac{1260}{4} = \boxed{315^\circ}$

Coterminal Angles 	Angles in standard position with the same terminal side are coterminal angles. Give two coterminal angles for each given angle, one positive and one negative: 10. 65° $65 + 360 = \boxed{425^\circ}$ $65 - 360 = \boxed{-295^\circ}$ 11. 540° $540 - 360 = \boxed{180^\circ}$ $180 - 360 = \boxed{-180^\circ}$ 12. $\frac{13\pi}{18}$ $\frac{13\pi}{18} + 2\pi = \boxed{\frac{49\pi}{18}}$ $\frac{13\pi}{18} - 2\pi = \boxed{-\frac{23\pi}{18}}$ 13. $\frac{14\pi}{9}$ $\frac{14\pi}{9} + 2\pi = \boxed{\frac{32\pi}{9}}$ $\frac{14\pi}{9} - 2\pi = \boxed{-\frac{4\pi}{9}}$								
Reference Angles 	For an angle θ in standard form, the reference angle is the positive acute angle form by the terminal side and the x-axis. Sketch and find the reference angles for each angle: 14. 225°  $225 - 180 = \boxed{45^\circ}$ 15. -310°  $-310 + 360 = \boxed{50^\circ}$ 16. $\frac{2\pi}{3}$ (120°)  $\pi - \frac{2\pi}{3} = \boxed{\frac{\pi}{3}}$								
Trig Functions 	Let θ be an angle in standard form and $P(x, y)$ be a point on the terminal side of θ. The distance from P to the the origin, r, can be found using the formula: $\underline{x^2 + y^2 = r^2} \quad (\text{The Pythagorean Theorem}).$ <table><tr><td>$\sin \theta = \frac{y}{r}$</td><td>$\cos \theta = \frac{x}{r}$</td><td>$\tan \theta = \frac{y}{x}$</td></tr><tr><td>$\csc \theta = \frac{r}{y}$</td><td>$\sec \theta = \frac{r}{x}$</td><td>$\cot \theta = \frac{x}{y}$</td></tr></table>			$\sin \theta = \frac{y}{r}$	$\cos \theta = \frac{x}{r}$	$\tan \theta = \frac{y}{x}$	$\csc \theta = \frac{r}{y}$	$\sec \theta = \frac{r}{x}$	$\cot \theta = \frac{x}{y}$
$\sin \theta = \frac{y}{r}$	$\cos \theta = \frac{x}{r}$	$\tan \theta = \frac{y}{x}$							
$\csc \theta = \frac{r}{y}$	$\sec \theta = \frac{r}{x}$	$\cot \theta = \frac{x}{y}$							
	17. $P(5, -2)$ is a point on the terminal side of θ in standard form. Find the exact values of the trigonometric functions of θ: $5^2 + 2^2 = r^2$$25 + 4 = r^2$$29 = r^2$ $x = 5$$y = -2$$r = \sqrt{29}$ <table><tr><td>$\sin \theta = \frac{-2}{\sqrt{29}} = -\frac{2\sqrt{29}}{29}$</td><td>$\cos \theta = \frac{5}{\sqrt{29}} = \frac{5\sqrt{29}}{29}$</td><td>$\tan \theta = -\frac{2}{5}$</td></tr><tr><td>$\csc \theta = -\frac{\sqrt{29}}{2}$</td><td>$\sec \theta = \frac{\sqrt{29}}{5}$</td><td>$\cot \theta = -\frac{5}{2}$</td></tr></table>			$\sin \theta = \frac{-2}{\sqrt{29}} = -\frac{2\sqrt{29}}{29}$	$\cos \theta = \frac{5}{\sqrt{29}} = \frac{5\sqrt{29}}{29}$	$\tan \theta = -\frac{2}{5}$	$\csc \theta = -\frac{\sqrt{29}}{2}$	$\sec \theta = \frac{\sqrt{29}}{5}$	$\cot \theta = -\frac{5}{2}$
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Name: _____

Unit 12: Trigonometry

Date: _____ Bell: _____

Homework 3: Angles and Angle Measure

**** This is a 2-page document! ******Directions:** Convert each measure to radians.1. 225°

$$225 \cdot \frac{\pi}{180} = \frac{225\pi}{180} = \boxed{\frac{5\pi}{4}}$$

2. 20°

$$20 \cdot \frac{\pi}{180} = \frac{20\pi}{180} = \boxed{\frac{\pi}{9}}$$

3. -255°

$$-255 \cdot \frac{\pi}{180} = \frac{-255\pi}{180} = \boxed{-\frac{17\pi}{12}}$$

4. -140°

$$-140 \cdot \frac{\pi}{180} = \frac{-140\pi}{180} = \boxed{-\frac{7\pi}{9}}$$

5. 75°

$$75 \cdot \frac{\pi}{180} = \frac{75\pi}{180} = \boxed{\frac{5\pi}{12}}$$

6. -300°

$$-300 \cdot \frac{\pi}{180} = \frac{-300\pi}{180} = \boxed{-\frac{5\pi}{3}}$$

Directions: Convert each measure to degrees.7. $\frac{23\pi}{12}$

$$\frac{23\pi}{12} \cdot \frac{180}{\pi} = \frac{4140}{12} = \boxed{345^\circ}$$

8. $-\frac{31\pi}{36}$

$$-\frac{31\pi}{36} \cdot \frac{180}{\pi} = \frac{-5580}{36} = \boxed{-155^\circ}$$

9. $\frac{\pi}{12}$

$$\frac{\pi}{12} \cdot \frac{180}{\pi} = \frac{180}{12} = \boxed{15^\circ}$$

10. $\frac{5\pi}{9}$

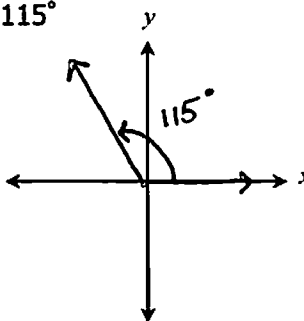
$$\frac{5\pi}{9} \cdot \frac{180}{\pi} = \frac{900}{9} = \boxed{100^\circ}$$

11. $-\frac{\pi}{2}$

$$-\frac{\pi}{2} \cdot \frac{180}{\pi} = \frac{-180}{2} = \boxed{-90^\circ}$$

12. $-\frac{7\pi}{6}$

$$-\frac{7\pi}{6} \cdot \frac{180}{\pi} = \frac{-1260}{6} = \boxed{-210^\circ}$$

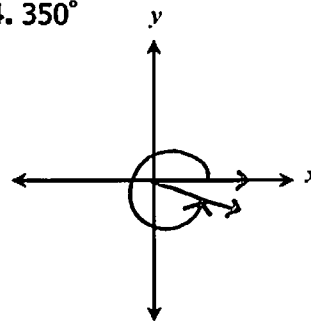
Directions: Sketch each angle. Then give two coterminal angles (one positive and one negative) and the reference angle.13. 115° Coterminal $\angle$'s:

$$115 + 360 = \boxed{475^\circ}$$

$$115 - 360 = \boxed{-245^\circ}$$

Reference $\angle$:

$$180 - 115 = \boxed{65^\circ}$$

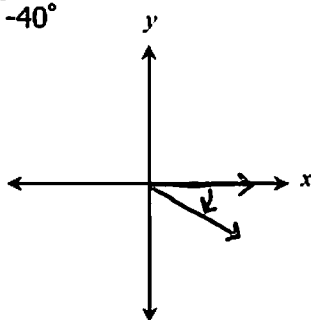
14. 350° Coterminal $\angle$'s:

$$350 + 360 = \boxed{710^\circ}$$

$$350 - 360 = \boxed{-10^\circ}$$

Reference $\angle$:

$$360 - 350 = \boxed{10^\circ}$$

15. -40° Coterminal $\angle$'s:

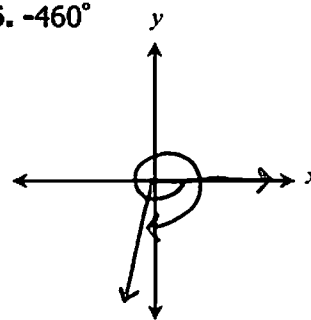
$$-40 + 360 = \boxed{320^\circ}$$

$$-40 - 360 = \boxed{-400^\circ}$$

Reference $\angle$:

$$(-40^\circ = 320^\circ)$$

$$360 - 320 = \boxed{40^\circ}$$

16. -460° Coterminal $\angle$'s:

$$-460 + 360 = \boxed{-100^\circ}$$

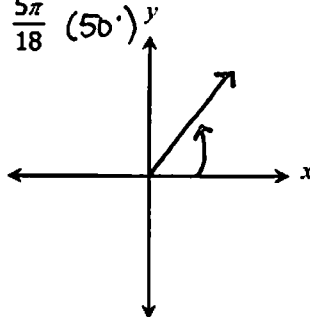
$$-100 + 360 = \boxed{260^\circ}$$

Reference $\angle$:

$$(-460^\circ = 260^\circ)$$

$$260 - 180 = \boxed{80^\circ}$$

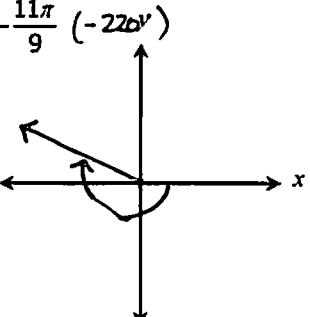
17. $\frac{5\pi}{18}$ (50°)



Coterminal $\angle$'s:
 $\frac{5\pi}{18} + 2\pi = \frac{41\pi}{18}$
 $\frac{5\pi}{18} - 2\pi = -\frac{31\pi}{18}$

Reference $\angle$:
 $\frac{5\pi}{18}$

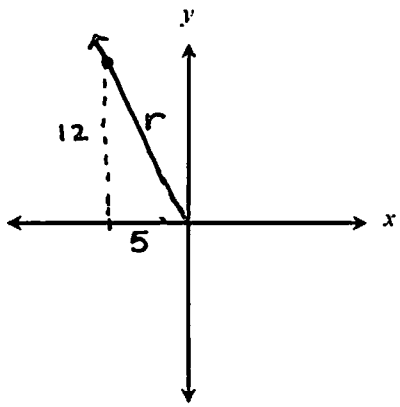
18. $-\frac{11\pi}{9}$ (-220°)

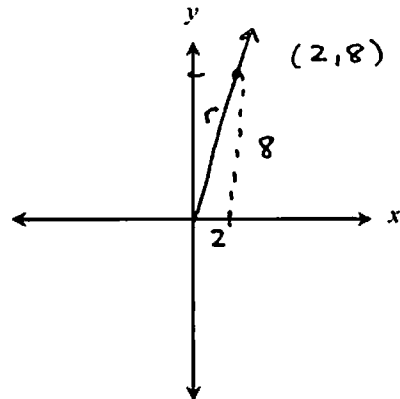


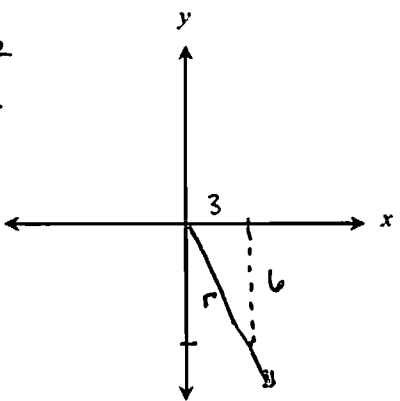
Coterminal $\angle$'s:
 $-\frac{11\pi}{9} + 2\pi = \frac{7\pi}{9}$
 $-\frac{11\pi}{9} - 2\pi = -\frac{29\pi}{9}$

Reference $\angle$:
 $(-\frac{11\pi}{9} = \frac{7\pi}{9})$
 $\pi - \frac{7\pi}{9} = \frac{2\pi}{9}$

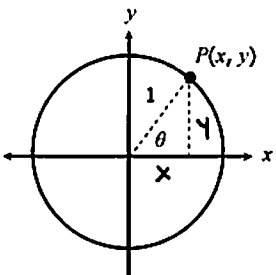
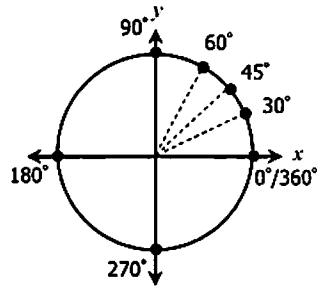
Directions: Sketch the angle in standard form whose terminal side passes through the given point. Find the exact value for each trigonometric function.

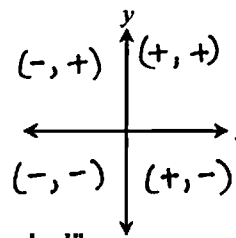
19. $(-5, 12)$ $5^2 + 12^2 = r^2$ $25 + 144 = r^2$ $169 = r^2$ $13 = r$ $x = -5$ $y = 12$ $r = 13$ 	$\sin \theta = \frac{12}{13}$	$\csc \theta = \frac{13}{12}$
	$\cos \theta = \frac{-5}{13}$	$\sec \theta = -\frac{13}{5}$
	$\tan \theta = -\frac{12}{5}$	$\cot \theta = -\frac{5}{12}$

20. $(2, 8)$ $2^2 + 8^2 = r^2$ $4 + 64 = r^2$ $68 = r^2$ $2\sqrt{17} = r$ $x = 2$ $y = 8$ $r = 2\sqrt{17}$ 	$\sin \theta = \frac{8}{2\sqrt{17}} = \frac{4\sqrt{17}}{17}$	$\csc \theta = \frac{2\sqrt{17}}{8} = \frac{\sqrt{17}}{4}$
	$\cos \theta = \frac{2}{2\sqrt{17}} = \frac{\sqrt{17}}{17}$	$\sec \theta = \frac{2\sqrt{17}}{2} = \sqrt{17}$
	$\tan \theta = \frac{8}{2} = 4$	$\cot \theta = \frac{2}{8} = \frac{1}{4}$

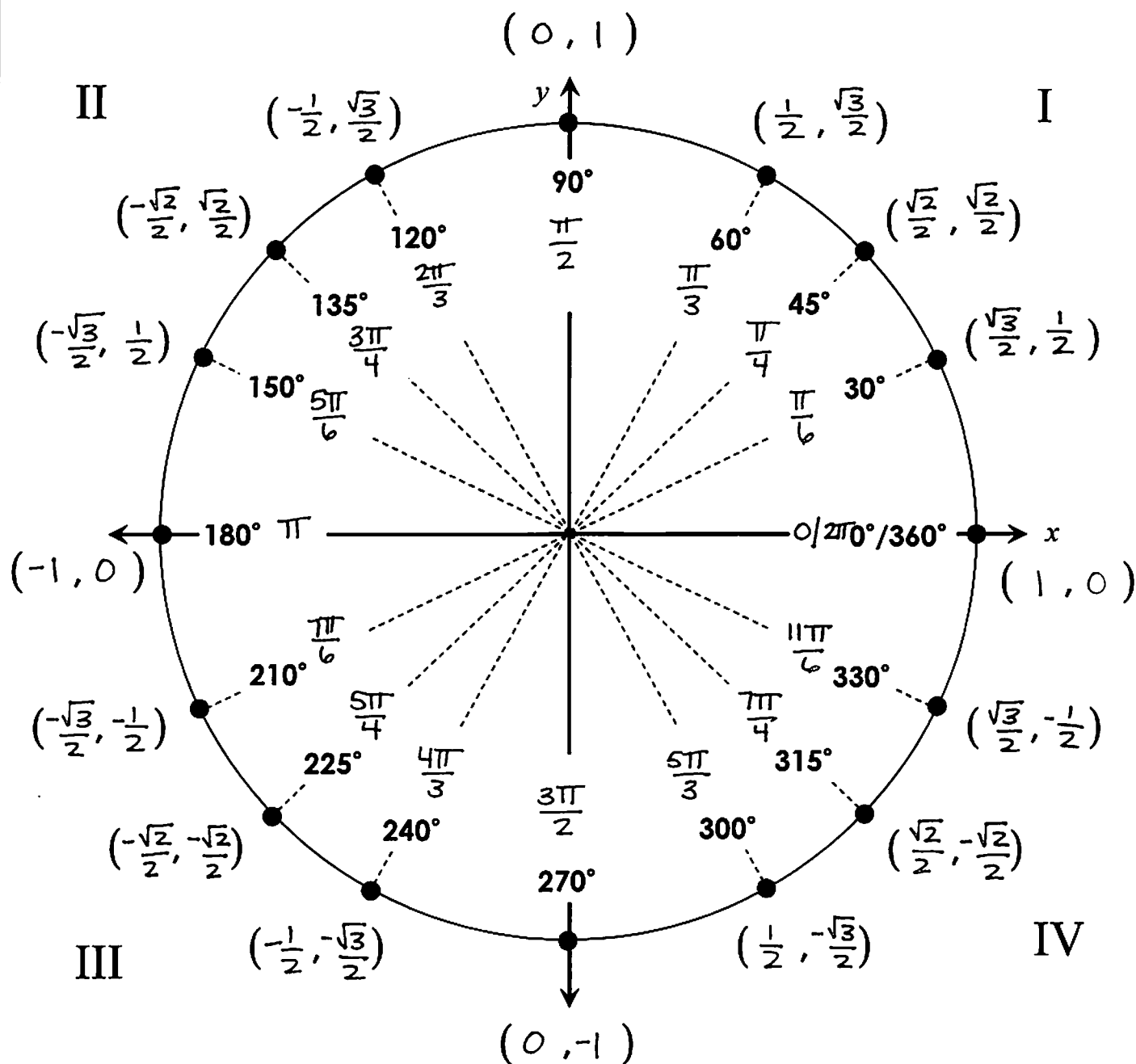
21. $(3, -6)$ $3^2 + 6^2 = r^2$ $9 + 36 = r^2$ $45 = r^2$ $3\sqrt{5} = r$ $x = 3$ $y = -6$ $r = 3\sqrt{5}$ 	$\sin \theta = \frac{-6}{3\sqrt{5}} = -\frac{2\sqrt{5}}{5}$	$\csc \theta = \frac{3\sqrt{5}}{-6} = -\frac{\sqrt{5}}{2}$
	$\cos \theta = \frac{3}{3\sqrt{5}} = \frac{\sqrt{5}}{5}$	$\sec \theta = \frac{3\sqrt{5}}{3} = \sqrt{5}$
	$\tan \theta = \frac{-6}{3} = -2$	$\cot \theta = \frac{3}{-6} = -\frac{1}{2}$

Name:	Class:
Topic:	Date:

Main Ideas/Questions		Notes			
The Unit Circle 		A unit circle is a circle with a radius of 1 unit. Because the value of r is 1 for each point $P(x, y)$ on the circle, the sine, cosine, and tangent values for θ are defined as:			
		$\sin \theta = \frac{y}{1} = y$	$\cos \theta = \frac{x}{1} = x$	$\tan \theta = \frac{y}{x}$	
		** The coordinates of P can be written as $(\cos \theta, \sin \theta)$ **			
Special Angles		The following angles are used frequently with the unit circle: $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ, 180^\circ, 270^\circ$, and 360° Because the terminal side of $0^\circ, 90^\circ, 180^\circ$ and 270° angles lie on an axis, they are called quadrantal angles . *Memorize these values!*			
Degrees	Radians	$\sin \theta$	$\cos \theta$	$\tan \theta$	$P(x, y)$
$0^\circ/360^\circ$	0	0	1	$\frac{0}{1} = 0$	$(1, 0)$
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$	$(\frac{\sqrt{3}}{2}, \frac{1}{2})$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}/2}{\sqrt{2}/2} = 1$	$(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{3}/2}{1/2} = \sqrt{3}$	$(\frac{1}{2}, \frac{\sqrt{3}}{2})$
90°	$\frac{\pi}{2}$	1	0	$\frac{1}{0} = \text{undefined}$	$(0, 1)$
180°	π	0	-1	$\frac{0}{-1} = 0$	$(-1, 0)$
270°	$\frac{3\pi}{2}$	-1	0	$\frac{-1}{0} = \text{undefined}$	$(0, -1)$
Labeling the Unit Circle		• Label the ordered pairs for each angle above.			
		• Use these angles as reference angles for the remaining angles around the circle.			
		• Adjust your coordinates by looking at which quadrant the point is located in.			

Signs of Trig Functions	Which trig functions are POSITIVE in: - Quadrant I? <u>All</u> - Quadrant II? <u>sine</u> - Quadrant III? <u>tangent</u> - Quadrant IV? <u>cosine</u> Remember the phrase "All Students Take Calculus!"  <table border="1"> <tr> <td>1. If $\tan \theta > 0$, which quadrant(s) could the terminal side of θ lie? <u>Q1 or Q3</u></td><td>2. If $\sec \theta < 0$, which quadrant(s) could the terminal side of θ lie? $(\sec \theta = \frac{1}{\cos \theta})$ <u>Q2 or Q3</u></td></tr> <tr> <td>3. If $\sin \theta > 0$, which quadrant(s) could the terminal side of θ lie? <u>Q1 or Q2</u></td><td>4. If $\cos \theta < 0$ and $\cot \theta > 0$, which quadrant(s) could the terminal side of θ lie? $\cos \theta < 0$: <u>Q2, Q3</u> $\cot \theta > 0$: <u>Q1, Q3</u> <u>Q3</u></td></tr> </table>	1. If $\tan \theta > 0$, which quadrant(s) could the terminal side of θ lie? <u>Q1 or Q3</u>	2. If $\sec \theta < 0$, which quadrant(s) could the terminal side of θ lie? $(\sec \theta = \frac{1}{\cos \theta})$ <u>Q2 or Q3</u>	3. If $\sin \theta > 0$, which quadrant(s) could the terminal side of θ lie? <u>Q1 or Q2</u>	4. If $\cos \theta < 0$ and $\cot \theta > 0$, which quadrant(s) could the terminal side of θ lie? $\cos \theta < 0$: <u>Q2, Q3</u> $\cot \theta > 0$: <u>Q1, Q3</u> <u>Q3</u>								
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Using Reference Angles to find Trig Values	Step 1: Find the measure of the reference angle. Step 2: Identify the trig function for the reference angle. Step 3: Adjust the sign, if needed. Directions: Give the exact value for each trigonometric function. <table border="1"> <tr> <td>5. $\sin 150^\circ \rightarrow 30^\circ$ $\frac{1}{2}$ </td><td>6. $\tan 300^\circ \rightarrow 60^\circ$ $-\sqrt{3}$ </td><td>7. $\cos 225^\circ \rightarrow 45^\circ$ $-\frac{\sqrt{2}}{2}$ </td></tr> <tr> <td>8. $\cot 315^\circ \rightarrow 45^\circ$ ($\tan 45 = 1$) -1 </td><td>9. $\tan 240^\circ \rightarrow 60^\circ$ $\sqrt{3}$ </td><td>10. $\sec 300^\circ \rightarrow 60^\circ$ ($\cos 60 = \frac{1}{2}$) 2 </td></tr> <tr> <td>11. $\cos \frac{11\pi}{6} \rightarrow \frac{\pi}{6}$ $\frac{\sqrt{3}}{2}$ </td><td>12. $\csc \frac{7\pi}{4} \rightarrow \frac{\pi}{4}$ ($\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$) $-\frac{2}{\frac{\sqrt{2}}{2}} = -\frac{2\sqrt{2}}{2} = -\sqrt{2}$ $-\sqrt{2}$ </td><td>13. $\sin \frac{4\pi}{3} \rightarrow \frac{\pi}{3}$ $-\frac{\sqrt{3}}{2}$ </td></tr> <tr> <td>14. $\tan \frac{3\pi}{4} \rightarrow 45^\circ$ -1 </td><td>15. $\sin \frac{7\pi}{6} \rightarrow 30^\circ$ $-\frac{1}{2}$ </td><td>16. $\sec \frac{\pi}{2}$ ($\cos \frac{\pi}{2} = 0$) undefined </td></tr> </table>	5. $\sin 150^\circ \rightarrow 30^\circ$ $\frac{1}{2}$	6. $\tan 300^\circ \rightarrow 60^\circ$ $-\sqrt{3}$	7. $\cos 225^\circ \rightarrow 45^\circ$ $-\frac{\sqrt{2}}{2}$	8. $\cot 315^\circ \rightarrow 45^\circ$ ($\tan 45 = 1$) -1	9. $\tan 240^\circ \rightarrow 60^\circ$ $\sqrt{3}$	10. $\sec 300^\circ \rightarrow 60^\circ$ ($\cos 60 = \frac{1}{2}$) 2	11. $\cos \frac{11\pi}{6} \rightarrow \frac{\pi}{6}$ $\frac{\sqrt{3}}{2}$	12. $\csc \frac{7\pi}{4} \rightarrow \frac{\pi}{4}$ ($\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$) $-\frac{2}{\frac{\sqrt{2}}{2}} = -\frac{2\sqrt{2}}{2} = -\sqrt{2}$ $-\sqrt{2}$	13. $\sin \frac{4\pi}{3} \rightarrow \frac{\pi}{3}$ $-\frac{\sqrt{3}}{2}$	14. $\tan \frac{3\pi}{4} \rightarrow 45^\circ$ -1	15. $\sin \frac{7\pi}{6} \rightarrow 30^\circ$ $-\frac{1}{2}$	16. $\sec \frac{\pi}{2}$ ($\cos \frac{\pi}{2} = 0$) undefined
5. $\sin 150^\circ \rightarrow 30^\circ$ $\frac{1}{2}$	6. $\tan 300^\circ \rightarrow 60^\circ$ $-\sqrt{3}$	7. $\cos 225^\circ \rightarrow 45^\circ$ $-\frac{\sqrt{2}}{2}$											
8. $\cot 315^\circ \rightarrow 45^\circ$ ($\tan 45 = 1$) -1	9. $\tan 240^\circ \rightarrow 60^\circ$ $\sqrt{3}$	10. $\sec 300^\circ \rightarrow 60^\circ$ ($\cos 60 = \frac{1}{2}$) 2											
11. $\cos \frac{11\pi}{6} \rightarrow \frac{\pi}{6}$ $\frac{\sqrt{3}}{2}$	12. $\csc \frac{7\pi}{4} \rightarrow \frac{\pi}{4}$ ($\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$) $-\frac{2}{\frac{\sqrt{2}}{2}} = -\frac{2\sqrt{2}}{2} = -\sqrt{2}$ $-\sqrt{2}$	13. $\sin \frac{4\pi}{3} \rightarrow \frac{\pi}{3}$ $-\frac{\sqrt{3}}{2}$											
14. $\tan \frac{3\pi}{4} \rightarrow 45^\circ$ -1	15. $\sin \frac{7\pi}{6} \rightarrow 30^\circ$ $-\frac{1}{2}$	16. $\sec \frac{\pi}{2}$ ($\cos \frac{\pi}{2} = 0$) undefined											

THE UNIT CIRCLE



Trig Values:

$\sin \theta =$	y
$\cos \theta =$	x
$\tan \theta =$	y/x

Summary of Signs in the Quadrants:

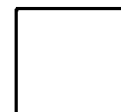
	I	II	III	IV
$\sin \theta$	+	+	-	-
$\cos \theta$	+	-	-	+
$\tan \theta$	+	-	+	-

Name: _____

Unit 12: Trigonometry

Date: _____ Bell: _____

Homework 4: The Unit Circle



1. Which trig functions are positive for angles terminating in Quadrant IV? Cosine	2. Which trig functions are negative for angles terminating in Quadrant II? Cosine and tangent
3. If $\cos \theta < 0$, which quadrant(s) could the terminal side of θ lie? Q 2 or Q 3	4. If $\csc \theta > 0$, which quadrant(s) could the terminal side of θ lie? ($\csc \theta = \frac{1}{\sin \theta}$) Q 1 and Q 2
5. If $\tan \theta < 0$ and $\sec \theta > 0$, which quadrant(s) could the terminal side of θ lie? ($\sec \theta = \frac{1}{\cos \theta}$) Q 4	6. If $\sin \theta < 0$ and $\cot \theta < 0$, which quadrant(s) could the terminal side of θ lie? Q 4

Directions: Find the exact value for each trigonometric function. Use reference angles when necessary.

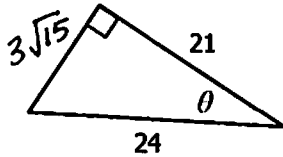
7. $\sin 60^\circ$ $\boxed{\frac{\sqrt{3}}{2}}$	8. $\cos 225^\circ$ $\boxed{-\frac{\sqrt{2}}{2}}$	9. $\tan 330^\circ$ $\frac{-\frac{1}{2}}{\frac{\sqrt{3}}{2}} = -\frac{1}{\sqrt{3}} = \boxed{-\frac{\sqrt{3}}{3}}$
10. $\cot 150^\circ$ $\frac{-\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \boxed{-\sqrt{3}}$	11. $\sec 135^\circ$ $\frac{-2}{\sqrt{2}} = \boxed{-\sqrt{2}}$	12. $\tan 210^\circ$ $\frac{-\frac{1}{2}}{-\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$
13. $\sec 270^\circ$ $\boxed{\text{undefined}}$	14. $\csc 315^\circ$ $-\frac{2}{\sqrt{2}} = \boxed{-\sqrt{2}}$	15. $\sin 240^\circ$ $\boxed{-\frac{\sqrt{3}}{2}}$
16. $\tan \frac{7\pi}{6}$ $\frac{-\frac{1}{2}}{-\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$	17. $\cos \frac{3\pi}{4}$ $\boxed{-\frac{\sqrt{2}}{2}}$	18. $\csc \frac{5\pi}{3}$ $-\frac{2}{\sqrt{3}} = \boxed{-\frac{2\sqrt{3}}{3}}$
19. $\sin \frac{\pi}{6}$ $\boxed{\frac{1}{2}}$	20. $\cot \frac{5\pi}{6}$ $\frac{-\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \boxed{-\sqrt{3}}$	21. $\sec \frac{11\pi}{6}$ $\frac{2}{\sqrt{3}} = \boxed{\frac{2\sqrt{3}}{3}}$

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 12: Trigonometry

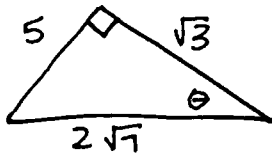
Quiz 12-1: Introduction to Trig; The Unit Circle1. Find the values of the six trigonometric functions for angle θ . Give answers in simplest form.

$$\begin{aligned} 21^2 + x^2 &= 24^2 \\ 441 + x^2 &= 576 \\ x^2 &= 135 \\ x &= 3\sqrt{15} \end{aligned}$$

$$\sin \theta = \frac{\sqrt{15}}{8} \quad \csc \theta = \frac{8\sqrt{15}}{15}$$

$$\cos \theta = \frac{7}{8} \quad \sec \theta = \frac{8}{7}$$

$$\tan \theta = \frac{\sqrt{15}}{7} \quad \cot \theta = \frac{7\sqrt{15}}{15}$$

2. If $\cot \theta = \frac{\sqrt{3}}{5}$, find the remaining trigonometric functions.

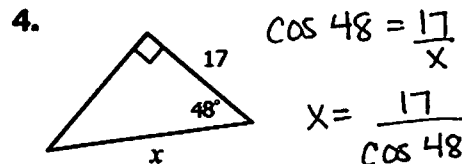
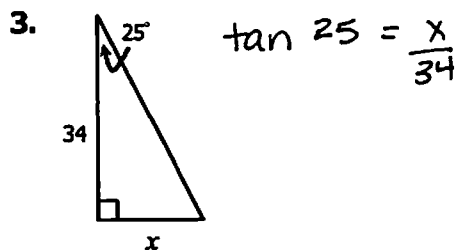
$$\begin{aligned} 5^2 + \sqrt{3}^2 &= x^2 \\ 25 + 3 &= x^2 \\ 28 &= x^2 \\ 2\sqrt{7} &= x \end{aligned}$$

$$\sin \theta = \frac{5\sqrt{7}}{14} \quad \csc \theta = \frac{2\sqrt{7}}{5}$$

$$\cos \theta = \frac{\sqrt{21}}{14} \quad \sec \theta = \frac{2\sqrt{21}}{3}$$

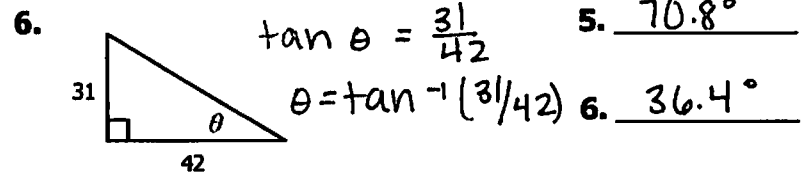
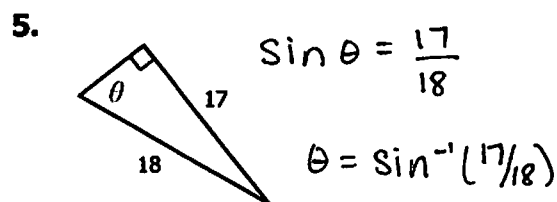
$$\tan \theta = \frac{5\sqrt{3}}{3}$$

Find the missing side lengths. Round all answers to the nearest tenth.



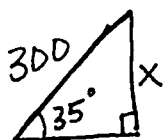
3. 15.9

4. 25.4

Find the measure of θ . Round all answers to the nearest tenth.

5. 70.8°

6. 36.4°

7. Beth is flying a kite with 300 feet of string. The kite gets stuck in a tree, so she ties the other end of the string to a rock on the ground making the string taut. The angle of elevation from the rock to kite is 35° . How far up the tree is the kite? Round your answer to the nearest tenth.

$$\sin 35 = \frac{x}{300}$$

$$x = 172.1$$

7. 172.1 ft

Give each degree measure in radians and each radian measure in degrees.

$$8. 130^\circ \cdot \frac{\pi}{180} = \frac{130\pi}{180}$$

$$9. -285^\circ \cdot \frac{\pi}{180} = -\frac{285\pi}{180}$$

$$8. \frac{13\pi}{18}$$

$$9. \frac{-19\pi}{12}$$

$$10. -\frac{17\pi}{9} \cdot \frac{180}{\pi} = -\frac{3060}{9}$$

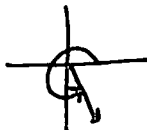
$$11. \frac{7\pi}{36} \cdot \frac{180}{\pi} = \frac{1260}{36}$$

$$10. -340^\circ$$

$$11. 35^\circ$$

Give two coterminal angles (one positive and one negative) and the reference angle.

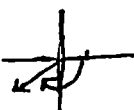
$$12. 290^\circ$$



$$12. \text{Coterminal } \angle\text{'s: } 650^\circ, -70^\circ$$

$$\text{Reference } \angle: 70^\circ$$

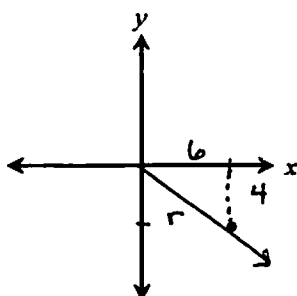
$$13. -\frac{4\pi}{5}$$



$$13. \text{Coterminal } \angle\text{'s: } \frac{6\pi}{5}, -\frac{14\pi}{5}$$

$$\text{Reference } \angle: \frac{\pi}{5}$$

14. The terminal side of an angle in standard form passes through the point (6, -4). Make a sketch, then find each function value.



$$\begin{aligned} 6^2 + 4^2 &= r^2 \\ 36 + 16 &= r^2 \\ 52 &= r^2 \\ 2\sqrt{13} &= r \\ x &= 6 \\ y &= -4 \\ r &= 2\sqrt{13} \end{aligned}$$

$$\sin \theta = \frac{-2\sqrt{13}}{13} \quad \csc \theta = \frac{\sqrt{13}}{-2}$$

$$\cos \theta = \frac{3\sqrt{13}}{13} \quad \sec \theta = \frac{\sqrt{13}}{3}$$

$$\tan \theta = \frac{-2}{3} \quad \cot \theta = \frac{-3}{2}$$

15. If $\tan \theta < 0$ and $\sec \theta < 0$ which quadrant(s) could the terminal side of θ lie?

$$15. Q2$$

Find the exact value for each trigonometric function.

$$16. \tan 135^\circ$$

$$\frac{\frac{\sqrt{2}}{2}}{-\frac{\sqrt{2}}{2}} = -1$$

$$17. \csc 240^\circ$$

$$\frac{-2}{\frac{\sqrt{3}}{2}} = -\frac{2\sqrt{3}}{3}$$

$$18. \sin 120^\circ$$

$$\frac{\sqrt{3}}{2}$$

$$16. -1$$

$$17. \frac{-2\sqrt{3}}{3}$$

$$18. \frac{\sqrt{3}}{2}$$

$$19. \frac{\sqrt{2}}{2}$$

$$20. -2$$

$$21. \sqrt{3}$$

$$19. \cos \frac{7\pi}{4}$$

$$\frac{\sqrt{2}}{2}$$

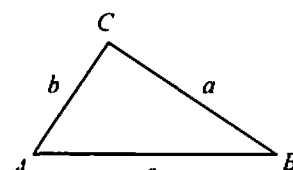
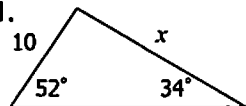
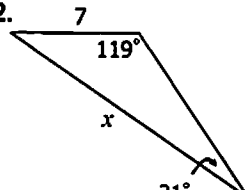
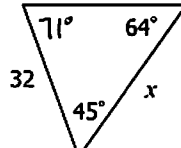
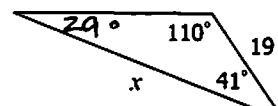
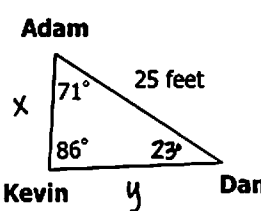
$$20. \sec \frac{2\pi}{3}$$

$$\frac{-2}{1} = -2$$

$$21. \cot \frac{\pi}{6}$$

$$\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Law of Sines	Given $\triangle ABC$: $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$  The Law of Sines can be used to find side and angle measures given two angles and any side (AAS or ASA) or two side lengths and an angle outside of them (SSA).
Finding Sides (AAS or ASA)	Directions: Find each missing measure. Round all answers to the nearest tenth. 1.  $\frac{\sin 34}{10} = \frac{\sin 52}{x}$ $x \cdot \sin 34 = 10 \cdot \sin 52$ $x = \frac{10 \cdot \sin 52}{\sin 34}$ x = 14.1 2.  $\frac{\sin 21}{7} = \frac{\sin 119}{x}$ $x \cdot \sin 21 = 7 \cdot \sin 119$ $x = \frac{7 \cdot \sin 119}{\sin 21}$ x = 17.1 3.  $\frac{\sin 71}{x} = \frac{\sin 64}{32}$ $x \cdot \sin 64 = 32 \cdot \sin 71$ $x = \frac{32 \cdot \sin 71}{\sin 64}$ x = 33.7 4.  $\frac{\sin 110}{x} = \frac{\sin 29}{19}$ $x \cdot \sin 29 = 19 \cdot \sin 110$ $x = \frac{19 \cdot \sin 110}{\sin 29}$ x = 36.8 Use for questions 5-6: Kevin, Adam, and Dan are passing a hockey puck. 5. How far is Kevin from Adam?  $\frac{\sin 86}{25} = \frac{\sin 23}{x}$ $x \cdot \sin 86 = 25 \cdot \sin 23$ $x = \frac{25 \cdot \sin 23}{\sin 86}$ x = 9.8 6. How far is Dan from Kevin? $\frac{\sin 86}{25} = \frac{\sin 71}{y}$ $y \cdot \sin 86 = 25 \cdot \sin 71$ $y = \frac{25 \cdot \sin 71}{\sin 86}$ y = 23.7

Finding Angles (SSA)

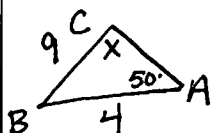
Watch out! When finding angles, it's possible to have one solution, two solutions, or no solution! This is called the **ambiguous case**.

Important Facts:

- The sine function always has a range between -1 and 1.
- If the sine function is positive, then the angle can lie in quad 1 or 2.
- To find the other possible angle, subtract from 180.
- The sum of the angle measures in a triangle is always 180.
- No triangle can have two obtuse angles.

Directions: Information about $\triangle ABC$ is given below. Check your answers and give all

7. $m\angle A = 50^\circ$, $AB = 4$, $BC = 9$, find $m\angle C$.



$$50 + 19.9 < 180 \checkmark$$

$$50 + 160.1 > 180 \times$$

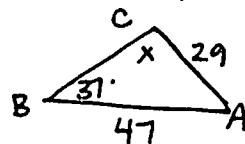
$$\frac{\sin 50}{4} = \frac{\sin x}{9}$$

$$9 \cdot \sin x = 4 \cdot \sin 50$$

$$\sin x = .3405$$

$$x = 19.9^\circ, 160.1^\circ$$

8. $m\angle B = 37^\circ$, $AB = 47$, $AC = 29$, find $m\angle C$.



$$37 + 77.3 < 180 \checkmark$$

$$37 + 102.7 < 180 \checkmark$$

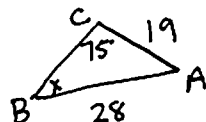
$$\frac{\sin 37}{29} = \frac{\sin x}{47}$$

$$29 \cdot \sin x = 47 \cdot \sin 37$$

$$\sin x = .9754$$

$$x = 77.3^\circ, 102.7^\circ$$

9. $m\angle C = 75^\circ$, $AB = 28$, $AC = 19$, find $m\angle B$.



$$75 + 41 < 180 \checkmark$$

$$75 + 139 > 180 \times$$

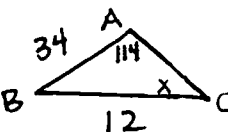
$$\frac{\sin 75}{28} = \frac{\sin x}{19}$$

$$28 \cdot \sin x = 19 \cdot \sin 75$$

$$\sin x = .6554$$

$$x = 41^\circ, 139^\circ$$

10. $m\angle A = 114^\circ$, $AB = 34$, $BC = 12$, find $m\angle C$.



$$\frac{\sin 114}{12} = \frac{\sin x}{34}$$

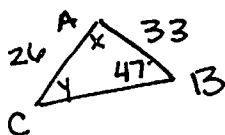
$$12 \cdot \sin x = 34 \cdot \sin 114$$

$$\sin x = 2.5884$$

$$x = \emptyset$$

No Solution

11. $m\angle B = 47^\circ$, $AB = 33$, $AC = 26$, find $m\angle A$.



$$x = 180 - (47 + 68.2)$$

$$= 64.8^\circ$$

$$x = 180 - (47 + 111.8)$$

$$= 21.2^\circ$$

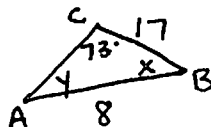
$$\frac{\sin 47}{26} = \frac{\sin y}{33}$$

$$26 \cdot \sin y = 33 \cdot \sin 47$$

$$\sin y = .9283$$

$$y = 68.2^\circ, 111.8^\circ$$

12. $m\angle C = 73^\circ$, $AB = 8$, $BC = 17$, find $m\angle B$.



$$\frac{\sin 73}{8} = \frac{\sin y}{17}$$

$$8 \cdot \sin y = 17 \cdot \sin 73$$

$$\sin y = 2.0321$$

$$y = \emptyset$$

No Solution

Name: _____

Unit 12: Trigonometry

Date: _____ Bell: _____

Homework 5: Law of Sines

**** This is a 2-page document! ******Directions:** Find each side length to the nearest tenth.

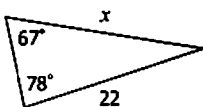
1.

$$\frac{\sin 67}{22} = \frac{\sin 78}{x}$$

$$x \cdot \sin 67 = 22 \cdot \sin 78$$

$$x = \frac{22 \cdot \sin 78}{\sin 67}$$

$$x = 23.4$$



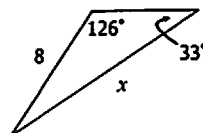
2.

$$\frac{\sin 33}{8} = \frac{\sin 126}{x}$$

$$x \cdot \sin 33 = 8 \cdot \sin 126$$

$$x = \frac{8 \cdot \sin 126}{\sin 33}$$

$$x = 11.9$$



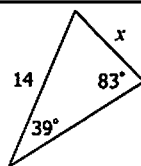
3.

$$\frac{\sin 83}{14} = \frac{\sin 39}{x}$$

$$x \cdot \sin 83 = 14 \cdot \sin 39$$

$$x = \frac{14 \cdot \sin 39}{\sin 83}$$

$$x = 8.9$$



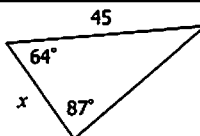
4.

$$\frac{\sin 87}{45} = \frac{\sin 29}{x}$$

$$x \cdot \sin 87 = 45 \cdot \sin 29$$

$$x = \frac{45 \cdot \sin 29}{\sin 87}$$

$$x = 21.8$$



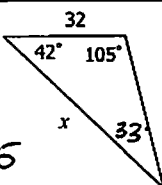
5.

$$\frac{\sin 33}{32} = \frac{\sin 105}{x}$$

$$x \cdot \sin 33 = 32 \cdot \sin 105$$

$$x = \frac{32 \cdot \sin 105}{\sin 33}$$

$$x = 56.8$$



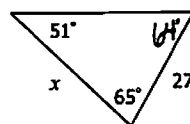
6.

$$\frac{\sin 51}{27} = \frac{\sin 64}{x}$$

$$x \cdot \sin 51 = 27 \cdot \sin 64$$

$$x = \frac{27 \cdot \sin 64}{\sin 51}$$

$$x = 31.2$$

**Directions:** Information about $\triangle ABC$ is given below. Give all possible answers.7. $m\angle C = 63^\circ$, $AB = 28$, $AC = 29$, find $m\angle B$.

$$\frac{\sin 63}{28} = \frac{\sin x}{29}$$

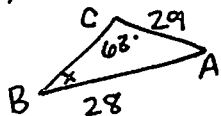
$$28 \cdot \sin x = 29 \cdot \sin 63$$

$$\sin x = .9228$$

$$x = 67.3^\circ, 112.7^\circ$$

$$63 + 67.3 < 180 \checkmark$$

$$63 + 112.7 < 180 \checkmark$$

8. $m\angle C = 36^\circ$, $AB = 31$, $BC = 24$, find $m\angle A$.

$$\frac{\sin 36}{31} = \frac{\sin x}{24}$$

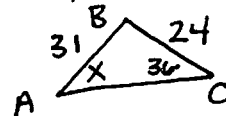
$$31 \cdot \sin x = 24 \cdot \sin 36$$

$$\sin x = .4551$$

$$x = 27.1^\circ, 152.9^\circ$$

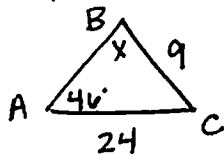
$$36 + 27.1 < 180 \checkmark$$

$$36 + 152.9 > 180 \times$$



9. $m\angle A = 46^\circ$, $BC = 9$, $AC = 24$, find $m\angle B$.

$$\frac{\sin 46}{9} = \frac{\sin x}{24}$$



$$9 \cdot \sin x = 24 \cdot \sin 46$$

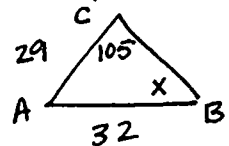
$$\sin x = 1.9182$$

$$x = \theta$$

No Solution!

10. $m\angle C = 105^\circ$, $AB = 32$, $AC = 29$, find $m\angle B$.

$$\frac{\sin 105}{32} = \frac{\sin x}{29}$$



$$32 \cdot \sin x = 29 \cdot \sin 105$$

$$\sin x = .8754$$

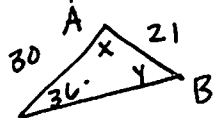
$$x = 61.1^\circ, 118.9^\circ$$

$$105 + 61.1 < 180 \checkmark$$

$$105 + 118.9 > 180 \times$$

11. $m\angle C = 36^\circ$, $AB = 21$, $AC = 30$, find $m\angle A$.

$$\frac{\sin 36}{21} = \frac{\sin y}{30}$$



$$21 \cdot \sin y = 30 \cdot \sin 36$$

$$\sin y = .8397$$

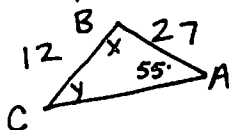
$$y = 57.1^\circ, 122.9^\circ$$

$$x = 180 - (36 + 57.1) = 85.9^\circ$$

$$x = 180 - (36 + 122.9) = 21.1^\circ$$

12. $m\angle A = 55^\circ$, $AB = 27$, $BC = 12$, find $m\angle B$.

$$\frac{\sin 55}{12} = \frac{\sin y}{27}$$



$$12 \cdot \sin y = 27 \cdot \sin 55$$

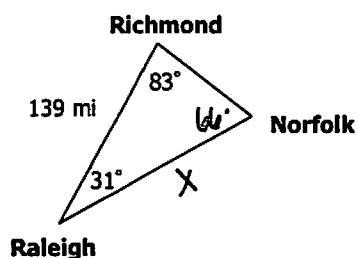
$$\sin y = 1.8431$$

$$y = \theta$$

No Solution!

Directions: Read each problem, then solve for the missing part(s).

13. A pilot flying over the Norfolk airport is told he will have to redirect and land in either Richmond or Raleigh. Due to air traffic, he is then told he must land in Raleigh. Find the distance he will travel.

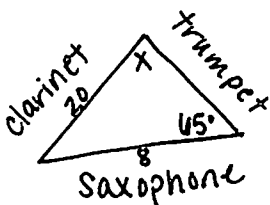


$$\frac{\sin 66}{139} = \frac{\sin 83}{x}$$

$$x \cdot \sin 66 = 139 \cdot \sin 83$$

$$x = 151 \text{ miles}$$

14. Part of a marching band formed a triangle with the trumpet players on one side, clarinet players on one side, and saxophone players on the third side. The angle formed by the trumpet and saxophone players measured 65° . If the clarinet line is 20 feet long and the saxophone line is 8 feet long, find the angle formed by the trumpet and clarinet players.



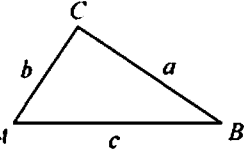
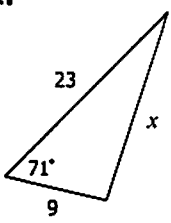
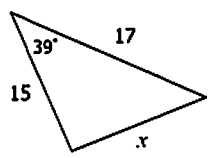
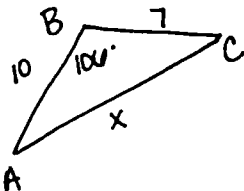
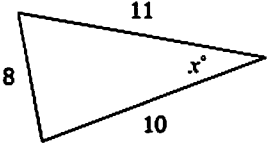
$$\frac{\sin x}{8} = \frac{\sin 65}{20}$$

$$20 \cdot \sin x = 8 \cdot \sin 65$$

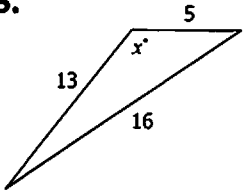
$$\sin x = .3625$$

$$x = 21.3^\circ, 158.7^\circ$$

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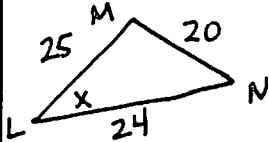
Main Ideas/Questions	Notes/Examples
Law of Cosines 	The Law of Cosines can be used to find side and angle measures given two sides and the included angle (SAS) or three sides (SSS). Given $\triangle ABC$: $a^2 = b^2 + c^2 - 2bc \cdot \cos A$ $b^2 = a^2 + c^2 - 2ac \cdot \cos B$ $c^2 = a^2 + b^2 - 2ab \cdot \cos C$
Finding Sides: Find each missing measure. Round all answers to the nearest tenth.	
1. 	$x^2 = 9^2 + 23^2 - 2(9)(23) \cdot \cos 71$ $x^2 = 610 - 414 \cdot \cos 71$ $x^2 = 475.2148$ $x = 21.8$
2. 	$x^2 = 15^2 + 17^2 - 2(15)(17) \cdot \cos 39$ $x^2 = 514 - 510 \cdot \cos 39$ $x^2 = 117.6556$ $x = 10.8$
3. In $\triangle ABC$, $m\angle B = 106^\circ$, $AB = 10$, and $BC = 7$. Find AC. 	$x^2 = 10^2 + 7^2 - 2(10)(7) \cdot \cos 106$ $x^2 = 149 - 140 \cdot \cos 106$ $x^2 = 187.5892$ $x = 13.7$
Finding Angles: Find each missing measure. Round all answers to the nearest tenth.	
4. 	$8^2 = 10^2 + 11^2 - 2(10)(11) \cdot \cos x$ $64 = 221 - 220 \cdot \cos x$ $-157 = -220 \cdot \cos x$ $.714 = \cos x$ $x = \cos^{-1}(.714)$ $x = 44.5^\circ$

5.



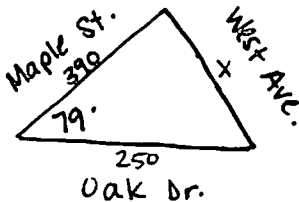
$$\begin{aligned}
 16^2 &= 13^2 + 5^2 - 2(13)(5) \cdot \cos X \\
 256 &= 194 - 130 \cdot \cos X \\
 62 &= -130 \cdot \cos X \\
 -.4769 &= \cos X \\
 X &= \cos^{-1}(-.4769)
 \end{aligned}$$

$$X = 118.5^\circ$$

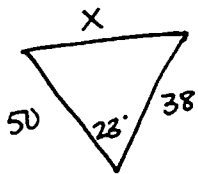
6. In $\triangle LMN$, $LM = 25$, $MN = 20$, and $LN = 24$. Find $m\angle L$.

$$\begin{aligned}
 20^2 &= 24^2 + 25^2 - 2(24)(25) \cdot \cos X \\
 400 &= 1201 - 1200 \cdot \cos X \\
 -801 &= -1200 \cdot \cos X \\
 .6675 &= \cos X \\
 X &= \cos^{-1}(.6675)
 \end{aligned}$$

$$X = 48.1^\circ$$

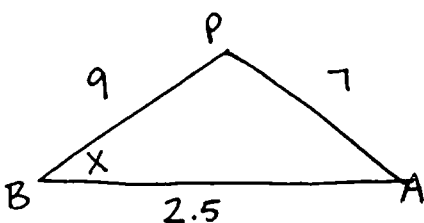
Applications: Give side measures to the nearest tenth and angle measures to the nearest degree.7. Maple Street, Oak Drive, and West Ave intersect to form a triangular piece of land. The angle formed by Maple Street and Oak Drive is 79° . If the Maple Street section of the triangle is 390 feet and the Oak Drive section is 250 feet, find the length of the West Ave section.

$$\begin{aligned}
 x^2 &= 390^2 + 250^2 - 2(390)(250) \cdot \cos 79 \\
 x^2 &= 214600 - 195000 \cdot \cos 79 \\
 x^2 &= 177392.2459 \\
 x &= 421.2 \text{ ft}
 \end{aligned}$$

8. Jack and Tom start at a tree and begin running apart with an angle of 23° between them. If Jack runs 50 meters and Tom runs 38 meters, find the distance between them.

$$\begin{aligned}
 x^2 &= 50^2 + 38^2 - 2(50)(38) \cdot \cos 23 \\
 x^2 &= 3944 - 3800 \cdot \cos 23 \\
 x^2 &= 446.0816 \\
 x &= 21.1 \text{ m}
 \end{aligned}$$

9. Two radar stations 2.5 miles apart are tracking an airplane. The straight-line distance between Station A and the plane is 9 miles. The straight-line distance between Station B and the plane is 7 miles. What is the angle of elevation from Station B to the plane?



$$\begin{aligned}
 7^2 &= 9^2 + 2.5^2 - 2(9)(2.5) \cdot \cos X \\
 49 &= 87.25 - 45 \cdot \cos X \\
 -38.25 &= -45 \cdot \cos X \\
 .85 &= \cos X \\
 X &= \cos^{-1}(.85)
 \end{aligned}$$

$$X = 32^\circ$$

Name: _____

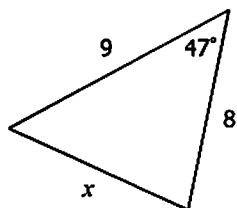
Unit 12: Trigonometry

Date: _____ Bell: _____

Homework 6: Law of Cosines

**** This is a 2-page document! ******Directions:** Find each missing measure. Round all answers to the nearest tenth.

1.



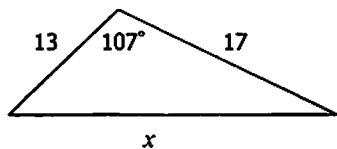
$$x^2 = 9^2 + 8^2 - 2(9)(8) \cdot \cos 47$$

$$x^2 = 145 - 144 \cdot \cos 47$$

$$x^2 = 46.7922$$

$$x = 6.8$$

2.

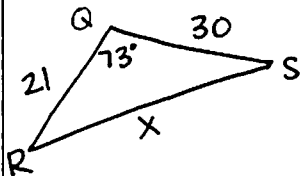


$$x^2 = 13^2 + 17^2 - 2(13)(17) \cdot \cos 107$$

$$x^2 = 458 - 442 \cdot \cos 107$$

$$x^2 = 587.2283$$

$$x = 24.2$$

3. In $\triangle QRS$, $m\angle Q = 73^\circ$, $QR = 21$, and $QS = 30$. Find RS .

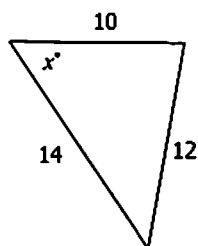
$$x^2 = 21^2 + 30^2 - 2(21)(30) \cdot \cos 73$$

$$x^2 = 1341 - 1260 \cdot \cos 73$$

$$x^2 = 972.6117$$

$$x = 31.2$$

4.



$$12^2 = 10^2 + 14^2 - 2(10)(14) \cdot \cos x$$

$$144 = 296 - 280 \cdot \cos x$$

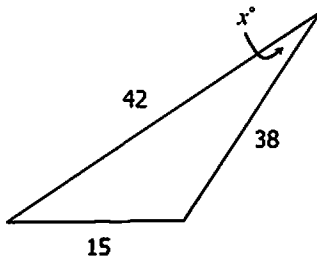
$$-152 = -280 \cdot \cos x$$

$$.5429 = \cos x$$

$$x = \cos^{-1}(.5429)$$

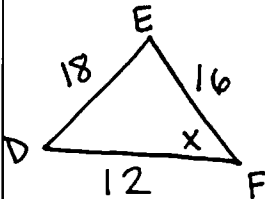
$$x = 57.1^\circ$$

5.



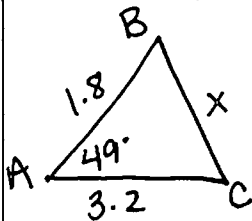
$$\begin{aligned}
 15^2 &= 38^2 + 42^2 - 2(38)(42) \cdot \cos X \\
 225 &= 3208 - 3192 \cdot \cos X \\
 -2983 &= -3192 \cdot \cos X \\
 .9345 &= \cos X \\
 X &= \cos^{-1}(.9345) \\
 \boxed{X &= 20.8^\circ}
 \end{aligned}$$

6. In $\triangle DEF$, $DE = 18$, $EF = 16$, and $DF = 12$. Find $m\angle F$.



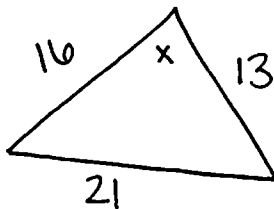
$$\begin{aligned}
 18^2 &= 16^2 + 12^2 - 2(16)(12) \cdot \cos X \\
 324 &= 400 - 384 \cdot \cos X \\
 -76 &= -384 \cdot \cos X \\
 .1979 &= \cos X \\
 X &= \cos^{-1}(.1979) \\
 \boxed{X &= 78.6^\circ}
 \end{aligned}$$

7. Three boats are out at sea: Boat A, Boat B, and Boat C. The crew of Boat A can see both Boat B and Boat C. The angle between the line of sight to Boat B and the line of sight to Boat C is 49° . If the distance between Boat A and Boat B is 1.8 miles and the distance between Boat A and Boat C is 3.2 miles, what is the distance between Boat B and Boat C?



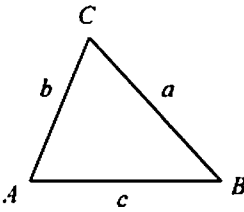
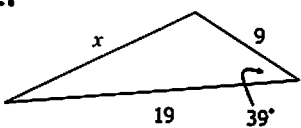
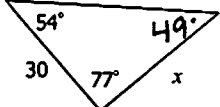
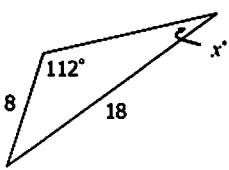
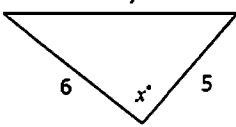
$$\begin{aligned}
 x^2 &= 1.8^2 + 3.2^2 - 2(1.8)(3.2) \cdot \cos 49 \\
 x^2 &= 13.48 - 11.52 \cdot \cos 49 \\
 x^2 &= 5.9222 \\
 \boxed{x &= 2.4 \text{ mi}}
 \end{aligned}$$

8. A farmer plants a 16-foot row of tomatoes, 13-foot row of carrots, and a 21-foot row of green peppers in a triangle. Find the angle formed by the tomatoes and carrots.



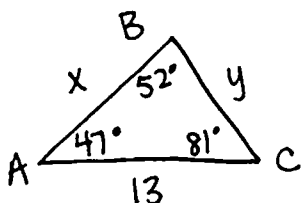
$$\begin{aligned}
 21^2 &= 16^2 + 13^2 - 2(16)(13) \cdot \cos X \\
 441 &= 425 - 416 \cdot \cos X \\
 16 &= -416 \cdot \cos X \\
 -.0385 &= \cos X \\
 X &= \cos^{-1}(-.0385) \\
 \boxed{X &= 92.2^\circ}
 \end{aligned}$$

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Main Ideas/Questions	Notes/Examples
LAW OF SINES <i>WJ</i> LAW OF COSINES 	Law of Sines $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$ Use when given... <u>2 angles, 1 side (ASA, AAS)</u> <u>2 sides, 1 angle (SSA)</u> Law of Cosines $a^2 = b^2 + c^2 - 2bc \cdot \cos A$ $b^2 = a^2 + c^2 - 2ac \cdot \cos B$ $c^2 = a^2 + b^2 - 2ab \cdot \cos C$ Use when given... <u>2 sides, 1 angle (SAS)</u> <u>3 sides (SSS)</u>
EXAMPLES: Indicate the law, then find each missing measure to the nearest tenth.	
1.  ☐ s ☒ c $x^2 = 9^2 + 19^2 - 2(9)(19)\cos 39$ $x^2 = 442 - 342 \cdot \cos 39$ $x^2 = 176.2161$ $x = 13.3$	2.  ☒ s ☐ c $\frac{\sin 49}{30} = \frac{\sin 54}{x}$ $x \cdot \sin 49 = 30 \cdot \sin 54$ $x = \frac{30 \cdot \sin 54}{\sin 49}$ $x = 32.2$
3.  ☒ s ☐ c $\frac{\sin x}{8} = \frac{\sin 112}{18}$ $18 \cdot \sin x = 8 \cdot \sin 112$ $\sin x = .4121$ $x = 24.3^\circ, 155.7^\circ$	4.  ☐ s ☒ c $7^2 = 5^2 + 6^2 - 2(5)(6) \cdot \cos x$ $49 = 61 - 60 \cdot \cos x$ $-12 = -60 \cdot \cos x$ $.2 = \cos x$ $x = \cos^{-1}(.2)$ $x = 78.5^\circ$

SOLVING TRIANGLES: To solve a triangle means to find ALL missing side and angle measures. Use the information given to solve each triangle. Round all answers to the nearest tenth.

5. In $\triangle ABC$, $m\angle A = 47^\circ$, $m\angle B = 52^\circ$ and $AC = 13$.



$$\frac{\sin 52}{13} = \frac{\sin 81}{x}$$

$$x \cdot \sin 52 = 13 \cdot \sin 81$$

$$\boxed{x = 16.3}$$

$$\frac{\sin 52}{13} = \frac{\sin 47}{y}$$

$$y \cdot \sin 52 = 13 \cdot \sin 47$$

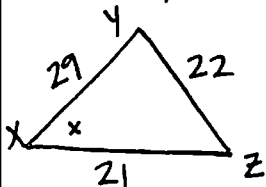
$$\boxed{y = 12.1}$$

$$m\angle C = 81^\circ$$

$$AB = 16.3$$

$$BC = 12.1$$

6. In $\triangle XYZ$, $XY = 29$, $YZ = 22$, and $XZ = 21$.



$$22^2 = 21^2 + 29^2 - 2(21)(29) \cdot \cos x$$

$$484 = 1282 - 1218 \cdot \cos x$$

$$-798 = -1218 \cos x$$

$$\boxed{49.1 = x}$$

$$\frac{\sin 49.1}{22} = \frac{\sin y}{21}$$

$$22 \cdot \sin y = 21 \cdot \sin 49.1$$

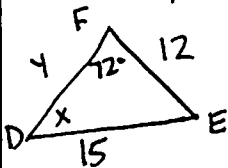
$$m\angle X = 49.1^\circ$$

$$m\angle Y = 46.2^\circ$$

$$m\angle Z = 84.7^\circ$$

$$\boxed{y = 46.2}$$

7. In $\triangle DEF$, $m\angle F = 72^\circ$, $DE = 15$, and $EF = 12$.



$$\frac{\sin 72}{15} = \frac{\sin x}{12}$$

$$15 \cdot \sin x = 12 \cdot \sin 72$$

$$\sin x = .7608$$

$$\boxed{x = 49.5^\circ}$$

$$\frac{\sin 72}{15} = \frac{\sin y}{12}$$

$$y \cdot \sin 72 = 15 \cdot \sin 58.5$$

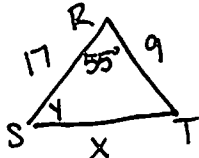
$$y = 13.4$$

$$m\angle D = 49.5^\circ$$

$$m\angle E = 58.5^\circ$$

$$DF = 13.4$$

8. In $\triangle RST$, $m\angle R = 55^\circ$, $RS = 17$, and $RT = 9$.



$$x^2 = 9^2 + 17^2 - 2(9)(17) \cdot \cos 55$$

$$x^2 = 370 - 306 \cdot \cos 55$$

$$x^2 = 194.4856$$

$$x = 13.9$$

$$\frac{\sin 55}{13.9} = \frac{\sin y}{9}$$

$$9 \cdot \sin 55 = 13.9 \cdot \sin y$$

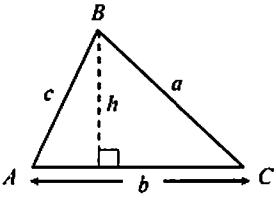
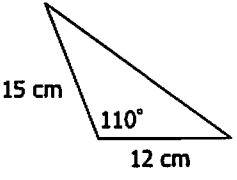
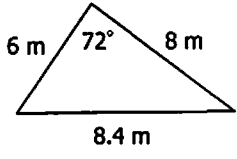
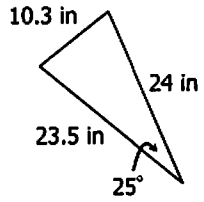
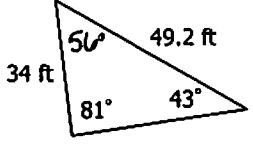
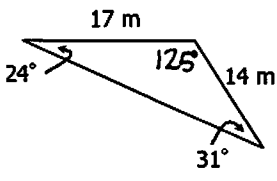
$$y = 32^\circ$$

$$m\angle S = 32^\circ$$

$$m\angle T = 93^\circ$$

$$ST = 13.9$$

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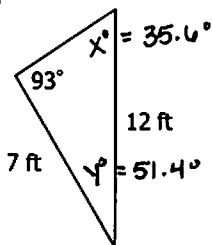
Main Ideas/Questions	Notes/Examples
Area of Triangles 	- What is the formula for the area of a triangle? $A = \frac{1}{2} b \cdot h$ - We know that $\sin C = \frac{h}{a}$ - Therefore, $h = a \cdot \sin C$ - Substitute this into the area formula for h: $A = \frac{1}{2} ab \cdot \sin C$
Examples	Directions: Find the area of each triangle to the nearest tenth.  $A = \frac{1}{2} (15)(12) \cdot \sin 110$ $A = 90 \cdot \sin 110$ $A = 84.6 \text{ cm}^2$  $A = \frac{1}{2} (6)(8) \cdot \sin 72$ $A = 24 \cdot \sin 72$ $A = 22.8 \text{ m}^2$  $A = \frac{1}{2} (23.5)(24) \cdot \sin 25$ $A = 282 \cdot \sin 25$ $A = 119.2 \text{ in}^2$  $A = \frac{1}{2} (34)(49.2) \cdot \sin 56$ $A = 836.4 \sin 56$ $A = 693.4 \text{ ft}^2$  $A = \frac{1}{2} (17)(14) \cdot \sin 125$ $A = 119 \cdot \sin 125$ $A = 97.5 \text{ m}^2$

* Work may vary, but areas should be very close to the same.

Using the Laws of Sines & Law of Cosines

In order to find the area, you will need two sides and an included angle (SAS).

6.



$$\frac{\sin 93}{12} = \frac{\sin X}{7}$$

$$12 \cdot \sin X = 7 \cdot \sin 93$$

$$\sin X = .5825$$

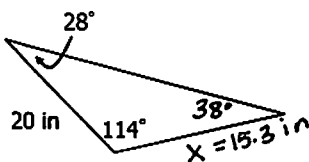
$$X = 35.6^\circ, 144.4^\circ$$

$$Y = 180 - (93 + 35.6) = 51.4^\circ$$

$$A = \frac{1}{2}(7)(12) \sin 51.4$$

$$A = 32.8 \text{ ft}^2$$

7.



$$\frac{\sin 38}{20} = \frac{\sin 28}{X}$$

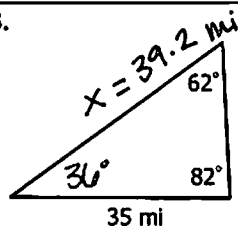
$$X \cdot \sin 38 = 20 \cdot \sin 28$$

$$X = 15.3 \text{ in}$$

$$A = \frac{1}{2}(20)(15.3) \sin 114$$

$$A = 139.8 \text{ in}^2$$

8.



$$\frac{\sin 62}{35} = \frac{\sin 82}{X}$$

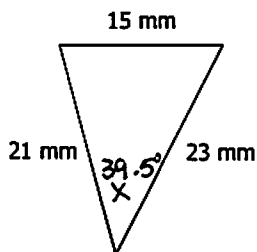
$$X \cdot \sin 62 = 35 \cdot \sin 82$$

$$X = 39.2$$

$$A = \frac{1}{2}(35)(39.2) \sin 36$$

$$A = 403.2 \text{ mi}^2$$

9.



$$15^2 = 21^2 + 23^2 - 2(21)(23) \cdot \cos X$$

$$225 = 970 - 966 \cdot \cos X$$

$$-745 = -966 \cdot \cos X$$

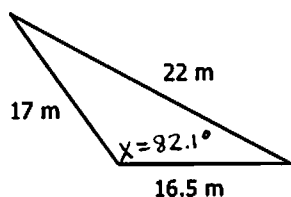
$$.7712 = \cos X$$

$$X = 39.5^\circ$$

$$A = \frac{1}{2}(21)(23) \sin 39.5$$

$$A = 153.6 \text{ mm}^2$$

10.



$$22^2 = 17^2 + 16.5^2 - 2(17)(16.5) \cdot \cos X$$

$$484 = 561.25 - 561 \cdot \cos X$$

$$-77.25 = -561 \cdot \cos X$$

$$.1377 = \cos X$$

$$X = 82.1^\circ$$

$$A = \frac{1}{2}(17)(16.5) \sin 82.1$$

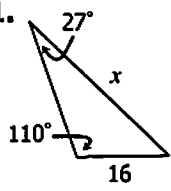
$$A = 138.9 \text{ m}^2$$

Name: _____

Unit 12: Trigonometry

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Homework 7: Law of Sines & Law of Cosines,
Area of Triangles**** This is a 2-page document! ******Directions:** Find each missing measure to the nearest tenth.

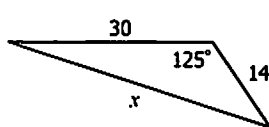
1. 

$$\frac{\sin 27}{16} = \frac{\sin 110}{x}$$

$$x \cdot \sin 27 = 16 \cdot \sin 110$$

$$x = \frac{16 \cdot \sin 110}{\sin 27}$$

$$x = 33.1$$

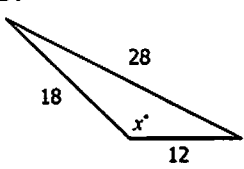
2. 

$$x^2 = 30^2 + 14^2 - 2(30)(14) \cdot \cos 125$$

$$x^2 = 1096 - 840 \cos 125$$

$$x^2 = 1577.8042$$

$$x = 39.7$$

3. 

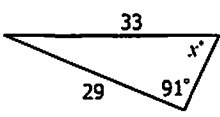
$$28^2 = 18^2 + 12^2 - 2(18)(12) \cdot \cos x$$

$$784 = 468 - 432 \cdot \cos x$$

$$316 = -432 \cdot \cos x$$

$$-.7315 = \cos x$$

$$x = 137^\circ$$

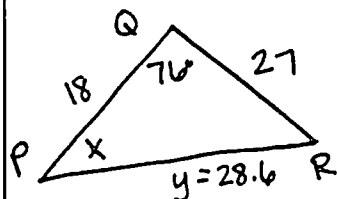
4. 

$$\frac{\sin 91}{33} = \frac{\sin x}{29}$$

$$33 \cdot \sin x = 29 \cdot \sin 91$$

$$\sin x = .8787$$

$$x = 61.5^\circ, 118.5^\circ$$

Directions: Use the information given to solve each triangle. Round to the nearest tenth.5. In $\triangle PQR$, $m\angle Q = 76^\circ$, $PQ = 18$, and $QR = 27$.

$$y^2 = 18^2 + 27^2 - 2(18)(27) \cdot \cos 76$$

$$y^2 = 1053 - 972 \cdot \cos 76$$

$$y^2 = 817.8519$$

$$y = 28.6$$

$$\frac{\sin 76}{28.6} = \frac{\sin x}{27}$$

$$28.6 \cdot \sin x = 27 \cdot \sin 76$$

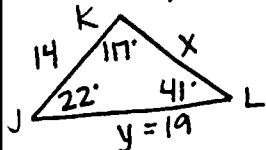
$$x = 66.4^\circ$$

$$m\angle P = 66.4^\circ$$

$$m\angle R = 37.6^\circ$$

$$PR = 28.6$$

6. In $\triangle JKL$, $m\angle K = 117^\circ$, $m\angle L = 41^\circ$, and $JK = 14$.



$$\frac{\sin 41}{14} = \frac{\sin 117}{y}$$

$$y \cdot \sin 41 = 14 \cdot \sin 117$$

$$y = 19$$

$$\frac{\sin 41}{14} = \frac{\sin 22}{x}$$

$$x \cdot \sin 41 = 14 \cdot \sin 22$$

$$x = 8$$

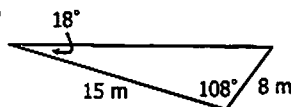
$$m\angle J = \underline{22^\circ}$$

$$JL = \underline{19}$$

$$KL = \underline{8}$$

Directions: Find the area of each triangle to the nearest tenth.

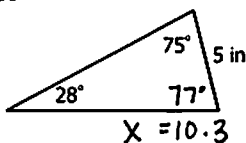
7.



$$A = \frac{1}{2}(15)(8) \cdot \sin 108$$

$$A = 57.1 \text{ m}^2$$

8.



$$\frac{\sin 75}{x} = \frac{\sin 28}{5}$$

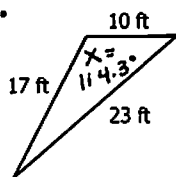
$$x \cdot \sin 28 = 5 \cdot \sin 75$$

$$x = 10.3 \text{ in}$$

$$A = \frac{1}{2}(10.3)(5) \sin 77$$

$$A = 25.1 \text{ in}^2$$

9.



$$23^2 = 17^2 + 10^2 - 2(17)(10) \cdot \cos x$$

$$529 = 389 - 340 \cdot \cos x$$

$$140 = -340 \cdot \cos x$$

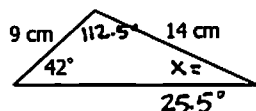
$$-.4118 = \cos x$$

$$x = 114.3^\circ$$

$$A = \frac{1}{2}(17)(10) \cdot \sin 114.3$$

$$A = 77.5 \text{ ft}^2$$

10.



$$\frac{\sin 42}{14} = \frac{\sin x}{9}$$

$$14 \cdot \sin x = 9 \cdot \sin 42$$

$$\sin x = .4302$$

$$x = 25.5^\circ, 154.5^\circ$$

$$A = \frac{1}{2}(9)(14) \cdot \sin 112.5$$

$$A = 58.2 \text{ cm}^2$$

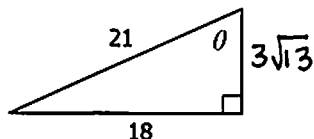
Unit 12 Test Study Guide (Trigonometry - Part 1)

Name: _____

Date: _____ Block: _____

Topic 1: Trigonometric Functions

1. Find the values of the six trigonometric functions for angle θ . Give answers in simplest form.



$$x^2 + 18^2 = 21^2$$

$$x^2 + 324 = 441$$

$$x^2 = 117$$

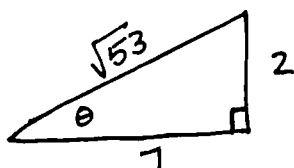
$$x = 3\sqrt{3}$$

$$\sin \theta = \frac{18}{21} = \frac{6}{7} \quad \csc \theta = \frac{21}{18} = \frac{7}{6}$$

$$\cos \theta = \frac{3\sqrt{3}}{21} = \frac{\sqrt{3}}{7} \quad \sec \theta = \frac{21}{3\sqrt{3}} = \frac{7\sqrt{3}}{3}$$

$$\tan \theta = \frac{18}{3\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3} \quad \cot \theta = \frac{3\sqrt{3}}{18} = \frac{\sqrt{3}}{6}$$

2. If $\tan \theta = \frac{2}{7}$, find the remaining trigonometric functions.



$$2^2 + 7^2 = x^2$$

$$4 + 49 = x^2$$

$$53 = x^2$$

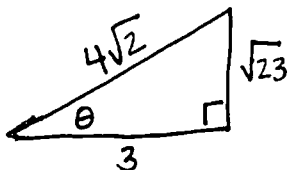
$$\sqrt{53} = x$$

$$\sin \theta = \frac{2}{\sqrt{53}} = \frac{2\sqrt{53}}{53} \quad \csc \theta = \frac{\sqrt{53}}{2}$$

$$\cos \theta = \frac{7}{\sqrt{53}} = \frac{7\sqrt{53}}{53} \quad \sec \theta = \frac{\sqrt{53}}{7}$$

$$\tan \theta = \frac{2}{7} \quad \cot \theta = \frac{7}{2}$$

3. If $\sec \theta = \frac{4\sqrt{2}}{3}$, find the remaining trigonometric functions.



$$3^2 + x^2 = (4\sqrt{2})^2$$

$$9 + x^2 = 32$$

$$x^2 = 23$$

$$x = \sqrt{23}$$

$$\sin \theta = \frac{\sqrt{23}}{4\sqrt{2}} = \frac{\sqrt{46}}{8} \quad \csc \theta = \frac{4\sqrt{2}}{\sqrt{23}} = \frac{4\sqrt{46}}{23}$$

$$\cos \theta = \frac{3}{4\sqrt{2}} = \frac{3\sqrt{2}}{8} \quad \sec \theta = \frac{4\sqrt{2}}{3}$$

$$\tan \theta = \frac{\sqrt{23}}{3} \quad \cot \theta = \frac{3}{\sqrt{23}} = \frac{3\sqrt{23}}{23}$$

Directions: Find each missing measure to the nearest tenth.

4. $\sin 42 = \frac{x}{14}$
 $x = 9.4$

5. $\sin 18 = \frac{x}{21}$
 $x = \frac{21}{\sin 18}$
 $x = 68$

6. $\cos \theta = \frac{19}{23}$
 $\theta = \cos^{-1}(19/23)$
 $\theta = 34.3^\circ$

7. $\sin \theta = \frac{34}{35}$
 $\theta = \sin^{-1}(34/35)$
 $\theta = 76.3^\circ$

Topic 2: Angles & Angle Measures

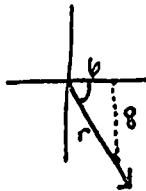
Directions: Give each degree measure in radians and radian measure in degrees.

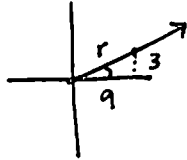
8. 165° $165 \cdot \frac{\pi}{180} = \frac{165\pi}{180} = \boxed{\frac{11\pi}{12}}$	9. -54° $-54 \cdot \frac{\pi}{180} = \frac{-54\pi}{180} = \boxed{-\frac{3\pi}{10}}$	10. 243° $243 \cdot \frac{\pi}{180} = \frac{243\pi}{180} = \boxed{\frac{27\pi}{20}}$
11. $\frac{9\pi}{5}$ $\frac{9\pi}{5} \cdot \frac{180}{\pi} = \frac{1620}{5} = \boxed{324^\circ}$	12. $-\frac{13\pi}{20}$ $-\frac{13\pi}{20} \cdot \frac{180}{\pi} = \frac{-2340}{20} = \boxed{-117^\circ}$	13. $-\frac{4\pi}{45}$ $-\frac{4\pi}{45} \cdot \frac{180}{\pi} = \frac{-720}{45} = \boxed{-16^\circ}$

Directions: Give two coterminal $\angle$'s (one positive and one negative) and a reference $\angle$ for each.

14. 305°  Coterminal $\angle$'s: <u>$665^\circ, -55^\circ$</u> Reference $\angle$: <u>55°</u>	15. $\frac{7\pi}{10}$ (126°)  Coterminal $\angle$'s: <u>$\frac{27\pi}{10}, -\frac{13\pi}{10}$</u> Reference $\angle$: <u>$\frac{3\pi}{10}$</u>
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Directions: The terminal side of θ in standard position contains each point. Find the exact values of each trigonometric function of θ .

16. (6, -8)  $6^2 + 8^2 = r^2$ $36 + 64 = r^2$ $100 = r^2$ $10 = r$ $x = 6$ $y = -8$ $r = 10$ $\sin \theta = \frac{-8}{10} = \frac{-4}{5}$ $\csc \theta = \frac{-5}{4}$ $\cos \theta = \frac{6}{10} = \frac{3}{5}$ $\sec \theta = \frac{5}{3}$ $\tan \theta = \frac{-8}{6} = \frac{-4}{3}$ $\cot \theta = \frac{-3}{4}$

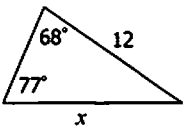
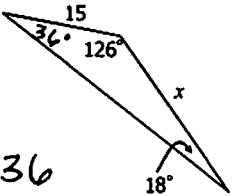
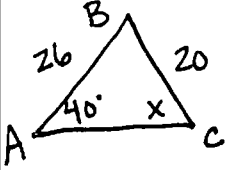
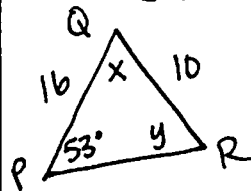
17. (9, 3)  $9^2 + 3^2 = r^2$ $81 + 9 = r^2$ $90 = r^2$ $3\sqrt{10} = r$ $x = 9$ $y = 3$ $r = 3\sqrt{10}$ $\sin \theta = \frac{3}{3\sqrt{10}} = \frac{\sqrt{10}}{10}$ $\csc \theta = \frac{\sqrt{10}}{3}$ $\cos \theta = \frac{9}{3\sqrt{10}} = \frac{3\sqrt{10}}{10}$ $\sec \theta = \frac{\sqrt{10}}{3}$ $\tan \theta = \frac{3}{9} = \frac{1}{3}$ $\cot \theta = 3$
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Topic 3: The Unit Circle

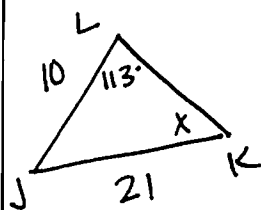
18. Which trigonometric functions are negative in quadrant IV? Sine and tangent	19. If $\cos \theta < 0$ and $\csc \theta < 0$ which quadrant(s) could the terminal side of θ lie? Q 3
--	--

Directions: Give the exact value of each trigonometric function.		
20. $\sin 45$ $\frac{\sqrt{2}}{2}$	21. $\cot 135^\circ$ $(\tan 135 = -1)$ -1	22. $\sec 330^\circ$ $(\cos 330 = \frac{\sqrt{3}}{2})$ $\frac{2\sqrt{3}}{3}$
23. $\tan \frac{4\pi}{3}$ $\sqrt{3}$	24. $\csc \frac{11\pi}{6}$ $(\sin \frac{11\pi}{6} = -\frac{1}{2})$ -2	25. $\cos \frac{2\pi}{3}$ $-\frac{1}{2}$
26. $\tan \frac{7\pi}{6}$ $\frac{\sqrt{3}}{3}$	27. $\sec \frac{5\pi}{4}$ $(\cos \frac{5\pi}{4} = -\frac{\sqrt{2}}{2})$ $-\sqrt{2}$	28. $\cot \pi$ $(\tan = 0)$ Undefined

Topic 4: Law of Sines

Directions: Find each missing length to the nearest tenth.	
29. $\frac{\sin 77}{12} = \frac{\sin 68}{x}$ $x \cdot \sin 77 = 12 \cdot \sin 68$ $x = 11.4$ 	30. $\frac{\sin 18}{15} = \frac{\sin 36}{x}$ $x \cdot \sin 18 = 15 \cdot \sin 36$ $x = 28.5$ 
Directions: Find each missing angle to the nearest tenth. Give all possible solutions.	
31. In $\triangle ABC$, $m\angle A = 40^\circ$, $AB = 26$, and $BC = 20$. Find $m\angle C$.  $\frac{\sin 40}{20} = \frac{\sin x}{26}$ $20 \cdot \sin x = 26 \cdot \sin 40$ $\sin x = .8356$ $x = 56.7^\circ, 123.3^\circ$	$40 + 56.7 < 180 \checkmark$ $40 + 123.3 < 180 \checkmark$
32. In $\triangle PQR$, $m\angle P = 53^\circ$, $PQ = 16$, and $QR = 10$. Find $m\angle Q$.  $\frac{\sin 53}{10} = \frac{\sin y}{16}$ $16 \cdot \sin 53 = 10 \cdot \sin y$ $\sin y = 1.2778$ $y = \emptyset$	No Solution

33. In $\triangle JKL$, $m\angle L = 113^\circ$, $JK = 21$, and $JL = 10$. Find $m\angle K$.



$$\frac{\sin 113}{21} = \frac{\sin x}{10}$$

$$10 \cdot \sin 113 = 21 \cdot \sin x$$

$$\sin x = .4383$$

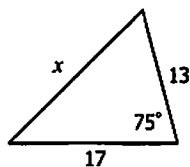
$$x = 26^\circ, 154^\circ$$

$$\boxed{26^\circ}$$

Topic 5: Law of Cosines

Directions: Find each missing measure to the nearest tenth.

34.



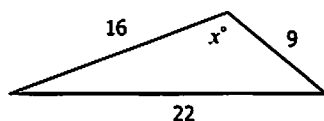
$$x^2 = 17^2 + 13^2 - 2(17)(13) \cdot \cos 75$$

$$x^2 = 458 - 442 \cos 75$$

$$x^2 = 343.6020$$

$$\boxed{x = 18.5}$$

35.



$$22^2 = 16^2 + 9^2 - 2(16)(9) \cdot \cos x$$

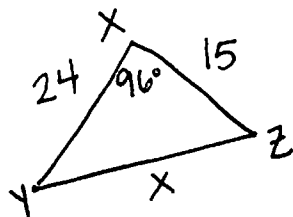
$$484 = 337 - 288 \cdot \cos x$$

$$147 = -288 \cdot \cos x$$

$$-.5104 = \cos x$$

$$\boxed{x = 120.7^\circ}$$

36. In $\triangle XYZ$, $m\angle X = 96^\circ$, $XZ = 15$, and $XY = 24$. Find YZ .



$$x^2 = 24^2 + 15^2 - 2(24)(15) \cdot \cos 96$$

$$x^2 = 801 - 720 \cdot \cos 96$$

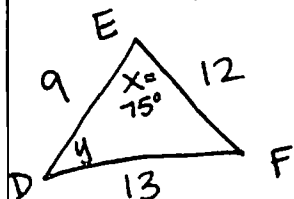
$$x^2 = 876.2605$$

$$\boxed{x = 29.6}$$

Topic 6: Solving Triangles

Directions: Solve each triangle. Round all answers to the nearest tenth.

37. In $\triangle DEF$, $DE = 9$, $EF = 12$, and $DF = 13$.



$$13^2 = 9^2 + 12^2 - 2(9)(12) \cdot \cos x$$

$$169 = 225 - 216 \cdot \cos x$$

$$-56 = -216 \cdot \cos x$$

$$.2593 = \cos x$$

$$x = 75^\circ$$

$$\frac{\sin 75}{13} = \frac{\sin y}{12}$$

$$12 \cdot \sin 75 = 13 \cdot \sin y$$

$$\sin y = .8916$$

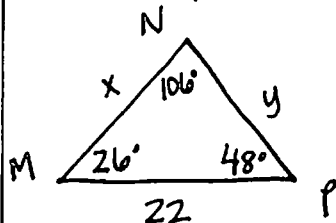
$$y = 63.1^\circ$$

$$m\angle D = 63.1^\circ$$

$$m\angle E = 75^\circ$$

$$m\angle F = 41.9^\circ$$

38. In $\triangle MNP$, $m\angle M = 26^\circ$, $m\angle P = 48^\circ$, and $MP = 22$.



$$\frac{\sin 106}{22} = \frac{\sin 48}{x}$$

$$x \cdot \sin 106 = 22 \cdot \sin 48$$

$$x = 17$$

$$\frac{\sin 106}{22} = \frac{\sin 26}{y}$$

$$y \cdot \sin 106 = 22 \cdot \sin 26$$

$$y = 10$$

$$m\angle N = 106^\circ$$

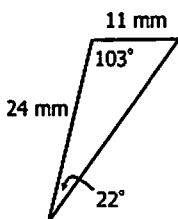
$$MN = 17$$

$$NP = 10$$

Topic 7: Area of Triangles

Directions: Find the area of each triangle to the nearest tenth.

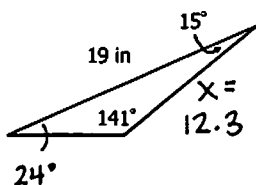
39.



$$A = \frac{1}{2} (24)(11) \cdot \sin 103$$

$$A = 128.6 \text{ mm}^2$$

40.



$$\frac{\sin 141}{19} = \frac{\sin 24}{x}$$

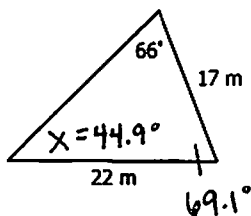
$$x \cdot \sin 141 = 19 \cdot \sin 24$$

$$x = 12.3$$

$$A = \frac{1}{2} (19)(12.3) \cdot \sin 15$$

$$A = 30.2 \text{ in}^2$$

41.



$$\frac{\sin 66}{22} = \frac{\sin x}{17}$$

$$22 \cdot \sin x = 17 \cdot \sin 66$$

$$\sin x = .7059$$

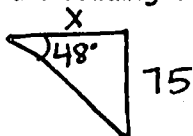
$$x = 44.9^\circ, 135.1$$

$$A = \frac{1}{2} (22)(17) \cdot \sin 69.1$$

$$A = 174.7 \text{ m}^2$$

Topic 8: Applications

42. From the top of a building, the angle of depression to a car parked on the street below is 48° . If the building is 75 feet tall, how far is the car parked from the base of the building?



$$\tan 48 = \frac{75}{x}$$

$$x = \frac{75}{\tan 48}$$

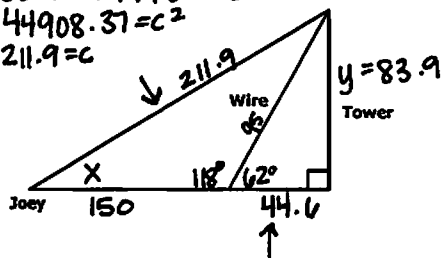
$$x = 67.5 \text{ ft}$$

43. A 95-foot wire attached from the top of a cell phone tower makes a 62° angle with the ground. Joey is standing 150 feet from the end of the wire, looking up at the tower. Find the angle of elevation from the point on the ground where Joey is standing to the top of the tower.

$$83.9^2 + 194.6^2 = c^2$$

$$44908.37 = c^2$$

$$211.9 = c$$



$$a^2 + 83.9^2 = 95^2$$

$$a^2 = 1985.79$$

$$a = 44.6$$

$$\frac{\sin 62}{y} = \frac{\sin 90}{95}$$

$$y \cdot \sin 90 = 95 \cdot \sin 62$$

$$y = 83.9$$

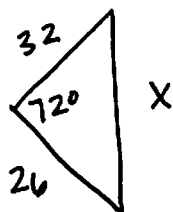
$$\frac{\sin 118}{211.9} = \frac{\sin X}{95}$$

$$211.9 \sin X = 95 \cdot \sin 118$$

$$\sin X = 0.3958$$

$$X = 23.3^\circ$$

44. Two trains leave from the same point and travel along straight tracks that differ in direction by 72° . If their speeds are 64 miles per hour and 52 miles per hour, approximate how far apart they will be after 30 minutes.



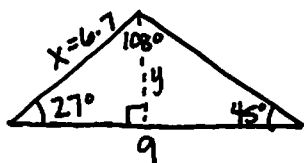
$$X^2 = 32^2 + 26^2 - 2(32)(26) \cdot \cos 72$$

$$X^2 = 1700 - 1664 \cdot \cos 72$$

$$X^2 = 1185.7957$$

$$X = 34.4 \text{ mi}$$

45. The angles of elevation to a balloon in the air from two points on level ground are 27° and 45° . If the points are 9 miles apart and the balloon is between the points, in the same vertical plane, approximate, to the nearest tenth of a mile, the height of the balloon above the ground.



$$\frac{\sin 45}{X} = \frac{\sin 108}{9}$$

$$X \cdot \sin 108 = 9 \cdot \sin 45$$

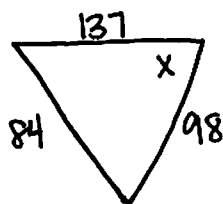
$$X = 6.7$$

$$\frac{\sin 90}{6.7} = \frac{\sin 27}{y}$$

$$y \cdot \sin 90 = 6.7 \cdot \sin 27$$

$$y = 3 \text{ mi}$$

46. A triangular lot has sides of 98 feet, 137 feet, and 84 feet. Find the area of the lot to the nearest tenth.



$$84^2 = 98^2 + 137^2 - 2(98)(137) \cos X$$

$$7056 = 28373 - 26852 \cos X$$

$$-21317 = -26852 \cos X$$

$$.7939 = \cos X$$

$$37.5^\circ = X$$

$$A = \frac{1}{2}(98)(137) \sin 37.5$$

$$A = 4086.6 \text{ ft}^2$$

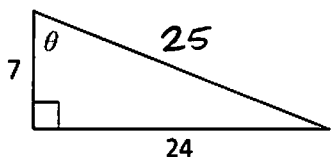
Name: _____

Date: _____ Bell: _____

Unit 12 Test (Part I)

Trigonometry

For questions 1-2, use the diagram below to give the exact trigonometric ratios.



1. $\tan \theta$

- A. $\frac{7}{24}$ B. $\frac{24}{7}$ C. $\frac{7}{25}$ D. $\frac{25}{7}$

B

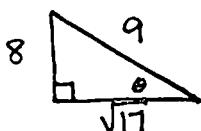
2. $\sec \theta$ $\cos \theta = \frac{7}{25}$

- A. $\frac{7}{25}$ B. $\frac{25}{7}$ C. $\frac{24}{25}$ D. $\frac{25}{24}$

B

3. If $\sin \theta = \frac{8}{9}$, find $\cot \theta$.

$$\tan \theta = \frac{8}{\sqrt{17}}$$



A. $\frac{9\sqrt{17}}{17}$

C. $\frac{8\sqrt{17}}{17}$

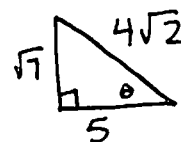
B. $\frac{\sqrt{17}}{9}$

D. $\frac{\sqrt{17}}{8}$

D

4. If $\tan \theta = \frac{\sqrt{7}}{5}$, find $\csc \theta$.

$$\sin \theta = \frac{\sqrt{7}}{4\sqrt{2}}$$



A. $\frac{4\sqrt{14}}{7}$

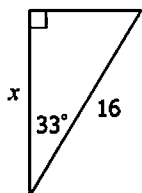
C. $\frac{5\sqrt{2}}{8}$

B. $\frac{\sqrt{14}}{8}$

D. $\frac{4\sqrt{2}}{5}$

A

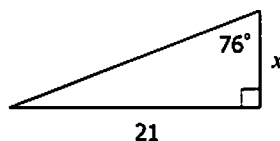
5. Find x to the nearest tenth.



$$\cos 33 = \frac{x}{16}$$

$$x = 13.4$$

6. Find x to the nearest tenth.

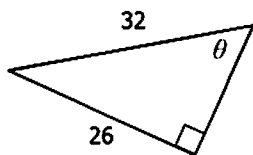


$$\tan 76 = \frac{21}{x}$$

$$x = \frac{21}{\tan 76}$$

$$x = 5.2$$

7. Find the measure of θ to the nearest tenth.

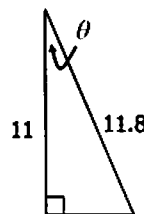


$$\sin \theta = \frac{26}{32}$$

$$\theta = \sin^{-1}\left(\frac{26}{32}\right)$$

$$\theta = 54.3^\circ$$

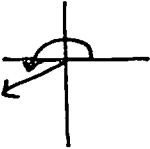
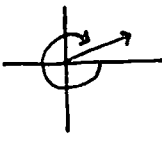
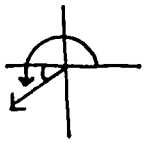
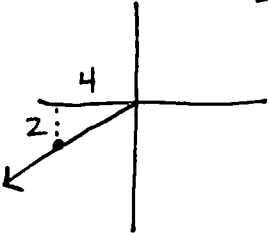
8. Find the measure of θ to the nearest tenth.



$$\cos \theta = \frac{11}{11.8}$$

$$\theta = \cos^{-1}\left(\frac{11}{11.8}\right)$$

$$\theta = 21.2^\circ$$

9. Convert 63° to radians. $\frac{63\pi}{180}$ $\frac{7\pi}{20}$	10. Convert -280° to radians. $\frac{-280\pi}{180}$ $-\frac{14}{9}\pi$
11. Convert $\frac{13\pi}{18}$ to degrees. $\frac{13\pi}{18} \cdot \frac{180}{\pi}$ 130°	12. Convert $\frac{19\pi}{10}$ to degrees. $\frac{19\pi}{10} \cdot \frac{180}{\pi}$ 342°
13. The terminal side of an angle measuring $\frac{13\pi}{12}$ radians lies in which quadrant? (195°) A. I B. II C. III D. IV  C	14. The terminal side of an angle measuring $-\frac{9\pi}{5}$ radians lies in which quadrant? (-324°) A. I B. II C. III D. IV  A
15. Which angle is coterminal to 128°? A. -52° B. 308° C. 232° D. 488° D	16. Which angle is coterminal to $\frac{2\pi}{5}$? A. $-\frac{3\pi}{5}$ B. $\frac{4\pi}{5}$ C. $-\frac{8\pi}{5}$ D. $\frac{7\pi}{5}$ C
17. Name the reference angle to 205°. A. 15° B. 25° C. 65° D. 75° $205 - 180$  B	18. Name the reference angle to $\frac{7\pi}{10}$. (126°) A. $\frac{\pi}{5}$ B. $\frac{3\pi}{10}$ C. $\frac{7\pi}{20}$ D. $\frac{3\pi}{20}$ $\pi - \frac{7\pi}{10}$  B
Use for questions 19-20: The terminal side of θ in standard position contains the point $(-4, -2)$. Find the exact value for each trigonometric function.  $2^2 + 4^2 = r^2$ $20 = r^2$ $2\sqrt{5} = r$ $x = -4$ $y = -2$ $r = 2\sqrt{5}$	19. $\sin \theta$ A. $\frac{2\sqrt{5}}{5}$ B. $-\frac{2\sqrt{5}}{5}$ C. $\frac{\sqrt{5}}{5}$ D. $-\frac{\sqrt{5}}{5}$ D
	20. $\cot \theta$ $\tan \theta = \frac{2}{4}$ A. $\frac{1}{2}$ B. $-\frac{1}{2}$ C. 2 D. -2 C

21. If $\cot \theta < 0$, in which quadrant(s) could the terminal side of angle θ lie?

- A. II and IV
B. II and III
C. III only
D. IV only

S⁺ | A⁺
T⁺ | C⁺

A

22. If $\sin \theta = \cos \theta$, in which quadrant(s) could the terminal side of angle θ lie?

- A. I only
B. III only
C. I and III
D. II and IV

C

For questions 23-26, give the exact value for each trigonometric function.

23. $\sin 120^\circ$

- A. $-\frac{\sqrt{3}}{2}$ C. $\frac{\sqrt{3}}{2}$
B. $-\frac{1}{2}$ D. $\frac{1}{2}$

C

24. $\csc 225^\circ$

$$\sin 225 = -\frac{\sqrt{2}}{2}$$

- A. $-\frac{\sqrt{2}}{2}$ C. $\frac{\sqrt{2}}{2}$
B. $-\sqrt{2}$ D. $\sqrt{2}$

B

25. $\tan \frac{11\pi}{6}$

- A. $\frac{\sqrt{3}}{3}$ C. $\sqrt{3}$
B. $-\frac{\sqrt{3}}{3}$ D. $-\sqrt{3}$

B

26. $\sec \frac{\pi}{2}$

$$\cos \frac{\pi}{2} = 0$$

- A. 0
B. 1
C. -1
D. undefined

D

27. What function has a value of $-\sqrt{3}$?

- A. $\tan \frac{\pi}{3}$ C. $\tan \frac{7\pi}{6}$
B. $\cot \frac{5\pi}{6}$ D. $\cot \frac{4\pi}{3}$

B

28. What function has a value of 2?

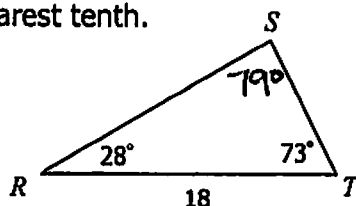
- A. $\csc \frac{11\pi}{6}$ C. $\sec \frac{5\pi}{3}$
B. $\csc \frac{7\pi}{6}$ D. $\sec \frac{4\pi}{3}$

C

29. Which functions have a value of -1? Check all that apply.

☒ $\cos \pi$	☒ $\tan \frac{3\pi}{4}$	☒ $\sin \frac{3\pi}{2}$	☐ $\csc \frac{\pi}{2}$
☐ $\sec \frac{3\pi}{2}$	☐ $\tan \frac{\pi}{2}$	☐ $\cot \pi$	☒ $\cot \frac{7\pi}{4}$

30. Find ST to the nearest tenth.



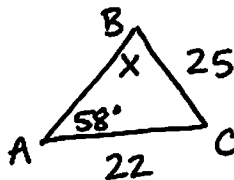
$$\frac{\sin 79}{18} = \frac{\sin 28}{x}$$

$$18 \cdot \sin 28 = x \cdot \sin 79$$

$$x = \frac{18 \cdot \sin 28}{\sin 79}$$

$$ST = 8.6$$

31. In $\triangle ABC$, $m\angle A = 58^\circ$, $AC = 22$, and $BC = 25$. Find $m\angle B$.



$$\frac{\sin 58}{25} = \frac{\sin x}{22}$$

$$22 \cdot \sin 58 = 25 \cdot \sin x$$

$$.7463 = \sin x$$

$$48^\circ = x$$

$$\cancel{132^\circ = x}$$

$$58 + 48 < 180 \checkmark$$

$$58 + 132 > 180 \times$$

(A) 48°

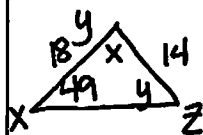
B. 74°

C. 48° or 132°

D. Not Possible

A

32. In $\triangle XYZ$, $m\angle X = 49^\circ$, $XY = 18$, and $YZ = 14$. Find $m\angle Y$.



$$\frac{\sin 49}{14} = \frac{\sin y}{18}$$

$$18 \cdot \sin 49 = 14 \cdot \sin y$$

$$.9703 = \sin y$$

$$y = 76^\circ, 104^\circ$$

$$x = 180 - (49 + 76) = 55^\circ$$

$$x = 180 - (49 + 104) = 27^\circ$$

A. 55°

B. 76°

C. 76° or 104°

(D) 55° or 27°

D

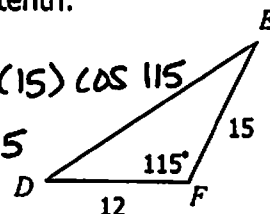
33. Find DE to the nearest tenth.

$$x^2 = 12^2 + 15^2 - 2(12)(15) \cos 115$$

$$x^2 = 369 - 360 \cdot \cos 115$$

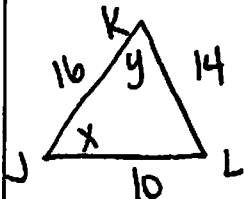
$$x^2 = 521.1426$$

$$x = 22.8$$



$$DE = 22.8$$

34. Given $\triangle JKL$ with $JK = 16$, $KL = 14$, and $JL = 10$ solve the triangle. Round all answers to the nearest tenth.



$$14^2 = 10^2 + 16^2 - 2(10)(16) \cos x$$

$$196 = 356 - 320 \cos x$$

$$-160 = -320 \cos x$$

$$.5 = \cos x$$

$$x = 60^\circ$$

$$\frac{\sin 60}{14} = \frac{\sin y}{10}$$

$$10 \cdot \sin 60 = 14 \cdot \sin y$$

$$.9620 = \sin y$$

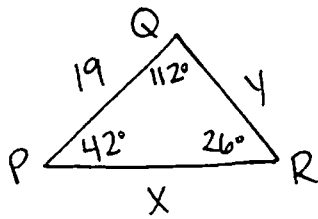
$$y = 38.2$$

$$m\angle J = 60^\circ$$

$$m\angle K = 38.2^\circ$$

$$m\angle L = 81.8^\circ$$

35. Given $\triangle PQR$ with $m\angle P = 42^\circ$, $m\angle R = 26^\circ$, and $PQ = 19$, solve the triangle. Round all answers to the nearest tenth.



$$\frac{\sin 26}{19} = \frac{\sin 112}{x}$$

$$x \cdot \sin 26 = 19 \cdot \sin 112$$

$$x = 40.2$$

$$\frac{\sin 26}{19} = \frac{\sin 42}{y}$$

$$y \cdot \sin 26 = 19 \cdot \sin 42$$

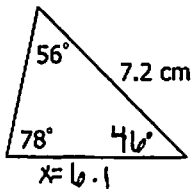
$$y = 29$$

$$m\angle Q = 112^\circ$$

$$QR = 29$$

$$PR = 40.2$$

36. Find the area of the triangle below to the nearest tenth.



$$\frac{\sin 78}{7.2} = \frac{\sin 56}{x}$$

$$x \cdot \sin 78 = 7.2 \cdot \sin 56$$

$$x = 6.1$$

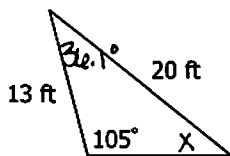
$$A = \frac{1}{2}(6.1)(7.2)\sin 46$$

$$A = 15.8 \text{ cm}^2$$

- A. 13.7 cm^2 C. 18.4 cm^2
B. 12.2 cm^2 D. 15.8 cm^2

D

37. Find the area of the triangle below to the nearest tenth.



$$\frac{\sin 105}{20} = \frac{\sin x}{13}$$

$$20 \cdot \sin x = 13 \cdot \sin 105$$

$$\sin x = .6279$$

$$x = 38.9^\circ$$

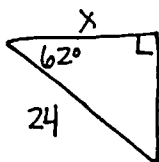
$$A = \frac{1}{2}(13)(20)\sin 36.1$$

$$A = 76.6 \text{ ft}^2$$

- A. 80.5 ft^2 C. 76.6 ft^2
B. 71.3 ft^2 D. 68.1 ft^2

C

38. A 24-foot ladder is placed against a building to reach a window on the second floor. The angle of depression from the top of the ladder to the bottom of the ladder is 62° . Find the distance between the bottom of the ladder and the base of the building to the nearest tenth of a foot.

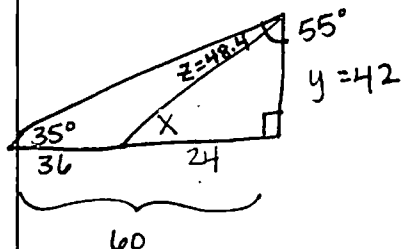


$$\cos 62 = \frac{x}{24}$$

$$x = 11.3$$

11.3 ft

39. Alex is standing 60 feet from the base of a flagpole. He measures the angle of elevation to the top of the pole as 35° . Vera is 36 feet closer to the base of the flagpole on a straight level path. Find the angle of elevation from the point Vera is standing to the top of the flagpole to the nearest tenth of a degree.



$$\frac{\sin 55}{60} = \frac{\sin 35}{y}$$

$$y \cdot \sin 55 = 60 \cdot \sin 35$$

$$y = 42$$

$$\begin{aligned} 24^2 + 42^2 &= z^2 \\ 2340 &= z^2 \\ z &= 48.4 \end{aligned}$$

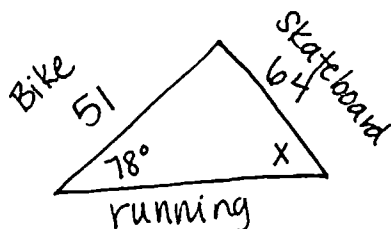
$$\frac{\sin 90}{48.4} = \frac{\sin x}{42}$$

$$\sin x = .8678$$

$$x = 60.2^\circ$$

$$60.2^\circ$$

40. A course in the shape of a triangle consists of a running path, biking path, and skateboarding path. The biking path is 51 meters long and the skateboarding path is 64 meters long. If the angle between the running and biking paths is 78° , find the angle between the running and skateboarding paths to the nearest tenth of a degree.



$$\frac{\sin x}{51} = \frac{\sin 78}{64}$$

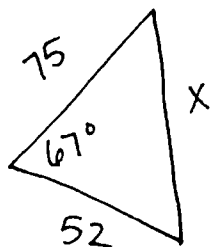
$$64 \cdot \sin x = 51 \cdot \sin 78$$

$$\sin x = .7795$$

$$x = 51.2^\circ$$

$$51.2^\circ$$

41. The distances from a boat to two buoys are 75 meters and 52 meters. If the angle between the two lines of sight is 67° , find the distance between the buoys. Round to the nearest tenth of a meter.



$$x^2 = 75^2 + 52^2 - 2(75)(52) \cdot \cos 67$$

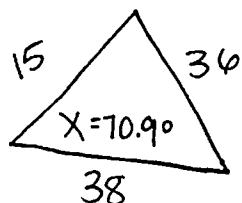
$$x^2 = 8329 - 7800 \cos 67$$

$$x^2 = 5281.2972$$

$$x = 72.7$$

$$72.7 \text{ m}$$

42. A triangular sail on a boat has side lengths that measure 15 feet, 36 feet, and 38 feet. Find the area of the sail to the nearest tenth of a foot.



$$36^2 = 15^2 + 38^2 - 2(15)(38) \cdot \cos x$$

$$1296 = 1669 - 1140 \cos x$$

$$-373 = -1140 \cos x$$

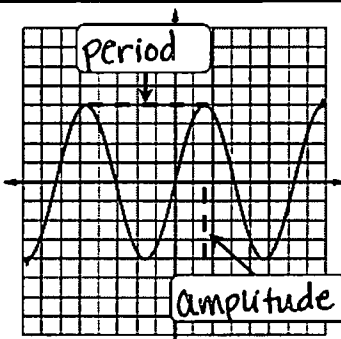
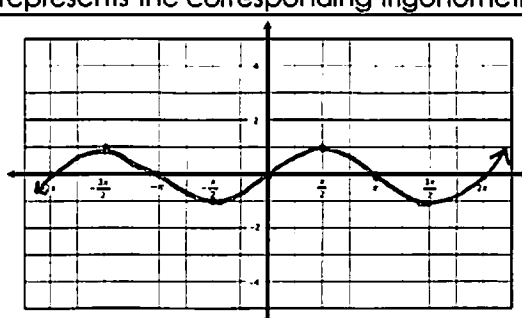
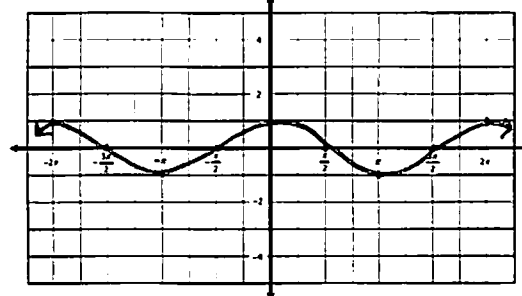
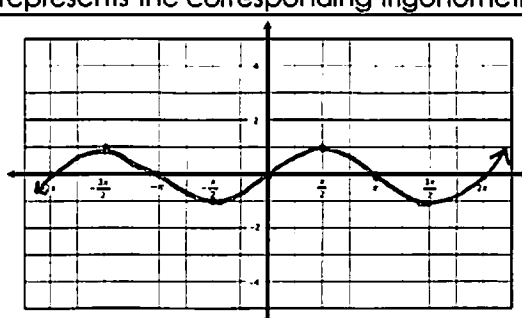
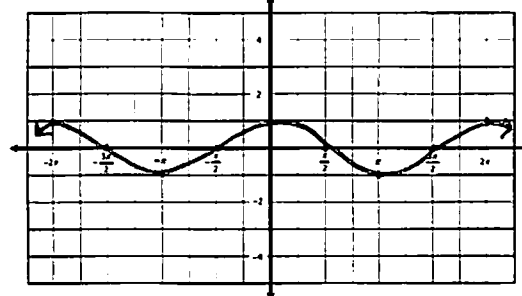
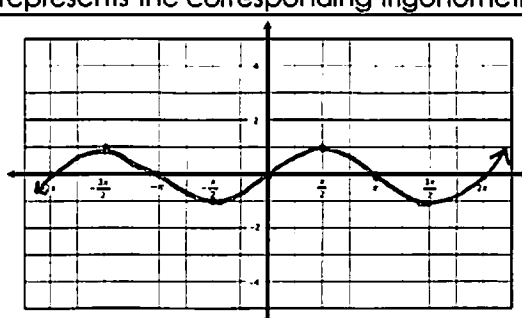
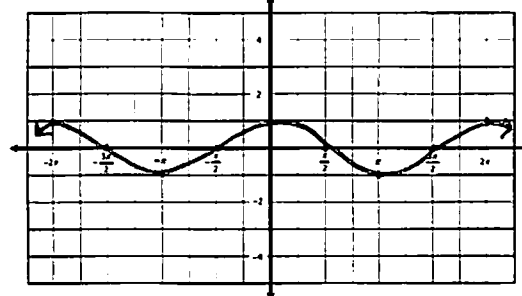
$$70.9^\circ = x$$

$$A = \frac{1}{2} (15)(38) \cdot \sin 70.9$$

$$A = 269.3$$

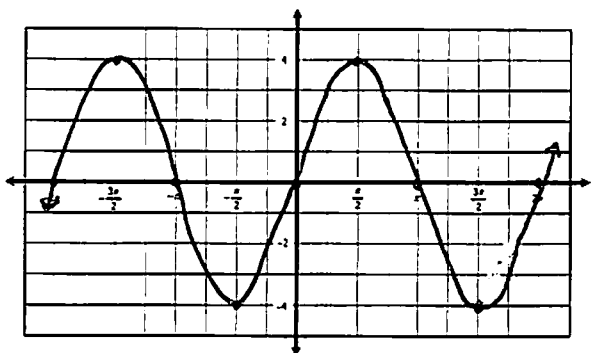
$$269.3 \text{ ft}^2$$

Name:	Date:
Topic:	Class:

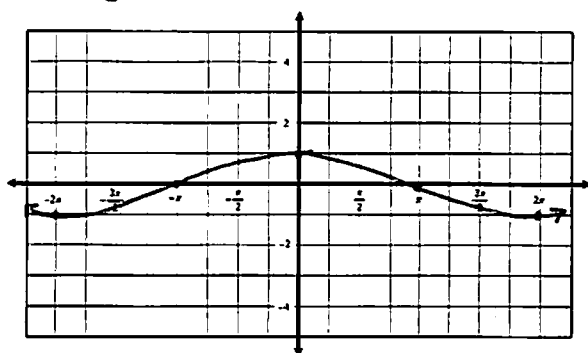
Main Ideas/Questions	Notes/Examples						
PERIODIC FUNCTIONS	- Periodic functions are functions that repeat exactly in regular intervals, called cycles. - Trigonometric functions are periodic! Period: the length of a cycle. Periods can start at any point on the graph. Amplitude: Half the distance between the minimum and maximum values. 						
GRAPHING SINE & COSINE	Use the unit circle values! x represents the angle θ in standard form and y represents the corresponding trigonometric value. <table style="width: 100%; border-collapse: collapse; margin-top: 10px;"> <tr> <td style="width: 25%; padding: 10px; vertical-align: top;"> SINE $(y = \sin \theta)$ </td><td style="width: 50%; padding: 10px; text-align: center;">  </td><td style="width: 25%; padding: 10px; vertical-align: top;"> Amplitude: 1 Period: 2π </td></tr> <tr> <td style="padding: 10px; vertical-align: top;"> COSINE $(y = \cos \theta)$ </td><td style="padding: 10px; text-align: center;">  </td><td style="padding: 10px; vertical-align: top;"> Amplitude: 1 Period: 2π </td></tr> </table> The graphs of sine and cosine are called "waves" because they have a wave-like rolling appearance.	SINE $(y = \sin \theta)$		Amplitude: 1 Period: 2π	COSINE $(y = \cos \theta)$		Amplitude: 1 Period: 2π
SINE $(y = \sin \theta)$		Amplitude: 1 Period: 2π					
COSINE $(y = \cos \theta)$		Amplitude: 1 Period: 2π					
Changing the Amplitude & Period	Given $y = a \cdot \sin b\theta$ or $y = a \cdot \cos b\theta$ where θ is in radians: the amplitude is <u>a</u> and the period is <u>$\frac{2\pi}{ b }$</u>.						
Directions: Identify the amplitude and period of each function.							
1. $y = 3 \cdot \sin 2\theta$	$\frac{2\pi}{2} = \pi$ Amp: <u>3</u> Period: <u>π</u>						
2. $y = \frac{3}{2} \cdot \sin \frac{1}{3}\theta$	$\frac{2\pi}{\frac{1}{3}} = 2\pi \cdot \frac{3}{1}$ Amp: <u>$\frac{3}{2}$</u> Period: <u>6π</u>						

Directions: Identify the amplitude and period, then graph each function.

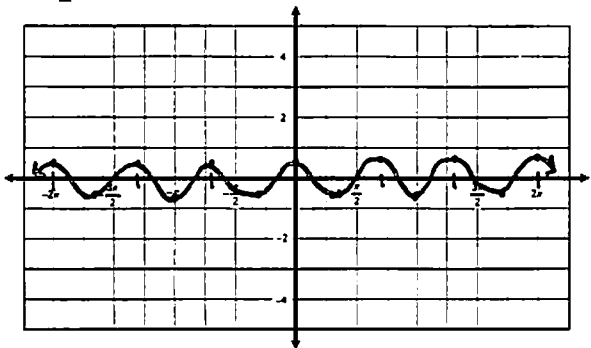
3. $y = 4 \cdot \sin \theta$ Amp: 4 Period: 2π



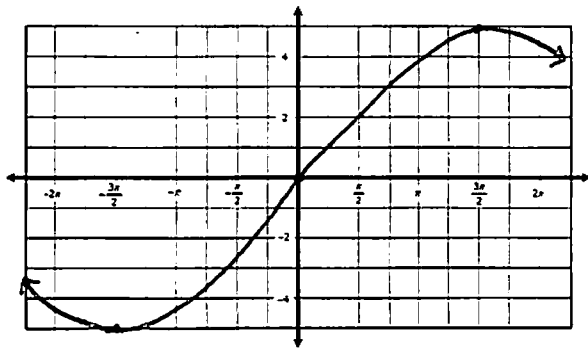
4. $y = \cos \frac{1}{2} \theta$ Amp: 1 Period: 4π



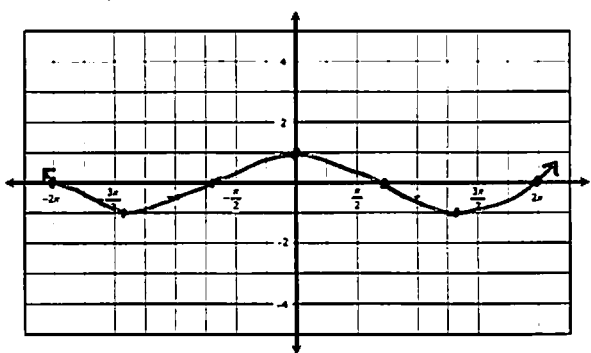
5. $y = \frac{1}{2} \cdot \cos 3\theta$ Amp: $\frac{1}{2}$ Period: $\frac{2\pi}{3}$



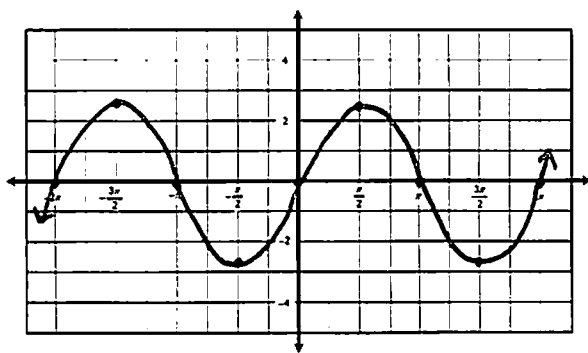
6. $y = 5 \cdot \sin \frac{1}{3} \theta$ Amp: 5 Period: 6π



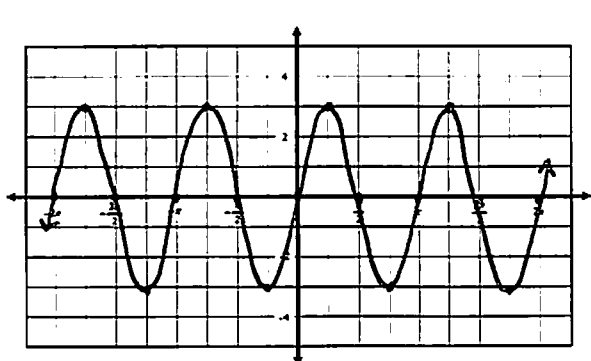
7. $y = \cos \frac{3}{4} \theta$ Amp: 1 Period: $\frac{8\pi}{3}$



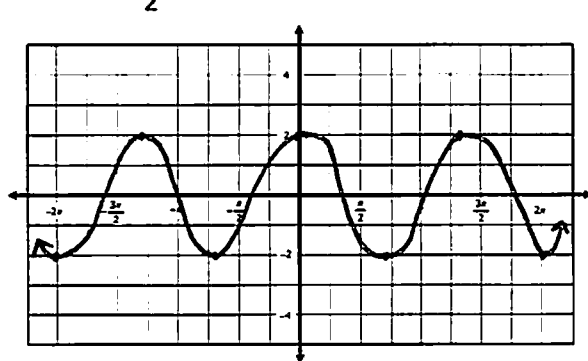
8. $y = \frac{5}{2} \cdot \sin \theta$ Amp: $\frac{5}{2}$ Period: 2π



9. $y = 3 \cdot \sin 2\theta$ Amp: 3 Period: π



10. $y = 2 \cdot \cos \frac{3}{2} \theta$ Amp: 2 Period: $\frac{4\pi}{3}$



Name: _____

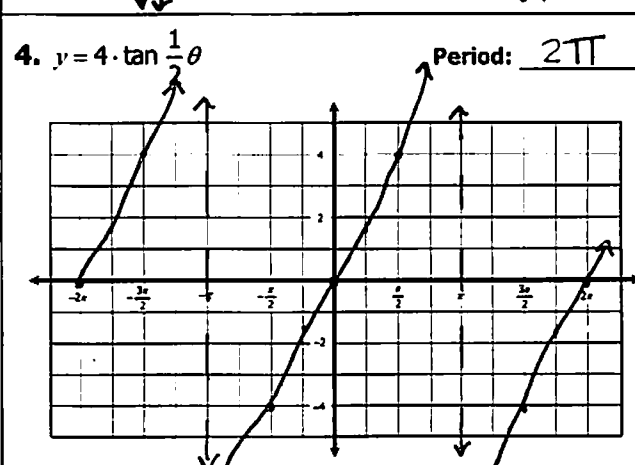
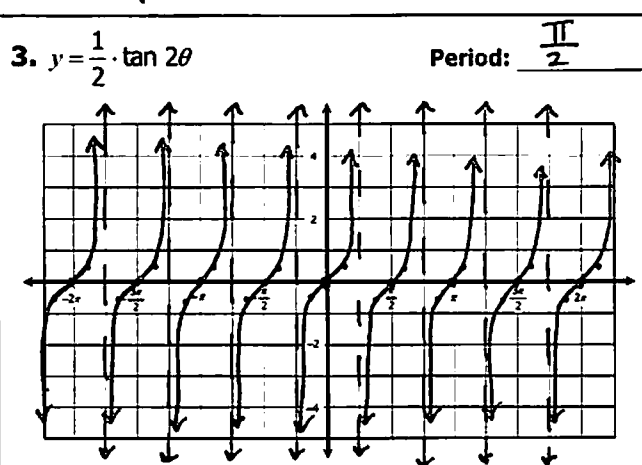
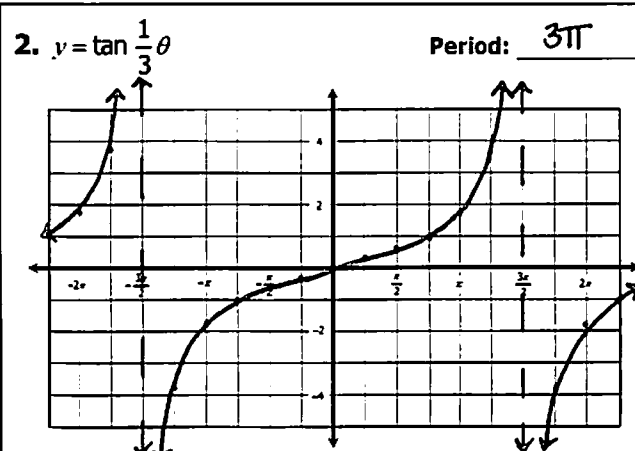
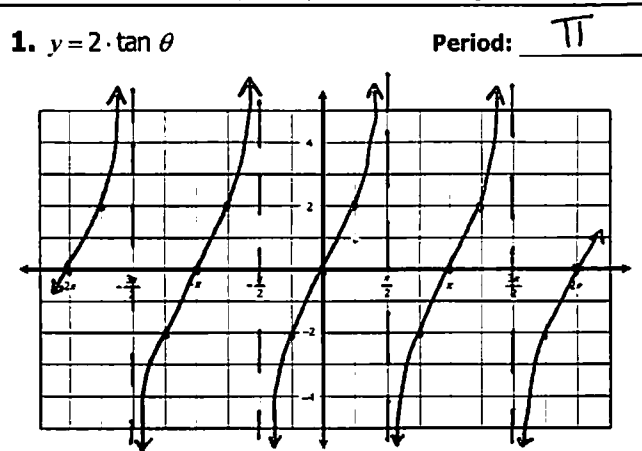
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Topic: _____

Class: _____

Main Ideas/Questions	Notes/Examples
TANGENT GRAPHS	The tangent function is one of the trigonometric functions whose graphs have asymptotes. Graph the function below and show the asymptotes. TANGENT ($y = \tan \theta$) Amplitude: Undefined Period: π
<i>Changing the Period & Asymptotes</i>	Given $y = a \cdot \tan b\theta$ where θ is in radians, the period is $\frac{\pi}{ b }$ and the asymptotes are <u>odd</u> multiples of $\frac{\pi}{2 b }$.

Directions: Identify the period, then graph each function.



Name: _____

Unit 12: Trigonometry

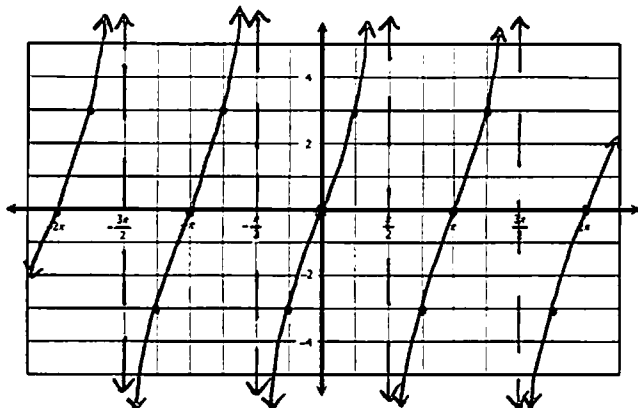
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Homework 8: Graphing Trigonometric Functions:
Sine, Cosine, and Tangent

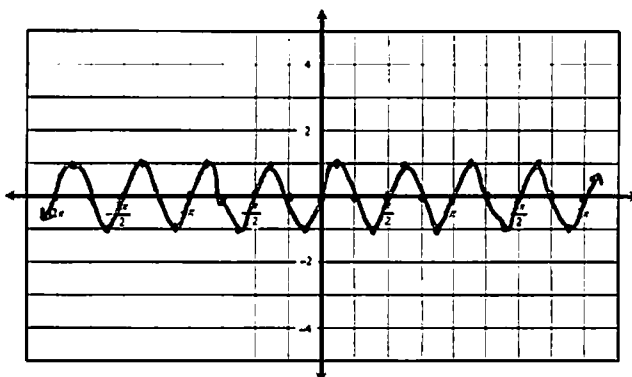
**** This is a 2-page document! ****

Directions: Identify the amplitude and period of each function, then graph.

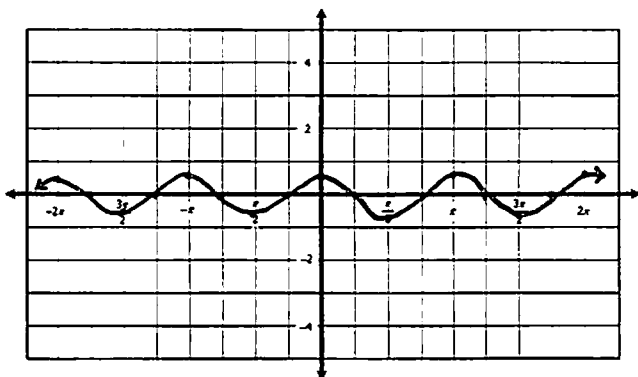
1. $y = 3 \cdot \tan \theta$ Amp: undef. Period: π



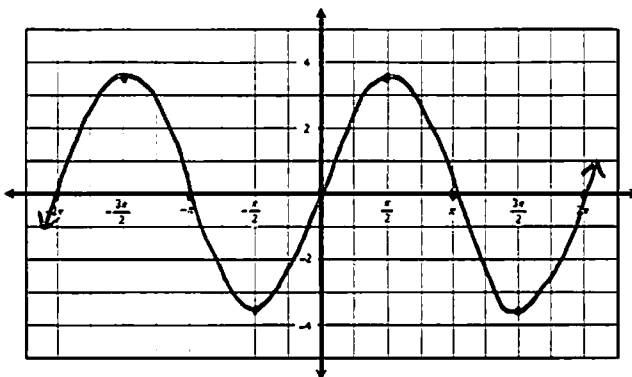
2. $y = \sin 4\theta$ Amp: 1 Period: $\frac{\pi}{2}$



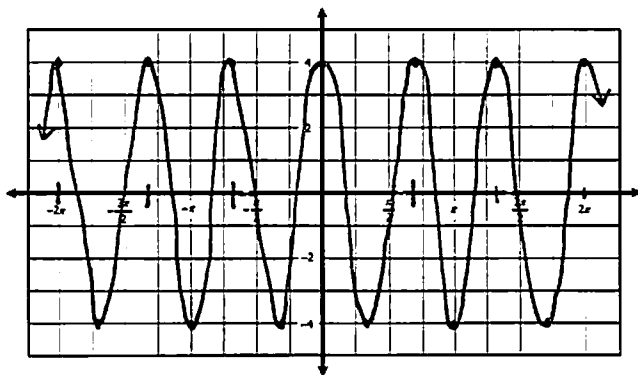
3. $y = \frac{1}{2} \cdot \cos 2\theta$ Amp: $\frac{1}{2}$ Period: π



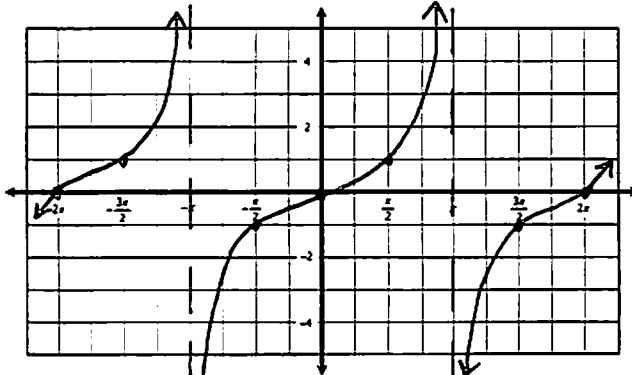
4. $y = \frac{7}{2} \cdot \sin \theta$ Amp: $\frac{7}{2}$ Period: 2π



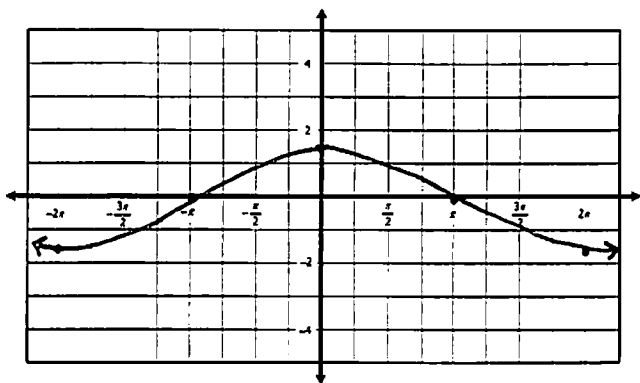
5. $y = 4 \cdot \cos 3\theta$ Amp: 4 Period: $\frac{2\pi}{3}$



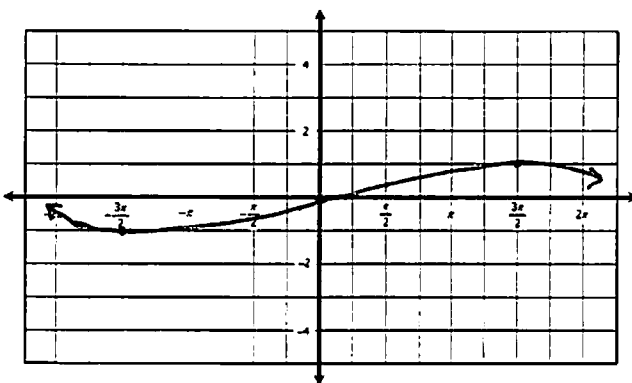
6. $y = \tan \frac{1}{2} \theta$ Amp: undef. Period: 2π



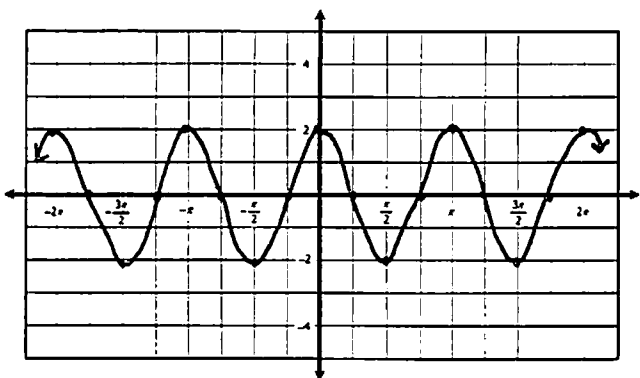
7. $y = \frac{3}{2} \cdot \cos \frac{1}{2} \theta$ Amp: $\frac{3}{2}$ Period: 4π



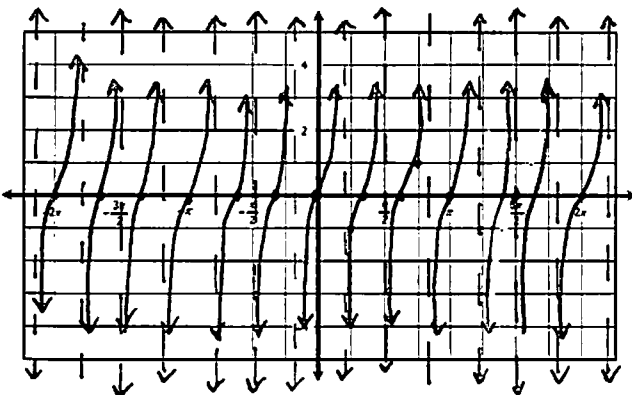
8. $y = \sin \frac{1}{3} \theta$ Amp: 1 Period: 6π



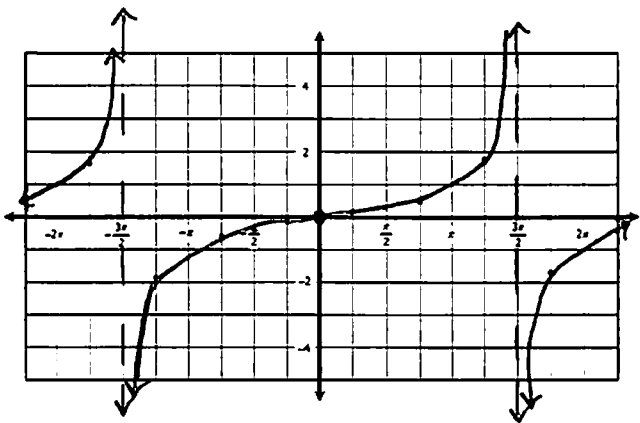
9. $y = 2 \cdot \cos 2\theta$ Amp: 2 Period: π



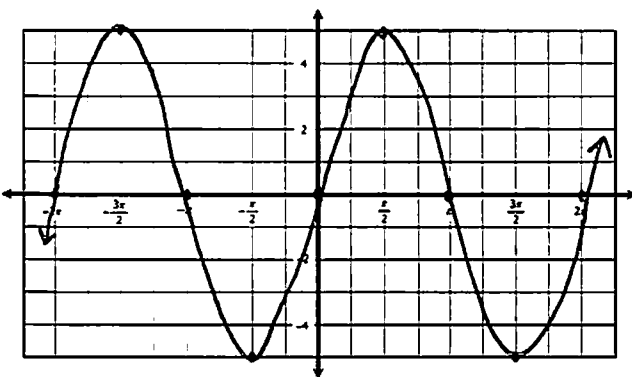
10. $y = \tan 3\theta$ Amp: undef Period: $\frac{\pi}{3}$



11. $y = \frac{1}{2} \cdot \tan \frac{1}{3} \theta$ Amp: undef Period: 3π



12. $y = 5 \cdot \sin \theta$ Amp: 5 Period: 2π



Name:	Date:
Topic:	Class:

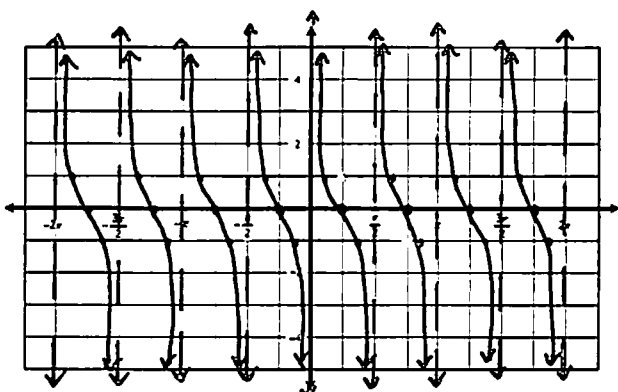
Main Ideas/Questions	Notes/Examples		
GRAPHING RECIPROCAL FUNCTIONS	COSECANT $(y = \csc \theta)$		Amplitude: undef.
	SECANT $(y = \sec \theta)$		Amplitude: undef.
	COTANGENT $(y = \cot \theta)$		Amplitude: undef.
	If given $y = a \cdot \csc b\theta$, $y = a \cdot \sec b\theta$, or $y = a \cdot \cot b\theta$, apply the changes to the amplitude, period, and/or asymptotes the same way you did for the other trigonometric functions.		

Directions: Identify the period, then graph each function.

1. $y = 3 \cdot \csc \theta$	Period: 2π
	2. $y = \sec \frac{1}{2} \theta$
	Period: 4π

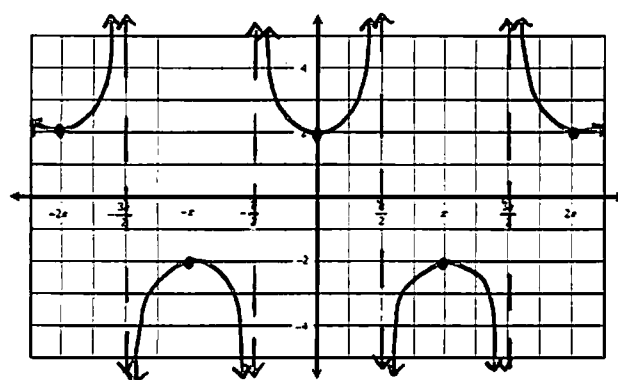
3. $y = \cot 2\theta$

Period: $\frac{\pi}{2}$



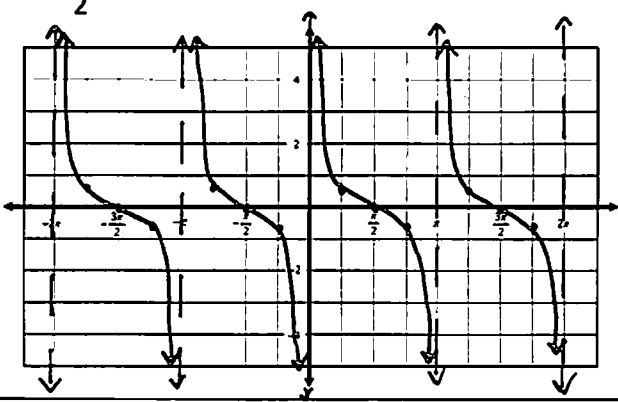
4. $y = 2 \cdot \sec \theta$

Period: 2π



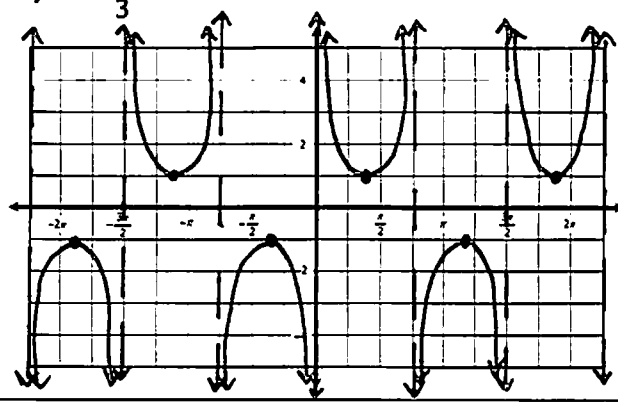
5. $y = \frac{1}{2} \cdot \cot \theta$

Period: π



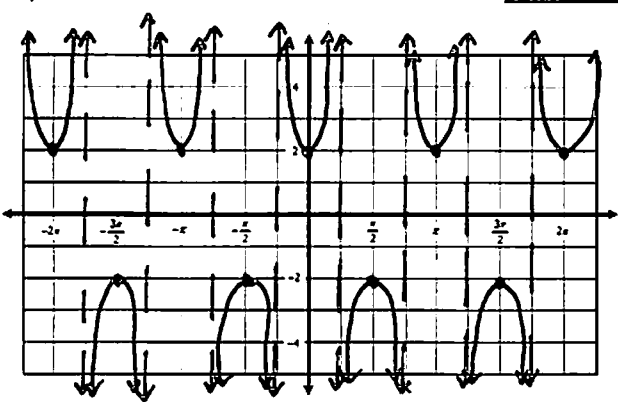
6. $y = \csc \frac{4}{3} \theta$

Period: $\frac{3\pi}{2}$



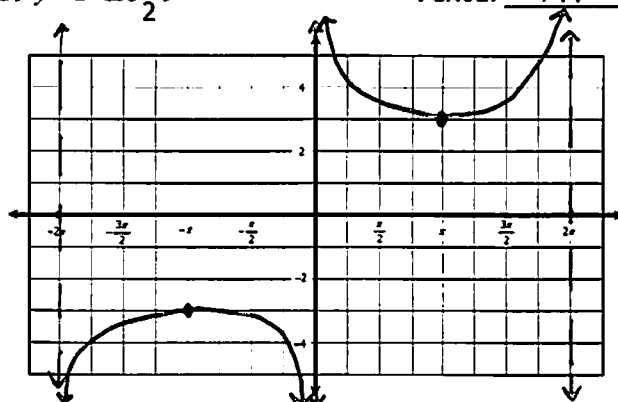
7. $y = 2 \cdot \sec 2\theta$

Period: π



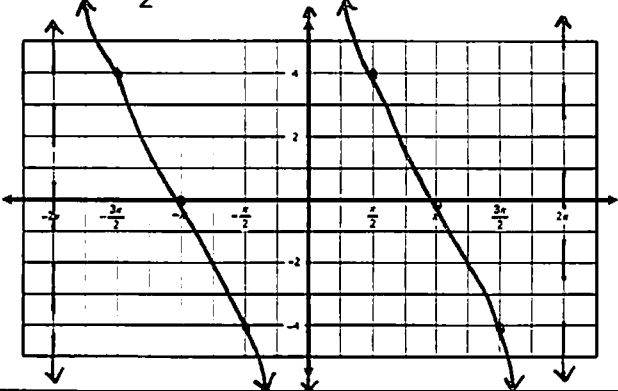
8. $y = 3 \cdot \csc \frac{1}{2} \theta$

Period: 4π



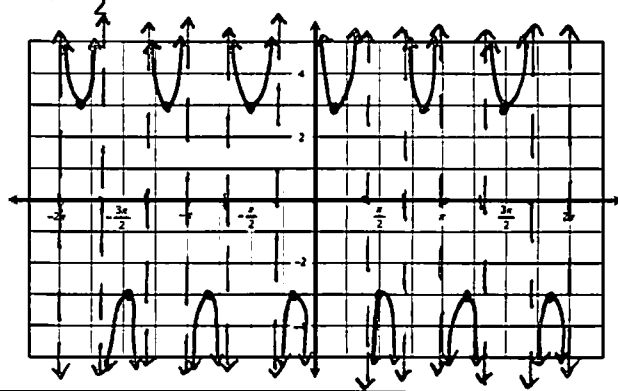
9. $y = 4 \cdot \cot \frac{1}{2} \theta$

Period: 2π



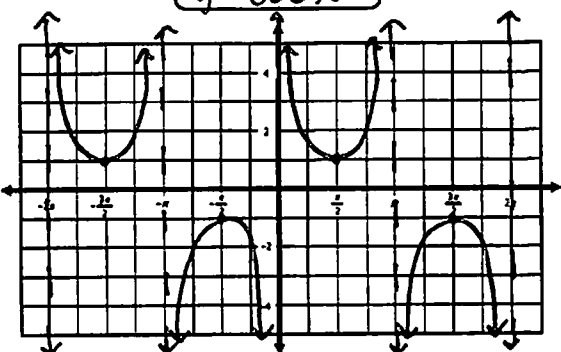
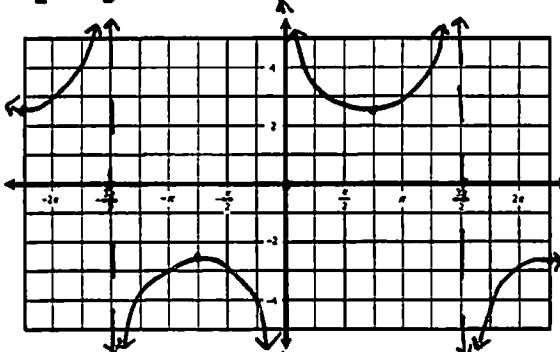
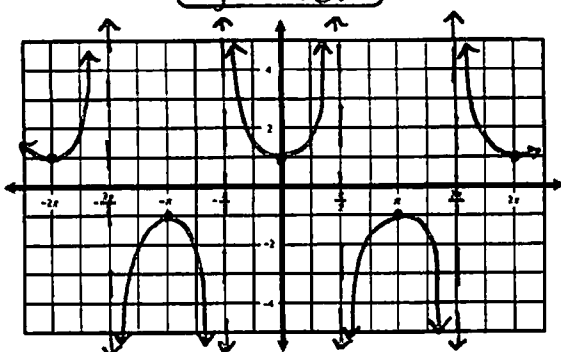
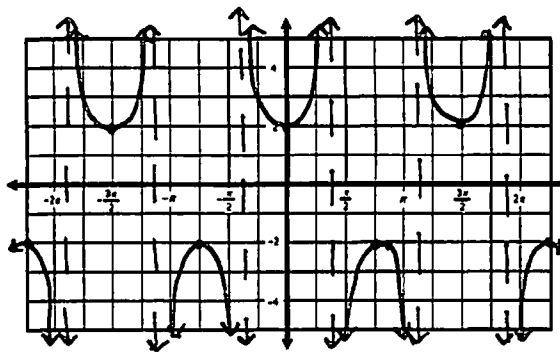
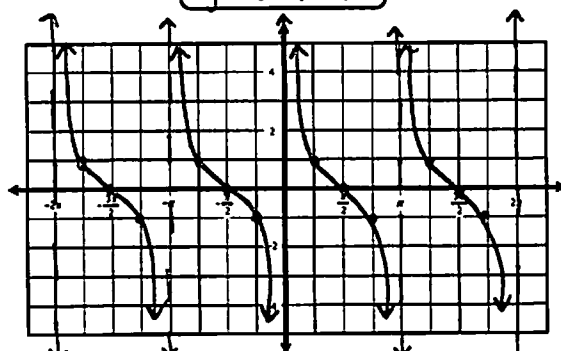
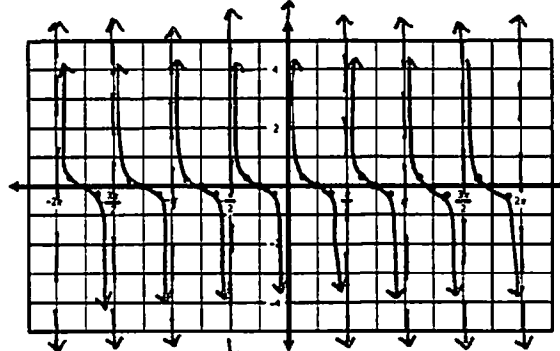
10. $y = \frac{5}{2} \cdot \csc 3\theta$

Period: $\frac{2\pi}{3}$



Graphing Trigonometric Functions

	Function & Graph	Amplitude & Period	Example
SINE	$y = \sin x$ 	$y = a \cdot \sin bx$ Amplitude = a Period = $\frac{2\pi}{b}$	$y = 4 \cdot \sin \frac{1}{2} \theta$ Amp: <u>4</u> Period: <u>4π</u>
COSINE	$y = \cos x$ 	$y = a \cdot \cos bx$ Amplitude = a Period = $\frac{2\pi}{b}$	$y = \frac{3}{2} \cdot \cos 2\theta$ Amp: <u>$\frac{3}{2}$</u> Period: <u>π</u>
TANGENT	$y = \tan x$ 	$y = a \cdot \tan bx$ Amplitude = <u>undef</u> Period = $\frac{\pi}{b}$	$y = 2 \cdot \tan \frac{1}{3} \theta$ Amp: <u>undef</u> Period: <u>3π</u>

COSECANT	$y = \csc x$ 	$y = a \cdot \csc bx$ Amplitude = <u>undef</u> Period = $\frac{2\pi}{b}$	$y = \frac{5}{2} \cdot \csc \frac{2}{3}\theta$ Amp: <u>undef</u> Period: 3π 
SECANT	$y = \sec x$ 	$y = a \cdot \sec bx$ Amplitude = <u>undef</u> Period = $\frac{2\pi}{b}$	$y = 2 \cdot \sec \frac{4}{3}\theta$ Amp: <u>undef</u> Period: $\frac{3\pi}{2}$ 
COTANGENT	$y = \cot x$ 	$y = a \cdot \cot bx$ Amplitude = <u>undef</u> Period = $\frac{\pi}{b}$	$y = \frac{1}{4} \cdot \cot 2\theta$ Amp: <u>undef</u> Period: $\frac{\pi}{2}$ 

Name: _____

Unit 12: Trigonometry

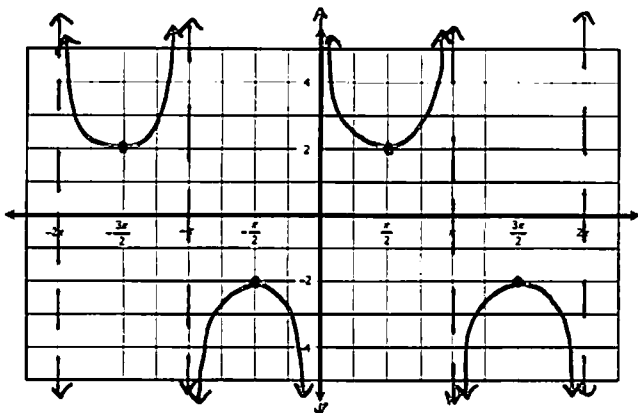
Date: _____ Bell: _____

Homework 9: Graphing Trigonometric Functions:
(All Six Functions!)

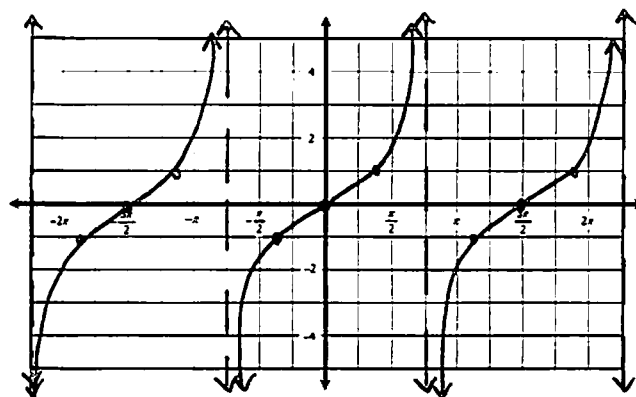
**** This is a 2-page document! ****

Directions: Identify the amplitude, if possible, and the period for each function. Then graph.

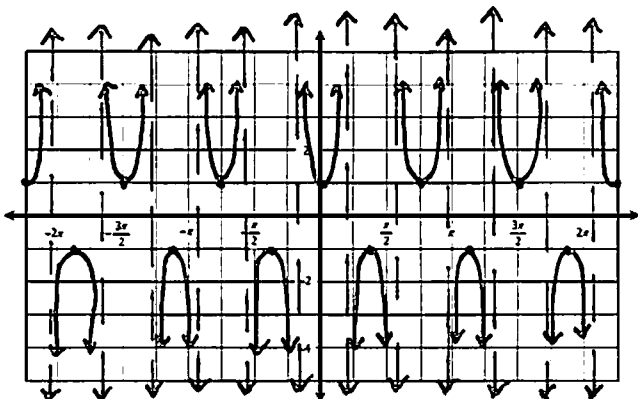
1. $y = 2 \cdot \csc \theta$ Amp: undef. Period: 2π



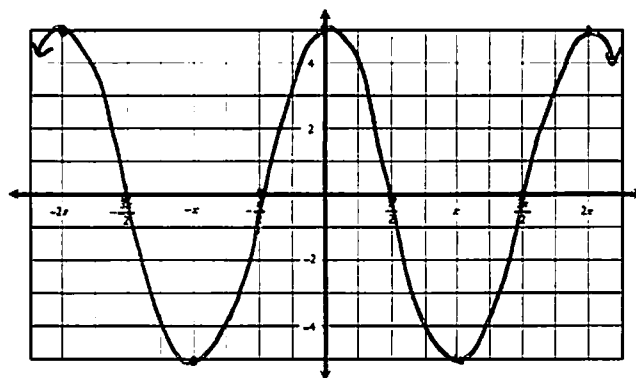
2. $y = \tan \frac{2}{3} \theta$ Amp: undef. Period: $\frac{3\pi}{2}$



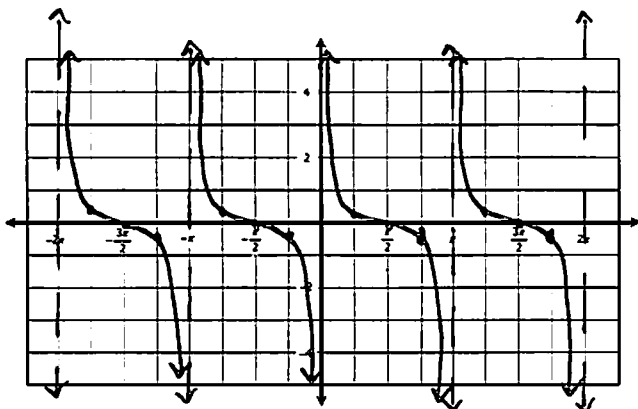
3. $y = \sec \frac{8}{3} \theta$ Amp: undef. Period: $\frac{3\pi}{4}$



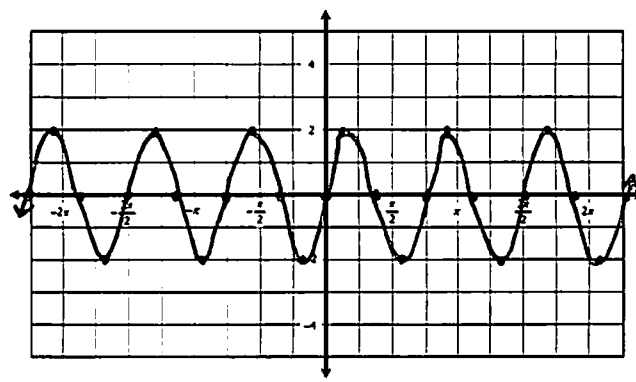
4. $y = 5 \cdot \cos \theta$ Amp: 5 Period: 2π



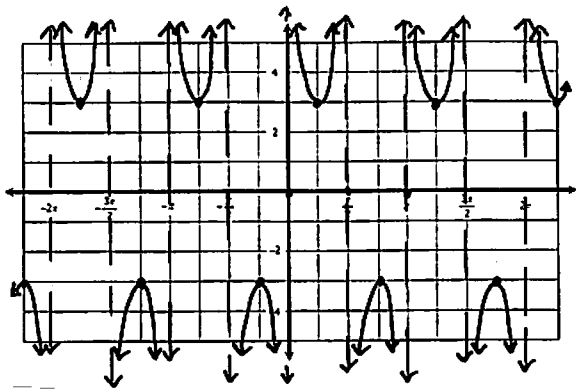
5. $y = \frac{1}{4} \cdot \cot \theta$ Amp: undef. Period: π



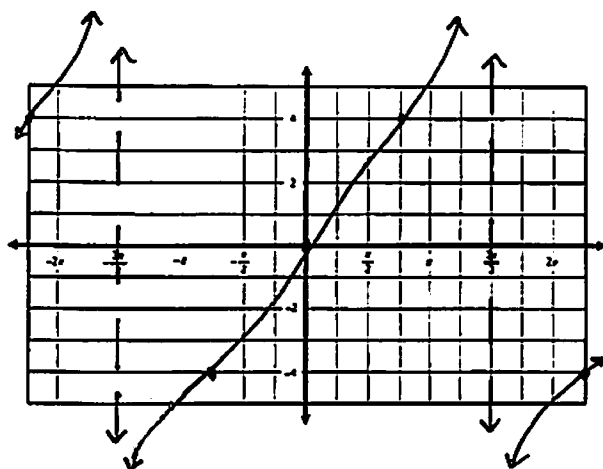
6. $y = 2 \cdot \sin \frac{8}{3} \theta$ Amp: 2 Period: $\frac{3\pi}{4}$



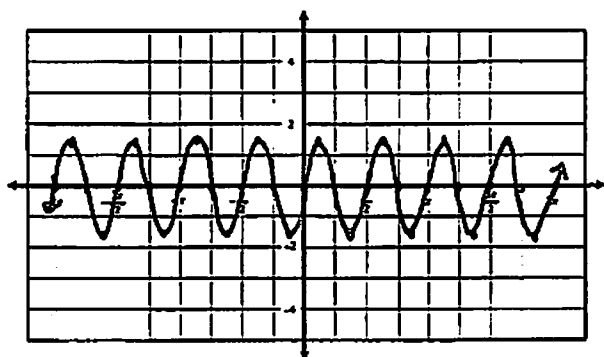
7. $y = 3 \cdot \csc 2\theta$ Amp: undef. Period: π



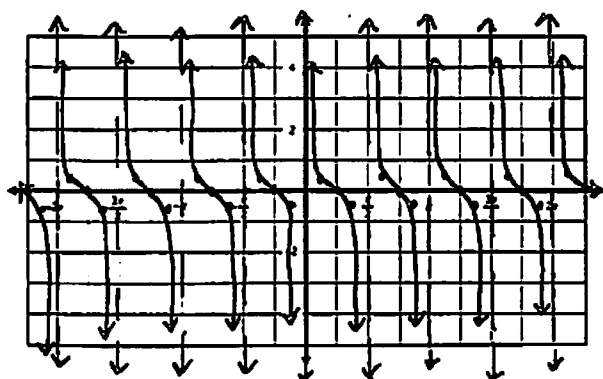
8. $y = 4 \cdot \tan \frac{1}{3}\theta$ Amp: undef. Period: 3π



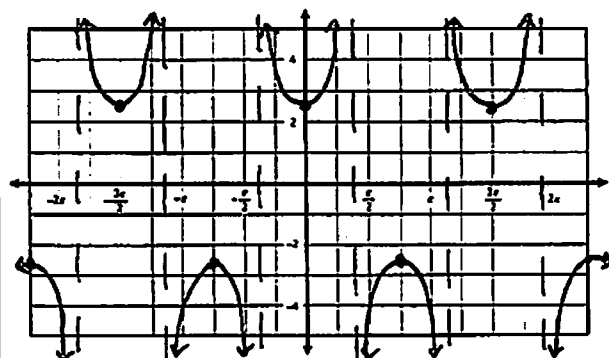
9. $y = \frac{3}{2} \cdot \sin 4\theta$ Amp: $3/2$ Period: $\pi/2$



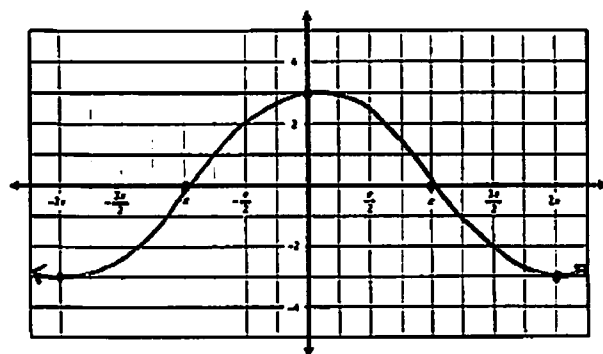
10. $y = \frac{1}{2} \cdot \cot 2\theta$ Amp: undef. Period: $\pi/2$



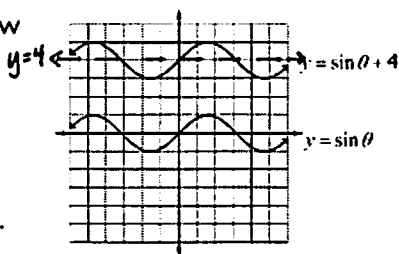
11. $y = \frac{5}{2} \cdot \sec \frac{4}{3}\theta$ Amp: undef. Period: $3\pi/2$



12. $y = 3 \cdot \cos \frac{1}{2}\theta$ Amp: 3 Period: 4π

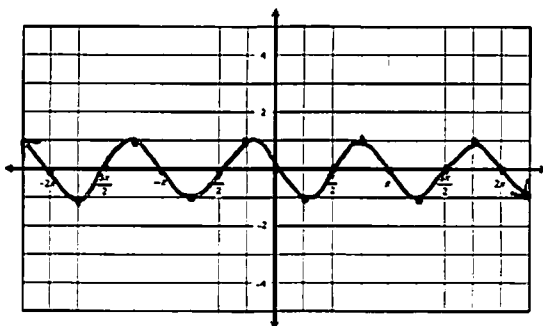


Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
TRANSFORMATIONS: Phase Shifts & Vertical Shifts $y = a \cdot \sin b(\theta - h) + k$ $y = a \cdot \cos b(\theta - h) + k$ $y = a \cdot \tan b(\theta - h) + k$	Like other functions, trigonometric functions can be translated. ➤ Phase Shifts: (Horizontal Shifts) • $+h$ shifts <u>left</u> and $-h$ shifts <u>right</u>. • Examples: (a) $y = \sin(\theta - \pi)$ shifts <u>π</u> units to the <u>right</u>. (b) $y = \tan\left(\theta + \frac{\pi}{4}\right)$ shifts <u>$\frac{\pi}{4}$</u> units to the <u>left</u>. ➤ Vertical Shifts: • $+k$ shifts <u>up</u> and $-k$ shifts <u>down</u>. • Examples: (c) $y = \tan \theta + 2$ shifts <u>2</u> units <u>up</u>. (d) $y = \cos \theta - 4$ shifts <u>4</u> units <u>down</u>.
PUTTING IT ALL TOGETHER	Using the sine function as an example, label each part: $y = a \cdot \sin b(\theta - h) + k$ amplitude ↓ phase shift ↓ ↑ period ↑ vertical shift
MIDLINE	• Shifting a graph vertically will create a new horizontal axis, called the <u>midline</u>. • This is the reference line to which a graph oscillates. • The equation for the midline is <u>$y = 4$</u>. 
DESCRIBING TRANSFORMATIONS	Directions: State the amplitude, period, phase shift, vertical shift, and midline for each function. 1. $y = \frac{1}{2} \sin 3\theta - 5$ Amplitude: <u>$\frac{1}{2}$</u> Period: <u>$2\pi/3$</u> Phase Shift: <u>none</u> Vertical Shift: <u>5, down</u> Midline: <u>$y = -5$</u> 2. $y = 4 \cdot \tan 2(\theta + 2\pi) + 1$ Amplitude: <u>undef.</u> Period: <u>$\pi/2$</u> Phase Shift: <u>2π, left</u> Vertical Shift: <u>1, up</u> Midline: <u>$y = 1$</u>

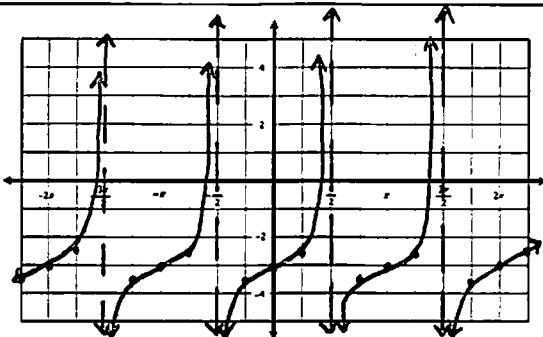
Directions: State the amplitude, period, phase shift, vertical shift, and midline for each function. Then graph each function.

3. $y = \sin 2\left(\theta + \frac{\pi}{2}\right)$



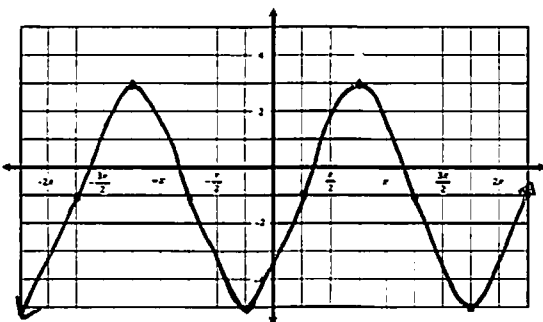
Amplitude: 1
 Period: π
 Phase Shift: $\pi/2$, left
 Vertical Shift: none
 Midline: $y=0$

4. $y = \frac{1}{2} \cdot \tan \theta - 3$



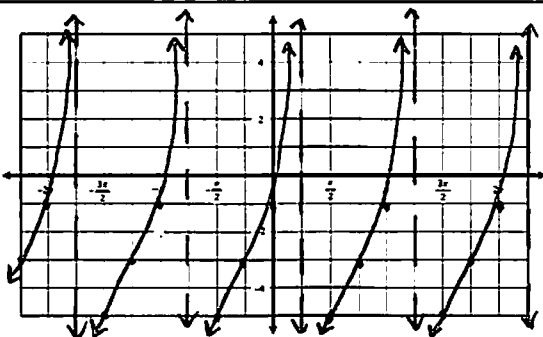
Amplitude: undef.
 Period: π
 Phase Shift: none
 Vertical Shift: down 3
 Midline: $y=-3$

5. $y = 4 \cdot \cos\left(\theta - \frac{3\pi}{4}\right) - 1$



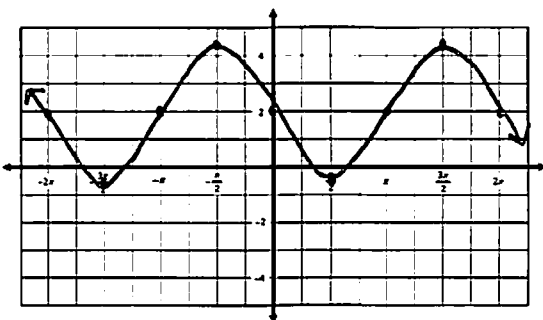
Amplitude: 4
 Period: 2π
 Phase Shift: $3\pi/4$, right
 Vertical Shift: down 1
 Midline: $y=-1$

6. $y = 2 \cdot \tan\left(\theta + \frac{\pi}{4}\right) - 3$



Amplitude: undef.
 Period: π
 Phase Shift: $\pi/4$, left
 Vertical Shift: down 3
 Midline: $y=-3$

7. $y = \frac{5}{2} \cdot \sin(\theta + \pi) + 2$



Amplitude: $5/2$
 Period: 2π
 Phase Shift: π , left
 Vertical Shift: up 2
 Midline: $y=2$

Name: _____

Unit 12: Trigonometry

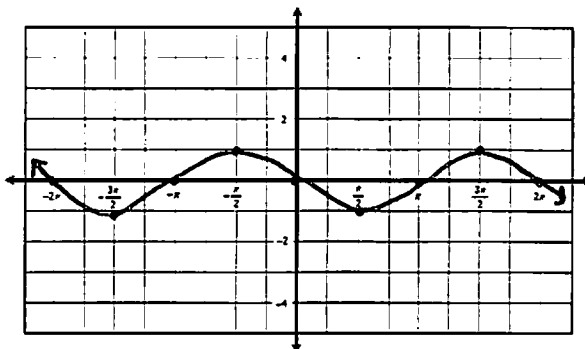
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Homework 10: Translating Trigonometric Graphs

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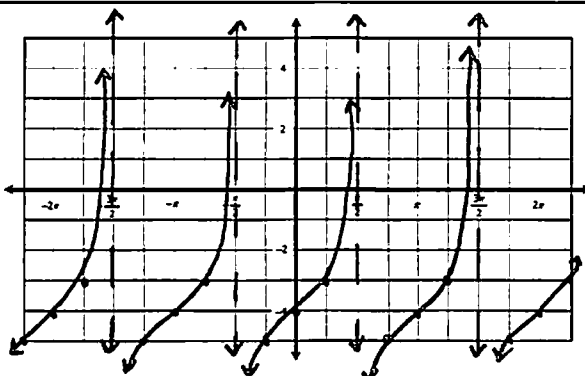
Directions: State the amplitude, period, phase shift, vertical shift, and midline for each function. Then graph.

1. $y = \sin(\theta + \pi)$



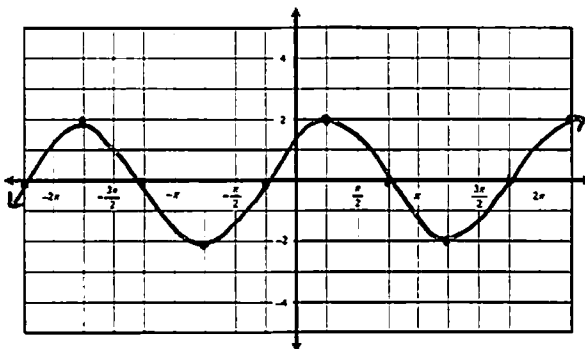
Amplitude: 1
 Period: 2π
 Phase Shift: π , left
 Vertical Shift: none
 Midline: $y = 0$

2. $y = \tan \theta - 4$



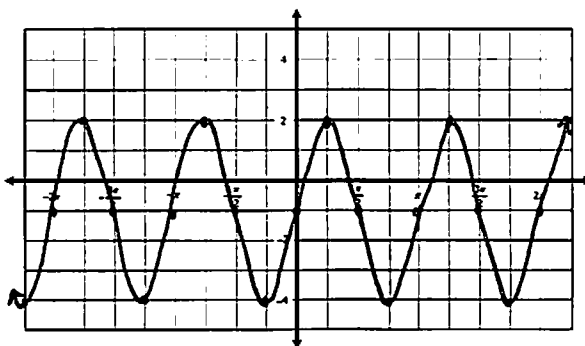
Amplitude: undef.
 Period: π
 Phase Shift: none
 Vertical Shift: down 4
 Midline: $y = -4$

3. $y = 2 \cdot \cos\left(\theta - \frac{\pi}{4}\right)$



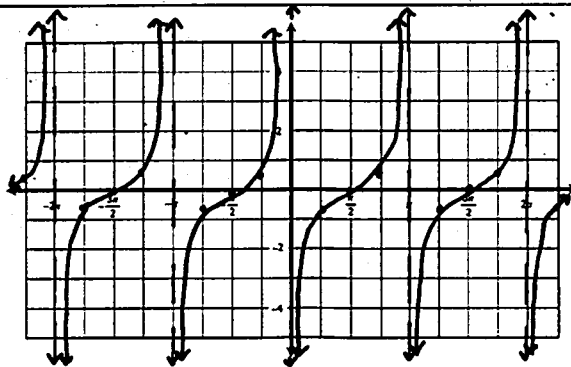
Amplitude: 2
 Period: 2π
 Phase Shift: $\pi/4$, right
 Vertical Shift: none
 Midline: $y = 0$

4. $y = 3 \cdot \sin 2\theta - 1$



Amplitude: 3
 Period: π
 Phase Shift: none
 Vertical Shift: down 1
 Midline: $y = -1$

5. $y = \frac{1}{2} \cdot \tan\left(\theta - \frac{\pi}{2}\right)$



Amplitude: undef.

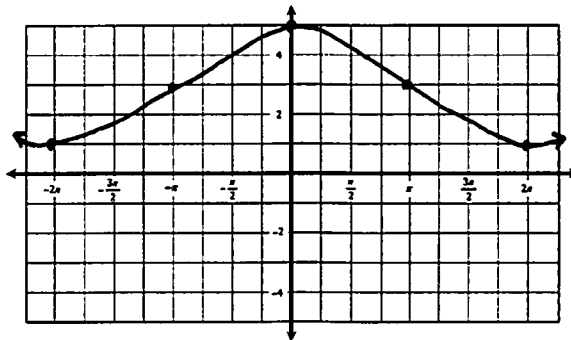
Period: π

Phase Shift: $\pi/2$, right

Vertical Shift: none

Midline: $y = 0$

6. $y = 2 \cdot \cos \frac{1}{2} \theta + 3$



Amplitude: 2

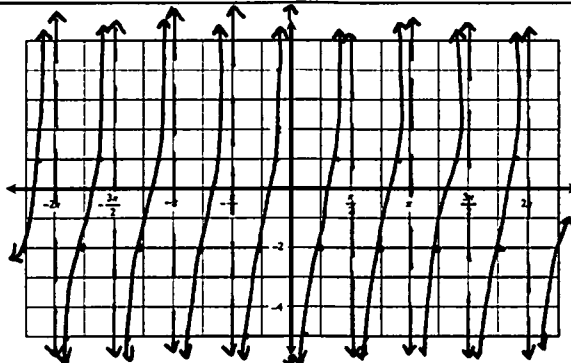
Period: 4π

Phase Shift: none

Vertical Shift: up 3

Midline: $y = 3$

7. $y = 3 \cdot \tan 2\left(\theta + \frac{5\pi}{4}\right) - 2$



Amplitude: undef.

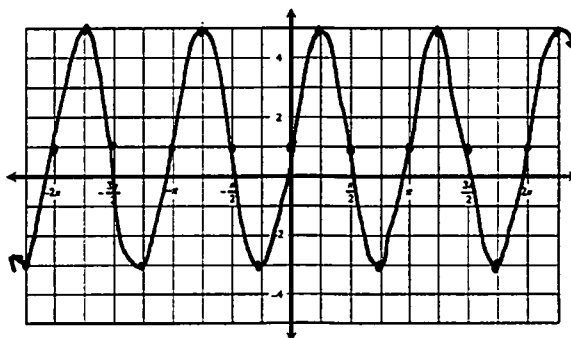
Period: $\pi/2$

Phase Shift: $5\pi/4$, left

Vertical Shift: down 2

Midline: $y = -2$

8. $y = 4 \cdot \sin 2(\theta - \pi) + 1$



Amplitude: 4

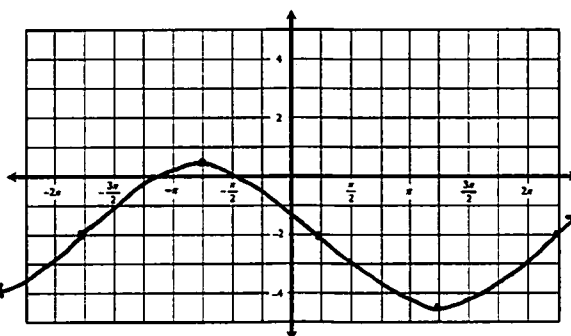
Period: π

Phase Shift: π , right

Vertical Shift: up 1

Midline: $y = 1$

9. $y = \frac{5}{2} \cdot \cos \frac{1}{2} \left(\theta + \frac{3\pi}{4}\right) - 2$



Amplitude: $5/2$

Period: 4π

Phase Shift: $3\pi/4$, left

Vertical Shift: down 2

Midline: $y = -2$

Name: _____

Algebra 2

Date: _____ Per: _____

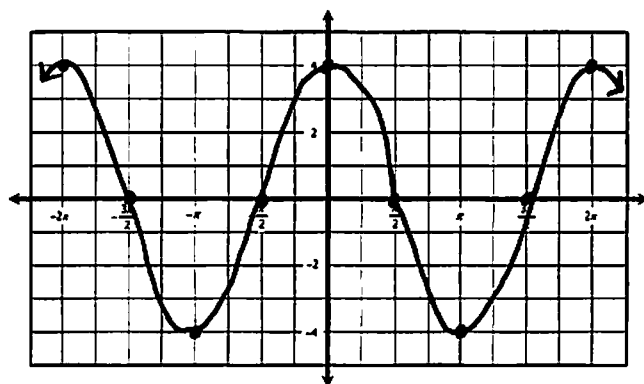
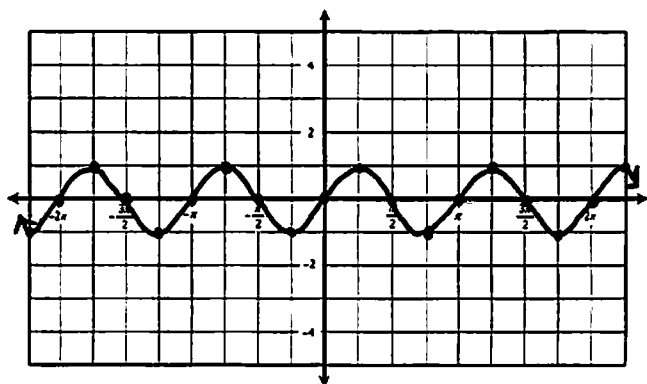
Unit 12: Trigonometry

Quiz 12-2: Graphing Trigonometric Functions

Directions: Identify the amplitude, if possible, and period for each function. Then graph.

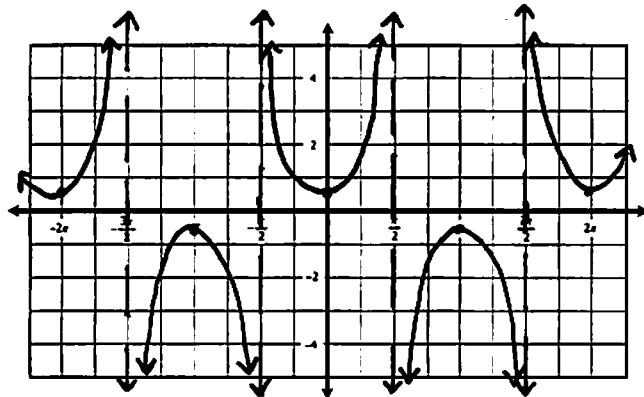
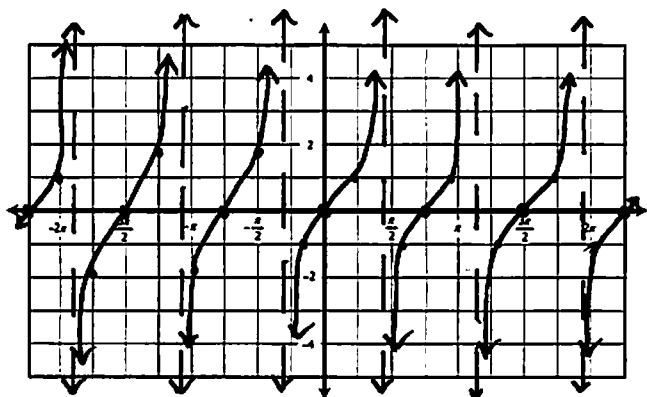
1. $y = \sin 2\theta$ Amp: 1 Per: π

2. $y = 4 \cdot \cos \theta$ Amp: 4 Per: 2π



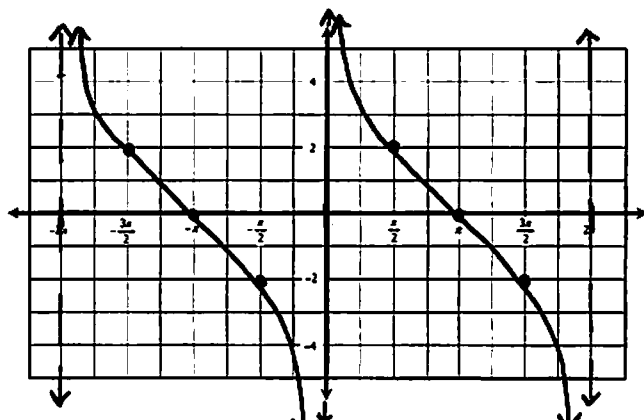
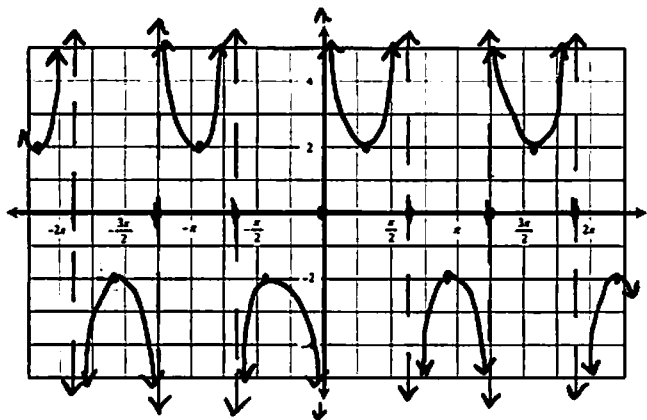
3. $y = \tan \frac{4}{3}\theta$ Amp: undef Per: $3\pi/4$

4. $y = \frac{1}{2} \cdot \sec \theta$ Amp: undef Per: 2π



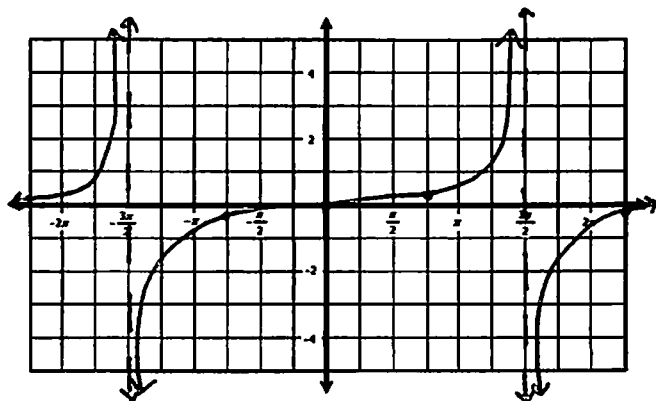
5. $y = 2 \cdot \csc \frac{8}{5}\theta$ Amp: undef Per: $5\pi/4$

6. $y = 2 \cdot \cot \frac{1}{2}\theta$ Amp: undef Per: 2π



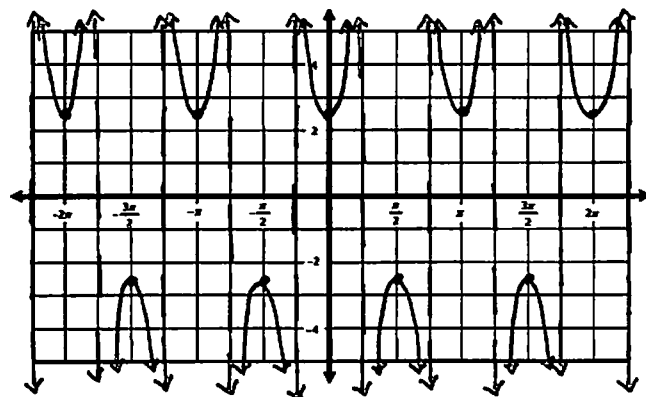
7. $y = \frac{1}{4} \cdot \tan \frac{1}{3} \theta$

Amp: undef Per: 3π



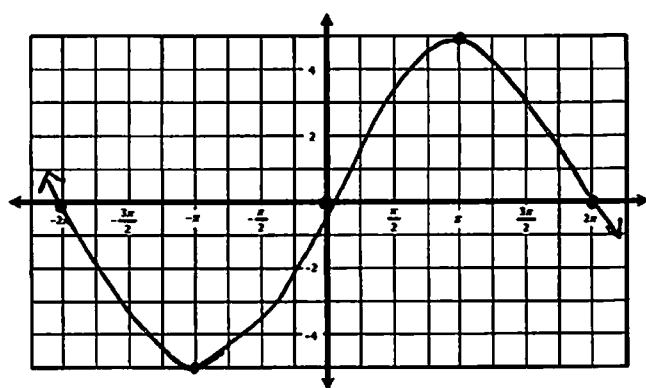
8. $y = \frac{7}{2} \cdot \sec 2\theta$

Amp: undef Per: π



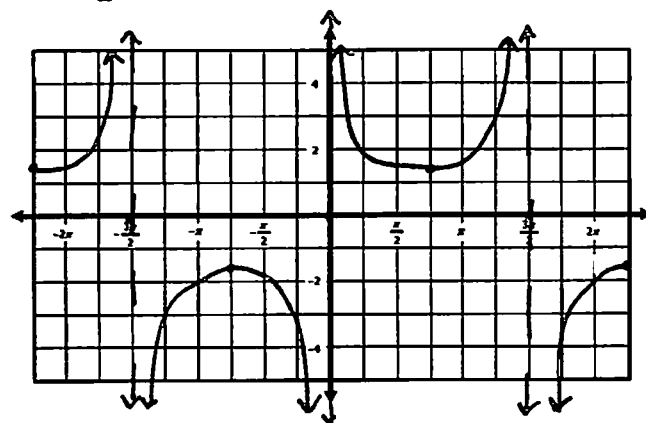
9. $y = 5 \cdot \sin \frac{1}{2} \theta$

Amp: 5 Per: 4π



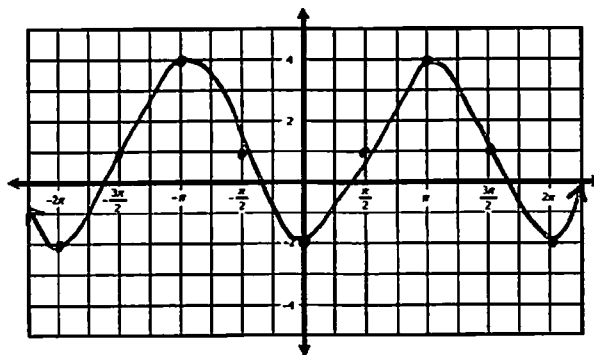
10. $y = \frac{3}{2} \cdot \csc \frac{2}{3} \theta$

Amp: undef Per: 3π



Directions: Identify the amplitude, if possible, period, translations, and midline. Then graph.

11. $y = 3 \cdot \cos(\theta - \pi) + 1$



Amplitude: 3

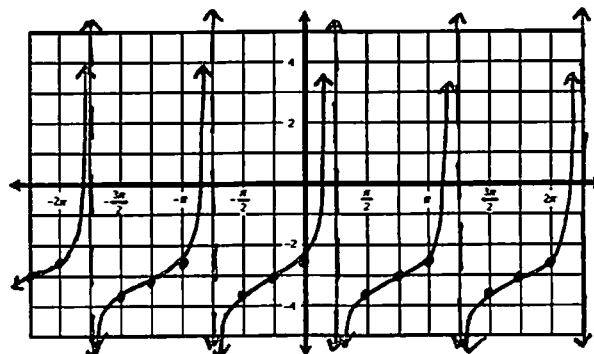
Period: 2π

Phase Shift: π , right

Vertical Shift: up 1

Midline: $y = 1$

12. $y = \frac{1}{2} \cdot \tan\left(\theta + \frac{\pi}{4}\right) - 3$



Amplitude: undef.

Period: π

Phase Shift: $\pi/4$, left

Vertical Shift: down 3

Midline: $y = -3$

Name:	Class:
Topic:	Date:

Main Ideas/Questions	Notes	
Trigonometric Identities	- A trigonometric identity is an equation involving trigonometric functions that is true for all values for which every expression in the equation is defined. - The identities help to evaluate functions, simplify expressions, prove identities, and solve equations.	
Fundamental Trigonometric Identities	Quotient Identities	$\tan \theta = \frac{\sin \theta}{\cos \theta}; \cos \theta \neq 0$ $\cot \theta = \frac{\cos \theta}{\sin \theta}; \sin \theta \neq 0$
	Reciprocal Identities	$\sin \theta = \frac{1}{\csc \theta}; \csc \theta \neq 0$ $\csc \theta = \frac{1}{\sin \theta}; \sin \theta \neq 0$ $\cos \theta = \frac{1}{\sec \theta}; \sec \theta \neq 0$ $\sec \theta = \frac{1}{\cos \theta}; \cos \theta \neq 0$ $\tan \theta = \frac{1}{\cot \theta}; \cot \theta \neq 0$ $\cot \theta = \frac{1}{\tan \theta}; \tan \theta \neq 0$
	Pythagorean Identities	$\cos^2 \theta + \sin^2 \theta = 1$ $\tan^2 \theta + 1 = \sec^2 \theta$ $\cot^2 \theta + 1 = \csc^2 \theta$
Finding Trig Function Values	1. Find the exact value of $\sin \theta$ if $\cos \theta = -\frac{4}{7}$ and $\frac{\pi}{2} < \theta < \pi$. $\sin^2 \theta + (-\frac{4}{7})^2 = 1$ $\sin^2 \theta + \frac{16}{49} = 1$ $\sin^2 \theta = \frac{33}{49}$ $\sin \theta = \frac{\sqrt{33}}{7}$	
	2. Find the exact value of $\cos \theta$ if $\cot \theta = \frac{5}{2}$ and $180 < \theta < 270^\circ$. $(\frac{5}{2})^2 + 1 = \sec^2 \theta$ $\frac{25}{4} + 1 = \sec^2 \theta$ $\frac{29}{4} = \sec^2 \theta$ $\sec \theta = \frac{\sqrt{29}}{2}$ $\cos \theta = -\frac{2\sqrt{29}}{29}$	
	3. Find the exact value of $\tan \theta$ if $\csc \theta = -2$ and $\frac{3\pi}{2} < \theta < 2\pi$. $\cot^2 \theta + 1 = (-2)^2$ $\cot^2 \theta + 1 = 4$ $\cot^2 \theta = 3$ $\cot \theta = \sqrt{3}$ $\tan \theta = -\frac{\sqrt{3}}{3}$	

Simplify Expressions

To **simplify a trigonometric expression** means to rewrite it as a numerical expression or in terms of a single trigonometric function, if possible. Use the identities to help you simplify the expressions below.

4. $\sin \theta \cdot \sec \theta$
 $\sin \theta \cdot \frac{1}{\cos \theta}$

$$= \frac{\sin \theta}{\cos \theta} = \boxed{\tan \theta}$$

5. $\tan \theta \cdot \cos \theta$

$$\frac{\sin \theta}{\cos \theta} \cdot \cos \theta$$

$$= \boxed{\sin \theta}$$

6. $2(\sec^2 \theta - \tan^2 \theta)$

$$2(1)$$

$$= \boxed{2}$$

$$\begin{cases} \tan^2 \theta + 1 = \sec^2 \theta \\ 1 = \sec^2 \theta - \tan^2 \theta \end{cases}$$

7. $\cos^2 \theta - \sin^2 \theta$

$$(1 - \sin^2 \theta) - \sin^2 \theta$$

$$= \boxed{1 - 2\sin^2 \theta}$$

$$\begin{cases} \cos^2 \theta + \sin^2 \theta = 1 \\ \cos^2 \theta = 1 - \sin^2 \theta \end{cases}$$

8. $\frac{\sin \theta \cdot \cos \theta}{\tan \theta}$

$$\frac{\sin \theta \cdot \cos \theta}{\frac{\sin \theta}{\cos \theta}} = \sin \theta \cdot \cos \theta \cdot \frac{\cos \theta}{\sin \theta}$$

$$= \boxed{\cos^2 \theta}$$

9. $\frac{2 - 2\sin^2 \theta}{\cos \theta}$

$$\frac{2(1 - \sin^2 \theta)}{\cos \theta}$$

$$= \frac{2 \cdot \cos^2 \theta}{\cos \theta} = \boxed{2 \cos \theta}$$

10. $\frac{\sec \theta \cdot \sin^2 \theta}{1 + \sec \theta}$

$$\frac{\frac{1}{\cos \theta} \cdot \sin^2 \theta}{1 + \frac{1}{\cos \theta}} = \frac{\frac{\sin^2 \theta}{\cos \theta}}{\frac{\cos \theta + 1}{\cos \theta}}$$

$$= \frac{1 - \cos^2 \theta}{\cos \theta + 1}$$

$$= \frac{(1 - \cos \theta)(1 + \cos \theta)}{\cos \theta + 1}$$

$$= \boxed{1 - \cos \theta}$$

11. $\sin^2 \theta - \cos^2 \theta \cdot \sin^2 \theta$

$$\sin^2 \theta \cdot (1 - \cos^2 \theta)$$

$$= \sin^2 \theta \cdot \sin^2 \theta$$

$$= \boxed{\sin^4 \theta}$$

Name:	Class:
Topic:	Date:

Main Ideas/Questions	Notes
PROVING IDENTITIES	- Proving an identity means to show that both sides of the equation are the same. - Simplify one side using the basic trigonometric identities of the equation until it is the same as the other side. Ideally, pick the side that is more complicated to work with.

Directions: Prove each identity below.

1. $\frac{\tan \theta}{\sec \theta} = \sin \theta$ $\frac{\sin \theta}{\cos \theta} = \sin \theta$ $\frac{1}{\cos \theta}$ $\frac{\sin \theta}{\cos \theta} \cdot \frac{\cos \theta}{1} = \sin \theta$ $\sin \theta = \sin \theta$	2. $1 + \sec^2 \theta \cdot \sin^2 \theta = \sec^2 \theta$ $1 + \frac{1}{\cos^2 \theta} \cdot \sin^2 \theta = \sec^2 \theta$ $1 + \frac{\sin^2 \theta}{\cos^2 \theta} = \sec^2 \theta$ $1 + \tan^2 \theta = \sec^2 \theta$ $\sec^2 \theta = \sec^2 \theta$
3. $\csc \theta = \frac{\cot \theta}{\sec \theta} + \sin \theta$ $\csc \theta = \frac{\cos \theta}{\sin \theta} + \sin \theta$ $\csc \theta = \frac{\cos^2 \theta}{\sin \theta} + \sin \theta$ $\csc \theta = \frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta}$ $\csc \theta = \csc \theta$	4. $\csc \theta - \cot \theta \cdot \cos \theta = \sin \theta$ $\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta} \cdot \cos \theta = \sin \theta$ $\frac{1 - \cos^2 \theta}{\sin \theta} = \sin \theta$ $\frac{\sin^2 \theta}{\sin \theta} = \sin \theta$ $\sin \theta = \sin \theta$
5. $\cos^2 \theta = \frac{\csc \theta \cdot \cos \theta}{\tan \theta + \cot \theta}$ $\cos^2 \theta = \frac{\frac{1}{\sin \theta} \cdot \cos \theta}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}}$ $\cos^2 \theta = \frac{\frac{\cos \theta}{\sin \theta}}{\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}}$ $\cos^2 \theta = \cos^2 \theta$	6. $\frac{\sec \theta}{\cos \theta} - \frac{\tan \theta}{\cot \theta} = 1$ $\frac{\frac{1}{\cos \theta}}{\cos \theta} - \frac{\frac{\sin \theta}{\cos \theta}}{\frac{\cos \theta}{\sin \theta}} = 1$ $\frac{1}{\cos^2 \theta} - \frac{\sin^2 \theta}{\cos^2 \theta} = 1$ $\sec^2 \theta - \tan^2 \theta = 1$ $1 = 1$

$$7. \cos^2 \theta - \sin^2 \theta = 1 - 2\sin^2 \theta$$

$$(1 - \sin^2 \theta) - \sin^2 \theta = 1 - 2\sin^2 \theta$$

$$\boxed{1 - 2\sin^2 \theta = 1 - 2\sin^2 \theta}$$

$$8. \sec^2 \theta \cdot \cot^2 \theta - 1 = \cot^2 \theta$$

$$\frac{1}{\cos^2 \theta} \cdot \frac{\cos^2 \theta}{\sin^2 \theta} - 1 = \cot^2 \theta$$

$$\frac{1}{\sin^2 \theta} - 1 = \cot^2 \theta$$

$$\csc^2 \theta - 1 = \cot^2 \theta$$

$$\boxed{\cot^2 \theta = \cot^2 \theta}$$

$$9. \sec^2 \theta = \frac{\csc^2 \theta}{\csc^2 \theta - 1}$$

$$\sec^2 \theta = \frac{\csc^2 \theta}{\cot^2 \theta}$$

$$\sec^2 \theta = \frac{1}{\sin^2 \theta} \cdot \frac{\sin^2 \theta}{\cos^2 \theta}$$

$$\sec^2 \theta = \frac{1}{\cos^2 \theta}$$

$$\boxed{\sec^2 \theta = \sec^2 \theta}$$

$$10. \cot \theta + 1 = \csc \theta \cdot (\cos \theta + \sin \theta)$$

$$\cot \theta + 1 = \frac{1}{\sin \theta} \cdot \cos \theta + \frac{1}{\sin \theta} \cdot \sin \theta$$

$$\cot \theta + 1 = \frac{\cos \theta}{\sin \theta} + 1$$

$$\boxed{\cot \theta + 1 = \cot \theta + 1}$$

$$11. \cos \theta + \sin \theta \cdot \tan \theta = \sec \theta$$

$$\cos \theta + \sin \theta \cdot \frac{\sin \theta}{\cos \theta} = \sec \theta$$

$$\frac{\cos^2 \theta + \sin^2 \theta}{\cos \theta} = \sec \theta$$

$$\frac{1}{\cos \theta} = \sec \theta$$

$$\boxed{\sec \theta = \sec \theta}$$

$$12. \frac{\cot \theta - \tan \theta}{\sin \theta \cdot \cos \theta} = \csc^2 \theta - \sec^2 \theta$$

$$\frac{\frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta}}{\sin \theta \cos \theta} = \csc^2 \theta - \sec^2 \theta$$

$$\frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta} \cdot \frac{1}{\sin \theta \cos \theta} = \csc^2 \theta - \sec^2 \theta$$

$$\frac{\cos^2 \theta - \sin^2 \theta}{\sin^2 \theta \cos^2 \theta} = \csc^2 \theta - \sec^2 \theta$$

$$\frac{\cos^2 \theta}{\sin^2 \theta \cos^2 \theta} - \frac{\sin^2 \theta}{\sin^2 \theta \cos^2 \theta} = \csc^2 \theta - \sec^2 \theta$$

$$\boxed{\csc^2 \theta - \sec^2 \theta = \csc^2 \theta - \sec^2 \theta}$$

Name: _____

Unit 12: Trigonometry

Date: _____ Bell: _____

Homework 11: Fundamental Trig Identities

**** This is a 2-page document! ******Directions:** Find the exact value of each expression if $90^\circ < \theta < 180^\circ$.

1. If $\cos \theta = -\frac{1}{2}$, find $\cot \theta$. $\sec \theta = -2$

$$\tan^2 \theta + 1 = (-2)^2$$

$$\tan^2 \theta + 1 = 4$$

$$\tan^2 \theta = 3$$

$$\tan \theta = \sqrt{3} \rightarrow \cot \theta = -\frac{\sqrt{3}}{3}$$

2. If $\csc \theta = \frac{4}{3}$, find $\sec \theta$. $\sin \theta = \frac{3}{4}$

$$\left(\frac{3}{4}\right)^2 + \cos^2 \theta = 1$$

$$\frac{9}{16} + \cos^2 \theta = 1$$

$$\cos^2 \theta = \frac{7}{16}$$

$$\cos \theta = -\frac{\sqrt{7}}{4}$$

$$\rightarrow \sec \theta = -\frac{4\sqrt{7}}{7}$$

Directions: Find the exact value of each expression if $270^\circ < \theta < 360^\circ$.

3. If $\tan \theta = -\frac{1}{4}$, find $\sin \theta$. $\cot \theta = -4$

$$(-4)^2 + 1 = \csc^2 \theta$$

$$17 = \csc^2 \theta$$

$$-\sqrt{17} = \csc \theta \rightarrow \sin \theta = -\frac{\sqrt{17}}{17}$$

4. If $\sec \theta = \frac{\sqrt{15}}{3}$, find $\csc \theta$. $\cos \theta = \frac{3}{\sqrt{15}}$

$$\left(\frac{3}{\sqrt{15}}\right)^2 + \sin^2 \theta = 1$$

$$\frac{9}{15} + \sin^2 \theta = 1$$

$$\sin^2 \theta = \frac{2}{5}$$

$$\sin \theta = \sqrt{\frac{2}{5}}$$

$$\rightarrow \csc \theta = -\frac{\sqrt{10}}{2}$$

Directions: Simplify each expression.

5. $\csc \theta \cdot \tan \theta$

$$\frac{1}{\sin \theta} \cdot \frac{\sin \theta}{\cos \theta} = \frac{1}{\cos \theta}$$

$$= \boxed{\sec \theta}$$

6. $\frac{(1 + \sin \theta)(1 - \sin \theta)}{\cot^2 \theta}$

$$\frac{1 - \sin^2 \theta}{\frac{\cos^2 \theta}{\sin^2 \theta}} = \cos^2 \theta \cdot \frac{\sin^2 \theta}{\cos^2 \theta}$$

$$= \boxed{\sin^2 \theta}$$

7. $\frac{1 - \cot^2 \theta \cdot \sec^2 \theta}{\cot^2 \theta}$

$$\frac{1 - \frac{\cos^2 \theta}{\sin^2 \theta} \cdot \frac{1}{\cos^2 \theta}}{\cot^2 \theta} = \frac{1 - \csc^2 \theta}{\cot^2 \theta}$$

$$= -\frac{\cot^2 \theta}{\cot^2 \theta} = \boxed{-1}$$

8. $\frac{\cos \theta \cdot \sin^2 \theta + \cos^3 \theta}{\sin \theta}$

$$\frac{\cos \theta (\sin^2 \theta + \cos^2 \theta)}{\sin \theta}$$

$$= \frac{\cos \theta}{\sin \theta} = \boxed{\cot \theta}$$

$$9. \csc^2 \theta \cdot \tan^2 \theta \cdot \cos \theta$$

$$\frac{1}{\sin^2 \theta} \cdot \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \cos \theta$$

$$= \frac{1}{\cos \theta} = \boxed{\sec \theta}$$

$$10. \frac{\sec^2 \theta - \tan^2 \theta}{\tan \theta \cdot \cos \theta}$$

$$\frac{(\tan^2 \theta + 1) - \tan^2 \theta}{\frac{\sin \theta}{\cos \theta} \cdot \cos \theta}$$

$$= \frac{1}{\sin \theta} = \boxed{\csc \theta}$$

Directions: Prove each identity.

$$11. \cos \theta \cdot \cot \theta + \sin^2 \theta \cdot \csc \theta = \csc \theta$$

$$\cos \theta \cdot \frac{\cos \theta}{\sin \theta} + \sin^2 \theta \cdot \frac{1}{\sin \theta} = \csc \theta$$

$$\frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta} = \csc \theta$$

$$\frac{1}{\sin \theta} = \csc \theta \quad \boxed{\csc \theta = \csc \theta}$$

$$12. \csc \theta - \sin \theta = \frac{\cot^2 \theta}{\csc \theta}$$

$$\frac{1}{\sin \theta} - \sin \theta = \frac{\cot^2 \theta}{\csc \theta}$$

$$\frac{1 - \sin^2 \theta}{\sin \theta} = \frac{\cot^2 \theta}{\csc \theta}$$

$$\frac{\cos^2 \theta}{\sin \theta} = \frac{\cot^2 \theta}{\csc \theta}$$

$$\frac{\sin \theta \cos^2 \theta}{\sin^2 \theta} = \frac{\cot^2 \theta}{\csc \theta} \quad \boxed{\frac{\cot^2 \theta}{\csc \theta} = \frac{\cot^2 \theta}{\csc \theta}}$$

$$13. \frac{\cos^2 \theta + \sin^2 \theta}{1 + \tan^2 \theta} = \cos^2 \theta$$

$$\frac{1}{\sec^2 \theta} = \cos^2 \theta$$

$$\boxed{\cos^2 \theta = \cos^2 \theta}$$

$$14. \tan^2 \theta = \csc^2 \theta \cdot \tan^2 \theta - 1$$

$$\tan^2 \theta = \frac{1}{\sin^2 \theta} \cdot \frac{\sin^2 \theta}{\cos^2 \theta} - 1$$

$$\tan^2 \theta = \frac{1}{\cos^2 \theta} - 1$$

$$\tan^2 \theta = \sec^2 \theta - 1$$

$$\boxed{\tan^2 \theta = \tan^2 \theta}$$

$$15. \frac{\csc \theta \cdot \cos \theta}{\tan \theta + \cot \theta} = \cos^2 \theta$$

$$\frac{\frac{1}{\sin \theta} \cdot \cos \theta}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}} = \cos^2 \theta$$

$$\frac{\frac{\cos \theta}{\sin \theta}}{\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}} = \cos^2 \theta$$

$$\frac{\cos \theta}{\sin \theta} \cdot \frac{\sin \theta \cos \theta}{1} = \cos^2 \theta$$

$$\boxed{\cos^2 \theta = \cos^2 \theta}$$

$$16. \frac{\cos \theta}{1 - \sin \theta} - \tan \theta = \sec \theta$$

$$\frac{\cos \theta}{1 - \sin \theta} - \frac{\sin \theta}{\cos \theta} = \sec \theta$$

$$\frac{\cos^2 \theta - \sin \theta + \sin^2 \theta}{\cos \theta (1 - \sin \theta)} = \sec \theta$$

$$\frac{1 - \sin \theta}{\cos \theta (1 - \sin \theta)} = \sec \theta$$

$$\boxed{\sec \theta = \sec \theta}$$

Name:

Class:

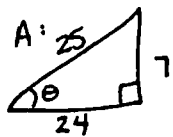
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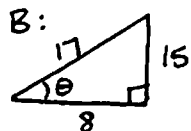
Main Ideas/Questions	Notes	
Sum and Difference of Angles Identities	Sum of Angles	Difference of Angles
	$\sin(A+B) = \sin A \cdot \cos B + \cos A \cdot \sin B$ $\cos(A+B) = \cos A \cdot \cos B - \sin A \cdot \sin B$ $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$	$\sin(A-B) = \sin A \cdot \cos B - \cos A \cdot \sin B$ $\cos(A-B) = \cos A \cdot \cos B + \sin A \cdot \sin B$ $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \cdot \tan B}$
Finding Trig Values	Directions: Find the exact value of each expression. Use the sum and difference identities to break the value apart into the unit circle angles!	
	1. $\sin 165^\circ \rightarrow \sin(135 + 30)$ $\sin 135 \cdot \cos 30 + \cos 135 \cdot \sin 30$ $= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(-\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$ $= \boxed{\frac{\sqrt{6} - \sqrt{2}}{4}}$	2. $\cos(-75^\circ) \rightarrow \cos(45 - 120)$ $\cos 45 \cos 120 + \sin 45 \sin 120$ $= \left(\frac{\sqrt{2}}{2}\right)\left(-\frac{1}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$ $= \boxed{\frac{-\sqrt{2} + \sqrt{6}}{4}}$
	3. $\tan \frac{\pi}{12} \rightarrow \tan(\pi/4 - \pi/6)$ $\frac{\tan \pi/4 - \tan \pi/6}{1 + \tan \pi/4 \cdot \tan \pi/6}$ $= \frac{1 - \frac{\sqrt{3}}{3}}{1 + 1\left(\frac{\sqrt{3}}{3}\right)} = \frac{\frac{3-\sqrt{3}}{3}}{\frac{3+\sqrt{3}}{3}} = \frac{3-\sqrt{3}}{3+\sqrt{3}} \cdot \frac{(3-\sqrt{3})}{(3-\sqrt{3})}$ $= \frac{9 - 6\sqrt{3} + 3}{9 - 3} = \frac{12 - 6\sqrt{3}}{6} = \frac{2 - \sqrt{3}}{1} = \boxed{2 - \sqrt{3}}$	4. $\cos\left(-\frac{19\pi}{12}\right) \rightarrow \cos(\pi/4 - 11\pi/6)$ $\cos \pi/4 \cos 11\pi/6 + \sin \pi/4 \cos 11\pi/6$ $\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(-\frac{1}{2}\right)$ $= \boxed{\frac{\sqrt{6} - \sqrt{2}}{4}}$
	5. $\sin \frac{7\pi}{12} \rightarrow \sin \pi/3 + \pi/4$ $\sin \pi/3 \cos \pi/4 + \cos \pi/3 \sin \pi/4$ $\left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$ $= \boxed{\frac{\sqrt{6} + \sqrt{2}}{4}}$	6. $\tan\left(-\frac{13\pi}{12}\right) \rightarrow \tan(\pi/6 - 5\pi/4)$ $\frac{\tan \pi/6 - \tan 5\pi/4}{1 + \tan \pi/6 \cdot \tan 5\pi/4}$ $= \frac{\frac{\sqrt{3}}{3} - 1}{1 + \frac{\sqrt{3}}{3}(1)} = \frac{\frac{\sqrt{3}-3}{3}}{\frac{3+\sqrt{3}}{3}} = \frac{\sqrt{3}-3}{3+\sqrt{3}} \cdot \frac{(3-\sqrt{3})}{(3-\sqrt{3})}$ $= \frac{-12 + 6\sqrt{3}}{6} = \frac{-2 + \sqrt{3}}{1} = \boxed{-2 + \sqrt{3}}$

Directions: Find the exact value of each expression by applying the identity.

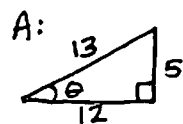
7. Given angles A and B in Quadrant I with $\tan A = \frac{7}{24}$ and $\tan B = \frac{15}{8}$, find $\tan(A+B)$.



$$\frac{\tan A + \tan B}{1 - \tan A \cdot \tan B} = \frac{\frac{7}{24} + \frac{15}{8}}{1 - \frac{7}{24} \left(\frac{15}{8} \right)} = \frac{\frac{13}{6}}{1 - \frac{35}{64}} = \boxed{\frac{416}{87}}$$

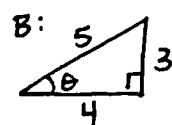


8. Given angles A and B in Quadrant I with $\sin A = \frac{5}{13}$ and $\cos B = \frac{4}{5}$, find $\sin(A+B)$.



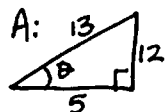
$$\sin A \cos B + \cos A \sin B$$

$$= \left(\frac{5}{13} \right) \left(\frac{4}{5} \right) + \left(\frac{12}{13} \right) \left(\frac{3}{5} \right)$$



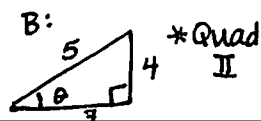
$$= \frac{20}{65} + \frac{36}{65} = \boxed{\frac{56}{65}}$$

9. Given angle A in Quadrant I with $\sin A = \frac{12}{13}$ and angle B in Quadrant II with $\sin B = \frac{4}{5}$, find $\cos(A-B)$.



$$\cos A \cos B + \sin A \sin B$$

$$= \left(\frac{5}{13} \right) \left(-\frac{3}{5} \right) + \left(\frac{12}{13} \right) \left(\frac{4}{5} \right)$$



$$= -\frac{15}{65} + \frac{48}{65} = \boxed{\frac{33}{65}}$$

Proving Identities

Directions: Use the sum and difference identities to verify each identity.

10. $\sin(\theta - \pi) = -\sin \theta$

$$\sin \theta \cos \pi - \cos \theta \sin \pi = -\sin \theta$$

$$\sin \theta (-1) - \cos \theta (0) = -\sin \theta$$

$$\boxed{-\sin \theta = -\sin \theta}$$

11. $\cos\left(\frac{\pi}{2} + \theta\right) = -\sin \theta$

$$\cos \frac{\pi}{2} \cos \theta - \sin \frac{\pi}{2} \sin \theta = -\sin \theta$$

$$0 \cdot \cos \theta - 1 \cdot \sin \theta = -\sin \theta$$

$$\boxed{-\sin \theta = -\sin \theta}$$

12. $\tan(\pi + \theta) = \tan \theta$

$$\frac{\tan \pi + \tan \theta}{1 - \tan \pi \cdot \tan \theta} = \tan \theta$$

$$\frac{0 + \tan \theta}{1 - 0 \cdot \tan \theta} = \tan \theta$$

$$\frac{\tan \theta}{1} = \tan \theta$$

$$\boxed{\tan \theta = \tan \theta}$$

13. $\cos\left(\frac{3\pi}{2} - \theta\right) = -\sin \theta$

$$\cos \frac{3\pi}{2} \cos \theta + \sin \frac{3\pi}{2} \sin \theta = -\sin \theta$$

$$0 \cdot \cos \theta + (-1) \cdot \sin \theta = -\sin \theta$$

$$\boxed{-\sin \theta = -\sin \theta}$$

Name: _____

Unit 12: Trigonometry

Date: _____ Bell: _____

Homework 12: Sum and Difference of Angles Identities

**** This is a 2-page document! ******Directions:** Find the exact value of each function.

1. $\cos 195^\circ \rightarrow \cos (135 + 60)$

$$\cos 135 \cdot \cos 60 - \sin 135 \cdot \sin 60$$

$$\left(-\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$$

$$\boxed{\frac{-\sqrt{2} - \sqrt{6}}{4}}$$

2. $\sin(-345^\circ) \rightarrow \sin(45 - 390)$

$$\sin 45 \cos 390 - \cos 45 \sin 390$$

$$\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$$

$$\boxed{\frac{\sqrt{6} - \sqrt{2}}{4}}$$

3. $\tan \frac{5\pi}{12} \rightarrow \tan \left(\frac{\pi}{4} + \frac{\pi}{6}\right)$

$$\frac{\tan \frac{\pi}{4} + \tan \frac{\pi}{6}}{1 - \tan \frac{\pi}{4} \cdot \tan \frac{\pi}{6}}$$

$$\frac{1 + \frac{\sqrt{3}}{3}}{1 - 1\left(\frac{\sqrt{3}}{3}\right)} = \frac{\frac{3 + \sqrt{3}}{3}}{\frac{3 - \sqrt{3}}{3}} = \frac{3 + \sqrt{3}}{3 - \sqrt{3}} \cdot \frac{(3 + \sqrt{3})}{(3 + \sqrt{3})} = \frac{9 + 6\sqrt{3} + 3}{9 - 3} = \boxed{2 + \sqrt{3}}$$

4. $\sin\left(-\frac{17\pi}{12}\right) \rightarrow \sin\left(\frac{\pi}{4} - \frac{5\pi}{3}\right)$

$$\sin \frac{\pi}{4} \cos \frac{5\pi}{3} - \cos \frac{\pi}{4} \sin \frac{5\pi}{3}$$

$$\left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(-\frac{\sqrt{3}}{2}\right)$$

$$\boxed{\frac{\sqrt{2} + \sqrt{6}}{4}}$$

5. $\cos \frac{\pi}{12} \rightarrow \cos \left(\frac{\pi}{4} - \frac{\pi}{6}\right)$

$$\cos \frac{\pi}{4} \cos \frac{\pi}{6} + \sin \frac{\pi}{4} \sin \frac{\pi}{6}$$

$$\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$$

$$\boxed{\frac{\sqrt{6} + \sqrt{2}}{4}}$$

6. $\tan\left(-\frac{11\pi}{12}\right) \rightarrow \tan\left(\frac{\pi}{4} - \frac{\pi}{6}\right)$

$$\frac{\tan \frac{\pi}{4} - \tan \frac{\pi}{6}}{1 + \tan \frac{\pi}{4} \cdot \tan \frac{\pi}{6}}$$

$$= \frac{1 - \frac{\sqrt{3}}{3}}{1 + 1\left(\frac{\sqrt{3}}{3}\right)} = \frac{\frac{3 - \sqrt{3}}{3}}{\frac{3 + \sqrt{3}}{3}} = \frac{3 - \sqrt{3}}{3 + \sqrt{3}} \cdot \frac{(3 - \sqrt{3})}{(3 - \sqrt{3})} = \frac{9 - 6\sqrt{3} + 3}{9 - 3} = \boxed{2 - \sqrt{3}}$$

7. Given angles A and B in Quadrant I with $\tan A = 8$ and $\tan B = \frac{1}{2}$, find $\tan(A + B)$.

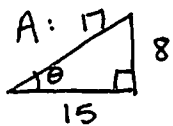
$$\frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$$

$$\frac{8 + \frac{1}{2}}{1 - 8\left(\frac{1}{2}\right)}$$

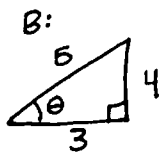
$$= \frac{\frac{17}{2}}{-3}$$

$$= \boxed{-\frac{17}{6}}$$

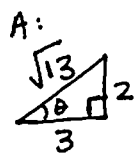
8. Given angles A and B in Quadrant I with $\sin A = \frac{8}{17}$ and $\cos B = \frac{3}{5}$, find $\sin(A+B)$.



$$\begin{aligned} \sin A \cos B + \cos A \sin B \\ = \left(\frac{8}{17}\right)\left(\frac{3}{5}\right) + \left(\frac{15}{17}\right)\left(\frac{4}{5}\right) \\ = \frac{24}{85} + \frac{60}{85} = \boxed{\frac{84}{85}} \end{aligned}$$

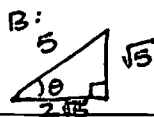


9. Given angles A and B in Quadrant I with $\tan A = \frac{2}{3}$ and $\sin B = \frac{\sqrt{5}}{5}$, find $\tan(A-B)$.



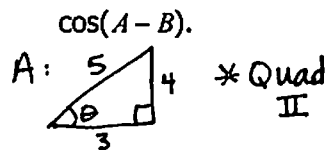
$$\frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$= \frac{\frac{2}{3} - \frac{1}{2}}{1 + \frac{2}{3}\left(\frac{1}{2}\right)}$$

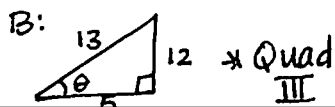


$$= \frac{\frac{1}{6}}{\frac{8}{6}} = \boxed{\frac{1}{8}}$$

10. Given angle A in Quadrant II with $\sin A = \frac{4}{5}$ and angle B in Quadrant III with $\cos B = -\frac{5}{13}$, find $\cos(A-B)$.



$$\begin{aligned} \cos A \cos B + \sin A \sin B \\ = \left(-\frac{3}{5}\right)\left(-\frac{5}{13}\right) + \left(\frac{4}{5}\right)\left(-\frac{12}{13}\right) \end{aligned}$$



$$= \frac{15}{65} - \frac{48}{65} = \boxed{\frac{-33}{65}}$$

Directions: Prove each identity.

11. $\sin\left(\frac{3\pi}{2} - \theta\right) = -\cos \theta$

$$\sin \frac{3\pi}{2} \cos \theta - \cos \frac{3\pi}{2} \sin \theta = -\cos \theta$$

$$-1 \cdot \cos \theta - 0 \cdot \sin \theta = -\cos \theta$$

$$\boxed{-\cos \theta = -\cos \theta}$$

12. $\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$

$$\cos \frac{\pi}{2} \cos \theta + \sin \frac{\pi}{2} \sin \theta = \sin \theta$$

$$0 \cdot \cos \theta + 1 \cdot \sin \theta = \sin \theta$$

$$\boxed{\sin \theta = \sin \theta}$$

13. $\tan(\pi - \theta) = -\tan \theta$

$$\frac{\tan \pi - \tan \theta}{1 + \tan \pi \cdot \tan \theta} = -\tan \theta$$

$$\frac{0 - \tan \theta}{1 + 0 \cdot \tan \theta} = -\tan \theta$$

$$\frac{-\tan \theta}{1} = -\tan \theta$$

$$\boxed{-\tan \theta = -\tan \theta}$$

14. $\sin\left(\frac{\pi}{2} + \theta\right) = \cos \theta$

$$\sin \frac{\pi}{2} \cos \theta + \cos \frac{\pi}{2} \sin \theta = \cos \theta$$

$$1 \cdot \cos \theta + 0 \cdot \sin \theta = \cos \theta$$

$$\boxed{\cos \theta = \cos \theta}$$

Name:	Class:
Topic:	Date:

Main Ideas/Questions	Notes		
Double-Angle Identities	SINE	COSINE	TANGENT
	$\sin 2\theta = 2 \cdot \sin \theta \cdot \cos \theta$	$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ $\cos 2\theta = 1 - 2 \cdot \sin^2 \theta$ $\cos 2\theta = 2 \cdot \cos^2 \theta - 1$	$\tan 2\theta = \frac{2 \cdot \tan \theta}{1 - \tan^2 \theta}$
Finding Trig Values	1. If $\sin \theta = \frac{5}{13}$ and $0 < \theta < \frac{\pi}{2}$, find $\cos 2\theta$. $1 - 2\sin^2 \theta = 1 - 2\left(\frac{5}{13}\right)^2 = 1 - 2\left(\frac{25}{169}\right) = 1 - \frac{50}{169} = \boxed{\frac{119}{169}}$		
	2. If $\cos \theta = -\frac{7}{25}$ and $\pi < \theta < \frac{3\pi}{2}$, find $\tan 2\theta$. $\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{2(24/7)}{1 - (24/7)^2} = \frac{48/7}{1 - 576/49} = \frac{48}{7} \cdot \frac{49}{-527} = \boxed{-\frac{336}{527}}$		
	3. If $\tan \theta = -\frac{20}{21}$ and $\frac{\pi}{2} < \theta < \pi$, find $\cos 2\theta$. $1 - 2\sin^2 \theta = 1 - 2\left(\frac{20}{29}\right)^2 = 1 - 2\left(\frac{400}{841}\right) = 1 - 800/841 = \boxed{41/841}$		
Half-Angle Identities	SINE	COSINE	TANGENT
	$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$	$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$	$\tan \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$
Finding Trig Values	4. If $\cos \theta = -\frac{4}{5}$ and $\frac{\pi}{2} < \theta < \pi$, find $\sin \frac{\theta}{2}$. $\pm \sqrt{\frac{1 - \cos \theta}{2}} = \sqrt{\frac{1 - (-4/5)}{2}} = \sqrt{\frac{9/5}{2}} = \sqrt{\frac{9}{10}} = \boxed{\frac{3\sqrt{10}}{10}}$		

5. If $\tan \theta = 2\sqrt{6}$ and $0 < \theta < \frac{\pi}{2}$, find $\cos \frac{\theta}{2}$.

$$(2\sqrt{6})^2 + 1 = \sec^2 \theta$$

$$24 + 1 = \sec^2 \theta$$

$$25 = \sec^2 \theta$$

$$5 = \sec \theta$$

$$\frac{1}{5} = \cos \theta$$

$$\pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$= \sqrt{\frac{1 + \frac{1}{5}}{2}}$$

$$= \sqrt{\frac{\frac{6}{5}}{2}}$$

$$= \sqrt{\frac{3}{5}} = \boxed{\frac{\sqrt{15}}{5}}$$

6. If $\sin \theta = -\frac{8}{17}$ and $\pi < \theta < \frac{3\pi}{2}$, find $\tan \frac{\theta}{2}$.

$$-\sqrt{\frac{1 - (-\frac{15}{17})}{1 + (-\frac{15}{17})}} = -\sqrt{\frac{\frac{32}{17}}{\frac{2}{17}}} = -\sqrt{16} = \boxed{-4}$$

Proving Identities

Directions: Prove each identity.

7. $\cos^2 \theta - \cos 2\theta = \sin^2 \theta$

$$\cos^2 \theta - (1 - 2\sin^2 \theta) = \sin^2 \theta$$

$$\cos^2 \theta - 1 + 2\sin^2 \theta = \sin^2 \theta$$

$$\cos^2 \theta + \sin^2 \theta + \sin^2 \theta - 1 = \sin^2 \theta$$

$$1 + \sin^2 \theta - 1 = \sin^2 \theta$$

$$\boxed{\sin^2 \theta = \sin^2 \theta}$$

8. $(\cos \theta + \sin \theta)^2 = 1 + \sin 2\theta$

$$\cos^2 \theta + 2\sin \theta \cos \theta + \sin^2 \theta$$

$$= 1 + \sin 2\theta$$

$$1 + 2\sin \theta \cos \theta = 1 + \sin 2\theta$$

$$\boxed{1 + \sin 2\theta = 1 + \sin 2\theta}$$

9. $\tan \theta = \frac{1 - \cos 2\theta}{\sin 2\theta}$

$$\tan \theta = \frac{1 - (1 - 2\sin^2 \theta)}{2\sin \theta \cos \theta}$$

$$\tan \theta = \frac{2\sin^2 \theta}{2\sin \theta \cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\boxed{\tan \theta = \tan \theta}$$

10. $\sin 2\theta = \frac{2 \cdot \tan \theta}{1 + \tan^2 \theta}$

$$\sin 2\theta = \frac{2 \cdot \frac{\sin \theta}{\cos \theta}}{1 + \frac{\sin^2 \theta}{\cos^2 \theta}}$$

$$\sin 2\theta = \frac{\frac{2\sin \theta}{\cos \theta}}{\frac{\cos^2 \theta + \sin^2 \theta}{\cos^2 \theta}}$$

$$\sin 2\theta = \frac{2\sin \theta}{\cos \theta} \cdot \frac{\cos^2 \theta}{1}$$

$$\sin 2\theta = 2\sin \theta \cos \theta$$

$$\boxed{\sin 2\theta = \sin 2\theta}$$

Name: _____

Unit 12: Trigonometry

Date: _____ Bell: _____

Homework 13: Double- and Half-Angle Identities

**** This is a 2-page document! ******Directions:** Use the information given to find the exact value of each trigonometric function.

1. $\cos \theta = -\frac{3}{5}$ and $\frac{\pi}{2} < \theta < \pi$; find $\sin 2\theta$.

$$\sin^2 \theta + (-3/5)^2 = 1$$

$$\sin^2 \theta + 9/25 = 1$$

$$\sin^2 \theta = 16/25$$

$$\sin \theta = 4/5$$

$$2 \sin \theta \cos \theta$$

$$2(4/5)(-3/5)$$

$$= \boxed{-\frac{24}{25}}$$

2. $\tan \theta = \frac{12}{13}$ and $\pi < \theta < \frac{3\pi}{2}$; find $\cos 2\theta$.

$$(12/13)^2 + 1 = \sec^2 \theta$$

$$144/169 + 1 = \sec^2 \theta$$

$$313/169 = \sec^2 \theta$$

$$\sqrt{313}/13 = \sec \theta \rightarrow \cos \theta = -13/\sqrt{313}$$

$$2 \cos^2 \theta - 1$$

$$2\left(-\frac{13}{\sqrt{313}}\right)^2 - 1$$

$$= \frac{388}{313} - 1$$

$$= \boxed{\frac{25}{313}}$$

3. $\sin \theta = -\frac{20}{29}$ and $\frac{3\pi}{2} < \theta < 2\pi$; find $\tan 2\theta$.

$$\cot^2 \theta + 1 = (-29/20)^2$$

$$\cot^2 \theta + 1 = 841/400$$

$$\cot^2 \theta = 441/400$$

$$\cot \theta = -21/20$$

$$\tan \theta = -20/21$$

$$\frac{2 \cdot \tan \theta}{1 - \tan^2 \theta}$$

$$\frac{2(-20/21)}{1 - (-20/21)^2}$$

$$= \frac{-40/21}{1 - 400/441}$$

$$= \frac{-40}{21} \cdot \frac{441}{41} = \boxed{-\frac{840}{41}}$$

4. $\sin \theta = \frac{7}{25}$ and $0 < \theta < \frac{\pi}{2}$; find $\cos \frac{\theta}{2}$.

$$(7/25)^2 + \cos^2 \theta = 1$$

$$\frac{49}{625} + \cos^2 \theta = 1$$

$$\cos^2 \theta = \frac{576}{625}$$

$$\cos \theta = 24/25$$

$$\pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\sqrt{\frac{1 + 24/25}{2}} = \sqrt{\frac{49/25}{2}} = \sqrt{49/50}$$

$$= \frac{7}{5\sqrt{2}} = \boxed{\frac{7\sqrt{2}}{10}}$$

5. $\cos \theta = -\frac{4}{5}$ and $\frac{\pi}{2} < \theta < \pi$; find $\tan \frac{\theta}{2}$.

$$\pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

$$\sqrt{\frac{1 + 4/5}{1 - 4/5}} = \sqrt{\frac{9/5}{1/5}} = \sqrt{9} = \boxed{3}$$

6. $\tan \theta = \sqrt{35}$ and $\pi < \theta < \frac{3\pi}{2}$; find $\sin \frac{\theta}{2}$.

$$(\sqrt{35})^2 + 1 = \sec^2 \theta$$

$$36 = \sec^2 \theta$$

$$-6 = \sec \theta \rightarrow \cos \theta = -\frac{1}{6}$$

$$\pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\sqrt{\frac{1 - (-1/6)}{2}} = \sqrt{\frac{7/6}{2}} = \sqrt{\frac{7}{12}}$$

$$= \frac{\sqrt{7}}{2\sqrt{3}} = \boxed{\frac{\sqrt{21}}{6}}$$

Directions: Prove each identity.

7. $\cot \theta = \frac{\sin 2\theta}{1 - \cos 2\theta}$

$$\cot \theta = \frac{2 \sin \theta \cos \theta}{1 - (1 - 2 \sin^2 \theta)}$$

$$\cot \theta = \frac{2 \sin \theta \cos \theta}{2 \sin^2 \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\boxed{\cot \theta = \cot \theta}$$

8. $\sin \theta \cdot \cos \theta + \sin 2\theta = 3 \cdot \sin \theta \cdot \cos \theta$

$$\sin \theta \cos \theta + 2 \sin \theta \cos \theta$$

$$= 3 \sin \theta \cos \theta$$

$$\boxed{3 \sin \theta \cos \theta = 3 \sin \theta \cos \theta}$$

9. $2 \sin^2 \theta + \cos 2\theta = 1$

$$2 \sin^2 \theta + 1 - 2 \sin^2 \theta = 1$$

$$\boxed{1 = 1}$$

10. $\sin 2\theta = 2 \cot \theta \cdot \sin^2 \theta$

$$\sin 2\theta = \frac{2 \cos \theta}{\sin \theta} \cdot \sin^2 \theta$$

$$\sin 2\theta = 2 \cos \theta \sin \theta$$

$$\boxed{\sin 2\theta = \sin 2\theta}$$

11. $\sin 2\theta - 2 \cdot \sin \theta = (2 \cdot \sin \theta)(\cos \theta - 1)$

$$\sin 2\theta - 2 \sin \theta = 2 \sin \theta \cos \theta - 2 \sin \theta$$

$$\boxed{\sin 2\theta - 2 \sin \theta = \sin 2\theta - 2 \sin \theta}$$

12. $\frac{2 \cdot \cos \theta}{\sin 2\theta} = \csc \theta$

$$\frac{2 \cos \theta}{2 \sin \theta \cos \theta} = \csc \theta$$

$$\frac{1}{\sin \theta} = \csc \theta$$

$$\boxed{\csc \theta = \csc \theta}$$

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 12: Trigonometry

Quiz 12-3: Trigonometric Identities**Pythagorean Identities:**

1. Given
- $\sin \theta = \frac{9}{41}$
- and
- $0 < \theta < \frac{\pi}{2}$
- , find
- $\cos \theta$
- .

1. $\frac{40}{41}$

$$\left(\frac{9}{41}\right)^2 + \cos^2 \theta = 1$$

$$\frac{81}{1681} + \cos^2 \theta = 1$$

$$\cos^2 \theta = \frac{1600}{1681}$$

$$\cos \theta = \frac{40}{41}$$

2. Given
- $\sec \theta = -\frac{6}{5}$
- and
- $\pi < \theta < \frac{3\pi}{2}$
- , find
- $\cot \theta$
- .

2. $\frac{5\sqrt{11}}{11}$

$$\tan^2 \theta + 1 = \left(-\frac{6}{5}\right)^2$$

$$\tan^2 \theta + 1 = \frac{36}{25}$$

$$\tan^2 \theta = \frac{11}{25}$$

$$\tan \theta = \frac{\sqrt{11}}{5}$$

$$\rightarrow \cot \theta = \frac{5}{\sqrt{11}} = \frac{5\sqrt{11}}{11}$$

Sum and Differences of Angles Identities:

3. Find the exact value of
- $\tan \frac{17\pi}{12}$
- .
- $\tan\left(\frac{7\pi}{6} + \frac{\pi}{4}\right)$

3. $2 + \sqrt{3}$

$$\frac{\frac{\sqrt{3}}{3} + 1}{1 - \frac{\sqrt{3}}{3}(1)}$$

$$= \frac{\frac{\sqrt{3} + 3}{3}}{\frac{3 - \sqrt{3}}{3}}$$

$$= \frac{\sqrt{3} + 3}{3 - \sqrt{3}} \cdot \frac{(3 + \sqrt{3})}{(3 + \sqrt{3})}$$

$$= \frac{9 + 6\sqrt{3} + 3}{9 - 3} = \frac{12 + 6\sqrt{3}}{6} = 2 + \sqrt{3}$$

$$\frac{-\sqrt{2} - \sqrt{6}}{4}$$

4. Find the exact value of
- $\sin\left(-\frac{5\pi}{12}\right)$
- .
- $\sin \frac{\pi}{4} - \frac{2\pi}{3}$

4. $\frac{4}{5}$

$$\sin \frac{\pi}{4} \cos \frac{2\pi}{3} - \cos \frac{\pi}{4} \sin \frac{2\pi}{3}$$

$$= \left(\frac{\sqrt{2}}{2}\right)\left(-\frac{1}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{-\sqrt{2} - \sqrt{6}}{4}$$

5. Given angles
- A
- and
- B
- in Quadrant I with
- $\sin A = \frac{12}{13}$
- and
- $\cos B = \frac{4}{5}$
- , find:

5a. $\frac{-16}{65}$

a) $\cos(A + B)$

$$\cos A \cos B - \sin A \sin B$$

$$= \left(\frac{5}{13}\right)\left(\frac{4}{5}\right) - \left(\frac{12}{13}\right)\left(\frac{3}{5}\right)$$

$$= \frac{20}{65} - \frac{36}{65} = \frac{-16}{65}$$

b) $\tan(A - B)$

$$\frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$= \frac{\left(\frac{12}{5}\right) - \left(\frac{3}{4}\right)}{1 + \left(\frac{12}{5}\right)\left(\frac{3}{4}\right)}$$

$$= \frac{\frac{33}{20}}{\frac{56}{20}} = \frac{33}{56}$$

b. $\frac{33}{56}$

Double-Angle and Half-Angle Identities:

6. Given $\cos \theta = \frac{1}{4}$ and $\frac{3\pi}{2} < \theta < 2\pi$, find $\sin 2\theta$.

$$\sin^2 \theta + (\frac{1}{4})^2 = 1$$

$$\sin^2 \theta + \frac{1}{16} = 1$$

$$\sin^2 \theta = \frac{15}{16}$$

$$\sin \theta = -\frac{\sqrt{15}}{4}$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2 \left(-\frac{\sqrt{15}}{4} \right) \left(\frac{1}{4} \right)$$

$$= -\frac{2\sqrt{15}}{16} = \boxed{-\frac{\sqrt{15}}{8}}$$

6. $\underline{-\frac{\sqrt{15}}{8}}$

7. Given $\sin \theta = \frac{\sqrt{5}}{5}$ and $0 < \theta < \frac{\pi}{2}$, find $\tan 2\theta$.

$$\cot^2 \theta + 1 = \left(\frac{5}{\sqrt{5}} \right)^2$$

$$\cot^2 \theta + 1 = 5$$

$$\cot^2 \theta = 4$$

$$\cot \theta = 2 \rightarrow \tan \theta = \frac{1}{2}$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$= \frac{2(\frac{1}{2})}{1 - (\frac{1}{2})^2} = \frac{1}{1 - \frac{1}{4}} = \frac{1}{\frac{3}{4}} = \boxed{\frac{4}{3}}$$

7. $\underline{\frac{4}{3}}$

8. Given $\sin \theta = -\frac{7}{25}$ and $\pi < \theta < \frac{3\pi}{2}$, find $\cos \frac{\theta}{2}$.

$$(-\frac{7}{25})^2 + \cos^2 \theta = 1$$

$$\frac{49}{625} + \cos^2 \theta = 1$$

$$\cos^2 \theta = \frac{576}{625}$$

$$\cos \theta = -\frac{24}{25}$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$= \sqrt{\frac{1 + (-\frac{24}{25})}{2}} = \sqrt{\frac{\frac{1}{25}}{2}} = \sqrt{\frac{1}{50}} = \frac{1}{5\sqrt{2}} = \boxed{\frac{\sqrt{2}}{10}}$$

8. $\underline{\frac{\sqrt{2}}{10}}$

9. Given $\tan \theta = -\frac{5}{12}$ and $\frac{\pi}{2} < \theta < \pi$, find $\tan \frac{\theta}{2}$.

$$\tan \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \sqrt{\frac{1 - (-\frac{12}{13})}{1 + (-\frac{12}{13})}} = \sqrt{\frac{\frac{25}{13}}{\frac{1}{13}}} = \sqrt{25} = \boxed{5}$$

9. $\underline{5}$

Simplifying Trigonometric Expressions:

10. $\frac{\sec \theta \cdot \sin \theta}{\tan \theta + \cot \theta}$

$$\frac{\frac{1}{\cos \theta} \cdot \sin \theta}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}}$$

$$= \frac{\frac{\sin \theta}{\cos \theta}}{\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}}$$

$$= \frac{\sin \theta}{\cos \theta} \cdot \frac{\sin \theta \cos \theta}{1}$$

$$= \boxed{\sin^2 \theta}$$

10. $\underline{\sin^2 \theta}$

$$11. \frac{\sec \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} = \frac{\frac{1}{\cos \theta}}{\sin \theta} - \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1}{\sin \theta \cos \theta} - \frac{\sin^2 \theta}{\sin \theta \cos \theta} = \frac{1 - \sin^2 \theta}{\sin \theta \cos \theta} = \frac{\cos^2 \theta}{\sin \theta \cos \theta} = \frac{\cos \theta}{\sin \theta} = \boxed{\cot \theta}$$

$$11. \cot \theta$$

$$12. \frac{\cos \theta}{1 + \sin \theta} + \tan \theta$$

$$\frac{\cos \theta}{1 + \sin \theta} + \frac{\sin \theta}{\cos \theta} = \frac{\cos^2 \theta}{\cos \theta (1 + \sin \theta)} + \frac{\sin^2 \theta + \sin \theta}{\cos \theta (1 + \sin \theta)} = \frac{1 + \sin \theta}{\cos \theta (1 + \sin \theta)} = \frac{1}{\cos \theta} = \boxed{\sec \theta}$$

$$12. \sec \theta$$

Proving Trigonometric Identities. Show all steps!

$$13. \csc \theta = \sin \theta + \sin \theta \cdot \cot^2 \theta$$

$$\csc \theta = \sin \theta + \sin \theta \cdot \frac{\cos^2 \theta}{\sin^2 \theta}$$

$$\csc \theta = \sin \theta + \frac{\cos^2 \theta}{\sin \theta}$$

$$\csc \theta = \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\boxed{\csc \theta = \csc \theta}$$

$$15. \frac{\tan \theta - \cot \theta}{\sin \theta \cdot \cos \theta} = \sec^2 \theta - \csc^2 \theta$$

$$\frac{\frac{\sin \theta}{\cos \theta} - \frac{\cos \theta}{\sin \theta}}{\sin \theta \cos \theta} = \sec^2 \theta - \csc^2 \theta$$

$$\frac{\frac{\sin^2 \theta - \cos^2 \theta}{\sin \theta \cos \theta}}{\sin \theta \cos \theta} = \sec^2 \theta - \csc^2 \theta$$

$$\frac{\sin^2 \theta - \cos^2 \theta}{\sin^2 \theta \cos^2 \theta} = \sec^2 \theta - \csc^2 \theta$$

$$\frac{\sin^2 \theta}{\sin^2 \theta \cos^2 \theta} - \frac{\cos^2 \theta}{\sin^2 \theta \cos^2 \theta} = \sec^2 \theta - \csc^2 \theta$$

$$\boxed{\sec^2 \theta - \csc^2 \theta = \sec^2 \theta - \csc^2 \theta}$$

$$14. \sec \theta + \tan \theta = \frac{\cos \theta}{1 - \sin \theta}$$

$$\frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} = \frac{\cos \theta}{1 - \sin \theta}$$

$$\frac{1 + \sin \theta}{\cos \theta} = \frac{\cos \theta}{1 - \sin \theta}$$

$$\frac{(1 - \sin \theta)(1 + \sin \theta)}{\cos \theta (1 - \sin \theta)} = \frac{\cos \theta}{1 - \sin \theta}$$

$$\frac{1 - \sin^2 \theta}{\cos \theta (1 - \sin \theta)} = \frac{\cos \theta}{1 - \sin \theta}$$

$$\frac{\cos^2 \theta}{\cos \theta (1 - \sin \theta)} = \frac{\cos \theta}{1 - \sin \theta}$$

$$\boxed{\frac{\cos \theta}{1 - \sin \theta} = \frac{\cos \theta}{1 - \sin \theta}}$$

$$16. \frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$$

$$\frac{1 - (1 - 2\sin^2 \theta)}{2\sin \theta \cos \theta} = \tan \theta$$

$$\frac{2\sin^2 \theta}{2\sin \theta \cos \theta} = \tan \theta$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta$$

$$\boxed{\tan \theta = \tan \theta}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples						
TRIGONOMETRIC EQUATIONS	- A trigonometric equation can have several solutions, depending on the defined interval. - To solve a trigonometric equation, you must isolate the trigonometric function(s). - Watch out for extraneous solutions!						
BASIC EXAMPLES	Directions: Solve each equation for the given interval. For questions 1-4, give the solution(s) in degrees. For questions 5-6, give the solution(s) in radians. <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; padding: 5px; vertical-align: top;"> 1. $6\sin\theta - 3 = 0; 0 \leq \theta \leq 360^\circ$ $6\sin\theta = 3$ $\sin\theta = \frac{1}{2}$ $\theta = 30^\circ, 150^\circ$ </td><td style="width: 50%; padding: 5px; vertical-align: top;"> 2. $4 = \cos\theta + 5; 0 \leq \theta \leq 180^\circ$ $-1 = \cos\theta$ $\theta = 180^\circ$ </td></tr> <tr> <td style="padding: 5px; vertical-align: top;"> 3. $-2\tan\theta = 2\sqrt{3}; 180^\circ \leq \theta \leq 360^\circ$ $\tan\theta = -\sqrt{3}$ $\theta = 120^\circ, 300^\circ$ $\theta = 300^\circ$ </td><td style="padding: 5px; vertical-align: top;"> 4. $5\cos\theta + \sqrt{2} = 3\cos\theta; 0^\circ \leq \theta \leq 270^\circ$ $2\cos\theta + \sqrt{2} = 0$ $2\cos\theta = -\sqrt{2}$ $\cos\theta = -\frac{\sqrt{2}}{2}$ $\theta = 135^\circ, 225^\circ$ </td></tr> <tr> <td style="padding: 5px; vertical-align: top;"> 5. $\sin\theta = 2 - \sin\theta; 0 \leq \theta \leq 2\pi$ $2\sin\theta = 2$ $\sin\theta = 1$ $\theta = \frac{\pi}{2}$ </td><td style="padding: 5px; vertical-align: top;"> 6. $3\tan\theta = \tan\theta + 2; \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$ $2\tan\theta = 2$ $\tan\theta = 1$ $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$ $\theta = \frac{5\pi}{4}$ </td></tr> </table>	1. $6\sin\theta - 3 = 0; 0 \leq \theta \leq 360^\circ$ $6\sin\theta = 3$ $\sin\theta = \frac{1}{2}$ $\theta = 30^\circ, 150^\circ$	2. $4 = \cos\theta + 5; 0 \leq \theta \leq 180^\circ$ $-1 = \cos\theta$ $\theta = 180^\circ$	3. $-2\tan\theta = 2\sqrt{3}; 180^\circ \leq \theta \leq 360^\circ$ $\tan\theta = -\sqrt{3}$ $\theta = 120^\circ, 300^\circ$ $\theta = 300^\circ$	4. $5\cos\theta + \sqrt{2} = 3\cos\theta; 0^\circ \leq \theta \leq 270^\circ$ $2\cos\theta + \sqrt{2} = 0$ $2\cos\theta = -\sqrt{2}$ $\cos\theta = -\frac{\sqrt{2}}{2}$ $\theta = 135^\circ, 225^\circ$	5. $\sin\theta = 2 - \sin\theta; 0 \leq \theta \leq 2\pi$ $2\sin\theta = 2$ $\sin\theta = 1$ $\theta = \frac{\pi}{2}$	6. $3\tan\theta = \tan\theta + 2; \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$ $2\tan\theta = 2$ $\tan\theta = 1$ $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$ $\theta = \frac{5\pi}{4}$
1. $6\sin\theta - 3 = 0; 0 \leq \theta \leq 360^\circ$ $6\sin\theta = 3$ $\sin\theta = \frac{1}{2}$ $\theta = 30^\circ, 150^\circ$	2. $4 = \cos\theta + 5; 0 \leq \theta \leq 180^\circ$ $-1 = \cos\theta$ $\theta = 180^\circ$						
3. $-2\tan\theta = 2\sqrt{3}; 180^\circ \leq \theta \leq 360^\circ$ $\tan\theta = -\sqrt{3}$ $\theta = 120^\circ, 300^\circ$ $\theta = 300^\circ$	4. $5\cos\theta + \sqrt{2} = 3\cos\theta; 0^\circ \leq \theta \leq 270^\circ$ $2\cos\theta + \sqrt{2} = 0$ $2\cos\theta = -\sqrt{2}$ $\cos\theta = -\frac{\sqrt{2}}{2}$ $\theta = 135^\circ, 225^\circ$						
5. $\sin\theta = 2 - \sin\theta; 0 \leq \theta \leq 2\pi$ $2\sin\theta = 2$ $\sin\theta = 1$ $\theta = \frac{\pi}{2}$	6. $3\tan\theta = \tan\theta + 2; \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$ $2\tan\theta = 2$ $\tan\theta = 1$ $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$ $\theta = \frac{5\pi}{4}$						

EXAMPLES WITH FACTORING

Solve for the interval $0^\circ \leq \theta \leq 360^\circ$. Give all solutions in degrees.

7. $\sin^2 \theta + \sin \theta = 0$

$$\sin \theta (\sin \theta + 1) = 0$$

$$\begin{array}{ll} \sin \theta = 0 & \sin \theta + 1 = 0 \\ \theta = 0^\circ, 180^\circ, 360^\circ & \sin \theta = -1 \\ & \theta = 270^\circ \end{array}$$

$$\theta = 0^\circ, 180^\circ, 270^\circ, 360^\circ$$

8. $\cos \theta \cdot \tan \theta + \cos \theta = 0$

$$\cos \theta (\tan \theta + 1) = 0$$

$$\begin{array}{ll} \cos \theta = 0 & \tan \theta + 1 = 0 \\ \theta = 90^\circ, 270^\circ & \tan \theta = -1 \\ & \theta = 135^\circ, 315^\circ \end{array}$$

$$\theta = 90^\circ, 135^\circ, 270^\circ, 315^\circ$$

9. $\sin^2 \theta + 3\sin \theta - 4 = 0$

$$(\sin \theta + 4)(\sin \theta - 1) = 0$$

$$\begin{array}{ll} \sin \theta + 4 = 0 & \sin \theta - 1 = 0 \\ \sin \theta = -4 & \sin \theta = 1 \\ \theta = \emptyset & \theta = 90^\circ \end{array}$$

$$\theta = 90^\circ$$

10. $\cos^2 \theta - 2\cos \theta - 3 = 0$

$$(\cos \theta - 3)(\cos \theta + 1) = 0$$

$$\begin{array}{ll} \cos \theta - 3 = 0 & \cos \theta + 1 = 0 \\ \cos \theta = 3 & \cos \theta = -1 \\ \theta = \emptyset & \theta = 180^\circ \end{array}$$

$$\theta = 180^\circ$$

11. $4\cos^2 \theta - 1 = 0$

$$(2\cos \theta - 1)(2\cos \theta + 1) = 0$$

$$\begin{array}{ll} 2\cos \theta - 1 = 0 & 2\cos \theta + 1 = 0 \\ 2\cos \theta = 1 & 2\cos \theta = -1 \\ \cos \theta = \frac{1}{2} & \cos \theta = -\frac{1}{2} \\ \theta = 60^\circ, 300^\circ & \theta = 120^\circ, 240^\circ \end{array}$$

$$\theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$$

12. $2\sin^2 \theta + \sin \theta - 1 = 0$

$$(2\sin \theta - 1)(\sin \theta + 1) = 0$$

$$\begin{array}{ll} 2\sin \theta - 1 = 0 & \sin \theta + 1 = 0 \\ 2\sin \theta = 1 & \sin \theta = -1 \\ \sin \theta = \frac{1}{2} & \theta = 270^\circ \\ \theta = 30^\circ, 150^\circ & \end{array}$$

$$\theta = 30^\circ, 150^\circ, 270^\circ$$

13. $2\cos^2 \theta - 5\cos \theta = 3$

$$2\cos^2 \theta - 5\cos \theta - 3 = 0$$

$$(2\cos \theta + 1)(\cos \theta - 3) = 0$$

$$\begin{array}{ll} 2\cos \theta + 1 = 0 & \cos \theta - 3 = 0 \\ 2\cos \theta = -1 & \cos \theta = 3 \\ \cos \theta = -\frac{1}{2} & \theta = \emptyset \\ \theta = 120^\circ, 240^\circ & \end{array}$$

$$\theta = 120^\circ, 240^\circ$$

14. $3\sin^2 \theta = 2\sin \theta + 1$

$$3\sin^2 \theta - 2\sin \theta - 1 = 0$$

$$(3\sin \theta + 1)(\sin \theta - 1) = 0$$

$$\begin{array}{ll} 3\sin \theta + 1 = 0 & \sin \theta - 1 = 0 \\ 3\sin \theta = -1 & \sin \theta = 1 \\ \sin \theta = -\frac{1}{3} & \theta = 90^\circ \\ \theta = 340.5^\circ, 199.5^\circ & \end{array}$$

$$\theta = 90^\circ, 199.5^\circ, 340.5^\circ$$

USING IDENTITIES

Common Identities

Pythagorean
Reciprocal
Double Angle

Solve for the interval $0^\circ \leq \theta \leq 180^\circ$. Give all solutions in degrees.

$$15. 4(\cos 2\theta + \sin^2 \theta) = 3$$

$$4(\cos^2 \theta - \sin^2 \theta + \sin^2 \theta) = 3$$

$$4 \cos^2 \theta = 3$$

$$\cos^2 \theta = \frac{3}{4}$$

$$\cos \theta = \pm \frac{\sqrt{3}}{2}$$

$$\theta = 30^\circ, 150^\circ$$

$$16. \sin 2\theta - \cos \theta = 0$$

$$2\sin \theta \cos \theta - \cos \theta = 0$$

$$\cos \theta (2\sin \theta - 1) = 0$$

$$\cos \theta = 0 \quad 2\sin \theta - 1 = 0$$

$$\theta = 90^\circ, 270^\circ \quad 2\sin \theta = 1$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ, 150^\circ$$

$$\theta = 30^\circ, 90^\circ, 150^\circ$$

$$17. \cos \theta \cdot \tan \theta - \sin^2 \theta = 0$$

$$\cos \theta \cdot \frac{\sin \theta}{\cos \theta} - \sin^2 \theta = 0$$

$$\sin \theta - \sin^2 \theta = 0$$

$$\sin \theta (1 - \sin \theta) = 0$$

$$\sin \theta = 0$$

$$1 - \sin \theta = 0$$

$$\theta = 0^\circ, 180^\circ, 360^\circ \quad -\sin \theta = -1$$

$$\sin \theta = 1$$

$$\theta = 90^\circ$$

$$\theta = 0^\circ, 90^\circ, 180^\circ$$

$$18. \cos^2 \theta - \cos \theta + 1 = \sin^2 \theta$$

$$\cos^2 \theta - \cos \theta + 1 = 1 - \cos^2 \theta$$

$$2\cos^2 \theta - \cos \theta = 0$$

$$\cos \theta (2\cos \theta - 1) = 0$$

$$\cos \theta = 0$$

$$\theta = 90^\circ, 270^\circ$$

$$2\cos \theta - 1 = 0$$

$$2\cos \theta = 1$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = 60^\circ, 300^\circ$$

$$\theta = 60^\circ, 90^\circ$$

$$19. \cos 2\theta + \cos \theta = 0$$

$$2\cos^2 \theta - 1 + \cos \theta = 0$$

$$2\cos^2 \theta + \cos \theta - 1 = 0$$

$$(2\cos \theta - 1)(\cos \theta + 1) = 0$$

$$2\cos \theta - 1 = 0$$

$$\cos \theta + 1 = 0$$

$$2\cos \theta = 1$$

$$\cos \theta = -1$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = 180^\circ$$

$$\theta = 60^\circ, 300^\circ$$

$$\theta = 60^\circ, 180^\circ$$

$$20. \cos 2\theta + 8\sin^2 \theta + 11\sin \theta + 3 = 0$$

$$1 - 2\sin^2 \theta + 8\sin^2 \theta + 11\sin \theta + 3 = 0$$

$$6\sin^2 \theta + 11\sin \theta + 4 = 0$$

$$(3\sin \theta + 4)(2\sin \theta + 1) = 0$$

$$3\sin \theta + 4 = 0$$

$$2\sin \theta + 1 = 0$$

$$3\sin \theta = -4$$

$$2\sin \theta = -1$$

$$\sin \theta = -\frac{4}{3}$$

$$\sin \theta = -\frac{1}{2}$$

$$\theta = \emptyset$$

$$\theta = 210^\circ, 330^\circ$$

No Solution

Name: _____

Unit 12: Trigonometry

Date: _____ Bell: _____

Homework 14: Trigonometric Equations

**** This is a 2-page document! ******Directions:** Solve each equation for the given interval. Give all solutions in **radians**.

1. $3\tan\theta = -3; 0 \leq \theta \leq 2\pi$

$$\tan\theta = -1$$

$$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

2. $-4\sqrt{3} = 8\cos\theta; 0 \leq \theta \leq \frac{3\pi}{2}$

$$-\frac{\sqrt{3}}{2} = \cos\theta$$

$$\frac{5\pi}{6}, \frac{7\pi}{6} = \theta$$

3. $-2\sin\theta = -\sqrt{2}; 0 \leq \theta \leq \pi$

$$\sin\theta = \frac{\sqrt{2}}{2}$$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}$$

4. $5\tan\theta = 2\sqrt{3} + 3\tan\theta; \pi \leq \theta \leq 2\pi$

$$2\tan\theta = 2\sqrt{3}$$

$$\tan\theta = \sqrt{3}$$

$$\theta = \frac{4\pi}{3}$$

5. $4\cos\theta = \sqrt{2} + 2\cos\theta; \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$

$$2\cos\theta = \sqrt{2}$$

$$\cos\theta = \frac{\sqrt{2}}{2}$$

$$\theta = \frac{\pi}{4}, \frac{7\pi}{4}$$

No Solution!

6. $6\sin\theta - 2 = 4\sin\theta; \frac{\pi}{2} \leq \theta \leq 2\pi$

$$2\sin\theta - 2 = 0$$

$$2\sin\theta = 2$$

$$\sin\theta = 1$$

$$\theta = \frac{\pi}{2}$$

Directions: Solve each equation for the interval $0^\circ \leq \theta \leq 360^\circ$. Give all solutions in **degrees**.

7. $2\cos^2\theta - \cos\theta = 0$

$$\cos\theta (2\cos\theta - 1) = 0$$

$$\cos\theta = 0$$

$$2\cos\theta - 1 = 0$$

$$\theta = 90^\circ, 270^\circ$$

$$2\cos\theta = 1$$

$$\cos\theta = \frac{1}{2}$$

$$\theta = 60^\circ, 300^\circ$$

$$\theta = 60^\circ, 90^\circ, 270^\circ, 300^\circ$$

8. $3\tan^2\theta + \sqrt{3}\tan\theta = 0$

$$\tan\theta (3\tan\theta + \sqrt{3}) = 0$$

$$\tan\theta = 0$$

$$\theta = 0^\circ, 180^\circ, 360^\circ$$

$$3\tan\theta + \sqrt{3} = 0$$

$$3\tan\theta = -\sqrt{3}$$

$$\tan\theta = -\frac{\sqrt{3}}{3}$$

$$\theta = 150^\circ, 330^\circ$$

$$\theta = 0^\circ, 150^\circ, 180^\circ, 330^\circ, 360^\circ$$

9. $3\sin^2\theta - 3 = 0$

$$3\sin^2\theta = 3$$

$$\sin^2\theta = 1$$

$$\sin\theta = \pm 1$$

$$\sin\theta = 1$$

$$\theta = 90^\circ$$

$$\sin\theta = -1$$

$$\theta = 270^\circ$$

$$\theta = 90^\circ, 270^\circ$$

10. $\cos^2\theta - 2\cos\theta = 3$

$$\cos^2\theta + 2\cos\theta - 3 = 0$$

$$(\cos\theta + 3)(\cos\theta - 1) = 0$$

$$\cos\theta + 3 = 0$$

$$\cos\theta = -3$$

$$\theta = \emptyset$$

$$\cos\theta - 1 = 0$$

$$\cos\theta = 1$$

$$\theta = 0^\circ, 360^\circ$$

$$\theta = 0^\circ, 360^\circ$$

$$11. \sin^2 \theta + 3 \sin \theta + 7 = 5$$

$$\sin^2 \theta + 3 \sin \theta + 2 = 0$$

$$(\sin \theta + 2)(\sin \theta + 1) = 0$$

$$\sin \theta + 2 = 0$$

$$\sin \theta + 1 = 0$$

$$\sin \theta = -2$$

$$\sin \theta = -1$$

$$\theta = \theta$$

$$\theta = 270^\circ$$

$$\theta = 270^\circ$$

$$12. 2 \cos^2 \theta - \cos \theta - 1 = 0$$

$$(2 \cos \theta + 1)(\cos \theta - 1) = 0$$

$$2 \cos \theta + 1 = 0$$

$$\cos \theta - 1 = 0$$

$$2 \cos \theta = -1$$

$$\cos \theta = 1$$

$$\cos \theta = -\frac{1}{2}$$

$$\theta = 0^\circ, 360^\circ$$

$$\theta = 120^\circ, 240^\circ$$

$$\theta = 0^\circ, 120^\circ, 240^\circ, 360^\circ$$

$$13. 2 \tan^2 \theta - \tan \theta = 3$$

$$2 \tan^2 \theta - \tan \theta - 3 = 0$$

$$(2 \tan \theta - 3)(\tan \theta + 1) = 0$$

$$2 \tan \theta - 3 = 0$$

$$\tan \theta + 1 = 0$$

$$2 \tan \theta = 3$$

$$\tan \theta = -1$$

$$\tan \theta = \frac{3}{2}$$

$$\theta = 135^\circ, 315^\circ$$

$$\theta = 56.3^\circ, 236.3^\circ$$

$$\theta = 56.3^\circ, 135^\circ, 236.3^\circ, 315^\circ$$

$$14. 4 \cos^2 \theta + 3 \cos \theta = 1$$

$$4 \cos^2 \theta + 3 \cos \theta - 1 = 0$$

$$(4 \cos \theta - 1)(\cos \theta + 1) = 0$$

$$4 \cos \theta - 1 = 0$$

$$\cos \theta + 1 = 0$$

$$4 \cos \theta = 1$$

$$\cos \theta = -1$$

$$\cos \theta = \frac{1}{4}$$

$$\theta = 180^\circ$$

$$\theta = 75.5^\circ, 284.5^\circ$$

$$\theta = 75.5^\circ, 180^\circ, 284.5^\circ$$

$$15. \sin^2 \theta = \cos \theta + 1$$

$$1 - \cos^2 \theta = \cos \theta + 1$$

$$0 = \cos^2 \theta + \cos \theta$$

$$0 = \cos \theta (\cos \theta + 1)$$

$$\cos \theta = 0$$

$$\cos \theta + 1 = 0$$

$$\theta = 90^\circ, 270^\circ$$

$$\cos \theta = -1$$

$$\theta = 180^\circ$$

$$\theta = 90^\circ, 180^\circ, 270^\circ$$

$$16. \sin 2\theta + \sin \theta = 0$$

$$2 \sin \theta \cos \theta + \sin \theta = 0$$

$$\sin \theta (2 \cos \theta + 1) = 0$$

$$\sin \theta = 0$$

$$2 \cos \theta + 1 = 0$$

$$\theta = 0^\circ, 180^\circ, 360^\circ$$

$$2 \cos \theta = -1$$

$$\cos \theta = -\frac{1}{2}$$

$$\theta = 120^\circ, 240^\circ$$

$$\theta = 0^\circ, 120^\circ, 180^\circ, 240^\circ, 360^\circ$$

$$17. \sec^2 \theta - \tan \theta \cot \theta = 1$$

$$\sec^2 \theta - \frac{\sin \theta}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta} = 1$$

$$\sec^2 \theta - 1 = 1$$

$$\tan^2 \theta = 1$$

$$\tan^2 \theta - 1 = 0$$

$$(\tan \theta + 1)(\tan \theta - 1) = 0$$

$$\tan \theta + 1 = 0$$

$$\tan \theta - 1 = 0$$

$$\tan \theta = -1$$

$$\tan \theta = 1$$

$$\theta = 135^\circ, 315^\circ$$

$$\theta = 45^\circ, 225^\circ$$

$$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$

$$18. 3 \cos 2\theta - 5 \cos \theta = 1$$

$$3(2 \cos^2 \theta - 1) - 5 \cos \theta = 1$$

$$6 \cos^2 \theta - 5 \cos \theta - 3 = 1$$

$$6 \cos^2 \theta - 5 \cos \theta - 4 = 0$$

$$(3 \cos \theta - 4)(2 \cos \theta + 1) = 0$$

$$3 \cos \theta - 4 = 0$$

$$2 \cos \theta + 1 = 0$$

$$\cos \theta = \frac{4}{3}$$

$$\cos \theta = -\frac{1}{2}$$

$$\theta = \theta$$

$$\theta = 120^\circ, 240^\circ$$

$$\theta = 120^\circ, 240^\circ$$

Unit 12 Test Study Guide (Trigonometry - Part 2)

Name: _____

Date: _____ Block: _____

Topic 1: Graphing Trigonometric Functions

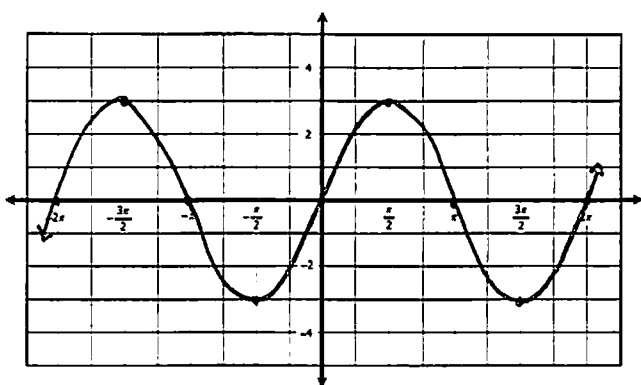
Given $y = a \cdot \sin b\theta$, $y = a \cdot \cos b\theta$, $y = a \cdot \tan b\theta$, $y = a \cdot \csc b\theta$, $y = a \cdot \sec b\theta$, or $y = a \cdot \cot b\theta$:

- The **amplitude** is $|a|$.

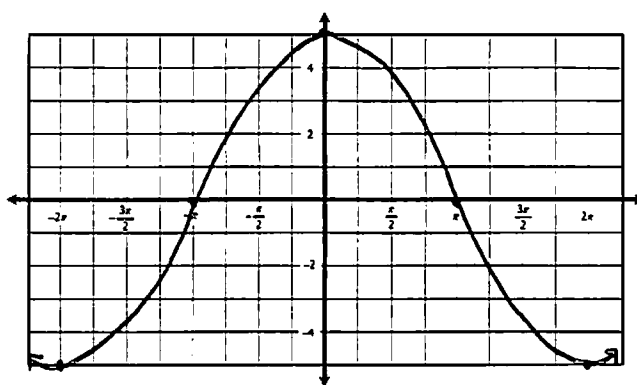
- For $\sin$, $\cos$, $\csc$, and $\sec$, the **period** is $\frac{2\pi}{b}$.
- For $\tan$ and $\cot$, the **period** is $\frac{\pi}{b}$.

Directions: Identify the amplitude, if possible, and period for each function. Then graph.

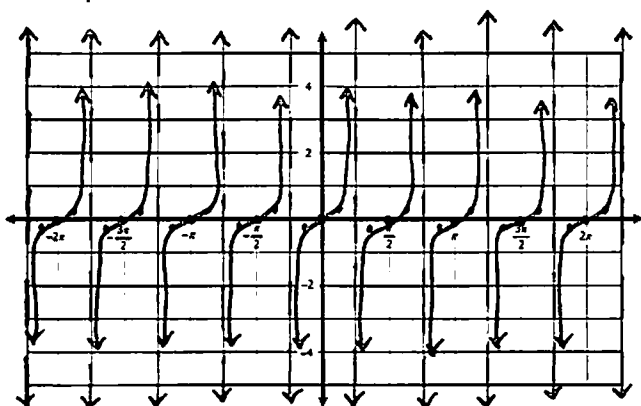
1. $y = 3 \cdot \sin \theta$ Amp: 3 Per: 2π



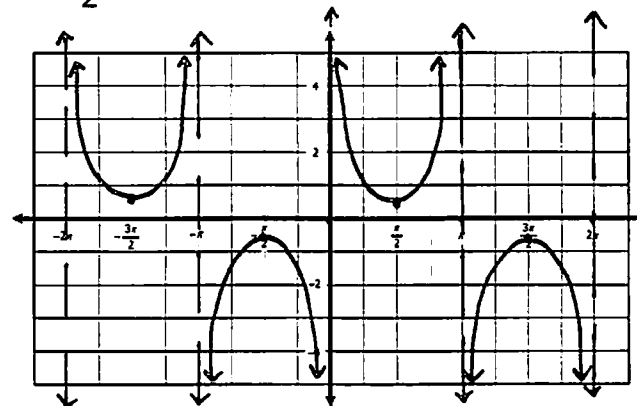
2. $y = 5 \cdot \cos \frac{1}{2} \theta$ Amp: 5 Per: 4π



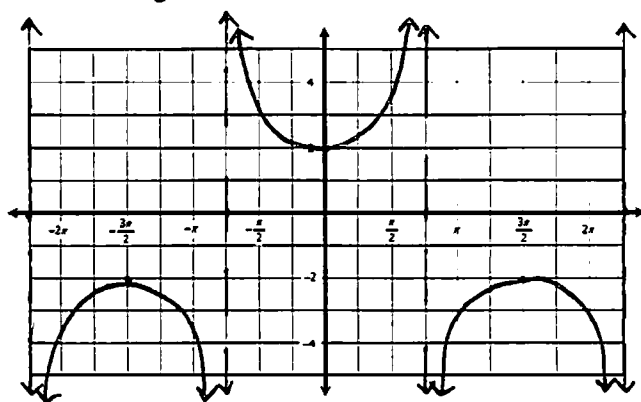
3. $y = \frac{1}{4} \cdot \tan 2\theta$ Amp: undef. Per: $\pi/2$



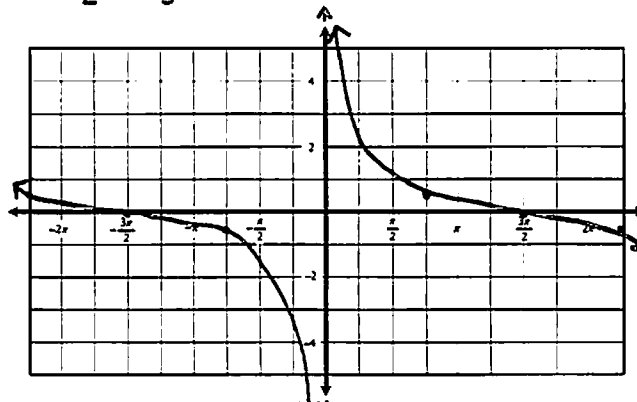
4. $y = \frac{1}{2} \cdot \csc \theta$ Amp: undef. Per: 2π



5. $y = 2 \cdot \sec \frac{2}{3} \theta$ Amp: undef. Per: 3π



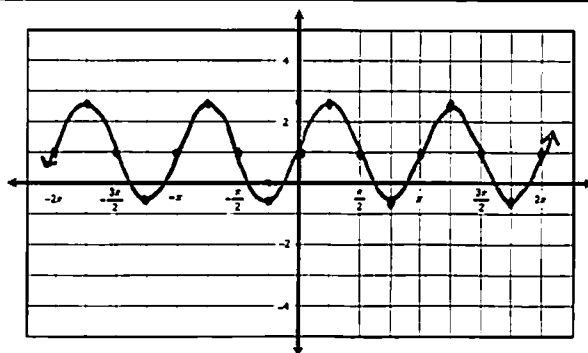
6. $y = \frac{1}{2} \cdot \cot \frac{1}{3} \theta$ Amp: undef. Per: 3π



Topic 2: Translating Trigonometric Graphs

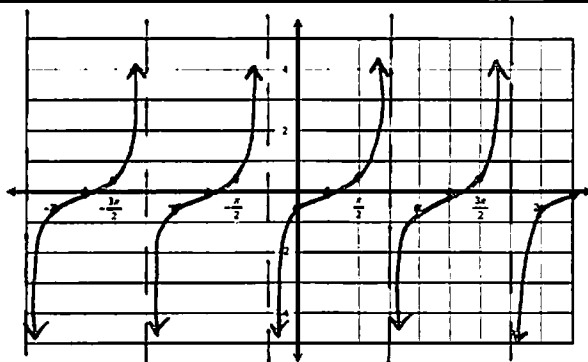
Directions: Identify the amplitude, if possible, period, translations, and midline. Then graph.

7. $y = \frac{3}{2} \cdot \sin 2\theta + 1$



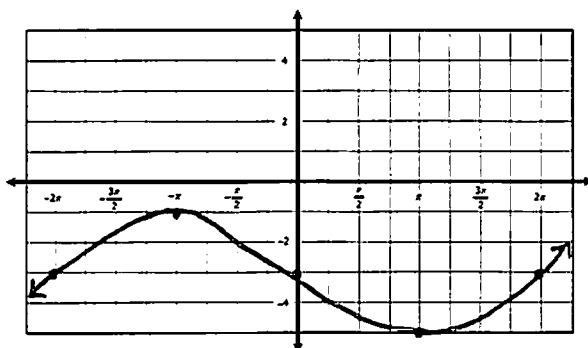
Amplitude: $\frac{3}{2}$
 Period: π
 Phase Shift: none
 Vertical Shift: up 1
 Midline: $y = 1$

8. $y = \frac{1}{2} \cdot \tan\left(\theta - \frac{\pi}{4}\right)$



Amplitude: undefined
 Period: π
 Phase Shift: $\pi/4$, right
 Vertical Shift: none
 Midline: $y = 0$

9. $y = 2 \cdot \cos \frac{1}{2}(\theta + \pi) - 3$



Amplitude: 2
 Period: 4π
 Phase Shift: π , left
 Vertical Shift: down 3
 Midline: $y = -3$

Topic 3: Fundamental Trigonometric Identities

Directions: Find the exact value of each function if $\frac{\pi}{2} < \theta < \pi$. (Quad II!)

10. Find $\sec \theta$ if $\sin \theta = \frac{1}{3}$.

$$\left(\frac{1}{3}\right)^2 + \cos^2 \theta = 1$$

$$\frac{1}{9} + \cos^2 \theta = 1$$

$$\cos^2 \theta = 8/9$$

$$\cos \theta = -\frac{2\sqrt{2}}{3}$$

$$\sec \theta = -\frac{3}{2\sqrt{2}} = \boxed{-\frac{3\sqrt{2}}{4}}$$

11. Find $\cot \theta$ if $\csc \theta = \frac{11}{5}$.

$$\cot^2 \theta + 1 = \left(\frac{11}{5}\right)^2$$

$$\cot^2 \theta + 1 = 121/25$$

$$\cot^2 \theta = 96/25$$

$$\cot \theta = -\frac{4\sqrt{6}}{5}$$

12. Find $\cos \theta$ if $\tan \theta = -\frac{4}{3}$.

$$\left(-\frac{4}{3}\right)^2 + 1 = \sec^2 \theta$$

$$16/9 + 1 = \sec^2 \theta$$

$$25/9 = \sec^2 \theta$$

$$\sec \theta = 5/3$$

$$\cos \theta = -3/5$$

13. Find $\tan \theta$ if $\sin \theta = \frac{3}{7}$. $\csc \theta = 7/3$

$$\cot^2 \theta + 1 = \left(\frac{7}{3}\right)^2$$

$$\cot^2 \theta + 1 = 49/9$$

$$\cot^2 \theta = 40/9$$

$$\cot \theta = -\frac{2\sqrt{10}}{3}$$

$$\tan \theta = -\frac{3}{2\sqrt{10}} = \boxed{-\frac{3\sqrt{10}}{20}}$$

Topic 4: Sum and Difference of Angles Identities

Directions: Find the exact value of each function.

14. $\tan 345^\circ \rightarrow \tan(300 + 45)$

$$\frac{\tan(300) + \tan(45)}{1 - \tan(300) \cdot \tan(45)}$$

$$= \frac{-\sqrt{3} + 1}{1 - (-\sqrt{3})(1)} = \frac{-\sqrt{3} + 1}{1 + \sqrt{3}} \cdot \frac{(1 - \sqrt{3})}{(1 - \sqrt{3})} = \frac{4 - 2\sqrt{3}}{1 - 3}$$

$$= \boxed{-2 + \sqrt{3}}$$

15. $\sin -195^\circ \rightarrow \sin(45 - 240)$

$$\sin(45) \cos(240) - \cos(45) \sin(240)$$

$$= \frac{\sqrt{2}}{2} \left(-\frac{1}{2}\right) - \frac{\sqrt{2}}{2} \left(-\frac{\sqrt{3}}{2}\right)$$

$$= -\frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4} = \boxed{\frac{-\sqrt{2} + \sqrt{6}}{4}}$$

16. $\cos\left(-\frac{\pi}{12}\right) \rightarrow \cos\left(\frac{\pi}{6} - \frac{\pi}{4}\right)$

$$\cos\left(\frac{\pi}{6}\right) \cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{6}\right) \sin\left(\frac{\pi}{4}\right)$$

$$= \frac{\sqrt{3}}{2} \left(\frac{\sqrt{2}}{2}\right) + \frac{1}{2} \left(\frac{\sqrt{2}}{2}\right)$$

$$= \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} = \boxed{\frac{\sqrt{6} + \sqrt{2}}{4}}$$

17. $\tan \frac{19\pi}{12} \rightarrow \tan\left(\frac{4\pi}{3} + \frac{\pi}{4}\right)$

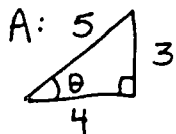
$$\frac{\tan\left(\frac{4\pi}{3}\right) + \tan\left(\frac{\pi}{4}\right)}{1 - \tan\left(\frac{4\pi}{3}\right) \tan\left(\frac{\pi}{4}\right)}$$

$$= \frac{\sqrt{3} + 1}{1 - (\sqrt{3})(1)} = \frac{\sqrt{3} + 1}{1 - \sqrt{3}} \cdot \frac{(1 + \sqrt{3})}{(1 + \sqrt{3})} = \frac{4 + 2\sqrt{3}}{1 - 3}$$

$$= \boxed{-2 - \sqrt{3}}$$

Directions: Find the exact value of each function if $\frac{3\pi}{2} < \theta < 2\pi$. (Quad IV!)

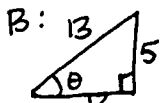
18. Find $\sin(A - B)$ if $\sin A = -\frac{3}{5}$ and $\cos B = \frac{12}{13}$.



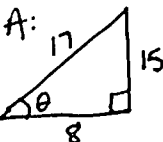
$$\sin A \cos B - \cos A \sin B$$

$$= \left(-\frac{3}{5}\right) \left(\frac{12}{13}\right) - \left(\frac{4}{5}\right) \left(-\frac{5}{13}\right)$$

$$= -\frac{36}{65} + \frac{20}{65} = \boxed{-\frac{16}{65}}$$



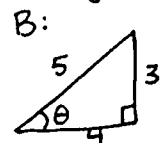
19. Find $\cos(A + B)$ if $\cos A = \frac{8}{17}$ and $\tan B = -\frac{3}{4}$.



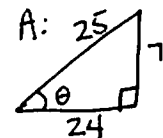
$$\cos A \cos B - \sin A \sin B$$

$$= \left(\frac{8}{17}\right) \left(\frac{4}{5}\right) - \left(-\frac{15}{17}\right) \left(-\frac{3}{5}\right)$$

$$= \frac{32}{85} - \frac{45}{85} = \boxed{-\frac{13}{85}}$$

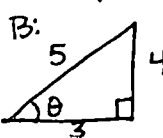


20. Find $\tan(A - B)$ if $\sin A = -\frac{7}{25}$ and $\sin B = -\frac{4}{5}$.



$$\frac{\left(-\frac{7}{24}\right) - \left(-\frac{4}{3}\right)}{1 + \left(-\frac{7}{24}\right) \left(-\frac{4}{3}\right)}$$

$$= \frac{25/24}{25/18} = \frac{25}{24} \cdot \frac{18}{25} = \boxed{\frac{3}{4}}$$



Topic 5: Double-Angle and Half-Angle Identities

Directions: Find the exact value of each function if $\pi < \theta < \frac{3\pi}{2}$. (Quad III!)

21. Find $\tan 2\theta$ if $\sin \theta = -\frac{\sqrt{2}}{4}$, $\csc \theta = -\frac{4}{\sqrt{2}}$

$$\cot^2 \theta + 1 = \left(-\frac{4}{\sqrt{2}}\right)^2$$

$$\cot^2 \theta + 1 = 8$$

$$\cot^2 \theta = 7$$

$$\cot \theta = \sqrt{7}$$

$$\tan \theta = \frac{1}{\sqrt{7}}$$

$$\tan \theta = \frac{\sqrt{7}}{7}$$

$$\frac{2\left(\frac{\sqrt{7}}{7}\right)}{1 - \left(\frac{\sqrt{7}}{7}\right)^2}$$

$$= \frac{\frac{2\sqrt{7}}{7}}{\frac{6}{7}} = \frac{2\sqrt{7}}{6} = \frac{\sqrt{7}}{3}$$

22. Find $\cos 2\theta$ if $\tan \theta = \frac{3}{5}$.

$$\left(\frac{3}{5}\right)^2 + 1 = \sec^2 \theta$$

$$\frac{9}{25} + 1 = \sec^2 \theta$$

$$\frac{34}{25} = \sec^2 \theta$$

$$-\frac{\sqrt{34}}{5} = \sec \theta$$

$$-\frac{5}{\sqrt{34}} = \cos \theta$$

$$2\left(-\frac{5}{\sqrt{34}}\right)^2 - 1$$

$$= \frac{25}{17} - 1$$

$$= \frac{8}{17}$$

23. Find $\sin 2\theta$ if $\cos \theta = -\frac{24}{25}$, $\sin \theta = -\frac{7}{25}$

$$2\left(-\frac{7}{25}\right)\left(-\frac{24}{25}\right)$$

$$= \frac{336}{625}$$

24. Find $\cos \frac{\theta}{2}$ if $\sin \theta = -\frac{7}{25}$, $\cos \theta = \frac{24}{25}$

$$-\sqrt{\frac{1 - \frac{24}{25}}{2}} = \sqrt{\frac{\frac{1}{25}}{2}} = \sqrt{\frac{1}{50}}$$

$$= -\frac{1}{5\sqrt{2}}$$

$$= -\frac{\sqrt{2}}{10}$$

25. Find $\tan \frac{\theta}{2}$ if $\sin \theta = -\frac{5}{13}$, $\cos \theta = -\frac{12}{13}$

$$-\sqrt{\frac{1 - \left(-\frac{12}{13}\right)}{1 + \left(-\frac{12}{13}\right)}} = -\sqrt{\frac{\frac{25}{13}}{\frac{1}{13}}} = -\sqrt{25}$$

$$= -5$$

26. Find $\sin \frac{\theta}{2}$ if $\tan \theta = \frac{4}{3}$, $\cos \theta = -\frac{3}{5}$

$$\sqrt{\frac{1 - \left(-\frac{3}{5}\right)}{2}} = \sqrt{\frac{\frac{8}{5}}{2}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$$

$$= \frac{2\sqrt{5}}{5}$$

Topic 6: Simplifying & Proving Trigonometric Identities

Directions: Simplify each expression.

27. $\sec \theta - \sin \theta \cdot \tan \theta$

$$\frac{1}{\cos \theta} - \sin \theta \cdot \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1 - \sin^2 \theta}{\cos \theta} = \frac{\cos^2 \theta}{\cos \theta} = \cos \theta$$

28. $\frac{\csc^2 \theta - 1}{\csc \theta}$

$$\frac{\cot^2 \theta}{\csc \theta} = \frac{\frac{\cos^2 \theta}{\sin^2 \theta}}{\frac{1}{\sin \theta}} = \frac{\cos^2 \theta}{\sin \theta} \cdot \frac{\sin \theta}{1}$$

$$= \frac{\cos^2 \theta}{1} = \cos^2 \theta$$

$$29. \cos \theta \cdot \cot \theta \cdot \tan \theta \cdot \sec \theta$$

$$\cos \theta \cdot \frac{\cos \theta}{\sin \theta} \cdot \frac{\sin \theta}{\cos \theta} \cdot \frac{1}{\cos \theta}$$

$$\boxed{= 1}$$

$$30. (1 + \cos \theta)(\csc \theta - \cot \theta)$$

$$\csc \theta - \cot \theta + \cos \theta \csc \theta - \cos \theta \cot \theta$$

$$\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta} + \cos \theta \cdot \frac{1}{\sin \theta} - \frac{\cos \theta \cdot \cos \theta}{\sin \theta}$$

$$\frac{1 - \cos \theta + \cos \theta - \cos^2 \theta}{\sin \theta} = \frac{1 - \cos^2 \theta}{\sin \theta} = \frac{\sin^2 \theta}{\sin \theta}$$

$$\boxed{= \sin \theta}$$

Directions: Prove each identity.

$$31. \frac{\csc \theta}{\sin \theta} - \frac{\cot \theta}{\tan \theta} = 1$$

$$\frac{\frac{1}{\sin \theta}}{\sin \theta} - \frac{\frac{\cos \theta}{\sin \theta}}{\frac{\cos \theta}{\sin \theta}} = 1$$

$$\frac{1}{\sin^2 \theta} - \frac{\cos^2 \theta}{\sin^2 \theta} = 1$$

$$\frac{\sin^2 \theta}{\sin^2 \theta} = 1$$

$$\boxed{1 = 1}$$

$$32. \tan \theta + \cot \theta = \sec \theta \cdot \csc \theta$$

$$\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \sec \theta \csc \theta$$

$$\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = \sec \theta \csc \theta$$

$$\frac{1}{\sin \theta \cos \theta} = \sec \theta \csc \theta$$

$$\boxed{\sec \theta \csc \theta = \sec \theta \csc \theta}$$

$$33. \cot \theta = \frac{\sin 2\theta}{1 - \cos 2\theta}$$

$$\cot \theta = \frac{2 \sin \theta \cos \theta}{1 - (1 - 2 \sin^2 \theta)}$$

$$\cot \theta = \frac{2 \sin \theta \cos \theta}{2 \sin^2 \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\boxed{\cot \theta = \cot \theta}$$

$$34. \tan \theta + \sin \theta = \frac{1 + \cos \theta}{\cot \theta}$$

$$\tan \theta + \sin \theta = \frac{1 + \cos \theta}{\frac{\cos \theta}{\sin \theta}}$$

$$\tan \theta + \sin \theta = (1 + \cos \theta) \cdot \frac{\sin \theta}{\cos \theta}$$

$$\tan \theta + \sin \theta = \frac{\sin \theta}{\cos \theta} + \frac{\sin \theta \cos \theta}{\cos \theta}$$

$$\boxed{\tan \theta + \sin \theta = \tan \theta + \sin \theta}$$

Topic 7: Trigonometric Equations

Directions: Solve each equation for the interval $0^\circ \leq \theta \leq 360^\circ$. Give all answers in **degrees**.

$$35. 9 \sin \theta + 8 = 3 \sin \theta + 5$$

$$6 \sin \theta + 8 = 5$$

$$6 \sin \theta = -3$$

$$\sin \theta = -1/2$$

$$\boxed{\theta = 210^\circ, 330^\circ}$$

$$36. 4 \cos^2 \theta - 3 = 0$$

$$4 \cos^2 \theta = 3$$

$$\cos^2 \theta = 3/4$$

$$\cos \theta = \pm \frac{\sqrt{3}}{2}$$

$$\boxed{\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ}$$

$$37. 2\cos\theta\sin\theta + \sqrt{2}\sin\theta = 0$$

$$\sin\theta (2\cos\theta + \sqrt{2}) = 0$$

$$\sin\theta = 0 \quad 2\cos\theta + \sqrt{2} = 0$$

$$\theta = 0^\circ, 180^\circ, 360^\circ \quad 2\cos\theta = -\sqrt{2}$$

$$\cos\theta = -\frac{\sqrt{2}}{2}$$

$$\theta = 135^\circ, 225^\circ$$

$$\theta = 0, 135^\circ, 180^\circ, 225^\circ, 360^\circ$$

$$38. 2\tan^2\theta + \sqrt{3}\tan\theta = \tan^2\theta$$

$$\tan^2\theta + \sqrt{3}\tan\theta = 0$$

$$\tan\theta (\tan\theta + \sqrt{3}) = 0$$

$$\tan\theta = 0 \quad \tan\theta + \sqrt{3} = 0$$

$$\theta = 0^\circ, 180^\circ \quad \tan\theta = -\sqrt{3}$$

$$\theta = 120^\circ, 300^\circ$$

$$\theta = 0^\circ, 120^\circ, 180^\circ, 300^\circ$$

$$39. \cos^2\theta - 2\cos\theta + 1 = 0$$

$$(\cos\theta - 1)(\cos\theta - 1) = 0$$

$$\cos\theta - 1 = 0 \quad \cos\theta - 1 = 0$$

$$\cos\theta = 1 \quad \cos\theta = 1$$

$$\theta = 0^\circ, 360^\circ \quad \theta = 0^\circ, 360^\circ$$

$$\theta = 0, 360^\circ$$

$$40. 2\sin^2\theta - 9\sin\theta = 5$$

$$2\sin^2\theta - 9\sin\theta - 5 = 0$$

$$(2\sin\theta + 1)(\sin\theta - 5) = 0$$

$$2\sin\theta + 1 = 0$$

$$\sin\theta - 5 = 0$$

$$2\sin\theta = -1$$

$$\sin\theta = 5$$

$$\sin\theta = -1/2$$

$$\theta = \emptyset$$

$$\theta = 210^\circ, 330^\circ$$

Directions: Solve each equation for the interval $0 \leq \theta \leq 2\pi$. Give all answers in **radians**.

$$41. \sin\theta \cdot 2\cot\theta = \sqrt{3}$$

$$\sin\theta \cdot \frac{2\cos\theta}{\sin\theta} = \sqrt{3}$$

$$2\cos\theta = \sqrt{3}$$

$$\cos\theta = \frac{\sqrt{3}}{2}$$

$$\theta = \frac{\pi}{6}, \frac{11\pi}{6}$$

$$42. 4\sin^4\theta + 4\sin^2\theta \cdot \cos^2\theta = 1$$

$$4\sin^2\theta (\sin^2\theta + \cos^2\theta) = 1$$

$$4\sin^2\theta (1) = 1$$

$$4\sin^2\theta = 1$$

$$\sin^2\theta = 1/4$$

$$\sin\theta = \pm 1/2$$

$$\theta = \pi/6, 5\pi/6, 7\pi/6, 11\pi/6$$

$$43. 3(\sec^2\theta - 1) = 1$$

$$3(\tan^2\theta) = 1$$

$$\tan^2\theta = 1/3$$

$$\tan\theta = \pm \frac{\sqrt{3}}{3}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$44. \sin^2\theta + \cos 2\theta - 3\cos\theta = 4$$

$$\sin^2\theta + (2\cos^2\theta - 1) - 3\cos\theta = 4$$

$$\sin^2\theta + \cos^2\theta + \cos^2\theta - 3\cos\theta = 4$$

$$1 + \cos^2\theta - 3\cos\theta - 1 = 4$$

$$\cos^2\theta - 3\cos\theta - 4 = 0$$

$$(\cos\theta - 4)(\cos\theta + 1) = 0$$

$$\cos\theta - 4 = 0$$

$$\cos\theta + 1 = 0$$

$$\cos\theta = 4$$

$$\cos\theta = -1$$

$$\theta = \emptyset$$

$$\theta = \pi$$

Name: _____

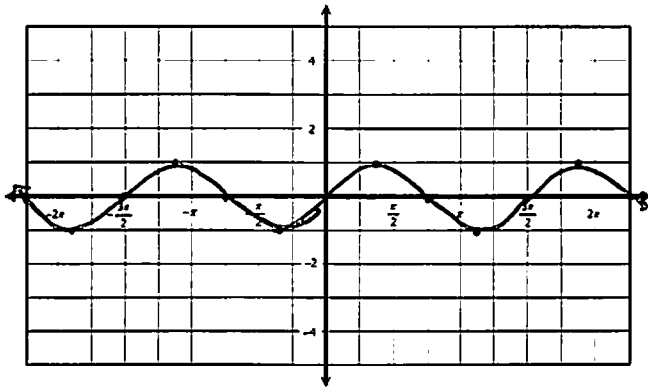
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Unit 12 Test (Part II)

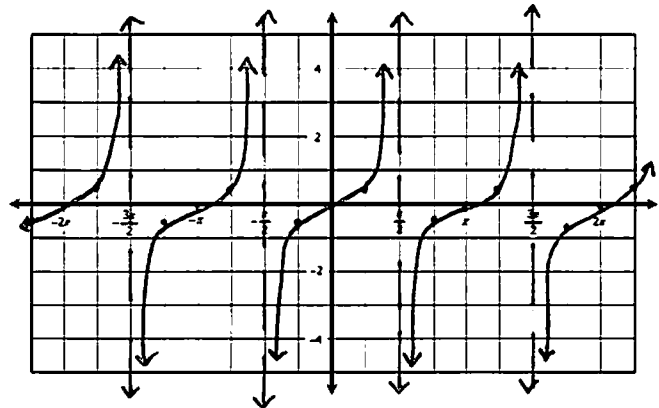
Trigonometry

Identify the amplitude (if possible) and period of each function, then graph.

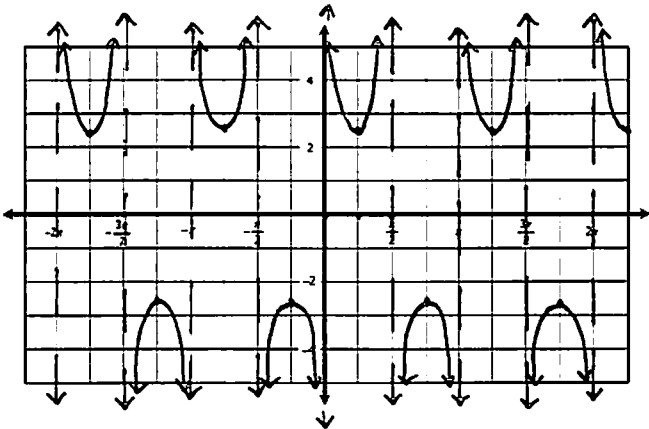
1. $y = \sin \frac{4}{3} \theta$ Amp: 1 Per: $\frac{3\pi}{2}$



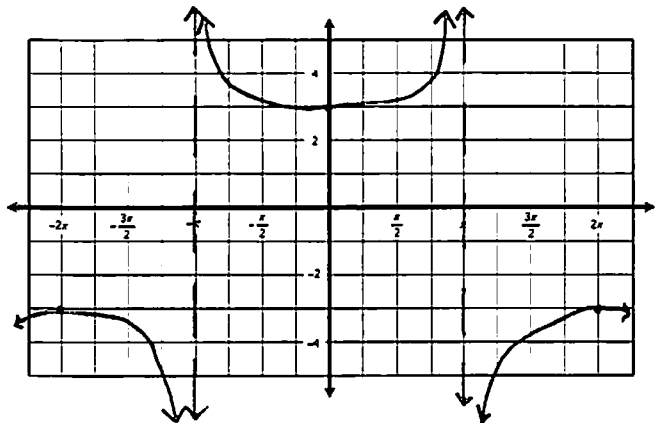
2. $y = \frac{1}{2} \cdot \tan \theta$ Amp: undef Per: π



3. $y = \frac{5}{2} \cdot \csc 4\theta$ Amp: undef Per: π

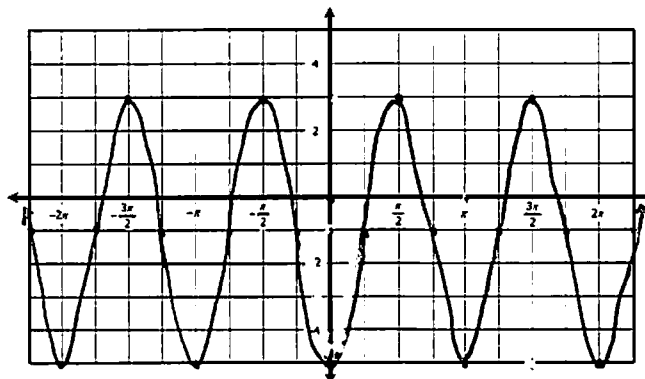


4. $y = 3 \cdot \sec \frac{1}{2} \theta$ Amp: undef Per: 4π



Identify the amplitude (if possible), period, translations, and midline. Then graph.

5. $y = 4 \cdot \cos 2\left(\theta + \frac{\pi}{2}\right) - 1$



Amplitude: 4

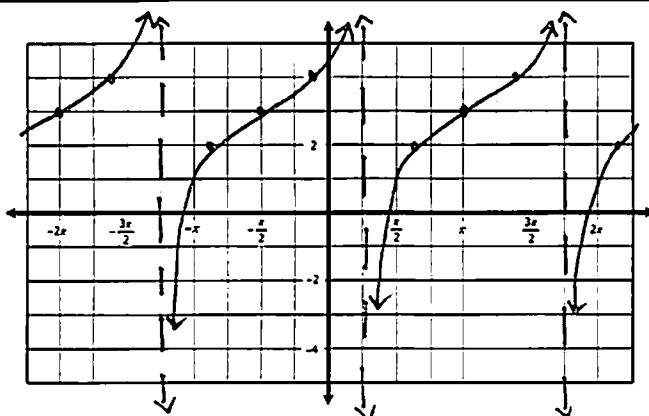
Period: π

Phase Shift: $\pi/2$, left

Vertical Shift: down 1

Midline: $y = -1$

6. $y = \tan \frac{2}{3}(\theta - \pi) + 3$



Amplitude: undef

Period: $3\pi/2$

Phase Shift: π , right

Vertical Shift: up 3

Midline: $y = 3$

7. Find $\sin \theta$ if $\cos \theta = \frac{5}{6}$ and $\frac{3\pi}{2} \leq \theta \leq 2\pi$. (quad 4)

$$\sin^2 \theta + (5/6)^2 = 1$$

$$\sin^2 \theta + 25/36 = 1$$

$$\sin^2 \theta = 11/36$$

$$\sin \theta = -\sqrt{11}/6$$

A. $\frac{6\sqrt{11}}{11}$

C. $\frac{\sqrt{11}}{6}$

B. $-\frac{6\sqrt{11}}{11}$

D. $-\frac{\sqrt{11}}{6}$

D

8. Find $\sin \theta$ if $\cot \theta = -\frac{\sqrt{15}}{5}$ and $\frac{\pi}{2} \leq \theta \leq \pi$. (quad 2)

$$\left(-\frac{\sqrt{15}}{5}\right)^2 + 1 = \csc^2 \theta$$

$$\frac{15}{25} + 1 = \csc^2 \theta$$

$$\frac{8}{5} = \csc^2 \theta \quad \frac{2\sqrt{2}}{\sqrt{5}} = \csc \theta \rightarrow \sin \theta = \frac{\sqrt{5}}{2\sqrt{2}} = \frac{\sqrt{10}}{4}$$

A. $\frac{2\sqrt{2}}{5}$

C. $\frac{\sqrt{10}}{4}$

B. $-\frac{2\sqrt{2}}{5}$

D. $-\frac{\sqrt{10}}{4}$

C

9. Find the exact value of $\sin 255^\circ$.

$$\sin(210 + 45)$$

$$\sin 210 \cos 45 + \cos 210 \sin 45$$

$$(-1/2)(\sqrt{2}/2) + (-\sqrt{3}/2)(\sqrt{2}/2)$$

$$-\frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4}$$

A. $\frac{\sqrt{6} + \sqrt{2}}{4}$

C. $\frac{\sqrt{2} - \sqrt{6}}{4}$

B. $\frac{\sqrt{6} - \sqrt{2}}{4}$

D. $-\frac{\sqrt{6} - \sqrt{2}}{4}$

D

10. Find the exact value of $\tan\left(\frac{5\pi}{12}\right)$.

$$\tan(\pi/4 + \pi/6)$$

$$\frac{\tan \pi/4 + \tan \pi/6}{1 - \tan \pi/4 \cdot \tan \pi/6} = \frac{1 + \frac{\sqrt{3}}{3}}{1 - 1(\frac{\sqrt{3}}{3})} = \frac{\frac{3 + \sqrt{3}}{3}}{\frac{3 - \sqrt{3}}{3}}$$

A. $2 + \sqrt{3}$

C. $2 - \sqrt{3}$

B. $\sqrt{3} - 2$

D. $-2 - \sqrt{3}$

$$= \frac{(3 + \sqrt{3})(3 + \sqrt{3})}{(3 - \sqrt{3})(3 + \sqrt{3})} = \frac{9 + 6\sqrt{3} + 3}{9 - 3} = \frac{12 + 6\sqrt{3}}{6}$$

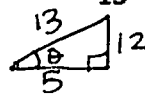
A

11. Find $(\cos A - B)$ if A and B are positive acute angles with $\sin A = \frac{12}{13}$ and $\cos B = \frac{3}{5}$.

$$\cos A \cos B + \sin A \sin B$$

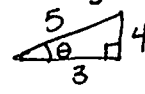
$$(5/13)(3/5) + (12/13)(4/5)$$

$$15/65 + 48/65$$



A. $\frac{56}{65}$

B. $\frac{63}{65}$



C. $-\frac{33}{65}$

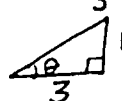
D. $-\frac{16}{65}$

B

12. Find $(\tan A + B)$ if A and B are in Quadrant II with $\tan A = -\frac{1}{3}$ and $\sin B = \frac{20}{29}$.

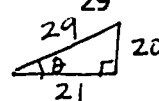
$$\frac{\tan A + \tan B}{1 - \tan A \cdot \tan B} = \frac{-1/3 + 20/21}{1 - (-1/3)(20/21)}$$

$$= \frac{-9/7}{43/43}$$



A. $-\frac{81}{83}$

B. $-\frac{81}{43}$



C. $\frac{39}{43}$

D. $\frac{39}{83}$

B

13. Find the exact value of $\sin 2\theta$ if $\tan \theta = -\frac{7}{24}$ and $\frac{\pi}{2} \leq \theta \leq \pi$.

$$2 \sin \theta \cos \theta$$

$$2(7/25)(-24/25)$$

A. $\frac{336}{625}$

B. $-\frac{336}{625}$

C. $\frac{14}{25}$

D. $-\frac{14}{25}$

B

14. Find the exact value of $\cos 2\theta$ if $\sin \theta = -\frac{7}{10}$ and $\pi \leq \theta \leq \frac{3\pi}{2}$.

$$1 - 2(-7/10)^2$$

$$1 - 2(49/100)$$

$$1 - 98/100$$

A. $-\frac{1}{50}$

B. $\frac{1}{50}$

C. $-\frac{99}{50}$

D. $\frac{99}{50}$

B

15. Find the exact value of $\tan 2\theta$ if $\cos \theta = \frac{3}{5}$ and $0 \leq \theta \leq \frac{\pi}{2}$.

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{2(4/3)}{1 - (4/3)^2} = \frac{8/3}{1 - 16/9}$$

$$= 8/3 \cdot -9/7$$

A. $\frac{7}{24}$

B. $\frac{24}{25}$

C. $-\frac{24}{7}$

D. $-\frac{25}{24}$

C

16. Find the exact value of $\cos \frac{\theta}{2}$ if $\sin \theta = -\frac{\sqrt{7}}{4}$ and $\pi \leq \theta \leq \frac{3\pi}{2}$. $\cos \theta = -3/4$

$$-\sqrt{\frac{1 - (-3/4)}{2}} = \sqrt{\frac{1/4}{2}} = -\sqrt{1/8}$$

$$= -\frac{1}{2\sqrt{2}} = -\frac{\sqrt{2}}{4}$$

A. $\frac{\sqrt{14}}{4}$

C. $\frac{\sqrt{2}}{4}$

B. $-\frac{\sqrt{14}}{4}$

D. $-\frac{\sqrt{2}}{4}$

D

17. Find the exact value of $\tan \frac{\theta}{2}$ if $\tan \theta = -\frac{12}{5}$ and $\frac{\pi}{2} \leq \theta \leq \pi$. $\cos \theta = -5/13$

$$\sqrt{\frac{1 - (-5/13)}{1 + (-5/13)}} = \sqrt{\frac{18/13}{8/13}} = \sqrt{\frac{9}{4}} = \frac{3}{2}$$

A. $\frac{3}{2}$

C. $-\frac{3}{2}$

B. 5

D. -5

A

Simplify each expression.

18. $\frac{\cot \theta \cdot \sin \theta}{\cot \theta \cdot \tan \theta - \sin^2 \theta}$

$$\frac{\frac{\cos \theta}{\sin \theta} \cdot \sin \theta}{\frac{\cos \theta}{\sin \theta} \cdot \frac{\sin \theta}{\cos \theta} - \sin^2 \theta} = \frac{\cos \theta}{1 - \sin^2 \theta} = \frac{\cos \theta}{\cos^2 \theta} = \sec \theta$$

- A. $\sin \theta$
- B. $\cos \theta$
- C. $\csc \theta$
- D. $\sec \theta$

D

19. $\cos \theta \cdot \csc \theta (\sec^2 \theta - 1)$

$$\cos \theta \cdot \frac{1}{\sin \theta} \cdot \tan^2 \theta = \cos \theta \cdot \frac{1}{\sin \theta} \cdot \frac{\sin^2 \theta}{\cos^2 \theta} = \frac{\sin \theta}{\cos \theta}$$

- A. $\tan \theta$
- B. $\cot \theta$
- C. $\cos \theta$
- D. $\sec \theta$

A

20. $\frac{\tan \theta}{\tan \theta + \cot \theta}$

$$\frac{\frac{\sin \theta}{\cos \theta}}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}} = \frac{\frac{\sin \theta}{\cos \theta} \cdot \frac{\cos \theta \sin \theta}{\cos \theta \sin \theta}}{\frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}} = \frac{\sin^2 \theta}{1} = \sin^2 \theta$$

- A. $\sec \theta$
- B. $\csc \theta$
- C. $\sin^2 \theta$
- D. $\cos^2 \theta$

C

21. $\frac{\sin 2\theta \cdot \sec \theta}{1 - \cos^2 \theta}$

$$\frac{2 \sin \theta \cos \theta \cdot \frac{1}{\cos \theta}}{\sin^2 \theta} = \frac{2 \sin \theta}{\sin^2 \theta} = \frac{2}{\sin \theta}$$

- A. $2 \sec \theta$
- B. $2 \csc \theta$
- C. $\sec \theta - 1$
- D. $\csc \theta - 1$

B

22. Which expressions are equivalent to $\cos \theta$? Check all that apply.

☐ $\frac{(\sec \theta + 1)(\sec \theta - 1)}{\sin^2 \theta \cdot \sec \theta}$	☐ $\frac{\csc \theta}{\cot^2 \theta + 1}$	☒ $(\cos 2\theta + \sin^2 \theta) \cdot \sec \theta$
☒ $\sin \theta \cdot \cot \theta (\sin^2 \theta + \cos^2 \theta)$	☒ $\sin\left(\theta + \frac{\pi}{2}\right)$	☐ $\sin \theta + \cot \theta \cdot \cos \theta$

Prove each identity.

23. $\tan \theta (\cos \theta + \cot \theta) = \sin \theta + 1$

$$\frac{\sin \theta}{\cos \theta} \cdot \cos \theta + \frac{\sin \theta}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta} = \sin \theta + 1$$

$$\sin \theta + 1 = \sin \theta + 1$$

24. $\frac{1 + \cos 2\theta}{\sin 2\theta} = \cot \theta$

$$\frac{1 + (2\cos^2 \theta - 1)}{2\sin \theta \cos \theta} = \cot \theta$$

$$\frac{2\cos^2 \theta}{2\sin \theta \cos \theta} = \cot \theta$$

$$\frac{\cos \theta}{\sin \theta} = \cot \theta$$

$$\cot \theta = \cot \theta$$

Solve each equation for the interval $0^\circ \leq \theta \leq 360^\circ$. Give all solutions in degrees.

25. $5\tan \theta - \sqrt{3} = 2\tan \theta$

$$3\tan \theta - \sqrt{3} = 0$$

$$3\tan \theta = \sqrt{3}$$

$$\tan \theta = \frac{\sqrt{3}}{3}$$

$$30^\circ, 210^\circ$$

26. $2\sin \theta \cos \theta + \sqrt{2} \sin \theta = 0$

$$\sin \theta (2\cos \theta + \sqrt{2}) = 0$$

$$\sin \theta = 0 \quad 2\cos \theta + \sqrt{2} = 0$$

$$\theta = 0, 180, 360 \quad \cos \theta = -\frac{\sqrt{2}}{2}$$

$$\theta = 135, 225$$

$$0^\circ, 135^\circ, 180^\circ, 225^\circ, 360^\circ$$

27. $\cos^2 \theta - 4 \cos \theta - 5 = 0$

$$(\cos \theta - 5)(\cos \theta + 1) = 0$$

$$\cos \theta = 5 \quad \cos \theta = -1$$

$$\theta = \theta$$

$$\theta = 180$$

$$180^\circ$$

28. $2 \sin^2 \theta + \sin \theta + 1 = 4 \sin \theta$

$$2 \sin^2 \theta - 3 \sin \theta + 1 = 0$$

$$(2 \sin \theta - 1)(\sin \theta - 1) = 0$$

$$\sin \theta = 1/2$$

$$\sin \theta = 1$$

$$\theta = 30, 150$$

$$\theta = 90$$

$$30^\circ, 90^\circ, 150^\circ$$

29. $\sin^2 \theta + \cos^2 \theta + 2 \cos \theta = 1 + \sqrt{3}$

$$1 + 2 \cos \theta = 1 + \sqrt{3}$$

$$2 \cos \theta = \sqrt{3}$$

$$\cos \theta = \frac{\sqrt{3}}{2}$$

$$30^\circ, 330^\circ$$

30. $6 \sin^2 \theta + \cos 2\theta - 2 = 0$

$$6 \sin^2 \theta + (1 - 2 \sin^2 \theta) - 2 = 0$$

$$4 \sin^2 \theta - 1 = 0$$

$$4 \sin^2 \theta = 1$$

$$\sin^2 \theta = 1/4$$

$$\sin \theta = \pm 1/2$$

$$30^\circ, 150^\circ, 210^\circ, 330^\circ$$

Directions: Solve each equation for the given interval. Give all solutions in radians.

31. $\tan^2 \theta - \tan \theta = 0; 0 \leq \theta \leq \pi$

$$\tan \theta (\tan \theta - 1) = 0$$

$$\tan \theta = 0$$

$$\tan \theta - 1 = 0$$

$$\theta = 0, \pi$$

$$\tan \theta = 1$$

$$\theta = \pi/4, \pi/4$$

$$0, \frac{\pi}{4}, \pi$$

32. $2 \sin \theta \cdot \cos \theta \cdot \cot = 1; \frac{\pi}{2} \leq \theta \leq 2\pi$

$$2 \sin \theta \cdot \cos \theta \cdot \frac{\cos \theta}{\sin \theta} = 1$$

$$2 \cos^2 \theta = 1$$

$$\cos^2 \theta = 1/2$$

$$\cos \theta = \frac{\sqrt{2}}{2}$$

$$\theta = \frac{7\pi}{4}, \frac{7\pi}{4}$$

$$\frac{7\pi}{4}$$

CREDITS

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